SSC JE
MECHANICAL ENGINEERING

Topper-Style Concept Notes

Paper-I (Part B) & Paper-II (Descriptive) • Concept + Formula + PYQ-Trend Revision

Engineering Mechanics • Strength of Materials • Theory of Machines • Machine Design • Thermodynamics • Heat Transfer • IC Engines • RAC • Power Plant • Fluid Mechanics • Hydraulic Machines • Automobile • Manufacturing • Materials • Metrology • Instrumentation • Industrial Engineering

Concept-based notes – no MCQ practice. "PYQ-Trend" boxes list concepts that SSC JE has repeatedly tested in past years. Always cross-check with official papers.

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INTRODUCTION

How to Use These Notes & Exam Blueprint

SSC JE Mechanical – Exam Structure

SSC JE (Junior Engineer) is conducted by the Staff Selection Commission for posts in CPWD, MES, BRO, CWC, Farakka Barrage, Naval/Defence establishments and others. Mechanical candidates take Paper-I (objective, computer-based) and Paper-II (now also objective in recent cycles; earlier conventional/descriptive). Always confirm the current notification, because SSC changes the pattern from cycle to cycle.

Paper	Part	Subject	Marks (typical)
Paper-I	A	General Intelligence & Reasoning	50
Paper-I	B	General Awareness	50
Paper-I	C	Mechanical Engineering	100
Paper-II	–	Mechanical Engineering (full syllabus, higher depth)	300

Official Syllabus Map (Mechanical)

- Theory of Machines & Machine Design: concept of simple machine, four-bar linkage, link motion, flywheels, governors, balancing, cams, gears & gear trains, belts & ropes, brakes, clutches, bearings, keys, couplings, joints (riveted/welded/bolted), shafts.
- Engineering Mechanics & Strength of Materials: equilibrium of forces, Newton's laws, friction, centroid & moment of inertia, work-energy, impulse-momentum, stress-strain, elastic constants, SFD & BMD, bending & shear stress, torsion, columns, thin cylinders, strain energy, principal stresses, springs.
- Thermal Engineering: properties of pure substances, 1st & 2nd laws, entropy, availability, air-standard cycles (Otto, Diesel, Dual, Brayton), IC engines (combustion, performance, cooling, lubrication), Rankine cycle, boilers, steam turbines, nozzles, condensers, refrigeration (VCR, VAR), psychrometry, air compressors, heat transfer basics.
- Fluid Mechanics & Machinery: properties of fluids, pressure measurement, hydrostatic forces, buoyancy, fluid kinematics, Bernoulli, flow measurement, viscous flow, pipe flow losses, dimensional analysis, impact of jets, hydraulic turbines, centrifugal & reciprocating pumps.
- Production Engineering: classification of steels, cast iron, heat treatment, casting, welding, metal forming, machining (lathe, milling, drilling, grinding), tool wear & life, non-traditional machining, metrology, limits-fits-tolerances, industrial engineering basics.

Weightage Pattern (from past papers – approximate)

Area	Paper-I share (of 100)	Priority
Thermal Engg (Thermo + IC + RAC + Steam + HT)	25–30	Very High
Fluid Mechanics & Machinery	15–20	Very High
SOM + Engg Mechanics	15–20	Very High
Theory of Machines + Machine Design	12–18	High
Production / Manufacturing + Materials	15–20	High

How the Notes Are Organised

- Each chapter = theory in compact points → formula box (yellow) → PYQ-Trend box (blue) → Topper Trick (green) → Common Mistake (red) → Descriptive Q (purple) for Paper-II style answers.
- PYQ-Trend boxes list the concepts SSC JE asks again and again (2007–2024 papers). They tell you what is examined. They are not reproduced questions.
- Read the chapter once. Make a one-page summary of each box. Revise only the boxes in the last 15 days.

90-Day Topper Plan

Days	Target
1–15	Engg Mechanics + SOM (build the base for TOM/MD)
16–25	Thermodynamics + IC Engines
26–35	Fluid Mechanics + Hydraulic Machines
36–45	TOM + Machine Design
46–55	RAC + Steam/Power Plant + Heat Transfer
56–68	Production + Materials + Metrology + IE
69–80	Formula sheet revision + PYQ concept boxes + mock tests
81–90	Weak areas, last-minute facts chapter, sleep well

✔ **TOPPER TRICK:** SSC JE repeats concepts heavily. A student who knows every PYQ-Trend box in this book perfectly usually scores more than one who has read five reference books halfway.

✖ **COMMON MISTAKE:** Don't ignore "facts" questions (inventor names, units, default values like the COP of the Carnot cycle or the compression ratio range of a diesel engine). They are 20–25% of the paper and the fastest marks.

CHAPTER 1

Engineering Mechanics

1.1 Basic Concepts

Engineering mechanics studies bodies at rest (statics) and in motion (dynamics). Dynamics has two parts: kinematics (motion without force) and kinetics (motion with force).

- Scalar: magnitude only (mass, time, work, energy, power, speed). Vector: magnitude + direction (force, displacement, velocity, acceleration, momentum, moment).
- Rigid body: the distance between any two particles stays constant. All statics in SSC JE assumes rigid bodies.
- Principle of transmissibility: a force may be moved anywhere along its line of action without changing its external effect on a rigid body. It applies only to rigid bodies and only to external effects.
- Particle: a body with mass but negligible dimensions, so rotation is ignored.
- Units: 1 N = 1 kg·m/s²; 1 kgf = 9.81 N; 1 dyne = 10^{-5} N; 1 J = 10^{7} erg; 1 hp (metric) = 735.5 W; 1 hp (British) = 746 W.

★ **PYQ-TREND:** Transmissibility applies to rigid bodies only. Definition of a newton or a dyne and their conversion. Scalar vs vector identification (work and energy are scalar; moment is vector; pressure is scalar).

1.2 Systems of Forces

System	Description	Example
Coplanar concurrent	All forces in one plane meeting at a point	Forces on a ring with strings
Coplanar parallel (like/unlike)	Lines of action parallel	Loads on a beam
Coplanar non-concurrent non-parallel	General planar	Ladder problem
Collinear	Same line of action	Tug of war
Non-coplanar concurrent	3D forces at a point	Tripod legs

Resultant of Two Forces (Parallelogram Law)

$R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$

$\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$ (α = angle of R with P)

$\theta = 0° \rightarrow R = P+Q$ (maximum); $\theta = 180° \rightarrow R = |P-Q|$ (minimum); $\theta = 90° \rightarrow R = \sqrt{(P^2+Q^2)}$

Two equal forces P at angle θ: $R = 2P\cos(\theta/2)$

- Triangle law: if two forces are represented by two sides of a triangle taken in order, the closing side taken in opposite order is the resultant.
- Polygon law: extension for more than two forces.
- Resolution: $F_x = F\cos\theta$, $F_y = F\sin\theta$. ΣF_x and ΣF_y give $R = \sqrt{(\Sigma F_x^2 + \Sigma F_y^2)}$, $\tan\alpha = \Sigma F_y/\Sigma F_x$.

★ **PYQ-TREND:** Two equal forces at 120° give R equal to each force. Two equal forces at 60° give R = √3 P. The resultant of two forces is maximum when they are collinear and like. "Magnitude of resultant of 3 N and 4 N at right angles = 5 N" is a typical conceptual check.

1.3 Equilibrium

- Conditions in a plane: ΣFx = 0, ΣFy = 0, ΣM = 0.
- Two-force member: forces must be equal, opposite and collinear.
- Three-force member: forces must be concurrent (or parallel) and coplanar.
- Lami's theorem: if three coplanar concurrent forces are in equilibrium, each force is proportional to the sine of the angle between the other two.

```
P/sin α = Q/sin β = R/sin γ (α = angle between Q and R, etc.)
```

- Free body diagram (FBD): isolate the body and show all external forces and reactions. Roller support gives 1 reaction (normal), hinge/pin gives 2 (H, V), fixed support gives 3 (H, V, M).
- Varignon's theorem (principle of moments): the moment of the resultant about any point equals the algebraic sum of the moments of the components about the same point.
- Couple: two equal, opposite, parallel non-collinear forces. Moment of couple = F × arm. It is the same about every point in the plane (free vector). A couple cannot be balanced by a single force, only by an equal and opposite couple.
- A force can be replaced by an equal parallel force at another point plus a couple (force–couple system).

★ **PYQ-TREND:** Lami's theorem is valid only for three concurrent forces. A couple's moment is independent of the reference point. Number of reactions at fixed, hinged and roller supports. Varignon's theorem relates to moments.

✔ **TOPPER TRICK:** When three forces act on a body and two meet at a point, the third must pass through the same point. This shortcut solves many ladder and hinge problems in seconds.

1.4 Friction

Friction is the tangential resistance at the contact of two surfaces. It always opposes the relative motion or the tendency of motion.

- Static friction is self-adjusting up to the limiting value. Limiting friction F = μN.
- Dynamic (kinetic) friction is slightly less than limiting friction. Rolling friction < sliding friction.
- Laws of dry (Coulomb) friction: F ∝ N; independent of apparent contact area; depends on nature of surfaces; kinetic friction roughly independent of velocity.

```
μ = F/N = tan φ (φ = angle of friction)
Angle of repose α = φ (max inclination at which a body rests on its own)
Body on incline (no applied force), tendency to slide down: equilibrium if tanθ ≤ μ
Pull P parallel to plane to move up: P = W(sinθ + μcosθ); to move down: P = W(μcosθ − sinθ)
Least force to drag a body on a horizontal plane: P_min = W sinφ, applied at angle φ to the horizontal
```

- Cone of friction: the cone generated by the resultant reaction with semi-vertex angle φ.
- Ladder problems: take friction at floor and wall, ΣFx=ΣFy=ΣM=0. If wall is smooth, only floor friction acts.

- Belt friction (see TOM): $T_1/T_2 = e^{(\mu\theta)}$.
- Screw jack: efficiency $\eta = \tan\alpha / \tan(\alpha+\varphi)$; max efficiency $= (1-\sin\varphi)/(1+\sin\varphi)$; self-locking if $\varphi > \alpha$, and then $\eta < 50\%$.

★ **PYQ-TREND:** Angle of repose = angle of friction. Efficiency of a self-locking screw is less than 50%. Force of friction is independent of the area of contact. The least force to drag a body on a rough horizontal plane is W sin φ. Coefficient of friction depends on the nature of surfaces.

✖ **COMMON MISTAKE:** μ can exceed 1 (e.g., rubber on dry concrete, clean metals in vacuum). Don't reject an answer just because μ > 1.

1.5 Centroid and Centre of Gravity

- Centre of gravity (CG): point through which the total weight acts. Centroid: geometric centre of an area/line/volume. They coincide for uniform density in a uniform gravity field.
- An axis of symmetry always contains the centroid.

Shape	Centroid location
Rectangle b×h	h/2 from base
Triangle	h/3 from base (2h/3 from apex)
Semicircle (area), radius r	4r/(3π) from diameter
Quarter circle	4r/(3π) from each straight edge
Semicircular arc (wire)	2r/π from diameter
Right circular solid cone	h/4 from base
Hollow cone (surface)	h/3 from base
Solid hemisphere	3r/8 from flat face
Hollow hemisphere (thin shell)	r/2 from flat face
Parabolic spandrel ($y = kx^2$), base a, height h	$\bar{x} = 3a/4$, $\bar{y} = 3h/10$

$\bar{x} = \Sigma(A_i x_i)/\Sigma A_i$; $\bar{y} = \Sigma(A_i y_i)/\Sigma A_i$ (subtract cut-out areas)

- Pappus–Guldinus: Area of surface of revolution = length of curve × distance travelled by its centroid. Volume of revolution = area × distance travelled by its centroid ($2\pi\bar{y}$).

★ **PYQ-TREND:** CG of a solid cone at h/4 from base, solid hemisphere at 3r/8, semicircle at 4r/3π, triangle at h/3. These four come almost every year.

1.6 Moment of Inertia (Area)

Second moment of area $I = \int y^2\, dA$. It measures resistance to bending (beam stiffness).

Parallel axis theorem: $I_{AB} = I_G + A h^2$
Perpendicular axis theorem (plane lamina): $I_{zz} = I_{xx} + I_{yy}$ (polar MOI J)
Radius of gyration $k = \sqrt{(I/A)}$

Section	I about centroidal axis
Rectangle b × d (about axis ∥ b)	$bd^3/12$
Rectangle about base	$bd^3/3$
Triangle (about centroidal axis ∥ base)	$bh^3/36$
Triangle about base	$bh^3/12$
Triangle about apex (∥ base)	$bh^3/4$

Section	I about centroidal axis
Circle dia d	$\pi d^4/64$ (polar $\pi d^4/32$)
Hollow circle	$\pi(D^4-d^4)/64$
Semicircle about diameter	$\pi r^4/8 = \pi d^4/128$
Semicircle about centroidal axis ∥ diameter	$0.11\ r^4$
Square side a about diagonal	$a^4/12$ (same as about centroidal axis)

- Mass MOI (dynamics): thin rod about centre $mL^2/12$, about end $mL^2/3$; solid disc $mR^2/2$; ring mR^2; solid sphere $2/5\ mR^2$; hollow sphere $2/3\ mR^2$.

★ **PYQ-TREND:** I of triangle about base $bh^3/12$ vs about centroid $bh^3/36$. MOI of a square about its diagonal equals that about the centroidal axis. Polar MOI of circle $\pi d^4/32$. Mass MOI of solid sphere $2mR^2/5$.

✔ **TOPPER TRICK:** For "about base" values, use $I_{base} = I_G + A(h/2)^2$ for a rectangle: $bh^3/12 + bh\cdot h^2/4 = bh^3/3$.

1.7 Kinematics of Particles

$v = u + at;\ s = ut + \frac{1}{2}at^2;\ v^2 = u^2 + 2as;\ s_{nth} = u + a(2n-1)/2$

- Free fall: $a = g = 9.81\ m/s^2$. Time of flight for vertical projection $= 2u/g$; max height $= u^2/2g$.

$\text{Projectile: } T = 2u\sin\theta/g;\ H = u^2\sin^2\theta/(2g);\ R = u^2\sin 2\theta/g;\ R_{max} = u^2/g \text{ at } \theta = 45°$
$\text{Complementary angles } (\theta, 90°-\theta) \text{ give equal range. Path is parabola: } y = x\tan\theta - gx^2/(2u^2\cos^2\theta)$

- Angular motion: $\omega = \omega_0 + \alpha t;\ \theta = \omega_0 t + \frac{1}{2}\alpha t^2;\ \omega^2 = \omega_0^2 + 2\alpha\theta$. $v = r\omega$, $a_t = r\alpha$, $a_n = v^2/r = r\omega^2$.
- Relative velocity: $v_{AB} = v_A - v_B$.

★ **PYQ-TREND:** Maximum range at 45°; at maximum range $H = R/4$. Equal range for 30° and 60°. Time to reach max height $= u\sin\theta/g$. Centripetal acceleration $\omega^2 r$ directed towards centre.

1.8 Kinetics – Newton's Laws, Work, Energy, Power

- First law: inertia. Second law: $F = dp/dt = ma$. Third law: action and reaction are equal and opposite and act on different bodies.
- D'Alembert's principle converts a dynamics problem into an equivalent statics problem by adding the inertia force $(-ma)$.
- Lift problems: going up with acceleration a → $R = m(g + a)$; going down → $R = m(g - a)$; free fall → $R = 0$.
- Work $W = F\cdot s\cos\theta$ (scalar, joule). Power $P = W/t = F\cdot v$. Energy: $KE = \frac{1}{2}mv^2$, $PE = mgh$; rotational $KE = \frac{1}{2}I\omega^2$.
- Work-energy principle: net work done = change in KE.
- Conservation of energy holds if only conservative forces (gravity, spring) do work.

$\text{Spring energy} = \frac{1}{2}kx^2;\ \text{springs in series } 1/k = 1/k_1 + 1/k_2;\ \text{in parallel } k = k_1 + k_2$
$\text{Rolling body KE} = \frac{1}{2}mv^2(1 + k^2/r^2);\ \text{acceleration down incline } a = g\sin\theta/(1 + k^2/r^2)$

- Rolling race down an incline (who reaches first): solid sphere ($k^2/r^2 = 2/5$) > solid cylinder (1/2) > hollow sphere (2/3) > ring (1).

★ **PYQ-TREND:** D'Alembert principle: inertia force. Apparent weight in a lift. Rolling-without-slipping order. Unit of power and conversion of hp. Spring constants in series/parallel.

1.9 Impulse, Momentum and Collisions

Impulse = $F \cdot \Delta t = \Delta(mv)$; Linear momentum $p = mv$

Coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$ = velocity of separation / velocity of approach

- Perfectly elastic: e = 1, KE conserved. Perfectly plastic (inelastic): e = 0, bodies move together, max KE loss. 0 < e < 1 for real collisions.
- Momentum is conserved in all collisions (no external impulse). KE is conserved only for e = 1.
- Ball dropped from height h, bounce height $h_1 = e^2h$; $e = \sqrt{(h_1/h)}$.

Loss of KE in direct impact = $m_1m_2(u_1-u_2)^2(1-e^2) / [2(m_1+m_2)]$

- Angular momentum L = Iω; conserved if no external torque (ice skater pulls in arms → ω increases).
- Recoil of gun: m_b v_b = m_g v_g.

★ **PYQ-TREND:** Value of e for perfectly elastic and plastic impact. Equal masses with elastic head-on collision exchange velocities. Bounce height $h_1 = e^2h$. Momentum conserved in all impacts.

1.10 Simple Machines

- Mechanical advantage MA = W/P (load/effort). Velocity ratio VR = distance moved by effort / distance moved by load. Efficiency η = MA/VR.
- Ideal machine: η = 100%, MA = VR. Law of machine: P = mW + C (linear). Max MA = 1/m; max efficiency = 1/(m·VR).
- Reversible machine: η > 50%. Self-locking (irreversible) machine: η ≤ 50% – the load does not run back when effort is removed.

Machine	Velocity Ratio
Simple wheel & axle	D/d
Differential wheel & axle	$2D/(d_1 - d_2)$
Weston differential pulley	2D/(D − d)
First system of pulleys (n movable pulleys)	2^n
Second system (n pulleys total)	n
Third system	$2^n - 1$
Simple screw jack	2πL/p
Differential screw jack	$2\pi L/(p_1 - p_2)$
Worm & worm wheel	(L·T)/(r·n) – depends on starts n
Single purchase crab winch	$(L/r)(T_2/T_1)$

★ **PYQ-TREND:** Condition for reversibility (η > 50%) is asked almost every year. Velocity ratio of the first/second/third system of pulleys and of the differential pulley block.

✎ **DESCRIPTIVE Q:** Explain the difference between a reversible and self-locking machine, and prove that a machine is self-locking when its efficiency is less than 50%.

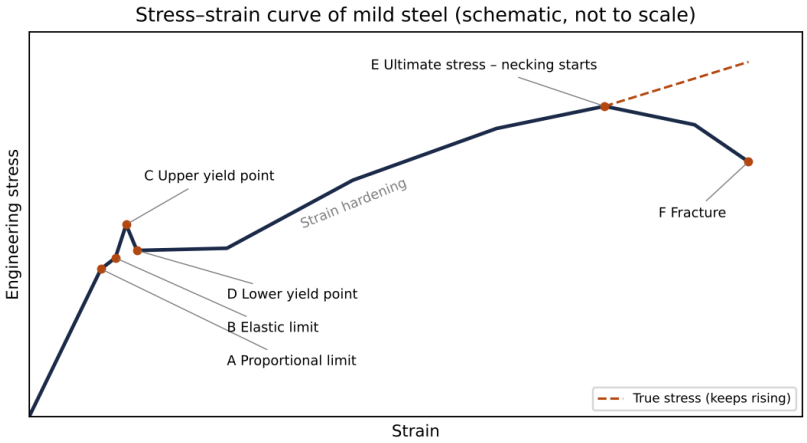
A machine is reversible if the load moves back when effort is removed. Let effort P lift load W through distance y, effort moving x. Input = Px, output = Wy, friction loss = Px − Wy. On reversal, the load W does work Wy, which must overcome the same friction loss Px − Wy. Reversal needs Wy > Px − Wy, i.e., Wy/Px > 1/2, i.e., η > 50%. So a machine with η ≤ 50% is self-locking.

CHAPTER 2

Strength of Materials

2.1 Stress and Strain

- Normal stress σ = P/A (N/mm² = MPa). Tensile is +ve, compressive is −ve. Shear stress τ = Q/A.
- Linear strain ε = δL/L (dimensionless). Shear strain γ = change in right angle (radians). Volumetric strain εv = δV/V.
- Hooke's law: within the proportional limit, stress ∝ strain. σ = Eε, τ = Gγ.



Stress–strain curve of mild steel (in order)

- Proportional limit → elastic limit → upper yield point → lower yield point → strain hardening → ultimate stress (max load, necking begins) → fracture (breaking point).
- Engineering stress uses original area, so the curve drops after ultimate point. True stress uses instantaneous area and keeps rising till fracture.
- Ductile materials (mild steel, copper, aluminium): large plastic deformation, % elongation > 5%. Brittle materials (cast iron, glass, concrete): fracture with little deformation, no yield point.
- Proof stress (0.2% offset) is used for materials without a clear yield point (aluminium, high-carbon steel).

Property	Meaning
Ductility	Can be drawn into wire (tensile deformation) – measured by % elongation & % reduction in area
Malleability	Can be hammered/rolled into sheets (compressive) – gold most malleable
Toughness	Energy absorbed up to fracture – area under full σ–ε curve; Izod/Charpy test
Resilience	Energy absorbed up to elastic limit – area under elastic part
Hardness	Resistance to indentation/scratch – Brinell, Rockwell, Vickers, Mohs

Property	Meaning
Creep	Slow permanent deformation under constant load at high temperature with time
Fatigue	Failure under fluctuating/repeated loads below ultimate stress
Stiffness	Resistance to deformation – measured by E

★ **PYQ-TREND:** Sequence of points on the stress–strain diagram. "Necking starts at ultimate stress". Toughness = area under the whole curve; resilience = area up to elastic limit. Lüders bands occur at yield in mild steel. Gold is the most malleable metal.

2.2 Elastic Constants

- Young's modulus E, shear (rigidity) modulus G, bulk modulus K, Poisson's ratio μ = −lateral strain/longitudinal strain.

$$E = 2G(1 + \mu)$$

$$E = 3K(1 - 2\mu)$$

$$E = 9KG/(3K + G);\ \mu = (3K - 2G)/(6K + 2G)$$

- Range of μ: 0 to 0.5 for ordinary materials. μ = 0.5 → incompressible (rubber ≈ 0.5, K → ∞). Cork ≈ 0. Steel ≈ 0.25–0.30. Cast iron ≈ 0.23–0.27. Concrete ≈ 0.1–0.2.
- E for steel ≈ 200 GPa, cast iron ≈ 100–120 GPa, aluminium ≈ 70 GPa, copper ≈ 110 GPa.
- An isotropic material has 2 independent elastic constants. Orthotropic: 9. Anisotropic: 21.

Volumetric strain (rod, axial load): $\varepsilon_v = \varepsilon(1 - 2\mu)$; (rectangular block) $\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$

Volumetric strain of a sphere = 3 × diametral strain; of a cylinder = $2\varepsilon_d + \varepsilon_L$

★ **PYQ-TREND:** E = 2G(1+μ) and E = 3K(1−2μ) are asked every cycle. Number of independent elastic constants for isotropic material = 2. If μ = 0.5, volume change is zero. Relation: when E = 2G, μ = 0; when K = E, μ = 1/3.

✔ **TOPPER TRICK:** For μ = 0.25, E = 2.5G; for μ = 0.3, E = 2.6G. Use these to verify options quickly.

2.3 Axially Loaded Members

Elongation $\delta = PL/(AE)$

Stepped bar: $\delta = (P/E)\ \Sigma(L_i/A_i)$

Tapered circular bar (d_1 to d_2): $\delta = 4PL/(\pi\ E\ d_1\ d_2)$

Tapered flat bar (width $b_1 \to b_2$, thickness t): $\delta = PL\ \ln(b_1/b_2)\ /\ [E\ t\ (b_1 - b_2)]$

Bar under self-weight (γ = weight density): $\delta = \gamma L^2/(2E) = WL/(2AE)$ (half of equal end load)

Cone under self-weight: $\delta = \gamma L^2/(6E)$

Bar of uniform strength: $A = A_0\ e^{(\gamma x/\sigma)}$

- Composite bars (in parallel, same strain): $P = P_1 + P_2$; $\delta_1 = \delta_2 \to \sigma_1/E_1 = \sigma_2/E_2$. Modular ratio $m = E_1/E_2$.
- Principle of superposition: valid in linear elastic range with small deflections.

Thermal stresses

Free expansion $\delta = \alpha L \Delta T$; if fully restrained, $\sigma = E\alpha\Delta T$ (independent of length and area)

Partial yielding of support by Δ: $\sigma = E(\alpha L\Delta T - \Delta)/L$

Composite bar heated ($\alpha_1 > \alpha_2$): material 1 in compression, material 2 in tension; $\sigma_1 A_1 = \sigma_2 A_2$

★ **PYQ-TREND:** Elongation due to self-weight is half that of the same load at the end. Thermal stress $E\alpha\Delta T$ does not depend on cross-section. In a steel–copper composite bar when heated, copper (higher α) is in compression. Elongation of a tapered rod $4PL/(\pi E d_1 d_2)$.

✖ **COMMON MISTAKE:** Students often use the average diameter in the tapered-rod formula. Use the product $d_1 d_2$, not the square of the average.

2.4 Strain Energy and Impact Loading

Strain energy $U = P^2L/(2AE) = \sigma^2/(2E) \times$ Volume

Modulus of resilience = $\sigma_e^2/(2E)$ (strain energy per unit volume at elastic limit)

Shear strain energy = $\tau^2/(2G) \times$ Volume; torsion $U = T^2L/(2GJ)$; bending $U = \int M^2 dx/(2EI)$

Sudden load: $\sigma = 2P/A$ (twice the gradual stress)

Impact (falling from h): $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$

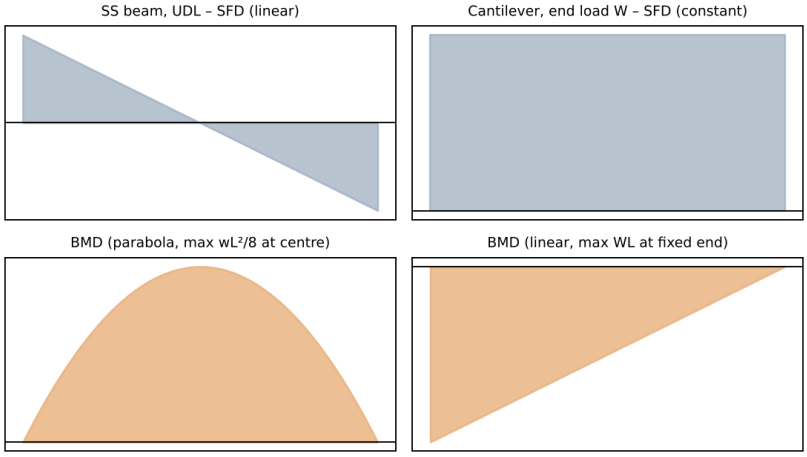
- Castigliano's theorem: $\delta = \partial U/\partial P$. Used for deflection of curved beams, springs and frames.
- Maxwell's reciprocal theorem: deflection at A due to load at B equals deflection at B due to the same load at A.

★ **PYQ-TREND:** Suddenly applied load gives double stress. Proof resilience vs modulus of resilience (per unit volume). Strain energy stored in solid shaft in torsion = $(\tau^2/4G) \times$ volume; hollow = $(\tau^2/4G)(D^2+d^2)/D^2 \times$ volume.

2.5 Shear Force and Bending Moment

- Sign convention: SF positive if it causes clockwise rotation (left upward); BM positive if sagging (concave up).
- Relations: $dM/dx = V$; $dV/dx = -w$. Hence BM is maximum where SF = 0 or changes sign.
- Point of contraflexure: where BM changes sign (BM = 0 inside the span). Occurs in overhanging and fixed beams, not in simply supported beams with downward loads.

Load on SF/BM	SFD shape	BMD shape
No load	Constant (horizontal)	Linear (inclined)
Point load	Sudden jump	Change in slope (kink)
UDL	Linear (inclined)	Parabola (2nd degree)
UVL (triangular)	Parabola (2nd degree)	Cubic (3rd degree)
Pure couple	No change	Sudden jump



Beam & Load	Max BM	Max deflection
Cantilever, point load W at free end	WL	$WL^3/(3EI)$
Cantilever, UDL w	$wL^2/2$	$wL^4/(8EI)$
Cantilever, UVL (zero at free end)	$wL^2/6$	$wL^4/(30EI)$
Cantilever, couple M at free end	M	$ML^2/(2EI)$
Simply supported, central point load	WL/4	$WL^3/(48EI)$
Simply supported, UDL	$wL^2/8$	$5wL^4/(384EI)$
SS, point load at a from A (b from B)	Wab/L	$Wa^2b^2/(3EIL)$ under load
Fixed–fixed, central point	WL/8 (at ends & centre)	$WL^3/(192EI)$
Fixed–fixed, UDL	$wL^2/12$ (ends), $wL^2/24$ (centre)	$wL^4/(384EI)$
Propped cantilever UDL	$wL^2/8$ (fixed end)	$wL^4/(185EI)$ approx; prop reaction 3wL/8

Slope at free end of cantilever: W at end $\theta = WL^2/(2EI)$; UDL $\theta = wL^3/(6EI)$

Slope at support of SS beam: point at centre $\theta = WL^2/(16EI)$; UDL $\theta = wL^3/(24EI)$

★ **PYQ-TREND:** SF/BM shapes for UDL & UVL. Max BM $wL^2/8$ and deflection $5wL^4/384EI$ for SS UDL. Ratio of central deflection of fixed to SS beam with UDL = 1:5; with central point load = 1:4. Prop reaction 3wL/8. Point of contraflexure in overhanging beam.

✔ **TOPPER TRICK:** Degree of BMD = degree of load + 2. Degree of SFD = degree of load + 1. A UDL (degree 0) gives a parabolic BMD (degree 2).

2.6 Bending Stresses

Bending equation: $M/I = \sigma/y = E/R$

Section modulus $Z = I/y_{max}$; $\sigma_{max} = M/Z$

- Assumptions: plane sections remain plane; material homogeneous, isotropic, obeys Hooke's law; E same in tension & compression; radius of curvature large compared with depth; pure bending.
- Neutral axis passes through the centroid; stress is zero there; it varies linearly with distance.

Section	Z
Rectangle b×d	$bd^2/6$

Section	Z
Circle d	$\pi d^3/32$
Hollow circle	$\pi(D^4 - d^4)/(32D)$
Square with diagonal vertical (side a)	$a^3/(6\sqrt{2}) = 0.1179a^3$
Triangle (base b, height h) about centroid	$bh^2/24$ (top fibre)

- Strongest rectangular beam cut from a circular log of dia D: d/b = √2 (b = D/√3). Stiffest: d/b = √3.
- Beam of uniform strength: depth or width varies so that σ_max is constant everywhere (e.g., leaf spring).
- Flitched beam: two materials fastened together; use equivalent section with modular ratio m.

★ **PYQ-TREND:** Strongest beam from a log: d/b = √2. Z for rectangular and circular sections. Neutral axis passes through centroid. A square beam with diagonal vertical is weaker (Z less) than with a side horizontal.

2.7 Shear Stresses in Beams

$\tau = V \cdot A \cdot \bar{y}/(I \cdot b)$ (A $\bar{y}$ = first moment of area above the level)

Section	τ_max / τ_avg	Location
Rectangle	1.5	Neutral axis
Circle (solid)	4/3 = 1.33	Neutral axis
Thin hollow circle	2.0	Neutral axis
Triangle	1.5	At h/2 from apex (not NA)
Triangle at NA	4/3 (i.e., τ_NA = 1.33 τ_avg)	NA
Square with diagonal vertical	9/8 = 1.125	At d/8 above/below NA

- Shear stress distribution in a rectangle is parabolic; in an I-section most of the shear is taken by the web (sudden jump at flange-web junction).

★ **PYQ-TREND:** Ratio τ_max/τ_avg = 1.5 for rectangle and 4/3 for circle – asked repeatedly. In a triangular section max shear stress is at mid-height, not at the NA.

2.8 Torsion of Shafts

$T/J = \tau/r = G\theta/L$

Solid shaft: $J = \pi d^4/32$; $T = (\pi/16)\,\tau\, d^3$; $Z_p = \pi d^3/16$

Hollow shaft: $J = \pi(D^4 - d^4)/32$; $T = (\pi/16)\,\tau\,(D^4 - d^4)/D$

Power: $P = 2\pi NT/60$ (W, with N in rpm and T in N·m)

Torsional stiffness = $T/\theta = GJ/L$

- Assumptions: circular section remains circular and plane; shear stress varies linearly from centre; material obeys Hooke's law.
- Shafts in series (same torque): $\theta = \theta_1 + \theta_2$. Shafts in parallel (same twist): $T = T_1 + T_2$.
- For the same weight, length and material, a hollow shaft is stronger and stiffer than a solid shaft.

Combined bending and torsion: equivalent torque $T_e = \sqrt{(M^2 + T^2)}$; equivalent BM $M_e = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$

- Te is used with max shear stress theory (ductile); Me with max principal stress theory (brittle).

Close-coiled helical spring: $\tau = 8WD/(\pi d^3) \times K_w$; deflection $\delta = 8WD^3n/(Gd^4)$; stiffness $k = Gd^4/(8D^3n)$

Wahl factor $K_w = (4C-1)/(4C-4) + 0.615/C$, spring index $C = D/d$

★ **PYQ-TREND:** Power transmitted 2πNT/60. Hollow shaft stronger than solid of same weight. Equivalent torque & bending moment. Torque transmission ratio of hollow to solid shafts. Spring stiffness ∝ $d^4/(D^3n)$. Cutting a spring in half doubles stiffness.

2.9 Principal Stresses and Mohr's Circle

$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$

$\tau_{max} = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]} = (\sigma_1 - \sigma_2)/2$

Principal plane angle: $\tan 2\theta_p = 2\tau xy/(\sigma x - \sigma y)$

Normal stress on oblique plane: $\sigma n = (\sigma x+\sigma y)/2 + (\sigma x-\sigma y)/2 \cos 2\theta + \tau xy \sin 2\theta$

- Principal planes have zero shear stress. Planes of max shear are at 45° to principal planes and carry normal stress $(\sigma_1+\sigma_2)/2$.
- Mohr's circle: centre at $((\sigma x+\sigma y)/2, 0)$, radius = τ_{max}. An angle θ on the element is 2θ on the circle.
- Pure shear: $\sigma_1 = +\tau$, $\sigma_2 = -\tau$ at 45° (why brittle shafts fail along a 45° helix in torsion).
- Uniaxial σ: $\tau_{max} = \sigma/2$ on 45° plane (why ductile specimens show cup-and-cone fracture).
- Hydrostatic ($\sigma x = \sigma y = \sigma$): Mohr's circle reduces to a point; no shear on any plane.

★ **PYQ-TREND:** Angle between principal planes = 90°; between principal plane and max shear plane = 45°. Mohr's circle for pure shear has centre at origin. Brittle material in torsion fails on a 45° helicoidal surface; ductile fails on plane perpendicular to the axis.

2.10 Theories of Failure

Theory	Proposed by	Criterion	Suited for
Max principal stress	Rankine	$\sigma_1 = \sigma yt$	Brittle
Max shear stress	Tresca/Guest	$\sigma_1 - \sigma_2 = \sigma yt$ ($\tau_{max} = \sigma yt/2$)	Ductile (conservative)
Max principal strain	St. Venant	$\sigma_1 - \mu\sigma_2 = \sigma yt$	Not used much
Max strain energy	Haigh	$\sigma_1^2 + \sigma_2^2 - 2\mu\sigma_1\sigma_2 = \sigma yt^2$	Ductile
Max distortion energy (shear strain energy)	von Mises–Hencky	$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 = \sigma yt^2$	Ductile (best)

- In pure shear: Tresca gives $\tau y = 0.5\sigma y$; von Mises gives $\tau y = 0.577\sigma y$.
- Shape in σ_1–σ_2 plane: Rankine – square; Tresca – hexagon; von Mises – ellipse; Haigh – ellipse; St. Venant – rhombus.

★ **PYQ-TREND:** Theory best for ductile = distortion energy (von Mises). Tresca hexagon, von Mises ellipse. Shear yield = 0.577 σy (von Mises).

2.11 Columns and Struts

Euler's critical load: $P_{cr} = \pi^2 E I_{min} / Le^2$

```
Slenderness ratio λ = Le/k_min
```

End condition	Effective length Le	P_cr relative to both-hinged
Both ends hinged	L	1
One fixed, other free	2L	1/4
Both ends fixed	L/2	4
One fixed, other hinged	L/√2	2

- Euler's formula is valid for long columns (λ > ~80 for mild steel) and ignores direct crushing.

```
Rankine–Gordon: 1/P = 1/Pc + 1/Pe → P = σc A / (1 + a λ²); a = σc/(π²E)
```

- Buckling occurs about the axis with minimum moment of inertia.
- Eccentric loading: secant formula. Core (kernel) of a section: load inside it causes no tension. Rectangle – rhombus with diagonals b/3 and d/3 (middle-third rule). Circle – concentric circle of dia d/4.

★ **PYQ-TREND:** Le for four end conditions (most-repeated SOM question). Ratio of buckling loads of fixed-fixed to fixed-free = 16. Middle-third rule (e ≤ b/6). Core of a circular section d/4.

✔ **TOPPER TRICK:** $P_{cr} \propto 1/Le^2$. Fixed-free Le = 2L → P = $P_0/4$; fixed-fixed Le = L/2 → $4P_0$. Ratio between them = 16.

2.12 Thin and Thick Cylinders

```
Thin cylinder: hoop stress σh = pd/(2t); longitudinal σl = pd/(4t) (σh = 2σl)
Thin sphere: σ = pd/(4t)
Hoop strain εh = (pd/4tE)(2 − μ); longitudinal strain εl = (pd/4tE)(1 − 2μ)
Volumetric strain (cylinder) = (pd/4tE)(5 − 4μ); sphere = (3pd/4tE)(1 − μ)
Max in-plane shear (cylinder) = (σh − σl)/2 = pd/(8t)
```

- Thin if d/t > 20 (t < d/20). Radial stress is neglected in thin shells.
- Longitudinal joint carries hoop stress → design longitudinal joint for hoop stress (it is weaker); circumferential joint for longitudinal stress.
- Thick cylinders – Lamé's equations: $\sigma_r = A - B/r^2$, $\sigma_h = A + B/r^2$; hoop stress is maximum at the inner surface. Compounding (shrink fit) and autofrettage reduce the hoop stress variation.

★ **PYQ-TREND:** Hoop stress is twice the longitudinal stress. Thin cylinder criterion d/t > 20. For a sphere, stress pd/4t. Lamé's theorem for thick cylinders. Volumetric strain (5−4μ) expression.

2.13 Springs (Leaf) and Rotating Discs

```
Leaf spring (semi-elliptic, n leaves): σ = 3WL/(2nbt²); δ = 3WL³/(8Enbt³)
Carriage spring with full-length leaves is a beam of uniform strength.
```

- Springs in series: total deflection adds, $1/k = \Sigma 1/k_i$. Springs in parallel: stiffness adds.
- Rotating thin ring: hoop stress $\sigma = \rho v^2 = \rho\omega^2 r^2$ (independent of cross-section).

★ **PYQ-TREND:** Stiffness of springs in series/parallel; hoop stress in a rotating ring ρv².

✎ **DESCRIPTIVE Q:** Derive the relation E = 2G(1 + μ).

Consider a square element ABCD of side a under pure shear τ. Diagonal BD elongates. Pure shear is equivalent to tension τ along one diagonal and compression τ along the other. Strain along the diagonal from shear deformation = γ/2 = τ/(2G). From the equivalent normal stresses, strain along the same diagonal = τ/E + μτ/E = (τ/E)(1+μ). Equating: τ/(2G) = (τ/E)(1+μ) → E = 2G(1 + μ).

✎ **DESCRIPTIVE Q:** State the assumptions in the theory of simple bending and derive M/I = σ/y = E/R.

Consider a beam bent into an arc of radius R. A fibre at distance y from the neutral axis has strain ε = y/R, so σ = Ey/R, giving σ/y = E/R. The resultant force on the section is zero (∫σdA = 0), so the NA passes through the centroid. Moment of resistance M = ∫σy dA = (E/R)∫y²dA = EI/R. Hence M/I = σ/y = E/R. Assumptions: initially straight, homogeneous, isotropic beam; plane sections remain plane; Hooke's law; same E in tension and compression; large R compared with depth; pure bending.

CHAPTER 3

Theory of Machines

3.1 Mechanisms – Basic Definitions

- Kinematic link (element): a resistant body that is part of a machine and has relative motion. Types: rigid, flexible (belt, rope, chain), fluid (hydraulic press).
- Kinematic pair: two links in contact with relative motion that is completely or successfully constrained.
- Constrained motion: completely constrained (piston in cylinder), incompletely constrained (shaft in round hole with free axial motion), successfully constrained (footstep bearing held by load, cam–follower held by spring, IC engine valve).

Pair by type of contact	Examples
Lower pair (surface contact)	Sliding pair, turning pair, screw pair, spherical pair, nut & bolt
Higher pair (point/line contact)	Cam–follower, gears, ball & roller bearings, wheel on rail
Wrapping pair	Belt–pulley, chain–sprocket
Pair by relative motion	Sliding, turning (revolute), rolling, screw (helical), spherical
Pair by mechanical closure	Self-closed (lower pair) / force-closed (cam with spring)

- Kinematic chain: links connected so that the motion of each relative to the others is definite. Mechanism: a chain with one link fixed. Machine: a mechanism that transmits force/work. Structure: no relative motion.
- Inversion: different mechanisms obtained by fixing different links of the same chain. A chain with n links gives n inversions (some may be identical).

Degrees of freedom (mobility)

```
Kutzbach (planar): F = 3(n − 1) − 2j − h (j = lower pairs, h = higher pairs)
Grübler (F = 1, only lower pairs): 3n − 2j − 4 = 0
Kinematic chain relation: l = 2p − 4; j = (3/2)l − 2
```

- F = 0 → structure (frame); F < 0 → statically indeterminate (super-structure); F = 1 → constrained mechanism; F > 1 → unconstrained.
- Grashof's law (4-bar): s + l ≤ p + q for continuous rotation of at least one link. Shortest link fixed → double-crank; adjacent to shortest fixed → crank-rocker; opposite to shortest fixed → double-rocker.

★ **PYQ-TREND:** Lower vs higher pair examples (ball bearing is a higher pair; nut & bolt is a lower pair). Number of inversions. Grübler/Kutzbach DOF calculation. Grashof condition. A cam & follower is a higher, force-closed pair.

3.2 Inversions of Standard Chains

Chain	Inversions
Four-bar chain	Beam engine (crank & lever), coupling rod of locomotive (double crank), Watt's indicator (double lever), pantograph, Ackermann steering (double rocker)
Single slider crank	Reciprocating engine/compressor (I), rotary/Gnome engine (II – crank fixed), Whitworth quick return (II), oscillating cylinder engine (III), crank & slotted lever QR (III), hand pump/pendulum pump (IV – slider fixed)
Double slider crank	Elliptical trammel, Scotch yoke, Oldham's coupling

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Quick-return ratio = time of cutting / time of return = (360° − α)/α (α = return-stroke crank angle)
Crank & slotted lever: stroke length = 2 × slotted lever length × (crank radius / distance between centres)
```

- Elliptical trammel: a point on the link (not midpoint) traces an ellipse; midpoint traces a circle.
- Oldham coupling connects two parallel shafts with small lateral offset; max sliding speed of the intermediate piece = ωδ (δ = offset).
- Scotch yoke produces simple harmonic motion of the slider.
- Pantograph: used to copy drawings to enlarged/reduced scale.
- Exact straight-line mechanisms: Peaucellier, Hart, Scott-Russell. Approximate: Watt, Grasshopper, Tchebicheff, Robert.
- Steering gears: Davis (exact, sliding pairs, wear more) and Ackermann (approximate, turning pairs, used in vehicles). Condition of correct steering: cot φ − cot θ = c/b.
- Hooke's (universal) joint connects two intersecting shafts. Velocity ratio ω_B/ω_A = cosα/(1 − cos²θ sin²α). Speed fluctuation is eliminated by a double Hooke's joint.

★ **PYQ-TREND:** Whitworth & crank-slotted lever QRM are inversions of the single slider crank chain. Elliptical trammel, Scotch yoke, Oldham coupling come from double slider chain. Peaucellier gives exact straight line. Ackermann steering condition. Hooke's joint used for intersecting shafts.

3.3 Velocity and Acceleration Analysis

- Instantaneous centre (I-centre): a point about which a link has pure rotation at an instant. Number of I-centres N = n(n − 1)/2.
- Aronhold–Kennedy theorem: three bodies with relative motion have three I-centres lying on a straight line.
- Types: fixed, permanent, neither fixed nor permanent.

```
Velocity of a point = ω × distance from I-centre
Acceleration components: centripetal (radial) a_r = ω²r = v²/r; tangential a_t = αr
Coriolis component = 2ωv (v = sliding velocity of slider on rotating link); direction: rotate v by 90° in the sense of ω
```

- Coriolis component appears only when a point slides on a rotating link (crank & slotted lever, Whitworth).

Slider-crank kinematics (n = l/r)

```
Piston velocity vp ≈ ωr(sinθ + sin2θ/(2n))
```

Piston acceleration $a_p \approx \omega^2 r(\cos\theta + \cos 2\theta/n)$

Angular velocity of connecting rod $\omega_c = \omega \cos\theta/n$

- Klein's construction gives the velocity and acceleration diagrams of the slider crank graphically.

★ **PYQ-TREND:** Number of I-centres for a 4-bar = 6, 6-link = 15. Kennedy's theorem (three centres collinear). Coriolis component 2ωv appears in quick-return mechanisms. Direction of Coriolis.

3.4 Friction Devices: Pivots, Collars, Clutches

Device	Uniform pressure (new)	Uniform wear (worn, used for design)
Flat pivot	$T = (2/3)\mu WR$	$T = (1/2)\mu WR$
Flat collar / single-plate clutch (per surface)	$T = (2/3)\mu W(R_1^3 - R_2^3)/(R_1^2 - R_2^2)$	$T = \mu W(R_1 + R_2)/2$
Conical pivot	$T = (2/3)\mu WR \operatorname{cosec}\alpha$	$T = (1/2)\mu WR \operatorname{cosec}\alpha$

- Uniform wear theory gives lower torque → safer, used for clutch design. Uniform wear: $p \cdot r$ = constant (max pressure at inner radius).
- Multi-plate clutch: $T = n\,\mu\,W\,R_{mean}$, n = number of pairs of contact surfaces = $n_1 + n_2 - 1$.
- Cone clutch: $T = \mu W R_m / \sin\alpha$. Centrifugal clutch engages automatically by speed.

★ **PYQ-TREND:** Uniform wear vs uniform pressure torque; uniform wear used for design since it is conservative. Number of friction surfaces in a multi-plate clutch.

3.5 Belt, Rope and Chain Drives

Velocity ratio $N_2/N_1 = d_1/d_2$ (with thickness t: $(d_1+t)/(d_2+t)$; with slip s%: × $(1 - s/100)$)

Ratio of tensions (flat belt): $T_1/T_2 = e^{(\mu\theta)}$

V-belt / rope (groove angle 2β): $T_1/T_2 = e^{(\mu\theta \operatorname{cosec}\beta)}$

Power $P = (T_1 - T_2)v$

Centrifugal tension $T_c = m v^2$ (m = mass per metre)

Max power condition: $T_c = T_{max}/3$, $v_{opt} = \sqrt{(T_{max}/(3m))}$

Initial tension $T_0 = (T_1 + T_2)/2$ (neglecting T_c); with T_c: $(T_1 + T_2 + 2T_c)/2$

Open belt length $L = \pi(r_1+r_2) + 2x + (r_1-r_2)^2/x$; Crossed belt $L = \pi(r_1+r_2) + 2x + (r_1+r_2)^2/x$

- Crossed belt length depends only on the sum of radii → stepped cone pulleys with crossed belt need $r_1 + r_2$ = constant.
- Angle of contact for open belt is decided by the smaller pulley ($\theta = 180° - 2\alpha$). Tight side should be the lower side in horizontal drives (to increase angle of wrap).
- Creep: unequal stretching of belt on tight and slack sides causes relative motion between belt and pulley – reduces speed slightly.
- Slip vs creep: slip is due to insufficient friction; creep is due to elasticity of the belt.
- Chain drive: positive drive, no slip. Chordal action (polygon effect) causes speed fluctuation; reduced by more teeth. Minimum teeth on small sprocket ~17. Silent (inverted tooth) chain for high speed.

★ **PYQ-TREND:** Max power transmitted when Tc = T_max/3. Centrifugal tension does not affect power. Crossed belt length depends on sum of radii. Creep definition. T_1/T_2 for V-belt is greater than flat belt (wedge action). Initial tension formula.

✖ **COMMON MISTAKE:** In e^(μθ), θ must be in radians, not degrees.

3.6 Brakes and Dynamometers

- Brakes absorb KE/PE and dissipate it as heat. Types: shoe (block), band, band & block, internal expanding shoe, disc.
- Single block brake: self-energising if the friction moment assists the applied force; self-locking if no external force is needed (undesirable).
- Band brake: braking torque = $(T_1 - T_2)r$; $T_1/T_2 = e^{(\mu\theta)}$. Simple band brake is self-locking when $T_2 \cdot b = T_1 \cdot a$.
- Effective μ for pivoted long shoe: $\mu' = 4\mu \sin\theta/(2\theta + \sin 2\theta)$.
- Absorption dynamometers: Prony brake, rope brake, hydraulic, eddy current. Transmission dynamometers: epicyclic train, belt transmission (Froude-Thornycroft), torsion dynamometer.

★ **PYQ-TREND:** Self-energising vs self-locking brakes. Prony and rope brake are absorption type. Torsion dynamometer is transmission type.

3.7 Gears

- Law of gearing: the common normal at the point of contact must always pass through the pitch point so that the angular velocity ratio stays constant.
- Involute profile: pressure angle constant (14.5° or 20°), centre distance can vary slightly without affecting velocity ratio, single curvature, easy to manufacture (rack). Cycloidal: no interference, less wear, stronger, used in watches/clocks; exact centre distance required.

Module m = D/T (mm); circular pitch pc = πm; diametral pitch Pd = T/D = 1/m (per mm or per inch)

Addendum = 1m; dedendum = 1.25m (20° full depth); clearance = 0.25m; whole depth = 2.25m

Velocity ratio = $\omega_1/\omega_2 = T_2/T_1 = D_2/D_1$

Path of contact = $\sqrt{(Ra^2 - R^2\cos^2\varphi)} + \sqrt{(ra^2 - r^2\cos^2\varphi)} - (R + r)\sin\varphi$

Arc of contact = path of contact / cos φ; contact ratio = arc of contact / circular pitch

Min teeth on pinion to avoid interference (rack): $T_{min} = 2a_w / \sin^2\varphi \rightarrow 32$ for 14.5°, 18 for 20° (≈17)

- Interference occurs when the tip of one gear tooth contacts the non-involute flank of the other. Remedies: increase pressure angle, undercut, stub teeth, increase number of teeth, profile modification.
- Contact ratio should be > 1 (typically 1.2–1.6) for continuous transmission.

Gear type	Shafts	Remarks
Spur	Parallel	Straight teeth, noisy at high speed
Helical	Parallel (or crossed)	Gradual engagement, axial thrust, quieter
Herringbone (double helical)	Parallel	Axial thrusts cancel
Bevel / Mitre (VR 1, 90°)	Intersecting	Mitre = equal bevel at 90°

Gear type	Shafts	Remarks
Spiral (crossed helical)	Non-parallel non-intersecting	Point contact
Worm & worm wheel	Non-intersecting perpendicular	High reduction, can be self-locking
Hypoid	Non-intersecting	Automobile differential
Rack & pinion	Rotary ↔ linear	Rack = gear of infinite radius

★ **PYQ-TREND:** Module definition and standard addendum (1 module). Minimum teeth to avoid interference (32 & 18). Contact ratio > 1. Herringbone eliminates axial thrust. Hypoid used in differential. Cycloidal teeth used in watches.

3.8 Gear Trains

- Simple train: VR = T_driven/T_driver; idlers do not change VR but change direction. Compound train: VR = product of driven teeth / product of driver teeth.
- Reverted train: first and last gears co-axial (clock hands, lathe back gear). Condition $r_1 + r_2 = r_3 + r_4$ (same module → $T_1 + T_2 = T_3 + T_4$).
- Epicyclic train: axis of at least one gear moves (planet on arm). Large speed ratio in compact space – automobile differential, wristwatch, hoists, Humpage's reduction gear.

```
Epicyclic tabular method: arm fixed → give +1 rotation to sun; add y to all, x to sun.
For sun–planet–ring: T_ring = T_sun + 2T_planet
Torque balance: T_in + T_out + T_fixed = 0
```

- Differential of an automobile allows the outer wheel to rotate faster on a turn.

★ **PYQ-TREND:** Purpose of idler gear. Reverted train condition. Epicyclic train applications (differential). Ring teeth relation.

3.9 Governors

- Function: control the mean speed of the engine over long periods by regulating fuel supply according to load. (Flywheel controls speed fluctuation within a cycle.)

Governor	Height / Feature
Watt (simple conical)	$h = 895/N^2$ m (N rpm); suits slow speeds only
Porter (loaded, central dead weight M)	$h = (m + M)/m \times 895/N^2$ (for equal arms)
Proell	Like Porter, balls fixed to extension of lower arms – more sensitive
Hartnell	Spring-loaded, centrifugal; can run at high speeds and in any position
Wilson-Hartnell, Pickering	Other spring-loaded types; Pickering for gramophones

- Sensitiveness = $(N_2 - N_1)/N_{mean}$ (smaller range → more sensitive).
- Stability: radius of rotation increases with speed. Isochronous: equilibrium speed constant for all radii (infinitely sensitive, practically unstable) – Porter with equal arms can't be isochronous but Hartnell can be designed so.
- Hunting: continuous fluctuation of speed above/below mean due to too sensitive a governor.
- Effort: mean force on the sleeve for a given % change of speed. Power: work done at sleeve for a given % change. Controlling force curve: Fc vs radius – stable if Fc/r increases with r

(line meets y-axis below origin).

- Coefficient of insensitiveness = $(N_1 - N_2)/N$ (due to friction).

★ **PYQ-TREND:** Height of Watt governor ∝ $1/N^2$. Isochronous governor is infinitely sensitive. Hunting definition. Hartnell is spring loaded. Governor controls mean speed, flywheel controls cyclic fluctuations – favourite comparison question.

3.10 Flywheel and Turning Moment Diagram

- Flywheel stores energy when supply > demand and releases it when supply < demand; reduces fluctuation of speed within a cycle. It does not control mean speed.

```
Coefficient of fluctuation of speed Cs = (ω₁ − ω₂)/ω_mean
Max fluctuation of energy ΔE = I ω² Cs = m k² ω² Cs = 2E·Cs
Coefficient of fluctuation of energy Ce = ΔE / (work done per cycle)
Rim stress in flywheel σ = ρv² (v = peripheral speed)
```

- Turning moment diagram (crank effort diagram): torque vs crank angle. Four-stroke single-cylinder has the largest fluctuation; multi-cylinder engines give smoother torque and need a smaller flywheel.
- Cs values: punching press ~0.1–0.2; IC engine ~0.003–0.03; generator drive ~0.002.

★ **PYQ-TREND:** $\Delta E = I\omega^2 Cs$. Function of flywheel vs governor. Four-stroke engine needs a heavier flywheel than two-stroke. Rim stress ρv^2.

3.11 Balancing

- Static balance: $\Sigma mr = 0$ (CG on axis). Dynamic balance: $\Sigma mr = 0$ and $\Sigma mrl = 0$ (no resultant couple). At least two balancing masses in different planes are required for dynamic balancing of a rotating shaft.
- Reciprocating masses: primary force = $m\,\omega^2\,r\cos\theta$; secondary force = $m\,\omega^2\,r\cos2\theta / n$. Secondary force has frequency twice the crank speed.
- Partial balancing of locomotives: fraction c (usually 2/3 or 3/4) of reciprocating mass balanced. Unbalanced effects: variation of tractive effort, swaying couple, hammer blow.
- In-line engines: 4-cylinder (in-line, 180° crank) – primary forces and couples balanced, secondary forces not. 6-cylinder in-line – completely balanced for primary & secondary forces and couples.
- V-twin, radial engines: use direct & reverse crank method.

★ **PYQ-TREND:** Conditions for static vs dynamic balance. Hammer blow, swaying couple, tractive effort variation. Secondary force frequency = 2ω. Complete reciprocating balance is impossible by rotating masses.

3.12 Cams and Followers

- Cam: rotating machine element giving reciprocating/oscillating motion to the follower by direct contact (higher pair).
- Followers: knife-edge (high wear), roller (most used, less wear), flat-faced/mushroom (used where space limited – automobile valves), spherical-faced. In-line (radial) or offset.
- Terms: base circle (smallest circle from cam centre to profile), prime circle, pitch curve, pressure angle (angle between direction of follower motion and common normal; keep < 30°), lift/stroke, dwell.

Follower motion	Characteristics
Uniform velocity	Infinite acceleration at start/end – not used without modification
Simple harmonic (SHM)	$v_{max} = \pi h\omega/(2\theta_0)$; $a_{max} = \pi^2 h\omega^2/(2\theta_0^2)$; finite but sudden jerk at ends
Uniform acceleration & retardation (parabolic)	Min a for given lift; $a = 4h\omega^2/\theta_0^2$
Cycloidal	Zero acceleration at start/end – best for high speeds, no jerk

- Undercutting of cam occurs when the radius of curvature of pitch curve is less than the roller radius.

★ **PYQ-TREND:** Cycloidal motion has no abrupt acceleration change → used at high speeds. Uniform velocity gives infinite acceleration. Pressure angle definition. Mushroom follower used in IC engine valves.

3.13 Vibrations

Natural frequency (undamped, longitudinal): $\omega_n = \sqrt{(k/m)}$; $f_n = (1/2\pi)\sqrt{(g/\delta)} = 0.4985/\sqrt{\delta}$ (δ in m)

Simple pendulum $T = 2\pi\sqrt{(L/g)}$; compound pendulum $T = 2\pi\sqrt{((k_G^2 + h^2)/(gh))}$

Damping ratio $\zeta = c/c_c$; critical damping $c_c = 2\sqrt{(km)} = 2m\omega_n$

Damped frequency $\omega_d = \omega_n \sqrt{(1 - \zeta^2)}$

Logarithmic decrement $\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1 - \zeta^2)}$

Transmissibility: isolation only when $\omega/\omega_n > \sqrt{2}$

Whirling (critical) speed of shaft = natural transverse frequency; $N_c = 60/(2\pi)\cdot\sqrt{(g/\delta)}$

Torsional vibration: $\omega_n = \sqrt{(q/I)}$, $q = GJ/L$

Two-rotor system: node located such that $l_A \cdot I_A = l_B \cdot I_B$

- $\zeta > 1$ overdamped (no oscillation), $\zeta = 1$ critically damped (fastest return without oscillation – used in gun recoil), $\zeta < 1$ underdamped.
- Resonance: forcing frequency = natural frequency → amplitude very large (limited by damping).
- Rayleigh's method (energy), Dunkerley's method (lower bound, $1/f^2 = \Sigma 1/f_i$) for transverse vibrations.

★ **PYQ-TREND:** $f_n = 0.4985/\sqrt{\delta}$. Critical damping definition and its use in recoil mechanisms. Vibration isolation requires frequency ratio > $\sqrt{2}$. Log decrement. Whirling speed equals natural transverse frequency.

✔ **TOPPER TRICK:** Static deflection δ in mm → $f_n \approx 15.76/\sqrt{\delta}$ Hz. Example: δ = 1 mm → ~15.8 Hz.

✎ **DESCRIPTIVE Q:** Differentiate between a governor and a flywheel.

A governor controls the mean speed over a long period by regulating fuel supply to match the load. It acts intermittently, only when the load changes, and is essential for engines with varying loads. A flywheel controls cyclic speed fluctuation within one cycle by storing and releasing kinetic energy. It works continuously, does not control fuel supply, and is needed even at constant load (e.g., punching press, IC engines). So a flywheel gives smoothness within a cycle and a governor gives regulation between cycles.

CHAPTER 4

Machine Design

4.1 Design Philosophy and Factor of Safety

- Factor of safety (FOS) = failure stress / working (allowable) stress. Ductile material under static load: FOS based on yield stress. Brittle: based on ultimate stress.
- Typical FOS: steel under static load 2–4; cast iron 4–6; dynamic/shock loads higher.
- Stress concentration: localised high stress near discontinuities (holes, notches, fillets, keyways). Theoretical stress concentration factor Kt = σ_max/σ_nominal. Plate with small circular hole in tension: Kt = 3. Elliptical hole: Kt = 1 + 2a/b.
- Notch sensitivity q = (Kf − 1)/(Kt − 1). q = 0 → insensitive (cast iron, due to internal flakes); q = 1 → fully sensitive.
- Reduce stress concentration: fillets, relief grooves, multiple small holes, gradual change in section.
- Preferred numbers (Renard series): R5, R10, R20, R40 – geometric progressions with ratios ≈ 1.58, 1.26, 1.12, 1.06.

★ **PYQ-TREND:** Kt = 3 for a small hole in an infinite plate. Notch sensitivity range 0–1. Cast iron is notch-insensitive. Renard series R10 ratio = 10^(1/10).

4.2 Fatigue Design

- Endurance limit Se: max stress amplitude that a polished standard specimen can withstand for infinite cycles (≈ 10^6–10^7). For steel Se' ≈ 0.5 Sut.
- S–N curve: ferrous metals show a knee (endurance limit); non-ferrous (aluminium) do not – use fatigue strength at a given life.

```
Se = ka·kb·kc·kd·ke·Se' (Marin factors: surface, size, reliability, temperature, misc.)
σ_mean = (σmax + σmin)/2; σ_amp = (σmax − σmin)/2; stress ratio R = σmin/σmax
```

- Fully reversed: R = −1 (σm = 0). Repeated (pulsating): R = 0.

Criterion	Equation (FOS n)
Soderberg (conservative, yield)	σa/Se + σm/Syt = 1/n
Goodman (ultimate, linear)	σa/Se + σm/Sut = 1/n
Gerber (parabolic)	nσa/Se + (nσm/Sut)² = 1

- Fatigue strength improvement: shot peening, surface rolling (compressive residual stress), nitriding, carburising, polishing. Decarburisation, plating (Cr, Ni) and corrosion reduce it.
- Miner's rule (cumulative damage): Σ n_i/N_i = 1.

★ **PYQ-TREND:** Soderberg is the most conservative line (joins Se and Syt). Goodman joins Se and Sut. Shot peening improves fatigue life. Endurance limit ≈ 0.5 × UTS for steels. Aluminium has no distinct endurance limit.

4.3 Riveted Joints

- Lap joint and butt joint (single/double strap). Chain riveting vs zig-zag (staggered) riveting.
- Failures: tearing of plate between rivets, shearing of rivet, crushing of rivet/plate, tearing at margin.

```
Tearing resistance Pt = (p − d) t σt
Shearing resistance Ps = n (π/4) d² τ (×2 for double shear; practice uses 1.875 for double shear in boiler code)
Crushing resistance Pc = n d t σc
Efficiency η = least of (Pt, Ps, Pc) / (p t σt)
Unwin's formula: d = 6√t (mm)
Margin m = 1.5d; pitch p ≥ 2d (min)
```

- Caulking and fullering make riveted boiler joints leak-proof. Fullering uses a tool as thick as the plate (more effective).
- Lozenge (diamond) joint used for structural butt joints in bridges – highest efficiency.

★ **PYQ-TREND:** Unwin's formula d = 6√t. Efficiency definition. Caulking & fullering for leak-proofing. Margin = 1.5d. Double-shear area of rivet.

4.4 Welded Joints

- Butt weld (tension): P = σt·t·l. Fillet weld: throat t = 0.707 × leg size s.

```
Transverse fillet (single): P = 0.707 s l σt; double transverse: P = 1.414 s l σt
Parallel fillet (shear): P = 1.414 s l τ (double)
Eccentric weld: primary shear τ₁ = P/A, secondary τ₂ = P·e·r/J; resultant by vector addition
```

- Advantages over riveting: lighter, leak-proof, no stress concentration from holes, 100% efficiency possible. Disadvantages: residual stresses, distortion, difficult inspection, brittle HAZ.

★ **PYQ-TREND:** Throat = 0.707 × leg. Strength of fillet weld is based on throat area. Parallel fillet weld is designed for shear.

4.5 Bolted / Threaded Joints

- Thread terminology: major dia, minor (root) dia, pitch, lead = n × pitch (n = starts), helix angle tanα = l/(πd_m).
- Thread forms: V/metric (fastening), square (power transmission, highest efficiency, screw jack), Acme (29°, lead screw of lathe), buttress (one-direction axial load – vice, gun breech), knuckle (rough use – railway coupling, bottle caps), whitworth (55°).
- Metric thread angle 60°. ISO designation M20 × 1.5 → nominal dia 20 mm, pitch 1.5 mm (fine). Coarse if pitch omitted.

```
Stress area: A = (π/4)·[(dp + dc)/2]² ≈ (π/4)dc²
Initial tightening load (Rotscher): Pi = 2840·d (N, d in mm)
Torque to tighten: T = 0.2 P d
Bolt of uniform strength – shank dia reduced to core dia (or axial hole) to increase shock-absorbing capacity
```

- Locking devices: jam/lock nut, castle nut with split pin, spring washer, Simmonds nut (nylon insert), set screw.

- Bolt under eccentric load: shear at each bolt = direct + secondary (proportional to distance from CG of bolt group).

★ **PYQ-TREND:** Square thread most efficient for power transmission; Acme used in lathe lead screw; buttress for one-way thrust. Metric thread angle 60°, Whitworth 55°. Designation M meaning. Initial tension 2840d.

4.6 Keys and Couplings

- Key: inserted between shaft and hub to prevent relative rotation; key is designed weaker than shaft so it fails first.

Key	Notes
Sunk rectangular	w = d/4, t = 2w/3; taper 1:100
Sunk square	w = t = d/4
Gib-head	Taper key with head for easy removal
Feather	Allows axial sliding of hub (gear shifting)
Woodruff	Semicircular, self-aligning, automobile/machine tools, tapered shafts
Saddle (flat / hollow)	Friction only, light loads
Kennedy	Two square keys at 90° – heavy duty
Splines	Integral keys, sliding + high torque (gearbox)

Key in shear: $T = l\ w\ \tau\ (d/2)$; in crushing: $T = l\ (t/2)\ \sigma c\ (d/2)$

Square key: equal strength in shear and crushing if $\sigma c = 2\tau$; length for equal strength to shaft ≈ 1.571d (≈1.5d)

- Couplings: rigid (sleeve/muff, split-muff/clamp, flange) – for collinear shafts; flexible (bushed-pin, Oldham, universal) – for misalignment.
- Bushed-pin type flexible coupling absorbs shock and misalignment through rubber bushes. Oldham – lateral misalignment. Hooke's – angular.
- Flange coupling: bolts in shear on pitch circle: $T = n\ (\pi/4)d_b^2\ \tau\ (D/2)$.

★ **PYQ-TREND:** Woodruff key is self-aligning. Feather key allows axial movement. Key length ≈ 1.5d. Coupling for lateral misalignment = Oldham; angular = universal. Width of rectangular key = d/4.

4.7 Shaft Design

ASME code: $d^3 = (16/\pi\tau)\ \sqrt{[(Km\ M)^2 + (Kt\ T)^2]}$

Max shear stress theory: $\tau_{max} = (16/\pi d^3)\ \sqrt{(M^2 + T^2)}$

Max normal stress theory: $\sigma_{max} = (16/\pi d^3)[M + \sqrt{(M^2 + T^2)}]$

- Shafts are designed on strength and rigidity (angle of twist typically ≤ 0.25°/m for line shafts; deflection limits for gears).
- Axle: non-rotating or rotating, carries bending only. Spindle: short rotating shaft (machine tools). Countershaft, jack shaft, line shaft.

★ **PYQ-TREND:** Axle subjected mainly to bending. ASME code combined shock factors. Equivalent torque concept.

4.8 Bearings

- Sliding contact (journal) bearings: full/partial journal, thrust (pivot/collar), hydrodynamic, hydrostatic. Rolling contact: ball, roller (cylindrical, tapered, spherical, needle).

```
Sommerfeld number S = (r/c)² (μN/p) (dimensionless; N in rps)
Bearing characteristic number = μN/p; Petroff's equation: f = 2π²(μN/p)(r/c)
Rolling bearing life L₁₀ = (C/P)^k million rev; k = 3 (ball), 10/3 (roller)
Equivalent load P = X V Fr + Y Fa
```

- McKee curve: μ vs μN/p. Minimum point = boundary between thin-film and thick-film lubrication. Bearing modulus (μN/p at min friction) – operate at ~3× to stay in hydrodynamic region.
- Hydrodynamic bearing needs: relative motion, wedge-shaped oil film (eccentricity), viscous fluid.
- Bearing materials: Babbitt (white metal – tin/lead based), bronze, copper-lead, PTFE. Need low friction, conformability, embeddability, good thermal conductivity.
- Rating life L_{10}: life that 90% of bearings attain (10% fail). Static load capacity C_0 (permanent deformation 0.0001 × ball dia).
- Needle bearings: small radial space, oscillating loads (wrist pin). Taper roller bearings: combined radial + axial (car wheel hubs). Self-aligning ball bearings: shaft misalignment.

★ **PYQ-TREND:** Life formula exponent 3 for ball bearing (life ∝ $1/P^3$). Sommerfeld number is dimensionless. Babbitt metal composition. Taper roller for combined loads. L_{10} meaning (90% reliability).

✔ **TOPPER TRICK:** If load on a ball bearing doubles, life becomes 1/8. If it halves, life increases 8 times.

4.9 Springs, Clutches, Brakes, Gears (Design Points)

- Helical spring design: spring index C = D/d typically 4–12; Wahl factor accounts for curvature + direct shear. Surge: resonance of spring coils with valve frequency – avoid by choosing a natural frequency 15–20 times the operating frequency.
- Spring ends: plain, plain-ground, squared, squared-ground. Active coils = total − inactive coils (squared & ground: n' = n + 2).
- Gear tooth strength – Lewis equation: F = σb·b·m·Y (Y = Lewis form factor). Buckingham equation for dynamic load. Wear strength: Fw = d_p·b·Q·K.
- Gear tooth failures: bending breakage, pitting (surface fatigue – most common in hardened gears), scoring (lubricant breakdown), abrasive wear.
- Belt selection: flat belts (long distance), V-belts (short, high reduction), timing belts (positive drive).

★ **PYQ-TREND:** Lewis equation assumes the tooth is a cantilever beam with full load at the tip. Pitting is surface fatigue. Spring index range. Surge in springs.

4.10 Pressure Vessels, Cotter & Knuckle Joints, Power Screws

- Cotter joint: connects two rods under axial tension/compression, no rotation (piston rod and crosshead, foundation bolt). Cotter taper 1 in 48 to 1 in 24. Types: socket & spigot, sleeve & cotter, gib & cotter.
- Knuckle joint: connects two rods under tensile load with some angular flexibility (tie bars of roof truss, valve rod, link of cycle chain). Pin in double shear.

- Power screw efficiency: $\eta = \tan\alpha/\tan(\alpha + \varphi)$; self-locking if $\varphi \geq \alpha$. Square thread most efficient; for Acme/trapezoidal use $\mu' = \mu/\cos\theta$.
- Overall efficiency of screw jack with collar friction is lower.

★ **PYQ-TREND:** Cotter joint transmits axial load only and is used for piston rod & crosshead. Knuckle joint pin is in double shear. Taper of cotter. Self-locking condition $\varphi > \alpha$.

✎ **DESCRIPTIVE Q:** Explain the difference between Soderberg and Goodman criteria with a sketch and state which is more conservative.

Both are drawn on the σm–σa plane with endurance limit Se on the σa axis. Soderberg's line joins Se to the yield strength Syt on the σm axis. Goodman's line joins Se to the ultimate strength Sut. Since Syt < Sut, the Soderberg line lies inside the Goodman line, so it permits lower stresses and is more conservative (safer). It also guards against yielding. Goodman is closer to test data for ductile materials. Gerber's parabola lies outside both and best fits experimental points, but it is not used directly for design.

CHAPTER 5

Engineering Thermodynamics

5.1 Basic Concepts

- System: quantity of matter or region under study. Closed (control mass – energy crosses, mass does not; piston-cylinder), Open (control volume – mass & energy cross; turbine, nozzle, compressor), Isolated (neither – universe, thermos flask approx.).
- Property: macroscopic characteristic. Intensive – independent of mass (P, T, ρ, specific volume v, specific properties). Extensive – depends on mass (V, U, H, S, mass). Ratio of two extensive properties is intensive.
- Property is a point function (exact differential). Heat and work are path functions (inexact differentials, δQ, δW).
- State, process, cycle. Quasi-static process: infinitely slow, system passes through a series of equilibrium states – only these can be shown on property diagrams.
- Thermodynamic equilibrium = thermal + mechanical + chemical equilibrium.
- Zeroth law: if A and B are each in thermal equilibrium with C, then A and B are in thermal equilibrium with each other. It is the basis of temperature measurement (thermometers).

```
T(K) = t(°C) + 273.15; t(°F) = 1.8 t(°C) + 32; −40 °C = −40 °F
1 bar = 10⁵ Pa = 100 kPa; 1 atm = 1.01325 bar = 760 mm Hg = 10.33 m of water
Absolute pressure = gauge + atmospheric; vacuum = atmospheric − absolute
```

- Thermometric properties: length of mercury column, resistance (RTD), EMF (thermocouple – Seebeck effect), pressure of a constant-volume gas thermometer (most accurate, standard). Triple point of water = 273.16 K, 0.611 kPa (single fixed point of the international scale).

★ **PYQ-TREND:** Intensive vs extensive identification. Heat & work as path functions. Zeroth law is the basis of temperature measurement. Temperature at which °C = °F (−40). Triple point of water 273.16 K. Turbine/compressor as open systems.

5.2 Work and Heat

```
Displacement work W = ∫p dV (only for quasi-static process in closed system)
```

Process	Law	Work W_{1-2} (closed)	Heat Q
Isochoric	V = c	0	m cv ($T_2 - T_1$)
Isobaric	p = c	p($V_2 - V_1$)	m cp ($T_2 - T_1$)
Isothermal	pV = c	$p_1V_1 \ln(V_2/V_1)$	Q = W
Adiabatic (reversible)	pV^γ = c	$(p_1V_1 - p_2V_2)/(\gamma - 1)$	0
Polytropic	pV^n = c	$(p_1V_1 - p_2V_2)/(n - 1)$	$W(\gamma - n)/(\gamma - 1)$

- Polytropic index n: n = 0 isobaric, n = 1 isothermal, n = γ adiabatic, n = ∞ isochoric.
- Free expansion (unresisted, into vacuum): W = 0, Q = 0, ΔU = 0 → T constant for ideal gas; irreversible.
- Sign convention (Clausius): heat supplied to system +ve, work done by system +ve.

- Flow work (open system) = pv; shaft work in steady flow = $-\int v\,dp$.

★ **PYQ-TREND:** Work in free expansion is zero. Work in isothermal process. Value of n for each process. Area under p–V curve = work; under T–s curve = heat. On p–V, the adiabatic curve is steeper than the isothermal (slope γ times).

5.3 First Law of Thermodynamics

- For a cycle: $\oint \delta Q = \oint \delta W$ (Joule's experiment; J = 4.186 J/cal).
- For a process: Q = ΔU + W. Internal energy U is a property. For an ideal gas U = f(T) only (Joule's law).
- PMM-1 (perpetual motion machine of first kind) – produces work without energy input – violates the 1st law.

Enthalpy $H = U + pV$; $dh = c_p\,dT$, $du = c_v\,dT$ (ideal gas)
$c_p - c_v = R$; $\gamma = c_p/c_v$; $c_v = R/(\gamma - 1)$; $c_p = \gamma R/(\gamma - 1)$
Air: $R = 0.287$ kJ/kg·K, $c_p = 1.005$, $c_v = 0.718$, $\gamma = 1.4$; universal $\bar{R} = 8.314$ kJ/kmol·K

- γ: monatomic 1.67, diatomic 1.4, triatomic ≈ 1.3.

Steady Flow Energy Equation (SFEE)

$h_1 + V_1^2/2 + g\,z_1 + q = h_2 + V_2^2/2 + g\,z_2 + w$ (per kg)

Device	SFEE simplification
Nozzle	$V_2 = \sqrt{(2(h_1 - h_2) + V_1^2)} \approx 44.72\sqrt{(\Delta h \text{ in kJ/kg})}$
Turbine (adiabatic)	$w = h_1 - h_2$
Compressor/pump	$w = -(h_2 - h_1)$
Throttling valve	$h_1 = h_2$ (isenthalpic)
Boiler / condenser / heat exchanger	$q = h_2 - h_1$ ($w = 0$)

- Throttling: irreversible, enthalpy constant, pressure drops, ideal gas temperature unchanged. Joule-Thomson coefficient $\mu_J = (\partial T/\partial p)_h$; for ideal gas $\mu_J = 0$. Below the inversion temperature real gases cool upon throttling (used in liquefaction).

★ **PYQ-TREND:** Throttling is isenthalpic. Nozzle velocity $44.72\sqrt{\Delta h}$. $c_p - c_v = R$. U of an ideal gas depends only on T. PMM-1 violates first law. Joule–Thomson coefficient zero for ideal gas.

✔ **TOPPER TRICK:** If the enthalpy drop is given in kJ/kg, the nozzle exit velocity in m/s is simply $44.72 \times \sqrt{\Delta h}$, assuming negligible inlet velocity.

5.4 Second Law of Thermodynamics

- Kelvin–Planck: it is impossible to construct a device operating in a cycle that produces work while exchanging heat with a single reservoir. (No 100% efficient heat engine; PMM-2 impossible.)
- Clausius: it is impossible to construct a device that, operating in a cycle, transfers heat from a colder to a hotter body without external work. (Refrigerator needs work.)
- Both statements are equivalent: violation of one implies violation of the other.

Heat engine $\eta = W/Q_1 = 1 - Q_2/Q_1$; Carnot $\eta = 1 - T_2/T_1$
Refrigerator $COP_R = Q_2/W = T_2/(T_1 - T_2)$; Heat pump $COP_{HP} = Q_1/W = T_1/(T_1 - T_2)$
$COP_{HP} = COP_R + 1$

- Carnot cycle: two reversible isothermals + two reversible adiabatics. Most efficient between two given temperatures; efficiency depends only on reservoir temperatures, not on the working fluid.
- Carnot theorem: no engine between two reservoirs can be more efficient than a reversible engine. Corollary: all reversible engines between the same reservoirs have equal efficiency.
- Thermodynamic (Kelvin) temperature scale: $Q_1/Q_2 = T_1/T_2$ – independent of working substance.
- Engines in series with equal work: $T = (T_1 + T_2)/2$. With equal efficiency: $T = \sqrt{(T_1T_2)}$.
- Sources of irreversibility: friction, free expansion, heat transfer through finite ΔT, mixing, unrestrained expansion, throttling, inelastic deformation, electric resistance.

★ **PYQ-TREND:** Kelvin–Planck vs Clausius statements. Carnot efficiency depends only on temperatures. COP_HP − COP_R = 1. Intermediate temperature for equal efficiency $\sqrt{(T_1T_2)}$. Carnot cycle not practical because isothermal heat transfer is too slow and adiabatic processes need very fast motion.

✖ **COMMON MISTAKE:** Always use absolute temperature (K) in Carnot formulas. 1 − 27/327 is wrong; use 1 − 300/600 = 0.5.

5.5 Entropy

$dS = \delta Q_{rev}/T$; Clausius inequality $\oint \delta Q/T \le 0$ (= 0 reversible, < 0 irreversible)

$\Delta S = m\,c_v \ln(T_2/T_1) + m R \ln(V_2/V_1) = m\,c_p \ln(T_2/T_1) - m R \ln(p_2/p_1)$

Isothermal $\Delta S = mR \ln(V_2/V_1)$; Isochoric $\Delta S = m\,c_v \ln(T_2/T_1)$; Isobaric $\Delta S = m\,c_p \ln(T_2/T_1)$

Polytropic: $\Delta S = m\,c_n \ln(T_2/T_1)$, where polytropic specific heat $c_n = c_v (n - \gamma)/(n - 1)$

Mixing / heat transfer $\Delta S_{universe} = Q(1/T_{cold} - 1/T_{hot}) > 0$

- Principle of increase of entropy: $\Delta S_{universe} \ge 0$. Entropy of an isolated system never decreases.
- Reversible adiabatic process = isentropic. An irreversible adiabatic process has $\Delta S > 0$. (Isentropic need not be adiabatic only if heat removal exactly cancels generation – but usually treated as reversible adiabatic.)
- On T–s diagram: constant-volume lines are steeper than constant-pressure lines (slope T/cv > T/cp).
- Third law (Nernst): entropy of a perfect crystal at absolute zero is zero.

★ **PYQ-TREND:** Reversible adiabatic = isentropic. Entropy of universe always increases for real processes. Slope of isochor vs isobar on T–s. Entropy change for isothermal expansion. Third law statement. Unit of entropy kJ/K.

5.6 Availability (Exergy) and Irreversibility

- Available energy (AE) = max work obtainable from heat Q at T with surroundings at T_0 = $Q(1 - T_0/T)$. Unavailable energy = $T_0\,\Delta S$.
- Availability of closed system: $\varphi = (U - U_0) + p_0(V - V_0) - T_0(S - S_0)$. Open system (flow): $\psi = (h - h_0) - T_0(s - s_0) + V^2/2 + gz$.
- Gouy–Stodola theorem: Irreversibility $I = T_0\,\Delta S_{gen} = W_{rev} - W_{actual}$.
- Second-law efficiency $\eta_{II} = \eta_{actual}/\eta_{rev}$.
- Helmholtz function $F = U - TS$; Gibbs function $G = H - TS$ (chemical equilibrium, phase change at constant T & p: dG = 0).

★ **PYQ-TREND:** Unavailable energy = $T_0\Delta S$. Gouy–Stodola theorem. Gibbs function used for chemical reactions at constant p, T.

5.7 Properties of Pure Substances (Steam)

- Pure substance: homogeneous and invariable chemical composition (water, steam-water mixture, air in single phase).
- Sensible heat, latent heat. At 1 atm water boils at 100 °C with hfg = 2257 kJ/kg.
- Critical point of water: 22.064 MPa (≈ 221.2 bar), 374.15 °C; hfg = 0 at critical point. Triple point: 0.01 °C, 0.611 kPa – solid, liquid and vapour coexist.

```
Dryness fraction x = m_vapour/(m_vapour + m_liquid)
h = hf + x hfg; v = vf + x vfg ≈ x vg; s = sf + x sfg
Superheated: h = hg + cps (T_sup − T_sat); degree of superheat = T_sup − T_sat
```

- Measurement of dryness: barrel calorimeter, separating calorimeter, throttling calorimeter (needs fairly dry steam), combined separating & throttling (for very wet steam).
- Mollier (h–s) diagram – used for steam turbine/nozzle calculations. On it, constant-pressure lines diverge and constant-temperature lines become horizontal in the superheated region far from saturation.
- Gibbs phase rule: F = C − P + 2. Triple point of pure substance: F = 0.

★ **PYQ-TREND:** Critical pressure 221.2 bar & temperature 374.15 °C. Latent heat zero at critical point. Throttling calorimeter for dryness. Triple point DOF = 0. Mollier = h–s chart.

5.8 Ideal and Real Gases

```
pV = mRT = nR¯T; R = R¯/M
Van der Waals: (p + a/v²)(v − b) = R¯T (a − intermolecular attraction, b − finite molecular volume)
Compressibility factor Z = pv/RT; Z = 1 for ideal gas; Zc ≈ 0.27 (most gases), vdW predicts 0.375
```

- Law of corresponding states: all gases at the same reduced pressure and reduced temperature have the same Z (generalised compressibility chart).
- Dalton's law of partial pressure: $p = \Sigma p_i$; Amagat's law of partial volumes.
- Avogadro: 1 kmol of any gas at NTP occupies 22.4 m^3.
- Real gas behaves ideally at low pressure and high temperature.

★ **PYQ-TREND:** Van der Waals constants meaning. Ideal behaviour at low p, high T. Z for ideal gas = 1. Value of R for air 287 J/kg·K. Dalton's law.

5.9 Thermodynamic Relations

- Maxwell's relations: $(\partial T/\partial v)s = -(\partial p/\partial s)v$; $(\partial T/\partial p)s = (\partial v/\partial s)p$; $(\partial p/\partial T)v = (\partial s/\partial v)T$; $(\partial v/\partial T)p = -(\partial s/\partial p)T$.
- Clausius–Clapeyron equation: dp/dT = hfg/(T vfg) – slope of the saturation curve; used to find hfg from p–T data.
- $cp - cv = -T(\partial v/\partial T)p^2(\partial p/\partial v)T \geq 0 \rightarrow cp \geq cv$ always; equal for incompressible substances and at 4 °C for water (max density).

★ **PYQ-TREND:** Clausius–Clapeyron relates latent heat and saturation slope. Maxwell relations derived from Gibbs/Helmholtz functions. cp = cv for water at 4 °C.

✎ **DESCRIPTIVE Q:** Prove that the Kelvin–Planck and Clausius statements of the second law are equivalent.

Suppose the Clausius statement is violated: a device transfers Q_2 from a cold reservoir to a hot one without work. Combine it with a heat engine that takes Q_1 from the hot reservoir, rejects Q_2 to the cold one and produces $W = Q_1 - Q_2$. The cold reservoir now has no net exchange. The composite device takes $(Q_1 - Q_2)$ from the hot reservoir alone and converts it fully into work, which violates Kelvin–Planck. Conversely, a Kelvin–Planck violating engine ($Q_1 \rightarrow W$) driving a refrigerator gives net heat transfer from cold to hot without external work, which violates Clausius. Hence the two statements are equivalent.

✎ **DESCRIPTIVE Q:** Show that the efficiency of a reversible engine operating between two reservoirs depends only on the reservoir temperatures.

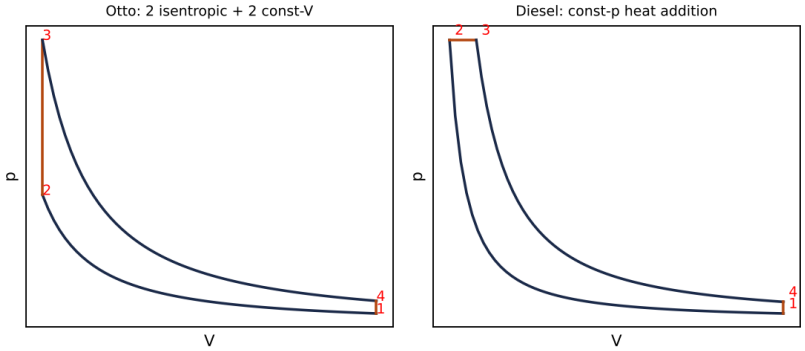
By Carnot's theorem all reversible engines between the same two reservoirs have the same efficiency, so η cannot depend on the working fluid or engine design. It must be a function of the reservoir temperatures only: $Q_1/Q_2 = f(\theta_1, \theta_2)$. Using three reservoirs and composing engines gives $f(\theta_1, \theta_2) = \psi(\theta_1)/\psi(\theta_2)$. Kelvin chose $\psi(\theta) = T$, so $Q_1/Q_2 = T_1/T_2$ and $\eta = 1 - T_2/T_1$.

CHAPTER 6

Air-Standard Cycles & IC Engines

6.1 Air-Standard Cycles

Assumptions: working fluid is air (ideal gas, constant specific heats); closed cycle; combustion replaced by heat addition; exhaust replaced by heat rejection; all processes internally reversible.



Cycle	Processes	Efficiency
Carnot	2 isothermal + 2 isentropic	$1 - T_2/T_1$
Otto (SI)	2 isentropic + 2 constant-volume	$1 - 1/r^{(\gamma-1)}$
Diesel (CI)	2 isentropic + const-p addition + const-v rejection	$1 - (1/r^{(\gamma-1)})\cdot(\rho^{\gamma} - 1)/(\gamma(\rho - 1))$
Dual (limited pressure)	2 isentropic + const-v & const-p addition + const-v rejection	$1 - (1/r^{(\gamma-1)})\cdot(\alpha\rho^{\gamma} - 1)/[(\alpha - 1) + \gamma\alpha(\rho - 1)]$
Brayton/Joule (gas turbine)	2 isentropic + 2 constant-pressure	$1 - 1/rp^{((\gamma-1)/\gamma)}$
Stirling	2 isothermal + 2 constant-volume (with regenerator)	= Carnot
Ericsson	2 isothermal + 2 constant-pressure (with regenerator)	= Carnot
Atkinson	2 isentropic + const-v addition + const-p rejection	Higher than Otto (longer expansion)
Lenoir	No compression; const-v addition, isentropic expansion, const-p rejection	Low (pulse jet)

- r = compression ratio V_1/V_2; ρ = cut-off ratio V_3/V_2; α = pressure ratio (explosion ratio) p_3/p_2; rp = pressure ratio of Brayton.
- Otto efficiency depends only on r (and γ). Increasing r is limited by knocking in SI engines (r = 6–10). CI engines: r = 14–22.
- Diesel efficiency decreases as cut-off ratio increases (more load → lower efficiency). Diesel efficiency → Otto efficiency as ρ → 1.

Comparison (very frequently asked)

Basis of comparison	Order of efficiency
Same compression ratio & heat input	Otto > Dual > Diesel
Same max pressure & temperature (and heat rejection)	Diesel > Dual > Otto
Same max pressure and heat input	Diesel > Dual > Otto

```
Mean effective pressure (Otto): MEP = p₁ r [(α − 1)(r^(γ−1) − 1)] / [(γ − 1)(r − 1)]
MEP = net work / swept volume
Brayton: optimum rp for max work = (T_max/T_min)^(γ/(2(γ−1))); then T₂ = T₄ = √(T₁T₃)
Work ratio (Brayton) = 1 − (T₁/T₃)·rp^((γ−1)/γ)
```

- Gas turbine improvements: regeneration (increases η, not W; effective only at low rp), reheating (increases W, reduces η unless with regeneration), intercooling (reduces compressor work, increases W). Back-work ratio in gas turbines is high (40–80%).

★ **PYQ-TREND:** Otto efficiency formula and its dependence only on r. Order of efficiencies for same r and for same max pressure. Stirling/Ericsson efficiency equals Carnot (with regenerator). Brayton cycle for gas turbines. Optimum pressure ratio for max work. Regeneration effect.

✔ **TOPPER TRICK:** For r = 8, γ = 1.4: r^0.4 = 2.297 → η_Otto = 56.5%. For r = 10: 60.2%. Remember 2^0.4 ≈ 1.32 for fast calculation.

6.2 IC Engine Classification and Working

Basis	SI engine	CI engine
Cycle	Otto	Diesel (practically dual)
Fuel	Petrol (high volatility, high SIT)	Diesel (low volatility, low SIT ~ 250 °C)
Charge introduction	Air–fuel mixture via carburettor / MPFI	Air only; fuel injected at end of compression
Ignition	Spark plug	Self-ignition by hot compressed air
Compression ratio	6–10 (12 for modern GDI)	14–22
Load control	Quantity governing (throttle)	Quality governing (fuel quantity)
Weight & cost	Lighter, cheaper	Heavier (higher pressure), costlier
Thermal efficiency	Lower	Higher
Knocking cause	Late-stage auto-ignition of end charge	Long ignition delay

- Four-stroke: one power stroke per 2 revolutions; valves; better volumetric efficiency and fuel economy, heavier flywheel needed.
- Two-stroke: one power stroke per revolution; ports (inlet/transfer/exhaust); scavenging; theoretically double power; higher fuel loss, lower thermal efficiency, more lubricating oil consumption; used in mopeds, large marine diesels.
- Scavenging types: cross, loop (reverse), uniflow (most efficient).
- Valve timing (4S SI typical): IVO 10–20° bTDC, IVC 30–50° aBDC, EVO 40–50° bBDC, EVC 10–15° aTDC. Valve overlap = both open near TDC (larger in CI and high-speed engines).
- Firing order: 4-cylinder in-line 1-3-4-2 (or 1-2-4-3); 6-cylinder 1-5-3-6-2-4.

★ **PYQ-TREND:** SI uses quantity governing, CI uses quality governing. CR ranges for SI & CI. Valve overlap meaning. Firing order of 4-cylinder engine 1-3-4-2. Two-stroke engine needs a lighter flywheel. Uniflow scavenging most efficient.

6.3 Engine Performance Parameters

Indicated power IP = pm L A n k / 60 (n = N/2 for 4-stroke, N for 2-stroke; k = cylinders)

Brake power BP = 2πNT/60

Friction power FP = IP − BP; mechanical efficiency ηm = BP/IP

Brake thermal efficiency ηbth = BP/(ṁf × CV); indicated thermal ηith = IP/(ṁf × CV)

Volumetric efficiency ηv = actual volume of air inducted / swept volume (at intake conditions)

bsfc = ṁf / BP (kg/kWh)

Swept volume Vs = (π/4)D²L; clearance volume Vc; r = (Vs + Vc)/Vc

Mean piston speed = 2LN/60

- Measurement of friction power: Willan's line (CI engines only – fuel consumption vs BP extrapolated to zero), Morse test (multi-cylinder engines, cut out one cylinder at a time), motoring test, retardation test.
- Heat balance sheet: BP ≈ 25–35%, cooling ≈ 25–35%, exhaust ≈ 30–40%, unaccounted the rest.
- Engine characteristic: bsfc minimum near full load; volumetric efficiency of NA engines 75–90%; superchargers increase density → power.

★ **PYQ-TREND:** Willan's line for CI engines; Morse test for multi-cylinder SI engines. IP formula with n = N/2 for four-stroke. Definitions of ηm, ηv. bsfc units. Supercharging increases power and is more suitable for CI engines (SI gets detonation).

6.4 Combustion in SI Engines

- Stages (Ricardo): ignition lag (preparation phase, chemical), flame propagation (rapid), after-burning.
- Detonation/knocking in SI: auto-ignition of end-gas ahead of flame front → pressure waves, pinging noise, damage.

Factor	Effect on SI knock	Effect on CI knock
Compression ratio ↑	Increases knock	Reduces knock
Inlet temperature ↑	Increases	Reduces
Engine speed ↑	Reduces	Increases (delay covers more crank angle)
Cylinder size ↑	Increases	Reduces
Fuel self-ignition temp ↑	Reduces (good)	Increases (bad)
Spark advance ↑	Increases	–
Turbulence ↑	Reduces	Reduces

- Antiknock rating SI: Octane number – % iso-octane (ON 100) in mixture with n-heptane (ON 0). Antiknock additives: TEL (banned), MTBE, ethanol.
- CI rating: Cetane number – % cetane (n-hexadecane, CN 100) in a mixture with α-methyl naphthalene (CN 0). Cetane improvers: amyl nitrate, ethyl nitrate. Diesel CN ≈ 40–55.

- Good SI fuel = high octane (paraffins (iso), aromatics). Good CI fuel = high cetane (straight-chain paraffins) – opposite characteristics.
- SI combustion chambers: T-head, L-head, I-head (overhead valve – best), F-head; aim for compact chamber, spark plug centrally placed, squish turbulence.
- Pre-ignition: ignition by hot spots before the spark. Run-on (dieseling): engine continues after switching off.
- Air–fuel ratio: stoichiometric for petrol ≈ 14.7:1. Max power at slightly rich (~12.5:1), max economy at slightly lean (~16:1). Cold start & idling need rich mixture.

★ **PYQ-TREND:** Octane number reference fuels (iso-octane & n-heptane), cetane number (cetane & α-methyl naphthalene). Factors which reduce knocking in SI increase it in CI. Stoichiometric A/F 14.7. Detonation in SI occurs at the end of combustion; diesel knock occurs at the start.

6.5 Combustion in CI Engines

- Stages: ignition delay (physical + chemical), period of rapid (uncontrolled) combustion, controlled combustion, after-burning.
- Diesel knock: due to long ignition delay – excessive fuel accumulates and burns suddenly. Reduce delay: high CR, high cetane fuel, high intake temperature, good atomisation.
- CI chambers: open (direct injection, quiescent – large engines), swirl chamber (Ricardo Comet), pre-combustion chamber, air-cell, energy cell (Lanova). Indirect injection chambers need glow plugs for cold starting.
- Fuel injection systems: air injection (obsolete), solid injection – individual pump, common rail (CRDI – high pressure 1500–2500 bar, electronic control), distributor type. Injection pressure ~100–300 bar in older systems.
- Nozzles: pintle (IDI), single-hole, multi-hole (DI).

★ **PYQ-TREND:** Delay period causes diesel knock. Glow plug used in IDI engines. CRDI full form. Higher CR used to reduce delay.

6.6 Fuel Supply, Ignition, Cooling, Lubrication

- Simple carburettor: float chamber, venturi, fuel jet, throttle valve, choke (rich mixture for cold start). Mixture gets richer at high speeds in simple carburettor – compensating devices (compensating jet, air bleed, emulsion tube). MPFI replaced carburettors for emission control.
- Ignition systems: battery (coil) ignition – battery, ignition coil, contact breaker, condenser (to prevent arcing), distributor, spark plug; magneto ignition – used in two-wheelers/racing (better at high speed); electronic ignition (CDI, transistorised).
- Spark plug gap ≈ 0.6–0.9 mm; voltage 10,000–25,000 V. Hot plug for low-speed engines, cold plug for high-speed/high CR engines.
- Cooling: air cooling (fins – motorcycles, aircraft) and water cooling (thermosyphon, pump circulation, pressurised radiator). Thermostat keeps engine at optimum temperature (opens ~70–85 °C). Antifreeze: ethylene glycol.
- Lubrication systems: petroil (two-stroke – oil mixed with petrol ~3–5%), splash, pressure (forced) feed, dry sump & wet sump. Viscosity index (VI): change of viscosity with temperature; high VI oils preferred. SAE grading (e.g., 20W-40 multi-grade; W = winter).
- Governing: quantity (SI), quality (CI), hit-and-miss (gas engines).

★ **PYQ-TREND:** Purpose of condenser in battery ignition (prevents arcing at contact points). Choke gives rich mixture during cold start. Petroil lubrication in two-stroke engines. Magneto ignition is better at high speed. Thermostat function. SAE 20W meaning. Cold plug for high-compression engines.

6.7 Emissions and Alternative Fuels

Pollutant	Cause	Control
CO	Incomplete combustion, rich mixture	Catalytic converter (oxidation)
HC (unburnt)	Flame quenching at walls, crevices, misfire	Catalytic converter
NOx	High combustion temperature	EGR, catalytic converter (reduction), retarding timing
Particulates/soot (CI)	Rich zones, incomplete combustion	DPF, high injection pressure

- Three-way catalytic converter (Pt, Pd, Rh): oxidises CO & HC and reduces NOx; works efficiently only near stoichiometric A/F ratio (closed-loop control with λ (oxygen) sensor). Lead poisons it.
- EGR reduces NOx by lowering peak temperature. SCR with urea (AdBlue) for diesel NOx.
- Bharat Stage norms: BS-VI implemented in India from April 2020 (BS-V skipped).
- Alternative fuels: CNG (methane, ON ≈ 120), LPG (propane + butane), ethanol (ON ≈ 108), methanol, biodiesel (transesterified vegetable oil – esters), hydrogen. Biogas ≈ 55–65% methane.

★ **PYQ-TREND:** EGR controls NOx. Three-way catalyst metals. Main constituent of CNG = methane, LPG = propane + butane. BS-VI from 2020. Biodiesel is produced by transesterification.

6.8 Supercharging and Turbocharging

- Supercharging: increases density of intake charge with a compressor driven from crankshaft (Roots blower, vane, centrifugal). Turbocharging uses an exhaust-driven turbine – recovers exhaust energy, better efficiency, turbo lag.
- Intercooler (aftercooler) cools compressed air to raise density further and to limit knock.
- Benefits: higher power per unit weight, compensates altitude loss (aircraft engines), better combustion in CI.

★ **PYQ-TREND:** Turbocharger is driven by exhaust gases. Supercharging is more suitable for CI engines. Roots blower is a positive displacement supercharger.

✎ **DESCRIPTIVE Q:** Compare SI and CI engines on the basis of cycle, fuel, ignition, compression ratio, governing and efficiency.

SI engines work on the Otto cycle with petrol. The air–fuel mixture is prepared outside (carburettor/port injection) and ignited by a spark. CR is limited to 6–10 by knock, and power is controlled by quantity governing through a throttle. CI engines work on the Diesel (practically dual) cycle. Only air is compressed (CR 14–22) and diesel is injected near TDC, so it self-ignites. Power is controlled by quality governing, varying fuel quantity per cycle. CI engines have higher thermal efficiency, are heavier and costlier, run at lower speeds and have higher torque. They are noisier and produce more NOx and particulates; SI engines produce more CO and HC.

CHAPTER 6B

Automobile Engineering Essentials

6B.1 Vehicle Layout and Chassis

- Chassis = frame + engine + transmission + suspension + steering + brakes + wheels (everything except the body). Frame types: conventional (ladder – trucks, SUVs), integral/unitised (monocoque – modern cars, body acts as frame), semi-integral.
- Frame cross-sections: channel, box, tubular. Frames are made of low-carbon steel (pressed).
- Drive layouts: front engine rear-wheel drive (FR – conventional), front engine front-wheel drive (FF – compact cars, transverse engine), rear engine rear-wheel drive (RR – some buses), four-wheel drive (4WD/AWD – transfer case).
- Resistances to vehicle motion: rolling resistance (≈ μ_r W), air (aerodynamic) resistance (½ρC_dAV²), gradient resistance (W sinθ). Tractive effort must exceed their sum to accelerate.

```
Total resistance R = μ_r W + ½ρC_d A V² + W sinθ
Power required = R × V; tractive effort = T_engine × G × η_t / r_wheel
```

★ **PYQ-TREND:** Monocoque/integral construction for cars. Air resistance ∝ V². Components of the chassis. Ladder frame for heavy vehicles.

6B.2 Clutch

- Function: connect/disconnect engine from transmission smoothly; allow gear change; permit gradual engagement.
- Requirements: torque capacity, gradual engagement, heat dissipation, low inertia of driven parts, damping of torsional vibration (torsional springs in clutch plate), ease of operation.
- Types: single-plate (cars – dry), multi-plate (two-wheelers – wet, in oil; heavy vehicles), cone, centrifugal (mopeds, automatic engagement by speed), semi-centrifugal, diaphragm-spring clutch (modern cars – uniform pressure, fewer parts), fluid coupling (automatic).
- Friction facings: asbestos (old), now organic/ceramic/sintered metal; μ ≈ 0.3–0.4.
- Clutch free play (pedal free travel) ensures full engagement and avoids slipping and release-bearing wear.
- Clutch slip causes: worn facing, oil on facing, weak springs, no free play. Clutch drag (spin): excessive free play, warped plate.

★ **PYQ-TREND:** Diaphragm spring clutch in modern cars. Wet multi-plate clutch in motorcycles. Centrifugal clutch in mopeds. Torsional damper springs in the clutch plate. Purpose of pedal free play.

6B.3 Gearbox (Transmission)

- Purpose: vary torque and speed to suit the load (starting, gradients, high speed); provide reverse; provide neutral.

- Sliding mesh (oldest – spur gears slide into mesh, noisy, double declutching), constant mesh (all gears always in mesh, dog clutches engage; helical gears – quieter), synchromesh (cone friction synchronisers equalise speeds before engagement – modern manual gearboxes), epicyclic (automatic transmissions – planetary sets with brake bands and clutches), CVT (continuously variable – belt & variable pulleys).
- Overdrive: output speed > input (gear ratio < 1) – economy at cruising. Direct drive: ratio 1:1 (top gear in many gearboxes).
- First gear gives maximum torque multiplication and lowest speed.
- Automatic transmission: torque converter + epicyclic gear sets + hydraulic/electronic control. AMT (automated manual) – manual gearbox with actuators.
- Transfer case in 4WD splits power to front and rear axles.

★ **PYQ-TREND:** Synchromesh avoids gear clash using cone synchronisers. Constant mesh uses dog clutches. Epicyclic gears in automatic transmissions. Overdrive ratio < 1. Double declutching needed in sliding mesh.

6B.4 Propeller Shaft, Differential and Axles

- Propeller (drive) shaft: transmits torque from gearbox to rear axle in FR layouts; hollow tubular steel (high torsional stiffness, light); universal joints (Hooke's) at each end allow angular movement; slip joint (splines) allows length change with suspension movement.
- For constant velocity in FWD drives: CV joints (Rzeppa – ball type, tripod).
- Final drive: reduces speed and turns drive through 90° (bevel/hypoid gears – crown wheel & pinion). Hypoid allows lower propeller shaft (lower floor), but needs EP (extreme pressure) lubricant.
- Differential: allows the outer wheel to rotate faster than the inner wheel on turns while transmitting equal torque. Sun (side) gears + planet (differential) pinions in a cage on the crown wheel. Limitation: if one wheel slips (mud), the other gets no torque → limited-slip differential (LSD) / differential lock.
- Rear axles: semi-floating (cars – axle shaft supports vehicle weight + torque), three-quarter floating, full floating (trucks – axle shaft carries only torque; weight on axle housing).
- Hotchkiss drive (leaf springs take torque reaction & driving thrust) vs torque tube drive.

★ **PYQ-TREND:** Function of the differential. Slip joint on the propeller shaft. CV joints in FWD. Hypoid gear needs EP oil. Full-floating axle shafts carry only torque. Crown wheel & pinion in the final drive.

6B.5 Steering System

- Ackermann principle: for true rolling on turns, the axes of all wheels must meet at a common point (instantaneous centre) on the rear axle extension: $\cot \varphi - \cot \theta = c/b$ (c = pivot distance, b = wheelbase).
- Steering gearboxes: rack & pinion (cars – direct, light), recirculating ball (heavy vehicles – low friction), worm & sector, worm & roller, cam & lever. Power steering: hydraulic or electric (EPS).
- Steering ratio = steering wheel angle / road-wheel angle (cars 14–20:1).

Wheel alignment (steering geometry)

Angle	Definition / effect
Camber	Inward/outward tilt of the wheel from vertical (front view); positive = top outward; affects tyre wear (one-side wear)
Castor (caster)	Backward tilt of the king pin from vertical (side view); gives directional stability & self-centring (positive castor 2–8°)
King pin inclination (KPI/SAI)	Inward tilt of king pin (front view); reduces steering effort, self-centring, reduces scrub radius
Toe-in / toe-out	Front of wheels closer (toe-in) or farther apart than the rear (top view); toe-in compensates for positive camber/rolling forces – prevents tyre scuffing
Toe-out on turns	Inner wheel turns more than outer (Ackermann)
Included angle	Camber + KPI
Scrub radius	Distance between the tyre contact centre and the KPI axis intersection with the ground

- Under-steer: front slip angle > rear (vehicle turns less than intended – safer). Over-steer: rear slip angle > front.

★ **PYQ-TREND:** Castor gives directional stability & self-righting. Camber definition. Toe-in purpose. Rack & pinion in cars. Ackermann condition. Under-steer vs over-steer. KPI reduces steering effort.

6B.6 Brakes

- Function: retard/stop the vehicle by converting KE into heat. Stopping distance = $v^2/(2\mu g)$ (ideal).
- Drum brakes (internal expanding shoes – leading & trailing shoes; leading shoe is self-energising) vs disc brakes (caliper & pads – better heat dissipation, less fade, self-adjusting, used on front wheels since ~60–70% of braking load shifts to the front).
- Brake fade: reduction of braking effectiveness due to heat (μ drops).
- Actuation: mechanical (handbrake/parking), hydraulic (Pascal's law – master cylinder & wheel cylinders; brake fluid glycol-based, high boiling point, non-corrosive to rubber), air/pneumatic (heavy vehicles – compressor, reservoir), vacuum servo (booster using intake manifold vacuum), electric/regenerative (EVs), exhaust & engine brakes, retarders (eddy-current) for heavy vehicles.
- ABS (anti-lock braking system): prevents wheel lock-up by modulating brake pressure (wheel speed sensors, ECU, hydraulic modulator) → steerability maintained during hard braking.
- Tandem master cylinder: dual circuit for safety.
- Weight transfer during braking: load on front axle increases.

★ **PYQ-TREND:** Disc brakes resist fade better. Hydraulic brakes work on Pascal's law. ABS prevents wheel locking and retains steering control. Air brakes for heavy vehicles. Leading shoe is self-energising. Brake fade definition.

6B.7 Suspension, Wheels and Tyres

- Suspension: isolates the body from road shocks, maintains tyre contact, supports weight, resists roll and pitch.

- Springs: leaf (semi-elliptic – trucks, also locates the axle), coil (cars), torsion bar, air (buses – adjustable height), rubber, hydro-pneumatic (Citroën).
- Shock absorber (damper): telescopic hydraulic – dissipates spring energy as heat (it is actually a damper, not a spring).
- Independent suspension (each wheel moves independently): MacPherson strut (most common front suspension in cars), double wishbone, multi-link, trailing arm. Rigid (beam) axle for trucks.
- Stabiliser (anti-roll) bar: torsion bar linking left & right suspension – reduces body roll on turns.
- Sprung mass (body) vs unsprung mass (wheels, axles, brakes) – low unsprung mass improves ride & road holding.
- Tyres: radial ply (plies at 90° to bead, steel belts – better grip, lower rolling resistance, longer life, standard today) vs cross/bias ply (plies at 30–40°, stiffer sidewall – cheaper, heavy duty). Tubeless tyres: self-sealing, safer on puncture.
- Tyre designation 195/65 R15 91H: width 195 mm, aspect ratio 65%, radial, rim diameter 15 inch, load index 91, speed rating H.
- Wheels: pressed steel disc, alloy (Al/Mg – light, better heat dissipation), wire spoke.

★ **PYQ-TREND:** MacPherson strut for front suspension. Shock absorber dissipates energy. Anti-roll bar reduces body roll. Radial tyres have lower rolling resistance. Tyre designation decoding. Leaf springs in heavy vehicles.

6B.8 Electrical, Electronic and Emission Systems

- Battery: lead–acid, 12 V (6 cells × 2.1 V); electrolyte dilute H_2SO_4 (specific gravity ~1.28 fully charged, measured by hydrometer). Capacity in ampere-hours.
- Charging system: alternator (AC generator with rectifier – diodes), voltage regulator. Starting: starter motor (DC series motor – high starting torque) with Bendix drive or solenoid-operated overrunning clutch engaging the flywheel ring gear.
- Electronic fuel injection (MPFI) sensors: MAP/MAF (air flow), throttle position, coolant temperature, oxygen (λ) sensor, crank position, knock sensor. ECU controls injection timing & duration.
- OBD (on-board diagnostics). CRDI in diesels.
- Hybrid vehicles: series, parallel, series-parallel (power-split). EVs: Li-ion battery, BLDC/PMSM/induction motors, regenerative braking.

★ **PYQ-TREND:** Starter motor is a DC series motor. Battery cell voltage 2 V (12 V = 6 cells). Hydrometer checks electrolyte specific gravity. Alternator needs rectifier diodes. Oxygen sensor for closed-loop A/F control. Regenerative braking in EVs.

✎ **DESCRIPTIVE Q:** Explain the need for a differential in an automobile and describe its working.

On a turn the outer driving wheel covers a larger arc than the inner one in the same time, so it must rotate faster. A solid axle would force equal speeds, making one wheel slip, wearing tyres and stressing the axle. In a differential, the crown wheel (driven by the propeller-shaft pinion) carries a cage with planet pinions that mesh with two sun (side) gears splined to the axle half-shafts. Driving straight, the cage, pinions and both sun gears rotate as one unit (pinions do not spin on their axes). On a turn, the inner wheel's resistance slows its sun gear; the pinions then rotate on their own axes and speed up the other sun gear by the same amount. The average of the wheel speeds equals the cage speed and equal torque goes to both wheels. Its

drawback is that a slipping wheel takes all the motion, which LSDs and differential locks overcome.

CHAPTER 7

Refrigeration & Air Conditioning

7.1 Fundamentals

- Refrigeration: removing heat from a space/substance to keep it below the surrounding temperature.

```
1 TR (ton of refrigeration) = 3.5167 kW ≈ 211 kJ/min = 12,660 kJ/h (≈ 50 kcal/min)
```

- 1 TR = heat to be removed to freeze 1 short ton (907 kg) of water at 0 °C into ice in 24 hours.

```
COP_ref = Refrigerating effect / Work input; Carnot COP = T_L/(T_H − T_L)
Relative COP = actual COP / theoretical COP
```

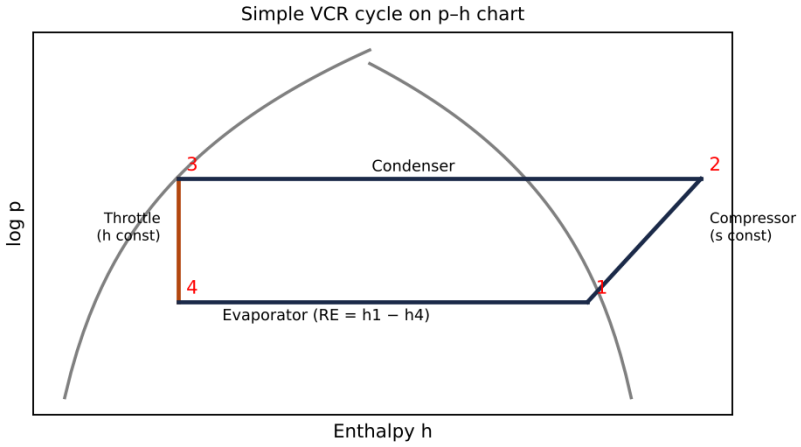
- Reversed Carnot cycle is the ideal but impractical (wet compression, isothermal heat transfer). Bell–Coleman (reversed Brayton/Joule) cycle is used in air refrigeration (aircraft cabins – light weight, but low COP).

```
Bell–Coleman COP = 1/(rp^((γ−1)/γ) − 1)
```

★ **PYQ-TREND:** 1 TR = 3.5 kW (211 kJ/min). Air refrigeration used in aircraft because of low weight per TR, despite low COP. Carnot COP decreases as the evaporator temperature decreases.

7.2 Vapour Compression Refrigeration (VCR) System

- Components & processes: compressor (isentropic compression, 1–2) → condenser (constant-pressure heat rejection, 2–3) → expansion device (throttling, isenthalpic, 3–4) → evaporator (constant-pressure heat absorption, 4–1).




```
Refrigerating effect RE = h₁ − h₄ (h₄ = h₃); Work W = h₂ − h₁; COP = (h₁ − h₄)/(h₂ − h₁)
Heat rejected in condenser = h₂ − h₃ = RE + W
Refrigerant mass flow ṁ = 211 × TR / RE (kg/min, RE in kJ/kg)
```

- p–h chart (Mollier chart for refrigerants) is used for VCR analysis; throttling is a vertical line on it.

- Subcooling of liquid (after condenser): increases RE and COP (no change in work).
- Superheating of vapour: increases RE but also work; COP may increase or decrease depending on refrigerant; useful superheat prevents liquid entering compressor (dry compression).
- Effect of lowering evaporator pressure/temperature: RE decreases, work increases, COP decreases. Raising condenser pressure: COP decreases.
- Compressors: reciprocating (small–medium), rotary, scroll (domestic AC), screw (large), centrifugal (very large capacity, > 200 TR). Hermetically sealed compressors in domestic refrigerators.
- Expansion devices: capillary tube (domestic refrigerator, small units – no moving parts, constant restriction), thermostatic expansion valve (TEV – maintains constant superheat), automatic (constant pressure) expansion valve, float valve (flooded evaporators), electronic expansion valve.
- Condensers: air-cooled (domestic), water-cooled (shell & tube), evaporative (cooling tower combination). Evaporators: flooded, dry expansion; bare tube, finned.
- Domestic refrigerator: hermetic compressor, wire-and-tube condenser at the back, capillary tube, evaporator in the freezer.

★ **PYQ-TREND:** Throttling in expansion valve is isenthalpic. Capillary tube in domestic refrigerators. Subcooling increases COP. Lowering evaporator temperature reduces COP. TEV maintains constant superheat. Centrifugal compressors for large capacity plants. COP from p–h chart values.

✔ **TOPPER TRICK:** In VCR numericals, $h_4 = h_3$ (throttling) always. Read h_1, h_2, h_3 from the given table and plug them into $COP = (h_1 - h_3)/(h_2 - h_1)$.

7.3 Refrigerants

Refrigerant	Formula/Code	Use / remark
Ammonia	R-717 (NH_3)	Cold storage, ice plants; highest RE per kg; toxic, attacks copper (use steel)
CO_2	R-744	Ships, dry ice; very high pressure; natural, ODP 0, GWP 1
Freon-12	R-12 (CCl_2F_2)	Old domestic refrigerators; CFC – banned (ozone depletion)
Freon-22	R-22 ($CHClF_2$)	Old window/split ACs; HCFC – being phased out
R-134a	$CH_2F–CF_3$	Domestic refrigerators, car AC; HFC, ODP = 0, GWP high
R-410A, R-32	HFC blends	Modern ACs
R-600a	Isobutane	Modern domestic refrigerators (HC, natural, flammable)
R-290	Propane	Natural refrigerant
R-718	Water	Absorption (LiBr–water), steam jet systems
R-11	CCl_3F	Centrifugal chillers (old)

- Numbering rule for halocarbons: R-(m−1)(n+1)(p) where m = C atoms, n = H atoms, p = F atoms. R-7xx: inorganic, xx = molecular weight (NH_3 = 17 → R-717; H_2O = 18 → R-718; CO_2

= 44 → R-744).

- Desirable properties: low boiling point, high latent heat, low specific volume of vapour, high critical temperature, low freezing point, non-toxic, non-flammable, non-corrosive, chemically stable, zero ODP, low GWP, easy leak detection, good miscibility with oil (for small systems).
- Leak detection: ammonia – sulphur candle (white fumes) or litmus; halocarbons – halide torch (green flame), electronic detector; soap solution for all.
- Montreal Protocol (1987) – phase-out of ozone-depleting CFC/HCFC. Kyoto Protocol / Kigali Amendment (2016) – phasing down HFCs (GWP).

★ **PYQ-TREND:** Refrigerant numbering (R-717 = NH_3, R-744 = CO_2, R-718 = water). Ammonia is used in cold storage, not with copper. R-134a has zero ODP. Halide torch for Freon leak. Montreal protocol relates to ozone.

7.4 Vapour Absorption Refrigeration (VAR)

- Compressor replaced by absorber + pump + generator (thermal compressor). Uses low-grade heat (waste heat, solar, steam). Pump work is very small.

System	Refrigerant	Absorbent	Use
Aqua-ammonia	NH_3	Water	Sub-zero (industrial, cold storage)
LiBr–water	Water	LiBr	Above 0 °C (large AC chillers)
Electrolux (3-fluid)	NH_3	Water (+ H_2 to equalise pressure)	Domestic, no pump/moving parts

- Analyser and rectifier remove water vapour from ammonia in aqua-ammonia systems. Heat exchanger between absorber and generator improves COP.

```
Max COP of VAR = [(T_g − T_o)/T_g] × [T_e/(T_o − T_e)] (heat engine × refrigerator)
```

- COP of VAR (≈ 0.5–0.7 single-effect) is lower than VCR but energy input is cheap heat.

★ **PYQ-TREND:** Electrolux refrigerator uses hydrogen as a third fluid and has no moving parts. In LiBr–water system water is the refrigerant. Ideal COP of VAR formula. Generator replaces compressor's high-pressure side function.

7.5 Other Refrigeration Systems

- Steam jet refrigeration: water as refrigerant, flash cooling under vacuum; for chilled water above 0 °C.
- Thermoelectric (Peltier effect): solid state, low COP – portable coolers, electronics.
- Vortex tube (Ranque–Hilsch): compressed air splits into hot and cold streams; spot cooling.
- Cryogenics (< −150 °C): Linde–Hampson (J-T throttling), Claude (expansion engine) systems for liquefying air/gases.
- Dry ice = solid CO_2 (sublimes at −78.5 °C at 1 atm).

★ **PYQ-TREND:** Peltier effect in thermoelectric refrigeration. Vortex tube has no moving parts. Dry ice temperature −78.5 °C.

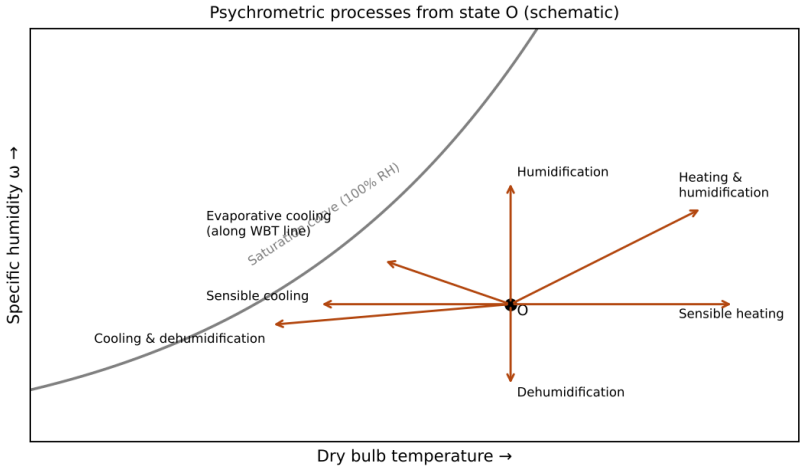
7.6 Psychrometry

- Psychrometry: study of moist air (dry air + water vapour). Dalton's law: p = pa + pv.

Term	Definition
Dry bulb temperature (DBT)	Temperature by ordinary thermometer
Wet bulb temperature (WBT)	Temperature by thermometer with wetted wick, air moving over it
Dew point temperature (DPT)	Temperature at which vapour starts condensing when cooled at constant pressure (saturation temp at pv)
Wet bulb depression	DBT − WBT (zero for saturated air)
Dew point depression	DBT − DPT
Specific humidity (humidity ratio) ω	Mass of vapour per kg dry air
Relative humidity φ	pv / ps (at same DBT)
Degree of saturation μ	ω/ωs

```
ω = 0.622 pv/(p − pv) kg/kg of dry air
φ = pv/ps; μ = φ(p − ps)/(p − pv)
Enthalpy of moist air h = 1.005 t + ω(2500 + 1.88 t) kJ/kg dry air
```

- For saturated air: DBT = WBT = DPT. Otherwise DBT > WBT > DPT.
- WBT ≈ adiabatic saturation temperature (thermodynamic WBT) for air–water mixtures (Lewis number ≈ 1).



Process	On psychrometric chart
Sensible heating / cooling	Horizontal line (ω constant)
Humidification / dehumidification (pure)	Vertical line (DBT constant)
Cooling & dehumidification	Down and to the left (summer AC) – towards ADP of coil
Heating & humidification	Up and to the right (winter AC)
Evaporative (adiabatic) cooling	Along constant WBT line (desert coolers)
Chemical dehumidification (adsorption with silica gel)	Along constant enthalpy line, DBT increases
Adiabatic mixing of two streams	Point on the straight line joining the two states (by mass ratio)

```
Sensible heat factor SHF = SH/(SH + LH); bypass factor BPF = (t₂ − t_ADP)/(t₁ − t_ADP);
contact factor = 1 − BPF
```

- Comfort conditions (human): DBT 21–25 °C, RH 40–60%, air velocity 6–9 m/min. Effective temperature: combined index of DBT, humidity, air motion.
- Desert cooler: evaporative cooling – effective in hot dry climate, fails in humid climate.
- Air-conditioning systems: window, split, packaged, central (all-air, all-water, air-water). Year-round AC includes heating & cooling.
- Cooling load sources: sensible (walls, roof, glass via solar radiation, people, lights, equipment, infiltration) and latent (people, infiltration). A sitting person gives roughly 100–120 W total.

★ **PYQ-TREND:** Order DBT > WBT > DPT; all equal at saturation. Specific humidity formula 0.622 pv/(p − pv). Sensible heating = horizontal line. Evaporative cooling follows constant WBT. Dew point definition. Comfort conditions. Bypass factor definition. Chemical dehumidification follows constant enthalpy.

✖ **COMMON MISTAKE:** Relative humidity is not the same as specific humidity. RH falls during sensible heating although ω is unchanged.

✎ **DESCRIPTIVE Q:** Explain the working of a simple vapour compression refrigeration system with a p–h diagram and state the effect of subcooling and superheating.

Low-pressure saturated (or slightly superheated) vapour enters the compressor and is compressed isentropically to condenser pressure, becoming superheated (1–2). In the condenser it rejects heat at constant pressure, is desuperheated and condensed to saturated or subcooled liquid (2–3). The liquid is throttled in the expansion valve at constant enthalpy to evaporator pressure, partly flashing to vapour (3–4). In the evaporator it absorbs heat from the space at constant pressure and evaporates (4–1). COP = $(h_1 - h_4)/(h_2 - h_1)$. Subcooling moves point 3 to the left, increasing RE without extra work, so COP rises. Superheating in the evaporator increases RE and also compressor work; it protects the compressor from liquid slugging, and the net effect on COP depends on the refrigerant.

CHAPTER 8

Steam Engineering, Power Plants & Compressors

8.1 Rankine Cycle

- Processes: pump (isentropic compression of liquid, 1–2) → boiler (constant-pressure heat addition, 2–3) → turbine (isentropic expansion, 3–4) → condenser (constant-pressure heat rejection, 4–1).

```
η_Rankine = [(h₃ − h₄) − (h₂ − h₁)] / (h₃ − h₂)
Pump work wp = v_f (p₂ − p₁) (small; often neglected)
Specific steam consumption SSC = 3600 / w_net (kg/kWh, w_net in kJ/kg)
Work ratio = w_net / w_turbine (close to 1 for Rankine — an advantage over gas cycles)
```

- Rankine is preferred over Carnot for steam plants: Carnot needs compression of a wet mixture (difficult) and gives a low work ratio.
- Methods to improve Rankine efficiency: raise boiler pressure (but increases moisture at exhaust), superheating (raises mean temperature and dryness), lower condenser pressure (more work but more moisture), reheating (mainly improves exhaust dryness; small gain in η), regeneration with feedwater heaters (raises mean temperature of heat addition – increases η, reduces work per kg and condenser size).
- Feedwater heaters: open (direct contact, deaerator – mixing) and closed (shell & tube).
- Exhaust dryness should not drop below ~0.88 to avoid blade erosion.
- Binary vapour cycle (mercury–steam), combined cycle (Brayton topping + Rankine bottoming via HRSG – efficiency 55–60%), cogeneration (power + process heat).

★ **PYQ-TREND:** Pump work is small due to low specific volume of water. Effect of superheat, reheat, regeneration on efficiency and dryness. Reheating mainly increases dryness of exhaust steam. Regeneration increases efficiency. SSC definition. Combined cycle uses HRSG.

8.2 Boilers (Steam Generators)

Basis	Type A	Type B
Contents in tube	Fire-tube (hot gases in tubes): Cochran, Cornish, Lancashire, Locomotive, Scotch marine	Water-tube: Babcock & Wilcox, Stirling, La Mont, Benson, Loeffler, Velox
Circulation	Natural (density difference): B&W, Lancashire	Forced: La Mont, Benson, Loeffler
Pressure	Sub-critical (most boilers)	Supercritical (Benson – above 221.2 bar, no drum)
Axis	Vertical: Cochran, simple vertical	Horizontal: Lancashire, Cornish, B&W
Firing	Internally fired: Cochran, Lancashire, Locomotive	Externally fired: Babcock & Wilcox

Boiler	Key fact
Cochran	Vertical, multi-tubular, fire-tube, internally fired, small capacity
Cornish	Single flue tube, horizontal, fire-tube
Lancashire	Two flue tubes, horizontal, fire-tube, internally fired; stationary, 15 bar
Locomotive	Multi-tubular fire-tube, high steam rate, draught by exhaust steam blast
Babcock & Wilcox	Horizontal straight-tube water-tube, externally fired, natural circulation
La Mont	Forced circulation, high-pressure, water-tube
Benson	Supercritical, once-through, drumless (no steam drum)
Loeffler	Steam used to evaporate water (superheated steam circulated); avoids salt deposition in tubes
Velox	Gas velocity exceeds sound velocity (supercharged combustion) – very high heat transfer

- Mountings (for safety & control – mounted on boiler): water level indicator (2 per boiler), pressure gauge (Bourdon), safety valves (2: dead-weight, lever, spring-loaded, high-steam low-water), fusible plug (melts if water level falls dangerously low – lead/tin alloy), steam stop valve, feed check valve, blow-off cock, manhole.
- Accessories (for efficiency): economiser (heats feedwater using flue gases), air preheater (heats combustion air), superheater, feed pump, injector (feeds water using steam jet – locomotive), steam separator, steam trap.
- Order of heat recovery in flue gas path: superheater → (reheater) → economiser → air preheater → chimney.
- Draught: natural (chimney) and artificial – forced (fan at base, positive pressure), induced (fan at chimney base, negative pressure), balanced (both). Steam jet draught in locomotives.

```
Chimney draught h_w = 353 H (1/T_a − ((m_a+1)/m_a)(1/T_g)) mm of water (H in m)
Equivalent evaporation = m_s(h − h_f)/2257 (from and at 100 °C)
Boiler efficiency = m_s(h − h_fw) / (m_f × CV)
```

- Boiler (IBR) – Indian Boiler Regulations 1950; Indian Boilers Act 1923.

★ **PYQ-TREND:** Fire-tube vs water-tube examples (B&W is water-tube; Cochran/Lancashire/Locomotive are fire-tube). Benson is supercritical and drumless. Fusible plug function. Mountings vs accessories (economiser is an accessory; safety valve is a mounting). Economiser heats feedwater; air preheater heats air. Loeffler avoids scale. Equivalent evaporation 2257 kJ/kg.

✔ **TOPPER TRICK:** Mountings = safety items mounted on the boiler. Accessories = efficiency boosters. If it saves fuel, it is an accessory.

8.3 Steam Nozzles

- Nozzle converts enthalpy into kinetic energy. Convergent (subsonic), convergent–divergent (supersonic exit), divergent (diffuser for subsonic is divergent).

```
Exit velocity V = 44.72 √(h₁ − h₂) (Δh in kJ/kg)
Critical pressure ratio p_c/p₁ = (2/(n+1))^(n/(n−1))
```

Steam/gas	n	Critical pressure ratio
Superheated steam	1.3	0.546
Dry saturated steam	1.135	0.577
Air (γ = 1.4)	1.4	0.528

- Choked flow: mass flow is maximum when throat pressure equals critical pressure; Mach number = 1 at the throat; reducing back pressure further does not increase flow.
- Supersaturated (metastable) flow: steam expands too fast to condense; actual discharge more than theoretical (by 2–5%); Wilson line ≈ 0.96–0.97 dryness.
- Nozzle efficiency = actual enthalpy drop / isentropic drop (0.90–0.95).

★ **PYQ-TREND:** Critical pressure ratio values 0.528 (air), 0.546 (superheated), 0.577 (saturated). At throat Mach = 1. Supersaturated flow increases discharge. Velocity formula 44.72√Δh.

8.4 Steam Turbines

Impulse turbine	Reaction turbine
Pressure drop only in nozzles; constant pressure across moving blades	Pressure drops in both fixed and moving blades
Blades symmetrical (profile)	Aerofoil blades
De Laval (simple impulse), Curtis (velocity compounded), Rateau (pressure compounded)	Parsons (50% reaction)
Partial admission possible	Full admission required
Smaller size for same power, fewer stages	More stages, larger
Efficiency lower	Efficiency higher

```
Blade (diagram) efficiency – impulse: η_b = 2(Vb/V₁)(cos α − Vb/V₁)(1 + k·cosβ₂/cosβ₁)
Optimum blade speed ratio – single-stage impulse: Vb/V₁ = cos α/2; max η = cos²α
Parsons (50% reaction): Vb/V₁ = cos α; max η = 2cos²α/(1 + cos²α)
Degree of reaction R = Δh_moving / Δh_stage
Work done per kg = Vb (Vw₁ ± Vw₂) (+ when the whirl components are in opposite directions)
```

- Compounding (reduces rotor speed): velocity compounding (Curtis – one nozzle set, several moving rows with guide blades), pressure compounding (Rateau – several nozzle-blade stages), pressure-velocity compounding.
- De Laval simple impulse turbine runs at very high speed (~30,000 rpm) – needs gearing.
- Governing: throttle governing, nozzle control governing, by-pass governing.
- Losses: nozzle friction, blade friction, leakage, disc friction, residual velocity, wetness, partial admission.

★ **PYQ-TREND:** Parsons turbine is 50% reaction. Optimum speed ratio cos α/2 for impulse, cos α for 50% reaction. Curtis = velocity compounded; Rateau = pressure compounded. De Laval is simple impulse. Degree of reaction definition.

8.5 Steam Condensers and Cooling Towers

- Purpose: lower exhaust pressure (increase work) and recover pure condensate. Types: jet (mixing – parallel, counter, ejector) and surface (shell & tube – condensate reused; used in large power plants).

```
Vacuum efficiency = actual vacuum / ideal vacuum (barometric pressure − partial pressure of steam)
Condenser efficiency = (t_w,out − t_w,in) / (t_sat − t_w,in)
```

- Air leakage reduces vacuum; removed by air extraction pump/steam ejector. Dalton's law used for air + steam partial pressures.
- Cooling towers: natural draught (hyperbolic – large plants), mechanical draught (forced/induced). Water cools by evaporation; approach = outlet water temp − WBT; range = inlet − outlet water temperature.

★ **PYQ-TREND:** Surface condenser used in thermal power plants. Vacuum efficiency definition. Approach and range of cooling tower. Hyperbolic shape for natural draught.

8.6 Power Plant Engineering Basics

Plant	Key points
Thermal (coal)	Rankine cycle; pulverised coal; ESP removes fly ash; efficiency 33–40%
Hydro	Renewable, low running cost, high initial cost; turbines by head
Nuclear	Fission of U-235; moderator (heavy water, graphite) slows neutrons; control rods (cadmium, boron) absorb neutrons; coolant (water, CO_2, liquid sodium)
Diesel	Quick start, small capacity, standby
Gas turbine	Brayton; peak load; quick start; combined cycle for base load

- Indian nuclear reactors: mostly PHWR (pressurised heavy water reactor) using natural uranium and D_2O moderator. Fast breeder reactor uses liquid sodium coolant and no moderator.
- Electrostatic precipitator (ESP): removes fly ash (> 99% efficiency). Cyclone separator for coarse particles. Coal handling: proximate analysis (moisture, volatile matter, fixed carbon, ash), ultimate analysis (C, H, O, N, S).
- Load curve terms: load factor = average load/max demand; plant capacity factor = average load/rated capacity; diversity factor = sum of individual max demands/max demand of system (> 1); plant use factor.
- Coal ranks (increasing carbon): peat → lignite → bituminous → anthracite (highest carbon, highest CV, burns with short smokeless flame).

★ **PYQ-TREND:** Moderator & control rod materials. PHWR uses heavy water. ESP for fly ash. Load factor and diversity factor definitions (diversity factor > 1). Anthracite has the highest carbon content. Proximate analysis components.

8.7 Air Compressors

- Reciprocating (positive displacement; high pressure, low flow) and rotary (Roots blower, vane, screw – positive displacement; centrifugal & axial – dynamic; high flow).

```
Work (single stage, polytropic, no clearance) W = (n/(n−1)) p₁V₁ [(p₂/p₁)^((n−1)/n) − 1]
Isothermal work W_iso = p₁V₁ ln(p₂/p₁) (minimum work)
Isothermal efficiency = isothermal work / actual work
```

```
Volumetric efficiency ηv = 1 + C − C (p₂/p₁)^(1/n) (C = clearance ratio Vc/Vs)
Multistage with perfect intercooling: optimum intermediate pressure p₂ = √(p₁p₃)
(2-stage); pressure ratio per stage = (p_final/p₁)^(1/stages); work equal in each stage
```

- Volumetric efficiency decreases as clearance or pressure ratio increases. Max pressure ratio for zero ηv: (1 + 1/C)^n.
- Multistaging advantages: less work (closer to isothermal), better ηv, lighter cylinders, lower delivery temperature, better lubrication.
- Free air delivery (FAD): volume delivered referred to atmospheric (intake) conditions.
- Centrifugal compressor: impeller + diffuser; slip factor, power input factor; surging (unstable flow at low mass flow) and choking (max flow when Mach = 1). Axial compressor: high efficiency, many stages, used in gas turbines/jet engines; degree of reaction usually 50%.
- Roots blower: no internal compression (efficiency drops at high pressure ratio). Vane compressor has internal compression.

★ **PYQ-TREND:** Isothermal compression requires minimum work. Intermediate pressure √(p₁p₃). Effect of clearance on volumetric efficiency. Surging and choking in centrifugal compressors. FAD definition. Roots blower has no internal compression.

8.8 Gas Turbines & Jet Propulsion

- Open cycle vs closed cycle gas turbines. Components: compressor, combustion chamber, turbine. Compressor consumes 50–60% of turbine work.
- Turbojet, turbofan (bypass – higher propulsive efficiency, quieter – modern airliners), turboprop (low speed aircraft), ramjet (no compressor; needs initial velocity), pulsejet (V-1 bomb), rocket (carries its own oxidiser – works in vacuum).

```
Thrust F = ṁ (Vj − Va); propulsive efficiency ηp = 2Va/(Va + Vj)
```

★ **PYQ-TREND:** Ramjet has no moving parts and cannot start from rest. Rocket carries oxidiser. Propulsive efficiency formula. Turbofan has higher propulsive efficiency.

✎ **DESCRIPTIVE Q:** Why is the Rankine cycle used in steam power plants instead of the Carnot cycle? Explain methods of improving Rankine efficiency.

The Carnot cycle with wet steam needs compression of a two-phase mixture up to saturated liquid. This is difficult to control and needs a large compressor. Its work ratio is very low, and superheating at constant temperature is impractical. Rankine condenses the steam completely, so a small pump handles liquid; the work ratio is near 1 and the plant is simpler. Efficiency improves by raising the mean temperature of heat addition (higher boiler pressure, superheating, reheating, regenerative feed heating) or lowering the condenser temperature (better vacuum). Reheating mainly keeps the exhaust steam dry. Regeneration gives the biggest practical rise in cycle efficiency.

CHAPTER 9

Heat Transfer

9.1 Modes of Heat Transfer

Mode	Law	Medium needed
Conduction	Fourier: $q = -kA\,dT/dx$	Yes (solid/fluid at rest)
Convection	Newton's law of cooling: $q = hA(T_s - T_\infty)$	Yes (moving fluid)
Radiation	Stefan–Boltzmann: $q = \varepsilon\sigma A(T_1^4 - T_2^4)$	No (fastest – speed of light)

$\sigma = 5.67 \times 10^{-8}\ W/m^2 \cdot K^4$

- Thermal conductivity k (W/m·K): silver 429 > copper 386 > gold 317 > aluminium 237 > iron 80 > steel ~45 > stainless ~16 > glass ~1 > water 0.6 > wood ~0.15 > air 0.026. Diamond ≈ 2000+ (highest of solids).
- k of metals decreases with temperature (mostly); k of gases increases with temperature; k of liquids decreases (except water up to ~150 °C).
- Thermal diffusivity $\alpha = k/(\rho c)$ m^2/s – rate of heat diffusion in transient conduction.

★ **PYQ-TREND:** Units of k, h, σ. Highest conductivity metal = silver. Fourier's law. k of gases increases with temperature. Radiation needs no medium. Thermal diffusivity definition.

9.2 Steady One-Dimensional Conduction

Plane wall: $Q = kA(T_1 - T_2)/L$; thermal resistance $R = L/(kA)$

Convection resistance $R = 1/(hA)$; radiation equivalent $h_r = \varepsilon\sigma(T_1^2 + T_2^2)(T_1 + T_2)$

Composite wall (series): $Q = \Delta T_{overall} / \Sigma R$

Cylinder: $R = \ln(r_2/r_1)/(2\pi kL)$; Sphere: $R = (r_2 - r_1)/(4\pi k\, r_1\, r_2)$

Overall heat transfer coefficient: $1/U = 1/h_1 + \Sigma(L/k) + 1/h_2$

Critical radius of insulation: cylinder $r_c = k/h$; sphere $r_c = 2k/h$

- Adding insulation to a pipe with $r < r_c$ increases heat loss (until $r = r_c$, where loss is maximum). Electric cables are deliberately kept near r_c to dissipate heat.
- Variable conductivity $k = k_0(1 + \beta T)$: use mean k at mean temperature for plane wall.
- Heat generation in plane wall (thickness 2L, both faces at Ts): $T_{max} = T_s + qL^2/(2k)$. Cylinder: $T_{max} = T_s + qR^2/(4k)$.

★ **PYQ-TREND:** Critical radius k/h (cylinder) and 2k/h (sphere) – very frequently asked. At critical radius heat loss is maximum. Series/parallel resistance concept. Overall U formula.

✖ **COMMON MISTAKE:** Insulating a small wire can increase heat loss. Insulation reduces heat loss only after the outer radius exceeds r_c.

9.3 Fins (Extended Surfaces)

$m = \sqrt{(hP/(kA_c))}$

```
Infinitely long fin: Q = √(hPkA_c) θ₀
Insulated tip: Q = √(hPkA_c) θ₀ tanh(mL)
Fin efficiency η_f = actual heat transfer / heat transfer if entire fin were at base temperature; long fin η = 1/(mL)
Fin effectiveness ε = Q_with fin / Q_without fin = √(kP/(hA_c)); use fins only if ε > 2
```

- Fins are most effective with high k material, low h (gas side), thin & closely spaced. Fins on the side of lower h (air side of radiator).
- Corrected length for convective tip: Lc = L + t/2 (rectangular), L + D/4 (pin fin).

★ **PYQ-TREND:** Fins placed on the side with lower heat transfer coefficient. Fin effectiveness should be > 2. Long fin heat flow formula. Corrected length.

9.4 Transient Conduction

```
Biot number Bi = hL_c/k_solid (L_c = V/A_s); lumped analysis valid if Bi < 0.1
Lumped: (T − T∞)/(T₀ − T∞) = exp(−hA_s t/(ρVc)) = exp(−Bi·Fo)
Fourier number Fo = αt/L_c² (dimensionless time)
Time constant τ = ρVc/(hA_s)
```

- L_c: plane wall half-thickness L; cylinder r/2; sphere r/3.
- Heisler charts for Bi > 0.1. Thermocouple response time (time constant) small for small bead and high h.

★ **PYQ-TREND:** Lumped capacity valid for Bi < 0.1. Biot number uses solid k; Nusselt uses fluid k. Characteristic length of sphere r/3.

9.5 Convection

Dimensionless number	Expression	Meaning
Reynolds Re	ρVL/μ	Inertia / viscous force
Prandtl Pr	μcp/k = ν/α	Momentum diffusivity / thermal diffusivity
Nusselt Nu	hL/k_fluid	Convective / conductive heat transfer (dimensionless h)
Grashof Gr	gβΔT L³/ν²	Buoyancy / viscous (free convection)
Stanton St	Nu/(Re·Pr) = h/(ρVcp)	Heat transferred / thermal capacity of fluid
Peclet Pe	Re·Pr	Advection / diffusion
Rayleigh Ra	Gr·Pr	Free convection onset
Lewis Le	α/D = Sc/Pr	Heat / mass diffusivity
Biot Bi	hL/k_solid	Internal conduction / surface convection resistance

- Pr: liquid metals ≪ 1 (0.01), gases ~0.7, water ~7 (at 20 °C), oils 100–10^5. Pr = 1 → hydrodynamic & thermal boundary layers equal thickness; δ_t/δ ≈ Pr^(−1/3).
- Forced convection: Nu = f(Re, Pr). Free (natural) convection: Nu = f(Gr, Pr).

```
Laminar flow in a tube, fully developed: Nu = 3.66 (constant wall temperature), 4.36 (constant heat flux)
```

```
Dittus–Boelter (turbulent tube flow): Nu = 0.023 Re^0.8 Pr^n; n = 0.4 heating, 0.3 cooling
Flat plate laminar: Nu_x = 0.332 Re_x^0.5 Pr^(1/3); δ = 5x/√Re_x; average Nu = 0.664 Re^0.5 Pr^(1/3)
Critical Re: flat plate 5 × 10⁵; pipe 2300
```

- Reynolds analogy: St = f/2 (for Pr = 1). Colburn analogy: St·Pr^(2/3) = f/2.
- Typical h (W/m²K): free convection gases 2–25; forced gases 25–250; forced liquids 50–20,000; boiling/condensation 2,500–100,000.

★ **PYQ-TREND:** Physical meaning of Nu, Pr, Gr, St (very frequently asked). Pr of liquid metals very low. Nu = 3.66/4.36. Grashof number in natural convection. Dittus-Boelter exponent n = 0.4 for heating. Prandtl number of air ≈ 0.7.

9.6 Boiling and Condensation

- Pool boiling curve (Nukiyama): natural convection → nucleate boiling (most efficient, up to critical heat flux / burnout point) → transition boiling (unstable) → film boiling (Leidenfrost point – vapour film insulates).
- Critical heat flux: max heat flux in nucleate boiling; exceeding it in heat-flux-controlled systems can melt the surface (burnout).
- Condensation: filmwise (Nusselt theory) and dropwise (5–10 times higher h; promoted by coatings, oils on surface). Condensers are designed for filmwise (conservative).

★ **PYQ-TREND:** Dropwise condensation gives higher h than filmwise. Nucleate boiling regime has max h. Leidenfrost point marks start of stable film boiling. Burnout point = critical heat flux.

9.7 Radiation

- Black body: absorbs all incident radiation (α = 1), ideal emitter; emissivity ε = 1. Grey body: ε < 1 constant over wavelengths. White body: reflects all (ρ = 1). Opaque: τ = 0 → α + ρ = 1.

```
α + ρ + τ = 1
Kirchhoff's law: at thermal equilibrium α = ε
Wien's displacement law: λ_max T = 2898 μm·K
Planck's law gives spectral distribution; total emissive power E_b = σT⁴
Two infinite parallel plates: q = σ(T₁⁴ − T₂⁴)/(1/ε₁ + 1/ε₂ − 1)
Small body in large enclosure: q = ε₁σA₁(T₁⁴ − T₂⁴)
n radiation shields (all same ε) between plates: heat transfer reduced to 1/(n + 1)
```

- Shape (view) factor F_{12}: fraction of radiation leaving surface 1 that reaches 2. Reciprocity: $A_1F_{12} = A_2F_{21}$. Summation: $\Sigma F_{1j} = 1$. Flat or convex surface $F_{11} = 0$; concave $F_{11} > 0$. Small sphere inside large sphere $F_{12} = 1$.
- Gases: CO_2 and H_2O emit/absorb in bands; O_2, N_2 are transparent. Greenhouse effect: glass transmits short-wave solar radiation and blocks long-wave re-radiation.
- Solar radiation is short-wave ($\lambda_{max} \approx 0.5$ μm); radiation from earth/room objects is long-wave (~10 μm).

★ **PYQ-TREND:** Kirchhoff's law α = ε. Wien's law constant 2898 μm·K. Radiation shields reduce heat flow to 1/(n+1). Shape factor reciprocity and summation rules. Black body emissivity 1. Stefan constant value.

9.8 Heat Exchangers

- Types: parallel flow, counter flow (most effective – outlet of cold can exceed outlet of hot), cross flow (car radiator), shell & tube, plate, regenerators (rotary).

```
Q = U A ΔT_LMTD; LMTD = (ΔT₁ − ΔT₂)/ln(ΔT₁/ΔT₂)
```

- For same terminal temperatures LMTD(counter) > LMTD(parallel) → counter flow needs less area.
- If $\Delta T_1 = \Delta T_2$ (counter flow with equal capacity rates), LMTD = ΔT (constant).
- Condensers/evaporators (one fluid changes phase, C_max → ∞): LMTD same for parallel & counter flow; ε = 1 − e^(−NTU).

```
Effectiveness ε = Q_actual/Q_max; Q_max = C_min (T_h,in − T_c,in)
NTU = UA/C_min; capacity ratio C = C_min/C_max
Parallel flow: ε = [1 − e^(−NTU(1+C))]/(1 + C); Counter flow: ε = [1 − e^(−NTU(1−C))]/[1 − C e^(−NTU(1−C))]
```

- Fouling factor (scale resistance) reduces U over time.

★ **PYQ-TREND:** Counter flow is more effective. LMTD formula. Effectiveness ε = 1 − e^(−NTU) when one fluid changes phase. NTU definition. Radiator of a car is a cross-flow heat exchanger.

✎ **DESCRIPTIVE Q:** Explain the concept of critical thickness of insulation for a cylinder and derive r_c = k/h.

Heat loss from an insulated pipe: $Q = 2\pi L(T_i - T_\infty) / [\ln(r/r_1)/k + 1/(hr)]$. Adding insulation raises the conduction resistance (ln r/k) but increases the outer surface area, which lowers the convection resistance 1/(hr). Setting dQ/dr = 0 means minimising the denominator: $d/dr[\ln(r/r_1)/k + 1/(hr)] = 1/(kr) - 1/(hr^2) = 0 \rightarrow r = k/h$. The second derivative shows the denominator is a minimum, so Q is a maximum at r_c = k/h. Insulation added below r_c increases heat loss. Above r_c it decreases. For a sphere the same method gives r_c = 2k/h.

CHAPTER 10

Fluid Mechanics

10.1 Fluid Properties

- Fluid: a substance that deforms continuously under any shear stress, however small.

Density ρ (kg/m³; water 1000); specific weight w = ρg (water 9810 N/m³); specific gravity S = ρ/ρ_water; specific volume = 1/ρ

Newton's law of viscosity: τ = μ (du/dy)

Kinematic viscosity ν = μ/ρ (m²/s; stoke = 1 cm²/s = 10^{-4} m²/s)

Dynamic viscosity unit: N·s/m² = Pa·s = 10 poise; 1 poise = 0.1 Pa·s; water at 20 °C ≈ 1 cP = 10^{-3} Pa·s

- Viscosity of liquids decreases with temperature (cohesion reduces). Viscosity of gases increases with temperature (molecular momentum exchange increases).

Fluid type	Behaviour (τ vs du/dy)	Example
Ideal	No viscosity, incompressible	Hypothetical
Newtonian	Linear through origin	Water, air, petrol, kerosene
Non-Newtonian – pseudoplastic (shear-thinning)	n < 1	Blood, paint, polymer solutions, milk
Dilatant (shear-thickening)	n > 1	Starch in water (oobleck), quicksand, rice starch
Bingham plastic	Yield stress then linear	Toothpaste, sewage sludge, drilling mud
Thixotropic	Viscosity decreases with time of shearing	Printer's ink, lipstick, some paints
Rheopectic	Viscosity increases with time	Gypsum paste, bentonite

- Bulk modulus K = −dp/(dV/V); compressibility = 1/K. Water K ≈ 2.1–2.2 GPa. For isothermal gas K = p; adiabatic K = γp.

Surface tension (N/m): pressure inside droplet Δp = 4σ/d; soap bubble Δp = 8σ/d (two surfaces); liquid jet Δp = 2σ/d

Capillary rise h = 4σ cosθ/(ρgd) (water–glass θ ≈ 0 → rise; mercury–glass θ ≈ 130–140° → depression)

- Cohesion > adhesion → convex meniscus and capillary fall (mercury). Adhesion > cohesion → concave meniscus, capillary rise (water).
- Vapour pressure: pressure at which liquid boils at a given temperature. Cavitation occurs if local pressure < vapour pressure. Mercury is used in barometers because of very low vapour pressure.

★ **PYQ-TREND:** Viscosity variation with temperature for liquids vs gases (asked almost every year). Non-Newtonian examples (blood – shear-thinning, toothpaste – Bingham). Pressure in soap bubble 8σ/d vs droplet 4σ/d. Capillary rise formula and ∝ 1/d. Unit of kinematic viscosity stoke. Why mercury is used in barometers.

10.2 Fluid Statics and Pressure Measurement

- Pascal's law: pressure at a point in a static fluid is equal in all directions. Hydrostatic law: dp/dz = −ρg → p = ρgh.

```
1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water
```

- Manometers: piezometer (cannot measure negative or high pressure, nor gas pressure), simple U-tube, single column (micromanometer), differential U-tube, inverted U-tube (small pressure difference of liquids), inclined manometer (small pressures – more sensitive).
- Mechanical gauges: Bourdon tube (most common, gauge pressure), diaphragm, bellows, dead-weight gauge (calibration).
- Barometer measures atmospheric (absolute) pressure; aneroid barometer has no liquid.

Hydrostatic forces on surfaces

```
Total pressure on plane surface F = ρ g A h̄ (h̄ = depth of centroid)
Centre of pressure (vertical/inclined): h* = h̄ + I_G sin²θ / (A h̄)
```

- Centre of pressure always lies below the centroid (except for horizontal surfaces where they coincide). As depth increases, the centre of pressure approaches the centroid.
- Vertical rectangle with top edge at free surface: h* = 2h/3. Triangle with apex at surface (base down): h* = 3h/4; base at surface: h* = h/2.
- Curved surface: horizontal component = force on vertical projection; vertical component = weight of fluid above (real or imaginary).

★ **PYQ-TREND:** Centre of pressure lies below centroid. Depth 2h/3 for a rectangular gate with top at surface. Piezometer limitations. Bourdon gauge measures gauge pressure. Inclined manometer for low pressures.

10.3 Buoyancy and Stability of Floating Bodies

- Archimedes' principle: buoyant force = weight of fluid displaced, acting through the centre of buoyancy (centroid of displaced volume).

```
Metacentric height GM = I/V − BG (I = least second moment of the waterplane area, V = displaced volume)
Experimentally: GM = w x / (W tanθ)
Time period of rolling: T = 2π √(k²/(g·GM))
```

- Stability of floating bodies: M above G → stable; M at G → neutral; M below G → unstable. Submerged bodies: B above G → stable.
- Large GM → stable but uncomfortable (rapid rolling) – cargo ships have small GM, warships larger GM.

★ **PYQ-TREND:** Stability conditions for floating vs submerged bodies. Metacentric height formula I/V − BG. Larger GM means a shorter rolling period. Centre of buoyancy = centroid of displaced volume.

10.4 Fluid Kinematics

- Steady (∂/∂t = 0) vs unsteady; uniform (∂/∂s = 0) vs non-uniform; laminar vs turbulent; compressible vs incompressible; rotational vs irrotational; 1D/2D/3D.
- Streamline: tangent gives the velocity direction at that instant. Pathline: actual path of a particle. Streakline: locus of particles passing through a fixed point (dye). Timeline. In steady

flow streamline = pathline = streakline.

```
Continuity (incompressible): A₁V₁ = A₂V₂; 3D: ∂u/∂x + ∂v/∂y + ∂w/∂z = 0
Acceleration: a = ∂V/∂t (local) + V·∇V (convective)
Rotation ωz = ½(∂v/∂x − ∂u/∂y); vorticity = 2ω; irrotational if ωz = 0
Velocity potential φ: u = −∂φ/∂x, v = −∂φ/∂y (exists only for irrotational flow); stream function ψ: u = −∂ψ/∂y, v = ∂ψ/∂x (sign convention varies; exists for any 2D incompressible flow)
```

- φ = constant (equipotential) lines and ψ = constant (streamlines) are orthogonal → flow net. Laplace equation $\nabla^2\varphi = 0$ for irrotational incompressible flow.
- Discharge between two streamlines = $\psi_2 - \psi_1$.
- Free vortex: V·r = constant (irrotational) – whirlpool, flow in a sink. Forced vortex: V = ωr (rotational) – rotating cylinder of liquid; free surface is a paraboloid with rise $h = \omega^2R^2/(2g)$.

★ **PYQ-TREND:** Streamline vs pathline vs streakline (equal in steady flow). Continuity equation for incompressible flow. Stream function exists for both rotational & irrotational flows; velocity potential exists only for irrotational. Free vs forced vortex examples. Paraboloid in forced vortex.

10.5 Fluid Dynamics – Bernoulli and Momentum

```
Bernoulli: p/(ρg) + V²/(2g) + z = constant (pressure head + velocity head + datum head)
```

- Assumptions: steady, incompressible, inviscid (frictionless), irrotational (or along a streamline), no shaft work/heat.
- Euler's equation (along streamline): dp/ρ + V dV + g dz = 0. Integrating gives Bernoulli.
- HGL (hydraulic gradient line) = p/ρg + z. TEL (total energy line) = HGL + V²/2g. TEL always slopes down in the flow direction for real fluids; HGL can rise if velocity decreases.
- Kinetic energy correction factor α: laminar pipe flow 2.0; turbulent ≈ 1.03–1.1. Momentum correction factor β: laminar 4/3; turbulent ≈ 1.01–1.04.

```
Momentum equation: ΣF = ṁ (V₂ − V₁)
Force on pipe bend: Fx = ṁ(V₁ − V₂cosθ) + p₁A₁ − p₂A₂cosθ
```

Flow measurement

Device	Formula / Feature
Venturimeter	$Q = Cd\ A_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$; Cd ≈ 0.95–0.98; low head loss; convergent angle ~20°, divergent 5–7°
Orificemeter	Same formula with Cd ≈ 0.6–0.65; cheap; high loss
Flow nozzle	Cd between venturi & orifice
Pitot tube	$V = Cv\sqrt{(2gh)}$ – measures stagnation pressure; point velocity
Pitot-static (Prandtl) tube	Velocity from stagnation – static
Rotameter	Variable area meter, float, vertical tube
Notch/weir (rectangular)	Q = (2/3)Cd L√(2g) H^(3/2)
V-notch (triangular)	Q = (8/15)Cd √(2g) tan(θ/2) H^(5/2) – accurate for low discharge
Cipolletti (trapezoidal, 1:4 side slope)	Compensates end contraction
Hot-wire anemometer	Velocity fluctuations in gas flow (turbulence)
Current meter	River/open-channel velocity

- Hydraulic coefficients of orifice: Cc = a_c/a (≈ 0.64), Cv = V_actual/V_th (≈ 0.97–0.99), Cd = Cc·Cv (≈ 0.62). Vena contracta: section of minimum jet area, about d/2 from the orifice.
- Mouthpieces: external cylindrical Cd = 0.855; Borda's re-entrant (running free) Cd = 0.5, (running full) 0.707; convergent-divergent Cd = 1.

Time to empty a tank through orifice (from H_1 to H_2): $T = 2A(\sqrt{H_1} - \sqrt{H_2})/(Cd\ a\ \sqrt{(2g)})$

★ **PYQ-TREND:** Bernoulli assumptions. Venturimeter Cd > orificemeter Cd. Divergent cone angle of venturi 5–7° (to avoid separation). Pitot tube measures stagnation (total) pressure. Cd = Cc × Cv. V-notch discharge ∝ H^(5/2), rectangular ∝ H^(3/2). Borda mouthpiece Cd 0.5. KE correction factor = 2 for laminar.

✔ **TOPPER TRICK:** Error in Q due to error in H: rectangular notch dQ/Q = 1.5 dH/H; triangular dQ/Q = 2.5 dH/H.

10.6 Viscous (Laminar) Flow

Hagen–Poiseuille (pipe): $\Delta p = 32\mu VL/D^2$; head loss $h_f = 32\mu VL/(\rho g D^2)$

Velocity profile parabolic: $u = u_{max}\ (1 - r^2/R^2)$; $V_{avg} = u_{max}/2$

Shear stress varies linearly: $\tau = (-dp/dx)\ r/2$ (zero at centre, max at wall)

Friction factor $f = 64/Re$ (Darcy); $f' = 16/Re$ (Fanning)

Flow between fixed parallel plates: $V_{avg} = (2/3)\ u_{max}$; $\Delta p = 12\mu VL/t^2$

Couette flow (one plate moving): linear velocity profile

- Laminar pipe flow: Re < 2000 (2300); transitional 2000–4000; turbulent > 4000.
- Measurement of viscosity: capillary tube (Poiseuille), falling sphere (Stokes' law F = 3πμDV), rotating cylinder, orifice-type (Saybolt, Redwood, Engler).

Stokes' law terminal velocity: $V = D^2(\rho s - \rho)g/(18\mu)$

- Dashpot mechanism and lubrication theory are applications of laminar flow.

★ **PYQ-TREND:** f = 64/Re for laminar flow (Darcy). V_avg = u_max/2 in pipe, (2/3)u_max between plates. Shear stress max at wall, zero at centre. Stokes' law drag 3πμDV. Critical Re 2000.

10.7 Turbulent Flow and Pipe Losses

Darcy–Weisbach: $h_f = f\ L\ V^2/(2gD) = f\ L\ Q^2/(12.1\ D^5)$

Blasius (smooth pipe, $Re < 10^5$): $f = 0.316/Re^{0.25}$

Chezy: $V = C\sqrt{(m\ i)}$; hydraulic mean depth $m = A/P$ (= D/4 for full pipe)

- Moody diagram: f vs Re with relative roughness ε/D. Fully rough zone: f depends only on ε/D. Laminar: f independent of roughness.
- Turbulent velocity profile flatter (1/7th power law); $u_{max} \approx 1.2\ V$.
- Hydrodynamically smooth if roughness k < laminar sub-layer δ'.

Minor loss	Formula
Sudden enlargement	$h = (V_1 - V_2)^2/(2g)$
Sudden contraction	$h = (1/Cc - 1)^2\ V_2^2/(2g) \approx 0.5\ V_2^2/(2g)$
Entrance (sharp)	$0.5\ V^2/(2g)$
Exit to reservoir	$V^2/(2g)$
Bend / fitting	$k\ V^2/(2g)$

- Pipes in series: Q same, h_f add; equivalent pipe (Dupuit): $L/D^5 = \Sigma L_i/D_i^5$. Pipes in parallel: h_f same, Q add.

```
Power transmitted through pipe P = ρgQ(H − h_f); maximum when h_f = H/3 (η_max = 66.67%)
Nozzle diameter for max power at end of pipe: d = (D⁵/(8fL))^(1/4)
Water hammer pressure rise (sudden closure, rigid pipe): Δp = ρVC; C = √(K/ρ); critical time 2L/C
```

- Siphon: summit pressure must stay above vapour pressure (practical limit ~7.6 m of water above the HGL).
- Surge tank protects penstock against water hammer in hydro plants.

★ **PYQ-TREND:** Darcy formula. Max power transmission when $h_f = H/3$ – asked very often. Loss at sudden enlargement formula. Pipes in series vs parallel. Water hammer and surge tank. Equivalent pipe (Dupuit equation). Exit loss = $V^2/2g$. Hydraulic mean depth D/4.

10.8 Boundary Layer, Drag and Lift

- Boundary layer: thin region near a wall where velocity changes from zero (no-slip) to free stream; viscous effects confined to it.
- Thickness definitions: δ (u = 0.99U), displacement thickness δ*, momentum thickness θ, energy thickness δ**. δ > δ* > θ. Shape factor H = δ*/θ (laminar Blasius ≈ 2.59; turbulent ≈ 1.3–1.4).

```
Laminar (Blasius): δ = 5x/√Re_x; δ* = 1.72x/√Re_x; θ = 0.664x/√Re_x; local Cf = 0.664/√Re_x
Turbulent: δ = 0.37x/Re_x^(1/5)
Linear velocity profile: δ*/δ = 1/2, θ/δ = 1/6
```

- Separation occurs under adverse pressure gradient ($dp/dx > 0$) where $\partial u/\partial y$ at wall = 0. Control: suction, blowing, streamlining, rotating cylinder, slotted wings, vortex generators.

```
Drag F_D = C_D (½ρV²) A; Lift F_L = C_L (½ρV²) A
```

- Drag = pressure (form) drag + skin-friction drag. Streamlined body: mostly friction drag. Bluff body: mostly pressure drag.
- Magnus effect: lift on a rotating cylinder/ball in uniform flow (swing of cricket/tennis ball). Kutta–Joukowski: $L = \rho V \Gamma$ per unit span.
- Golf ball dimples make the boundary layer turbulent earlier → delays separation → smaller wake → less drag.

★ **PYQ-TREND:** Separation occurs with adverse pressure gradient. Displacement thickness definition. $\delta \propto \sqrt{x}$ for laminar. Magnus effect. Bluff bodies – pressure drag dominates. Golf ball dimples reduce drag. Order $\delta > \delta^* > \theta$.

10.9 Dimensional Analysis and Similitude

- Methods: Rayleigh (≤ 3–4 variables), Buckingham π-theorem: number of π terms = n − m (n = variables, m = fundamental dimensions).

Number	Ratio	Significant in
Reynolds	Inertia / viscous	Pipe flow, submarines, aircraft (subsonic), fully submerged bodies

Number	Ratio	Significant in
Froude	Inertia / gravity (V/√(gL))	Open channels, ship models, spillways, free-surface flows
Euler	Pressure / inertia	Cavitation, pressure-driven flows
Weber	Inertia / surface tension	Droplets, capillary flows, thin films
Mach	Inertia / elastic (V/C)	Compressible flow, high-speed aircraft

- Model laws: Reynolds model (Vr Lr/νr = 1), Froude model (Vr = √Lr → Qr = Lr^2.5, time ratio Tr = √Lr, force ratio Fr = Lr³).
- Similitude: geometric, kinematic and dynamic similarity. Distorted models for rivers (different horizontal and vertical scales).
- Mach number ranges: incompressible < 0.3, subsonic < 1, transonic ~0.8–1.2, supersonic > 1, hypersonic > 5.

Speed of sound $C = \sqrt{(\gamma RT)} = \sqrt{(K/\rho)}$; air at 20 °C ≈ 343 m/s; Mach angle $\sin \alpha = 1/M$

Stagnation temperature $T_0 = T(1 + (\gamma-1)M^2/2)$

- Normal shock: supersonic → subsonic; entropy increases; static pressure & temperature increase; stagnation pressure decreases; stagnation temperature constant.

★ **PYQ-TREND:** Froude model law for ships/spillways; Reynolds for pipes/submarines. Discharge ratio Lr^2.5 in Froude models. Number of π-terms = n − m. Meaning of Weber and Mach numbers. Mach angle. Across normal shock stagnation temperature constant.

10.10 Open Channel Flow (Basics)

Chezy $V = C\sqrt{(mi)}$; Manning $V = (1/n)\ R^{(2/3)}\ S^{(1/2)}$

Specific energy $E = y + V^2/(2g)$; critical depth (rectangular) $y_c = (q^2/g)^{(1/3)}$; $E_{min} = 1.5\ y_c$

Froude number $Fr = V/\sqrt{(gy)}$: Fr < 1 subcritical, = 1 critical, > 1 supercritical

- Most economical sections: rectangle b = 2y (m = y/2); trapezoid – half hexagon (sides inclined 60°), m = y/2; triangle – 90° vertex.
- Hydraulic jump: supercritical → subcritical; energy dissipated (below spillways). Sequent depth $y_2/y_1 = ½(-1 + \sqrt{(1 + 8Fr_1^2)})$.

★ **PYQ-TREND:** Most economical rectangular channel b = 2y. Hydraulic jump converts supercritical to subcritical flow and dissipates energy. Critical depth relation.

✎ **DESCRIPTIVE Q:** State Bernoulli's theorem, list its assumptions and explain the working of a venturimeter.

For steady, incompressible, frictionless flow along a streamline, the sum of pressure head, velocity head and datum head is constant: $p/\rho g + V^2/2g + z$ = constant. A venturimeter has a convergent cone, a throat and a long divergent cone. Applying continuity ($A_1V_1 = A_2V_2$) and Bernoulli between inlet and throat, the drop in pressure head h (from a differential manometer) gives the theoretical discharge $Q = A_1A_2\sqrt{(2gh)}/\sqrt{(A_1^2 - A_2^2)}$. Multiply by Cd (0.95–0.98) for actual discharge. The divergent cone is made long (5–7° angle) to recover pressure gradually and avoid flow separation.

CHAPTER 11

Hydraulic Machines

11.1 Impact of Jets

Force of jet on fixed vertical flat plate: $F = \rho a V^2$

Moving vertical plate (single): $F = \rho a(V - u)^2$; series of plates (wheel): $F = \rho a V(V - u)$

Fixed curved vane, jet deflected by angle: $F = \rho a V^2(1 + \cos\theta)$ (θ measured from opposite direction; for semicircular vane $F = 2\rho a V^2$)

Efficiency of series of flat plates: $\eta = 2u(V - u)/V^2$; max $\eta = 50\%$ at $u = V/2$

Series of curved vanes (Pelton bucket, 180°): max $\eta = 100\%$ at $u = V/2$ (theoretically); with deflection φ: $\eta_{max} = (1 + \cos\varphi)/2$

Single moving plate: max efficiency at $u = V/3$, $\eta_{max} = 8/27$ (29.6%)

★ **PYQ-TREND:** Max efficiency of a series of flat plates = 50% at u = V/2. Single moving plate u = V/3. Force on fixed flat plate $\rho a V^2$. Semicircular vane doubles force.

11.2 Hydraulic Turbines – Classification

Basis	Types
Energy at inlet	Impulse (Pelton – only KE) / Reaction (Francis, Kaplan, Propeller – KE + pressure)
Direction of flow	Tangential (Pelton), Radial inward (old Francis, Thomson), Mixed (modern Francis), Axial (Kaplan, Propeller)
Head	High (> 250 m: Pelton), Medium (60–250 m: Francis), Low (< 60 m: Kaplan/Propeller)
Specific speed (SI, kW)	Pelton 8.5–30 (single jet), Francis 50–340, Kaplan 255–860

Specific speed of turbine $N_s = N\sqrt{P} / H^{(5/4)}$

Specific speed of pump $N_s = N\sqrt{Q} / H^{(3/4)}$

Unit quantities: $N_u = N/\sqrt{H}$; $Q_u = Q/\sqrt{H}$; $P_u = P/H^{(3/2)}$

- Specific speed: speed of a geometrically similar turbine producing unit power under unit head. Low Ns → high head; high Ns → low head.
- Efficiencies: hydraulic η_h = power developed by runner / water power ($\rho g Q H$); mechanical η_m = shaft power / runner power; volumetric η_v; overall $\eta_o = \eta_h \times \eta_m$.

★ **PYQ-TREND:** Pelton for high head, Kaplan for low head – the most asked HM question. Specific speed formula (with H^(5/4)). Specific speed ranges. Kaplan is axial-flow reaction. Modern Francis is mixed-flow.

✔ **TOPPER TRICK:** "Pelton → High Head → Low Ns; Kaplan → Low Head → High Ns". Head and Ns always move opposite.

11.3 Pelton Wheel

- Tangential flow impulse turbine; nozzle with spear (needle) valve regulates flow; buckets with splitter (double hemispherical cups) deflect jet by ~165°; casing only prevents splashing (no hydraulic function); breaking jet stops the wheel.

```
Jet velocity V = Cv √(2gH) (Cv ≈ 0.98)
Bucket speed u = φ√(2gH), speed ratio φ ≈ 0.43–0.47
Jet ratio m = D/d (≈ 12 for good design, min 6); number of buckets z = 15 + m/2 (Taygun)
Work per kg = (V − u)(1 + k cosφ) u (k = bucket friction factor)
```

- Governing: spear (needle) valve + deflector (to avoid water hammer during sudden load reduction).
- Pelton runs in air at atmospheric pressure; no draft tube.

★ **PYQ-TREND:** Jet ratio ~12. Number of buckets formula. Splitter divides jet. Deflection angle 165°. Casing has no hydraulic function. Spear valve controls discharge.

11.4 Francis and Kaplan Turbines

- Francis: spiral casing (uniform velocity distribution), stay vanes, guide vanes (wicket gates – adjustable, regulate flow), runner (mixed flow), draft tube. Flow ratio ψ = Vf/√(2gH) ≈ 0.15–0.30; speed ratio 0.6–0.9.
- Kaplan: axial flow propeller with adjustable runner blades (double regulation – guide vanes + runner blades) → high part-load efficiency. Propeller turbine has fixed blades. Hub-to-tip ratio ≈ 0.35–0.6. Few blades (3–8).
- Bulb turbine: Kaplan variant for very low head tidal plants.
- Euler's turbine equation: work per kg = $Vw_1u_1 \pm Vw_2u_2$. Max work when outlet whirl is zero (radial discharge, $Vw_2 = 0$).

Draft tube

- Function: allows the turbine to be placed above tail race without loss of head, and recovers part of the exit kinetic energy (converts it into pressure energy) – creates suction at runner outlet.
- Types: conical (straight divergent, η ~90%, angle < 8°), simple elbow, elbow with varying cross-section, Moody spreading.

```
Draft tube efficiency = (V₂² − V₃² − 2g·h_f)/V₂²
```

- Cavitation: formation and collapse of vapour bubbles where local pressure < vapour pressure → pitting, noise, vibration, efficiency drop. Most likely at runner outlet of reaction turbines and inlet (eye) of centrifugal pumps. Thoma's cavitation factor σ = (H_atm − H_v − H_s)/H; σ > σ_critical to avoid cavitation. Use stainless steel/cavitation-resistant runners.

★ **PYQ-TREND:** Kaplan has adjustable runner blades. Draft tube is used only in reaction turbines. Draft tube converts KE to pressure energy. Cavitation definition and Thoma's coefficient. Wicket gates regulate flow in Francis.

11.5 Performance and Governing of Turbines

- Characteristic curves: main (constant head), operating (constant speed), muhel (constant efficiency/iso-efficiency) curves.
- Pelton efficiency curve is flat (good part-load efficiency); Francis steeper; Kaplan flat due to double regulation; Propeller poor part-load.

- Governing: maintain constant speed (synchronous for generator). Oil pressure governor with servomotor, relief valve, deflector.
- Runaway speed: max speed under no load with full gate opening (Pelton ~1.8 N, Kaplan ~2.5–3 N).
- Synchronous speed N = 120f/p (p = number of poles).

★ **PYQ-TREND:** Kaplan efficiency at part load is high. Synchronous speed formula. Iso-efficiency curves. Runaway speed definition.

11.6 Centrifugal Pumps

- Converts mechanical energy into pressure energy by centrifugal action (forced vortex). Components: impeller (closed, semi-open, open), casing (volute – gradually expanding; vortex; diffuser with guide blades), suction pipe with foot valve and strainer, delivery pipe with delivery valve.
- Priming: filling the suction pipe, casing and impeller with liquid to remove air before starting – essential because the head developed depends on density (air head is useless).
- Backward-curved blades ($\beta_2 < 90°$) are used: stable, non-overloading power characteristic, high efficiency.

```
Work per kg (radial entry Vw₁ = 0): W = Vw₂ u₂ / g (head)
Manometric efficiency ηmano = g Hm / (Vw₂ u₂)
Minimum starting speed: N = 120 ηmano Vw₂ D₂ / (π (D₂² − D₁²)) (from u₂²/2g − u₁²/2g ≥ Hm)
Static head Hs = hs + hd; manometric head Hm = Hs + h_fs + h_fd + Vd²/2g
NPSH = (p_atm/ρg) − hs − h_fs − (p_v/ρg) (must exceed NPSH required to avoid cavitation)
```

- Affinity (similarity) laws for same pump: $Q \propto N$, $H \propto N^2$, $P \propto N^3$; for geometrically similar pumps: $Q \propto ND^3$, $H \propto N^2D^2$, $P \propto N^3D^5$.
- Pumps in series → increase head (multistage); in parallel → increase discharge.
- Characteristic curves: H–Q falls with Q; power rises; efficiency peaks at BEP (best efficiency point).
- Specific speed: radial (low Ns 10–80), mixed (80–160), axial (> 160 – propeller pumps, high Q low H).
- Shut-off head: head at zero discharge. Starting with delivery valve closed reduces starting power for centrifugal (radial) pumps; axial pumps are started with valve open.

★ **PYQ-TREND:** Priming is needed for centrifugal pumps. Backward-curved vanes are used. Affinity laws ($P \propto N^3$). Pumps in series for head, parallel for discharge. Manometric efficiency formula. Cavitation & NPSH. Centrifugal pump started with delivery valve closed. Volute casing converts KE into pressure.

✖ **COMMON MISTAKE:** Do not apply turbine specific speed (√P / H^(5/4)) to pumps. Pump Ns uses √Q / H^(3/4).

11.7 Reciprocating Pumps

- Positive displacement; suitable for high head, low discharge (oil drilling, boiler feed (older)). Single-acting: one delivery per revolution; double-acting: two.

```
Theoretical discharge: single-acting Q = ALN/60; double-acting Q = 2ALN/60
```

Slip = Q_th − Q_act; % slip = (1 − Cd) × 100. Negative slip occurs with long suction pipe, short delivery, high speed (Q_act > Q_th)

Power = ρg Q (hs + hd)/1000 kW

Pressure head due to acceleration: ha = (L/g)(A/a)ω²r cosθ

- Air vessels (on suction & delivery sides near the cylinder): give uniform flow, reduce acceleration head, allow higher speed, save work (≈ 84.8% saving for single-acting, 39.2% for double-acting due to friction reduction), prevent separation (cavitation).
- Indicator diagram: area ∝ work done; acceleration head tilts suction and delivery lines; friction gives a parabolic curve.
- Separation in suction pipe occurs when pressure drops below ~2.5 m of water (abs) at the beginning of suction stroke.

★ **PYQ-TREND:** Negative slip conditions. Purpose of air vessel. Reciprocating pump does not need priming (self-priming) but has pulsating flow. Discharge formula. Acceleration head max at start/end of stroke.

11.8 Other Hydraulic Devices

Device	Principle / Use
Hydraulic press	Pascal's law – small force on small plunger gives large force on ram; MA = A/a
Hydraulic accumulator	Stores pressure energy (energy = pAL) for intermittent demand (lifts, cranes)
Hydraulic intensifier	Increases pressure of liquid ($p_2 = p_1$ A/a)
Hydraulic ram (hydram)	Uses water hammer of a large flow to lift a small quantity to a higher level; no external power; D'Aubuisson efficiency
Hydraulic crane / lift	Uses jigger, direct acting ram
Fluid coupling	Transmits torque between shafts; torque output = input; slip 2–3%
Torque converter	Has stator (fixed guide) – output torque can exceed input (multiplication); used in automatic transmission
Jet pump	Momentum transfer; no moving parts
Air lift pump	Compressed air mixed with water in well; for oil/water wells
Gear pump, vane pump, screw pump	Rotary positive displacement – lubrication & hydraulic systems
Submersible pump	Multi-stage centrifugal pump with motor immersed – borewells
Hydraulic coupling vs converter	Coupling: T_out = T_in. Converter: T_out ≠ T_in (stator reaction)

★ **PYQ-TREND:** Hydraulic ram works on water hammer. Accumulator stores energy. Intensifier increases pressure. Torque converter has a stator; fluid coupling has none. Gear pump is a positive displacement pump.

✎ **DESCRIPTIVE Q:** Explain why a draft tube is used in a reaction turbine and why it is not used in a Pelton wheel.

In a reaction turbine water leaves the runner with appreciable velocity and the runner is completely filled with water under pressure. A draft tube is a gradually expanding pipe from the runner exit to below tail-race level. It produces a negative (suction) head at runner exit, so the turbine can be set above tail race without losing static head. It also recovers much of the exit KE as pressure energy, which raises the net head and the efficiency. In a Pelton wheel, the jet works at atmospheric pressure in open air and the buckets must be above tail-race water level, so a draft tube cannot create suction or recover energy and is not used. The small head between wheel and tail race is simply lost.

CHAPTER 12

Manufacturing I – Casting, Welding, Forming

12.1 Casting – Patterns

- Casting: molten metal poured into a mould cavity and allowed to solidify. Sand casting is the most widely used process.
- Pattern materials: wood (teak, pine – cheap, small runs), metal (aluminium, CI – mass production), plastic, wax (investment casting), plaster, polystyrene (full mould / lost foam).

Pattern allowance	Purpose / fact
Shrinkage (contraction)	Liquid-to-solid and solid contraction; CI ≈ 10 mm/m (lowest due to graphite expansion), steel ≈ 20 mm/m, brass ≈ 15 mm/m. Pattern made larger
Machining (finish)	Extra material for machining; more for CI & large castings
Draft (taper)	1–3° on vertical faces for easy withdrawal; more on inner surfaces
Distortion (camber)	For U/V shaped castings that warp during cooling
Shake (rapping)	Negative allowance – pattern made smaller as the cavity enlarges on rapping

- Only the shake allowance is negative.
- Pattern types: single piece, split (two-piece – most common), loose piece, match plate (mass production with moulding machines), gated, sweep (large symmetrical – bells), skeleton (large castings), cope & drag, follow board, segmental.
- Pattern colour code: black – surfaces to be left unmachined; red – to be machined; yellow – core prints; red stripes on yellow – loose pieces.
- Core: placed in mould to form internal cavities. Core prints support cores. Chaplets support cores where prints are not possible (they fuse into casting). Chills increase cooling rate locally (directional solidification).
- Core sand binders: linseed oil, dextrin, molasses, resin (CO_2 process uses sodium silicate hardened by CO_2 gas).

★ **PYQ-TREND:** Shake allowance is negative. Draft allowance purpose. CI has the least shrinkage. Match-plate pattern for mass production. Chaplets support cores. Colour code of patterns. Sweep pattern for bells/large symmetrical castings.

12.2 Moulding Sand and Gating

- Green sand: silica sand + clay (bentonite 5–10%) + water (2–8%). Dry sand moulds baked for strength. Loam sand (50% clay) for large castings. Facing sand (in contact with casting), backing/floor sand, parting sand (to prevent sticking).

- Properties: refractoriness, permeability (escape of gases), green & dry strength, cohesiveness, adhesiveness, flowability, collapsibility (for contraction of casting), thermal stability.

```
Permeability number = V H / (p A t) (V = 2000 cm³ air, H height of specimen, p pressure in g/cm², A area, t time in min)
```

- AFS grain fineness number. Standard specimen 50.8 mm dia × 50.8 mm height (2 in × 2 in).
- Gating system: pouring basin → sprue (tapered to avoid aspiration) → sprue well → runner → ingates → mould cavity. Riser supplies molten metal during solidification shrinkage.

```
Sprue taper (continuity): A₁/A₂ = √(h₂/h₁)
Gating ratio sprue : runner : ingate = 1:2:2 (unpressurised, Al) or 1:0.75:0.5 (pressurised, ferrous)
Pouring time (top gate) t = A_m h_m / (A_g √(2g h_t)); bottom gate t = 2A_m(√h_t − √(h_t − h_m))/(A_g√(2g))
Chvorinov's rule: solidification time t_s = B (V/A)^n, n ≈ 2
```

- Riser must solidify after the casting: (V/A)_riser > (V/A)_casting. Optimum riser shape: cylinder with H = D (sphere best theoretically). Blind risers and open risers; exothermic sleeves/insulation help.
- Directional solidification: from far end toward the riser – use chills, padding, correct riser location.
- Top gate: fast filling, turbulence/erosion. Bottom gate: smooth but unfavourable temperature gradient. Parting-line gate: compromise.

★ **PYQ-TREND:** Chvorinov's rule ($t \propto (V/A)^2$) – most frequent casting question. Riser with H = D. Sprue tapered to avoid aspiration. Bentonite as binder. Permeability number formula. Collapsibility prevents hot tears. Gating ratio pressurised vs unpressurised.

12.3 Special Casting Processes

Process	Key features / products
Shell moulding (Croning)	Resin-bonded sand shell on heated metal pattern; good finish & accuracy; crankshafts, camshafts
Investment (lost wax)	Wax pattern + ceramic slurry; best finish, complex shapes, high melting alloys; turbine blades, jewellery, dental
Permanent mould (gravity die)	Metal mould, non-ferrous; pistons
Pressure die casting – hot chamber	Zn, Sn, Pb (low MP); fast; gooseneck
Pressure die casting – cold chamber	Al, Mg, Cu alloys (higher MP)
Centrifugal casting – true	Hollow symmetric parts without core: pipes, liners
Semi-centrifugal	Symmetric solid parts – wheels, pulleys
Centrifuge casting	Small non-symmetric parts arranged around a sprue
Slush casting	Hollow decorative items – toys, lamp bases (low MP alloys)
Continuous casting	Billets, slabs, blooms directly from molten steel
CO_2 moulding	Sodium silicate + CO_2 hardening; cores and moulds
Full mould / lost foam (EPC)	Polystyrene pattern vaporises; no parting line
Squeeze casting	Combination of casting and forging

Process	Key features / products
Plaster mould	Non-ferrous, good finish

★ **PYQ-TREND:** Hot chamber for Zn (low MP), cold chamber for Al. Investment casting for turbine blades, best finish. True centrifugal casting for pipes (no core). Slush casting for hollow ornaments. Shell moulding (Croning process).

12.4 Casting Defects

Defect	Cause / description
Blow holes	Gas entrapment – low permeability, excess moisture; smooth rounded cavities below surface
Pin holes	Small gas holes from dissolved gases (hydrogen)
Porosity	Gas or shrinkage
Shrinkage cavity	Insufficient feeding (riser)
Hot tear (pull)	Restricted contraction (low collapsibility) while metal weak at high temperature
Cold shut	Two streams of metal meeting without fusing (low fluidity/temperature)
Misrun	Metal solidifies before filling cavity
Scab / rat tail / buckle	Sand expansion – patch of sand breaks away
Swell	Mould wall enlarges under metallostatic pressure (soft ramming)
Drop	Portion of cope sand falls into cavity
Mould shift / core shift	Misalignment of cope & drag / core
Fins (flash)	Metal enters parting gaps
Fusion	Sand fuses to casting (low refractoriness)
Metal penetration	Metal enters sand grains (coarse sand, high permeability)
Inclusions	Slag/sand particles
Run out	Metal leaks out of mould

★ **PYQ-TREND:** Cold shut vs misrun. Blow hole due to low permeability/high moisture. Hot tear due to poor collapsibility. Scab from sand expansion. Swell due to soft ramming.

12.5 Welding – Fundamentals and Arc Welding

- Welding: permanent joining by heat with or without pressure and filler. Fusion welding (gas, arc, thermit, EB, laser) vs solid-state (forge, friction, ultrasonic, explosive, diffusion, cold).

Process	Key facts
SMAW (MMA, stick)	Flux-coated consumable electrode; flux gives shielding gas + slag; most versatile
GTAW (TIG)	Non-consumable tungsten electrode; inert gas (Ar/He); thin sheets, Al, Mg, stainless; DCSP for steel, AC for Al (cleaning action)
GMAW (MIG/MAG)	Consumable bare wire electrode fed continuously; inert (MIG) or active CO_2 (MAG); DCRP

Process	Key facts
SAW (submerged arc)	Arc under granular flux; deep penetration, high deposition, flat position only, thick plates, ship building
Electroslag (ESW)	Vertical, very thick plates (>50 mm); resistance heating of molten slag
Plasma arc (PAW)	Constricted arc, very high temperature (~33,000 °C), keyhole mode
Atomic hydrogen	Arc between two tungsten electrodes in H_2; molecular H_2 dissociates and recombines releasing heat
Stud welding	Studs to plates

- Polarity (DC): DCSP/DCEN (electrode −ve, work +ve) – 2/3 heat at work → deeper penetration, faster; DCRP/DCEP (electrode +ve) – more heat at electrode → used for thin sheets, non-ferrous, and gives cleaning action (oxide removal) in Al.

```
Heat input H = η V I / v (J/mm); arc length ≈ electrode diameter
Arc voltage V = a + b·l (Ayrton equation); arc power maximum when the arc characteristic and power source characteristic are tangent
```

- Power source characteristics: drooping/constant current (SMAW, TIG – arc length changes little affect current) and constant voltage/flat (MIG, SAW – self-regulating arc).
- Electrode coating functions: arc stabilisation (potassium/sodium silicate), shielding gas (cellulose), slag formation (deoxidisers, ferro-manganese), alloying. Electrode classification E-6013: E – electrode, 60 – tensile strength 60 ksi, 1 – all positions, 3 – coating/current type.

★ **PYQ-TREND:** TIG uses non-consumable tungsten electrode, MIG uses consumable wire. DCSP gives deeper penetration. AC used for aluminium in TIG. SAW only in flat position. Atomic hydrogen welding principle. Electrode coating functions.

12.6 Gas, Resistance & Other Welding

- Oxy-acetylene: hottest flame ≈ 3200–3300 °C (inner cone tip). Flames: neutral (O_2 : C_2H_2 = 1:1; steel, CI, Cu, Al), oxidising (excess O_2; brass, bronze – hottest), carburising/reducing (excess acetylene; high-carbon steel, Monel, non-ferrous like Ni, hard-facing).
- Acetylene is stored dissolved in acetone in cylinders (porous filler); max pressure 15 psi (≈ 1 bar) free acetylene. Cylinder colours: oxygen black, acetylene maroon (India).
- Leftward (forehand) welding for thin plates (< 5 mm); rightward (backhand) for thick.
- Resistance welding: $H = I^2Rt$; high current, low voltage; spot (sheet metal, car bodies), seam (leak-tight containers, fuel tanks), projection, butt, flash butt (rails, pipes), percussion. Electrodes of copper alloys (Cu–Cr, Cu–W).
- Thermit welding: Al powder + iron oxide → exothermic (~2500 °C); repair of rails, heavy sections.
- Solid-state: friction welding (rotating, dissimilar metals – aluminium to steel), friction stir welding (non-consumable rotating tool – Al alloys), ultrasonic (foils, wires, electronics), explosive (cladding), diffusion (aerospace), forge welding (blacksmith).
- High energy: electron beam (vacuum needed, deep narrow welds, high depth-to-width ratio), laser beam (no vacuum, small HAZ).
- Soldering (< 450 °C filler, Sn–Pb; base metal not melted), brazing (> 450 °C filler like brass, below base metal MP; capillary action), braze welding (filler deposited like welding, no

capillary action).

- Welding defects: porosity, slag inclusion, lack of fusion, incomplete penetration, undercut, overlap, cracks (hot & cold – hydrogen induced), spatter, distortion.
- HAZ: region of base metal whose microstructure changes but does not melt.
- Weldability: low-carbon steel best; high carbon/CI difficult (preheat). Carbon equivalent CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15; CE < 0.4 → good weldability.

★ **PYQ-TREND:** Neutral flame for most metals; oxidising for brass; carburising for high-carbon steel/Monel. Spot welding for car bodies; seam for tanks; flash butt for rails. Thermit welding for rails. Soldering vs brazing temperature 450 °C. Acetylene dissolved in acetone. Electron beam needs vacuum. Friction welding for dissimilar metals.

12.7 Metal Forming

- Hot working: above recrystallisation temperature (≈ 0.4–0.5 Tm in K) – no strain hardening, lower forces, poor finish & tolerance, oxidation, grain refinement. Cold working: below recrystallisation – strain hardening, better finish & accuracy, higher strength, needs more force, lower ductility. Warm working in between.
- Recrystallisation temperature: lead and tin below room temperature (so working at room temperature is hot working for Pb); steel ≈ 500–700 °C.

Rolling

```
Max reduction (draft) Δh_max = μ²R; bite condition μ ≥ tan α; contact length L = √(RΔh)
Neutral (no-slip) point: roll velocity = strip velocity; forward slip ahead of it
Roll separating force F ≈ L w σ_avg; torque per roll T ≈ F·L/2; power (both rolls) = 2 × T × ω
```

- Mills: two-high, three-high, four-high (small work rolls backed by large backup rolls – reduce deflection), cluster (Sendzimir – very thin foils), tandem, planetary.
- Ring rolling, thread rolling (stronger threads – grain flow), Mannesmann process (piercing seamless tubes).
- Rolling defects: wavy edges (roll bending), zipper cracks, edge cracks, alligatoring.

Forging

- Open die (smith), closed (impression) die, drop forging (hammer), press forging (slow squeeze – deeper penetration), upset forging (heading – bolts), roll forging, swaging, rotary swaging.
- Forging operations: fullering (reduce cross-section, spreading), edging, drawing out, upsetting, bending, punching, cogging, swaging, flattening.
- Flash and gutter in closed-die forging; flash ensures die filling.

```
Forging force (cylinder, sticking/friction): F = σ_f π r² (1 + 2μr/(3h))
```

Extrusion

- Direct (forward – billet moves with ram, more friction), indirect (backward – less force), impact (toothpaste tubes, collapsible tubes; Pb, Sn, Al), hydrostatic (brittle materials), cold extrusion.

```
Extrusion force F = A₀ σ_f ln(A₀/A_f) × k (ideal); extrusion ratio R = A₀/A_f
```

- Defects: centre burst (chevron), piping (tailpipe/fishtailing), surface cracking (speckle).

Drawing

- Wire drawing (pull through die), tube drawing (mandrel/plug), deep drawing (cup).

```
Max reduction per pass in wire drawing (ideal, no friction): ln(A₀/A_f) = 1 → 63%
```

- Lubricants: soap, borax; patenting heat treatment for high-carbon wire.

Sheet metal operations

```
Blanking/punching force F = L t τ (L = perimeter, τ = shear strength)
Clearance (per side) c = 0.0032 t √τ (Indian texts) — typically 2–8% of t
```

- Blanking: die size = blank size, clearance on punch. Punching (piercing): punch size = hole size, clearance on die.
- Shear provided on punch/die reduces force (punch for punching, die for blanking).

```
Deep drawing: blank diameter D = √(d² + 4dh); limiting draw ratio LDR ≈ 2 (max D/d)
```

- Spring-back: elastic recovery after bending; compensated by overbending. Bending allowance BA = α(R + kt).
- Other: coining (closed die, fine detail – coins, medals; no flash), embossing (raised design, thin sheet – both sides), lancing, notching, nibbling, perforating, trimming, shaving, spinning, hydroforming, stretch forming, explosive forming, electromagnetic forming.
- Presses: progressive die (several operations at successive stations), compound die (several operations at one station, one stroke), combination die (cutting + non-cutting at one station).

★ **PYQ-TREND:** Hot vs cold working definitions via recrystallisation temperature. Max draft $\mu^2 R$. Four-high mill backup rolls. Mannesmann for seamless tubes. Clearance on punch for blanking and on die for punching – very frequent. Blanking force $Lt\tau$. Compound vs progressive die. Impact extrusion for collapsible tubes. Coining vs embossing. Thread rolling gives stronger threads. Lead worked at room temperature is hot working.

✖ **COMMON MISTAKE:** "Blanking – desired part is the blank, so die = blank size". "Punching – desired part is the sheet with hole, so punch = hole size". Students mix these up most often.

12.8 Powder Metallurgy

- Steps: powder production (atomisation – most common, reduction, electrolysis, crushing), blending, compacting (green compact), sintering (below melting point of main constituent, 70–90% of Tm, protective atmosphere), secondary operations (sizing, coining, infiltration, impregnation).
- Products: self-lubricating porous bronze bearings (oil impregnated), cemented carbide tools (WC + Co binder), filters, friction materials, tungsten filaments, magnets.
- Advantages: near net shape, minimal waste, controlled porosity, materials that can't be cast (refractory metals). Limitations: high powder cost, size limits, lower strength/ductility.

★ **PYQ-TREND:** Sintering is done below melting point. Self-lubricating bearings by PM. Cemented carbide with cobalt binder. Atomisation for powder production.

CHAPTER 13

Manufacturing II – Machining & Machine Tools

13.1 Mechanics of Metal Cutting

- Orthogonal cutting: cutting edge perpendicular to cutting velocity (2D – shaping a wide plate with a straight tool). Oblique cutting: edge inclined (3D – most practical operations, e.g., turning).
- Tool angles (ASA): back rake, side rake, end relief, side relief, end cutting edge angle, side cutting edge angle, nose radius → signature: αb–αs–θe–θs–Ce–Cs–R.
- Positive rake: lower forces, weaker edge (HSS, ductile materials). Negative rake: strong edge, used for carbide/ceramic tools, hard materials, interrupted cuts.
- Relief (clearance) angle prevents rubbing of the flank on the machined surface.

```
Chip thickness ratio r = t/t_c = L_c/L = V_c/V (< 1); cutting ratio
Shear angle: tan φ = r cos α / (1 − r sin α)
Merchant's theory: 2φ + β − α = 90° (β = friction angle)
Shear strain γ = cot φ + tan(φ − α); velocity of shear Vs = V cos α / cos(φ − α); chip velocity Vc = V sin φ / cos(φ − α)
Friction coefficient μ = (Fc tan α + Ft)/(Fc − Ft tan α)
Cutting power P = Fc·V; specific cutting energy u = Fc/(b·t)
```

- Merchant's circle: forces – cutting force Fc, thrust Ft, shear force Fs, normal to shear Fn, friction F, normal N. All meet at the tool tip; resultant R is the diameter.
- Lee–Shaffer: $\varphi + \beta - \alpha = 45°$.

Chip type	Conditions
Continuous	Ductile material, high speed, large rake, small feed, good lubrication – best finish
Continuous with built-up edge (BUE)	Ductile, low speed, small rake, high friction, no coolant – poor finish
Discontinuous (segmented)	Brittle materials (CI, brass), low speed, small/negative rake, large feed
Serrated/saw-tooth	Low thermal conductivity hard metals (Ti alloys)

- Heat distribution: majority goes into chip (~80%), tool ~10%, work ~10% at normal speeds. Primary shear zone generates the most heat; max temperature on the rake face a little away from the edge (crater location).

★ **PYQ-TREND:** Merchant's equation $2\varphi + \beta - \alpha = 90°$ – asked every cycle. Chip thickness ratio < 1. BUE formation at low speed in ductile materials. Discontinuous chips in cast iron. Negative rake for carbide tools. Maximum heat goes to chips. Orthogonal vs oblique cutting difference.

13.2 Tool Wear, Tool Life and Machinability

- Flank wear (on clearance face; due to abrasion; decides tool life – VB criterion, e.g., 0.3 mm). Crater wear (on rake face; due to diffusion at high temperature; carbide tools at high speeds). Notch wear, chipping, plastic deformation.

```
Taylor's tool life equation: V Tⁿ = C
Extended Taylor: V Tⁿ f^a d^b = C
```

- n values: HSS 0.08–0.2 (≈ 0.125), carbide 0.2–0.5 (≈ 0.25), ceramic 0.5–0.7.
- Tool life is most affected by cutting speed, then feed, then depth of cut.

```
Optimum tool life for min cost: T_o = (1/n − 1)(C_t/C_m + T_c); for max production rate: T_o = (1/n − 1) T_c
Optimum speed for minimum cost < optimum speed for maximum production (V_mc < V_mp)
```

- Machinability: ease of machining – assessed by tool life, surface finish, cutting forces/power, chip form. Free-cutting steel contains S, Pb (MnS inclusions act as chip breakers). Machinability index of free-cutting steel AISI 1112 = 100%.

Tool material	Hot hardness / remark
Plain carbon tool steel	Up to ~250 °C, low speeds
HSS (18-4-1: W-Cr-V)	Up to ~600 °C; drills, taps, form tools
Cast alloys (Stellite – Co-Cr-W)	Up to ~800 °C
Cemented carbides (WC–Co)	Up to ~1000 °C; P (steel, blue), M (multi), K (CI, red) grades; inserts; coated (TiN, TiC, Al_2O_3)
Ceramics (Al_2O_3, Si_3N_4)	Up to ~1200 °C; brittle; negative rake; high speeds
CBN (cubic boron nitride)	Second hardest; hardened steels
Diamond (PCD)	Hardest; non-ferrous, plastics; not for steel (carbon diffuses into iron)
UCON	Refractory alloy (Nb-Ti-W) tool

- Order of hardness: diamond > CBN > ceramic > carbide > cast alloy > HSS > carbon steel.
- Cutting fluids: cool, lubricate, flush chips, prevent corrosion. Water-soluble oils for high speed (cooling); straight oils for low speed (lubrication); dry cutting for CI (and ceramics).

★ **PYQ-TREND:** Taylor's equation and n values for HSS vs carbide. Flank wear decides tool life; crater wear due to diffusion. Diamond not used for ferrous metals. HSS composition 18-4-1. Free-cutting steel contains S and Pb. Cutting speed most affects tool life. CI is machined dry.

✔ **TOPPER TRICK:** From $V T^n = C$: if n = 0.5 and speed is doubled, tool life becomes 1/4. In general $T_2/T_1 = (V_1/V_2)^{(1/n)}$.

13.3 Lathe and Turning

- Lathe parts: bed, headstock (spindle, back gears), tailstock (dead centre), carriage (saddle, cross-slide, compound rest, tool post, apron), lead screw (thread cutting only, Acme thread), feed rod.
- Size of lathe: swing over bed and distance between centres (also bed length).
- Work-holding: three-jaw (self-centring – round/hexagonal), four-jaw (independent – irregular), collet (bar stock, high precision), faceplate (irregular), centres with carrier/dog, mandrel (holding bored jobs), steady rest (long jobs, fixed to bed) & follower rest (moves with carriage).

```
Cutting speed V = πDN/1000 (m/min)
```

```
Machining time t = L/(fN) per pass
Material removal rate MRR = π D_avg d f N ≈ V f d
Taper turning: tan(α/2) = (D − d)/(2L); tailstock set-over = L_total (D − d)/(2L)
Thread cutting: change gear ratio = pitch of job / pitch of lead screw = driver/driven
```

- Taper methods: form tool (short, steep), compound rest swivel (short, steep – internal & external), tailstock set-over (long, small taper, external only), taper turning attachment (long, internal & external), combined feed.
- Operations: facing, turning, step turning, chamfering, knurling (rough grip – forming, not cutting), parting (grooving), boring, drilling, reaming, threading, eccentric turning.
- Capstan & turret lathes: mass production; capstan (ram-type, light work, bar stock) vs turret (saddle-type, heavy chucking work). Automatic lathes; Swiss-type (sliding headstock). CNC lathes.

```
Surface roughness (theoretical, nose radius r): h = f²/(8r)
```

★ **PYQ-TREND:** Lead screw has Acme threads. Size of lathe specification. Tailstock set-over method for long tapers. Knurling is a forming operation. Peak-to-valley roughness f²/8r. Cutting speed formula. Steady rest vs follower rest. Four-jaw chuck for irregular jobs.

13.4 Drilling, Boring, Reaming

- Drill: twist drill (HSS, 118° point angle standard, helix angle 30°, lip clearance 8–12°). Chisel edge (dead centre) – does not cut, only extrudes → high thrust.

```
Drilling time t = (L + 0.3D)/(fN); MRR = (πD²/4) f N
```

- Drilling machines: sensitive/bench (small holes, hand feed), upright, radial (large heavy jobs – arm swings), gang (several spindles in a row), multi-spindle, deep-hole (gun drill).
- Operations: reaming (finishing a drilled hole, small allowance), boring (enlarging, correcting position), counterboring (stepped hole for bolt head), countersinking (conical for flat-head screw), spot facing (flat seat), tapping (internal thread), trepanning (large hole by removing annular ring).
- Jig: holds work and guides tool (drill bushes). Fixture: holds/locates work only (milling). 3-2-1 principle of location: 6 points restrict 12 degrees of freedom (6 locators + clamps).

★ **PYQ-TREND:** Point angle 118°. Radial drilling machine for heavy jobs. Countersinking vs counterboring. Jig guides tool, fixture does not. 3-2-1 principle. Chisel edge causes large thrust. Trepanning definition.

13.5 Milling

- Up (conventional) milling: cutter rotation opposite to feed; chip thickness zero → max; tends to lift work; suitable for older machines with backlash, castings with hard scale. Down (climb) milling: rotation along feed; chip max → zero; better finish & tool life; needs backlash eliminator; pushes work down.

```
Feed per minute fm = fz × Z × N
Approach (slab milling) = √(d(D − d)); machining time t = (L + approach)/fm
Max chip thickness (up milling) ≈ 2fz √(d/D)
```

- Cutters: plain/slab, side & face, staggered tooth, end mill, slot, T-slot, woodruff key-seat, form, fly cutter, angle, slitting saw, gang & straddle milling arrangements.

- Indexing (dividing head, 40:1 worm ratio): simple indexing crank turns = 40/N. Differential indexing for numbers not available by simple. Plate holes: Brown & Sharpe plates.
- Milling of gears: form milling with involute cutter (8 cutters per module set). Gear generation (hobbing – spur & helical, continuous, most productive; shaping – internal gears & clusters). Gear finishing: shaving (soft gears), grinding & lapping (hardened gears), burnishing.

★ **PYQ-TREND:** Up vs down milling – climb milling gives better finish but needs backlash eliminator. Simple indexing 40/N. Hobbing cannot cut internal gears; gear shaping can. Straddle milling uses two side cutters. Number of cutters in a set = 8.

13.6 Shaper, Planer, Slotter, Broaching

- Shaper: tool reciprocates, work feeds; quick return mechanism (crank & slotted lever / Whitworth / hydraulic); small jobs. Planer: work reciprocates on table, tool feeds; large heavy jobs. Slotter: vertical shaper – keyways, internal splines.

```
Shaper cutting time per stroke: L(1 + m)/(1000V) where m = return/cutting time ratio
```

- Broaching: multi-tooth tool with progressively increasing teeth (rise per tooth); one stroke completes roughing, semi-finishing & finishing; keyways, splines, internal profiles; high production; costly tool. Pull and push broaches.

★ **PYQ-TREND:** In a shaper the tool reciprocates; in a planer the workpiece reciprocates. Broaching for internal keyways and splines. Quick return ratio in shaper.

13.7 Grinding and Finishing

- Abrasives: aluminium oxide (A – for steels, high tensile materials), silicon carbide (C – for CI, non-ferrous, carbides, low tensile), CBN, diamond.
- Grinding wheel designation (IS 551): 51 A 36 L 5 V 23 → manufacturer's symbol – abrasive type (A) – grain size (36, coarse 10–24, medium 30–60, fine 70–180, very fine 220–600) – grade (hardness A soft → Z hard; L medium) – structure (1 dense → 15 open) – bond (V vitrified, S silicate, B resinoid, R rubber, E shellac, M metal) – manufacturer's record.
- Rule: hard material → soft wheel; soft material → hard wheel (to allow self-sharpening/self-dressing). Vitrified bond most common.
- Glazing: wheel surface becomes smooth/shiny (dull grains not breaking – wheel too hard, speed high). Loading: chips clog pores (soft materials, wheel too fine). Remedy: dressing. Truing: restoring geometry/concentricity.
- Grinding operations: cylindrical, surface, internal, centreless (no centres; regulating wheel rubber-bonded, inclined 2–8° gives axial feed; through-feed & in-feed).

```
Specific energy in grinding is very high (due to size effect & ploughing, negative rake grits)
```

- Finishing processes (increasing finish): honing (bores – abrasive sticks, cross-hatch – cylinder liners), lapping (loose abrasive, flat precise surfaces, gauges), superfinishing, buffing & polishing (lustre).
- Surface finish Ra: rough turning 6–25 µm; grinding 0.1–1.6 µm; lapping 0.05–0.4 µm.

★ **PYQ-TREND:** Grinding wheel marking order – almost every year. Soft wheel for hard material. SiC for cast iron, Al_2O_3 for steel. Glazing vs loading; dressing vs truing. Centreless grinding regulating wheel. Honing for engine cylinder bores. Lapping gives the best finish.

13.8 Non-Traditional Machining (NTM)

Process	Mechanism / Medium	Applications / facts
USM (ultrasonic)	Abrasive slurry hammered by tool vibrating at 20–30 kHz (magnetostrictive transducer)	Hard brittle: glass, ceramics, gems; MRR highest for brittle
AJM (abrasive jet)	High velocity gas jet (150–300 m/s) with Al_2O_3/SiC; air, CO_2, N_2 (not O_2)	Deburring, cutting thin brittle sections, etching
WJM / AWJM	Water jet ~ 400 MPa; with abrasive garnet	Soft materials (food, paper, rubber); AWJM for metals, composites
ECM (electrochemical)	Anodic dissolution (Faraday's law); electrolyte NaCl/$NaNO_3$; tool cathode; no tool wear	Hard conductive alloys, turbine blades, complex shapes; highest MRR among NTM
EDM (electric discharge)	Sparks in dielectric (kerosene, deionised water); tool = Cu/graphite; workpiece +ve usually	Dies, moulds, hard conductive materials; RC relaxation circuit; heat-affected recast layer
Wire EDM	Moving brass wire, deionised water	Through-cut profiles, punches/dies
LBM (laser)	Coherent light melts/vaporises (CO_2, Nd:YAG)	Fine holes, cutting, any material, no vacuum
EBM (electron beam)	High-velocity electrons in vacuum	Micro holes, very fine cuts, high precision
PAM (plasma arc)	Ionised gas at 10,000–30,000 °C	Cutting stainless, aluminium plates
CHM (chemical milling)	Etchant with maskant	Aircraft panels (weight reduction), PCB
ECG (electrochemical grinding)	ECM + grinding (90% electrochemical)	Carbide tools grinding

- EDM: MRR ∝ energy per spark ½CV^2; tool wear ratio; surface finish depends on spark energy & frequency.
- ECM: MRR = $I \cdot A/(Z \cdot F \cdot \rho)$ (Faraday) – independent of hardness; only conductive materials.
- Electrically non-conductive materials: use USM, AJM, LBM, EBM (not EDM/ECM).

★ **PYQ-TREND:** USM for glass/ceramics; ECM based on Faraday's law, no tool wear; EDM dielectric kerosene; EBM requires vacuum; LBM doesn't. AJM does not use oxygen (fire hazard). ECM highest MRR among NTMs. Non-conductive materials can't be machined by EDM/ECM. Magnetostriction in USM.

13.9 CNC, CAD/CAM and Automation Basics

- NC → CNC (computer) → DNC (direct, central computer controls many machines). Machining centre: ATC (automatic tool changer), multi-axis.
- Axes: Z-axis along the spindle; X horizontal (largest travel); Y by right-hand rule.
- Part programming codes: G-codes (preparatory): G00 rapid traverse, G01 linear interpolation, G02 circular CW, G03 circular CCW, G04 dwell, G20/G21 inch/mm, G90 absolute, G91 incremental, G41/G42 cutter radius compensation left/right, G40 cancel. M-codes (miscellaneous): M00 program stop, M02 end of program, M03 spindle CW, M04 CCW, M05 spindle stop, M06 tool change, M08 coolant on, M09 coolant off, M30 end &

rewind.

- Open loop (stepper motor) vs closed loop (servo with feedback – encoder/resolver). APT is a part-programming language.
- FMS (flexible manufacturing system), CIM, group technology (GT – Opitz/MICLASS coding), rapid prototyping (SLA, SLS, FDM, 3D printing – additive manufacturing).

★ **PYQ-TREND:** G02/G03 meanings, M03/M05/M06. Z-axis is along the spindle. DNC definition. FDM/SLA are additive manufacturing. Group technology coding systems.

13.10 Machining Economics & Unconventional Notes

- Total cost per piece = machining cost + tool cost + tool changing cost + non-productive cost.
- Higher-efficiency approaches: high-speed machining, hard turning (with CBN), dry & MQL machining.
- Cutting forces increase with feed and depth roughly linearly; with speed, decrease slightly (temperature softening).

★ **PYQ-TREND:** Optimum speed for max production is higher than for minimum cost. Feed affects surface finish the most ($h = f^2/8r$).

✎ **DESCRIPTIVE Q:** Explain Merchant's circle diagram and derive the relation $2\varphi + \beta - \alpha = 90°$.

In orthogonal cutting the resultant force R on the chip is shown three ways on one circle of diameter R: (i) cutting force Fc and thrust Ft along and normal to the cutting velocity; (ii) shear force Fs and normal force Fn along and normal to the shear plane; (iii) friction force F and normal N along and normal to the rake face. Geometry gives $Fs = R\cos(\varphi + \beta - \alpha)$. Shear stress on the shear plane $\tau = Fs\sin\varphi/(bt) = R\cos(\varphi + \beta - \alpha)\sin\varphi/(bt)$, and $Fc = R\cos(\beta - \alpha)$. Merchant assumed the shear plane forms at the angle φ that minimises Fc for a given τ, i.e., maximises $\cos(\varphi + \beta - \alpha)\sin\varphi$. Differentiating with respect to φ and setting the result to zero gives $\cos(2\varphi + \beta - \alpha) = 0$, so $2\varphi + \beta - \alpha = 90°$.

CHAPTER 14

Engineering Materials & Heat Treatment

14.1 Crystal Structure

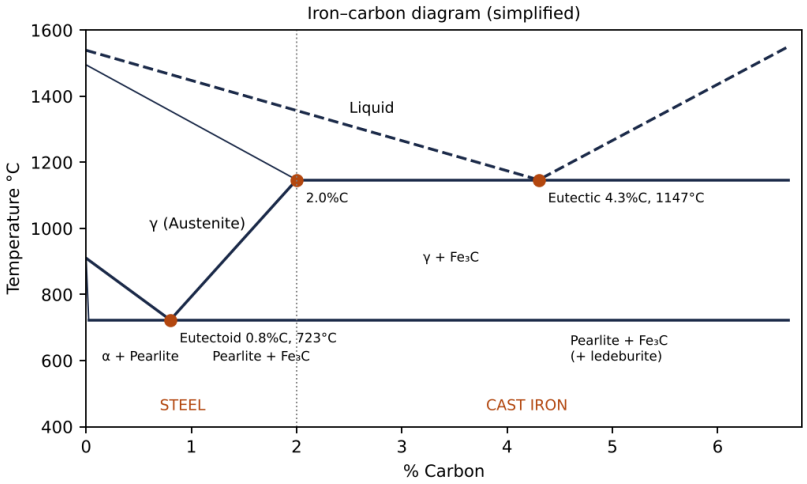
Structure	Atoms/unit cell	Coordination no.	APF	Examples
Simple cubic (SC)	1	6	0.52	Polonium
BCC	2	8	0.68	α-Fe (ferrite), Cr, W, Mo, V, Na
FCC	4	12	0.74	γ-Fe (austenite), Al, Cu, Ni, Ag, Au, Pb
HCP	6	12	0.74	Zn, Mg, Ti (α), Cd, Be, Co

`BCC: a = 4r/√3; FCC: a = 2√2 r = 4r/√2`

- FCC metals are most ductile (12 slip systems on close-packed planes {111}). BCC has 48 slip systems but not close packed → ductile-to-brittle transition. HCP has few slip systems (3) → less ductile.
- Miller indices: reciprocals of intercepts, cleared of fractions. Close-packed plane FCC {111}, BCC {110}.
- Imperfections: point (vacancy, interstitial, substitutional, Frenkel, Schottky), line (edge & screw dislocations – Burgers vector perpendicular to line in edge, parallel in screw), surface (grain boundaries, twin boundaries, stacking faults), volume (voids, inclusions).
- Plastic deformation occurs by slip (dislocation movement) and twinning. Strengthening: grain refinement (Hall–Petch $\sigma y = \sigma_0 + k\, d^{(-1/2)}$), solid solution, strain hardening, precipitation hardening, dispersion.
- Allotropy (polymorphism): iron – δ-Fe (BCC, 1394–1539 °C), γ-Fe (FCC, 910–1394 °C), α-Fe (BCC, < 910 °C). Curie temperature of iron 768 °C (magnetic change, no structure change).

★ **PYQ-TREND:** APF of FCC/HCP 0.74, BCC 0.68. Number of atoms per unit cell. Coordination numbers. Examples of BCC/FCC/HCP metals. Allotropic temperatures of iron. Edge vs screw dislocation Burgers vector. Hall–Petch relation.

14.2 Iron–Carbon Equilibrium Diagram



Reaction	Temp & %C	Equation
Peritectic	1495 °C, 0.16–0.18% C	L + δ → γ
Eutectic	1147 °C (1130–1148), 4.3% C	L → γ + Fe_3C (ledeburite)
Eutectoid	723 °C (727), 0.8% C (0.76–0.8)	γ → α + Fe_3C (pearlite)

- Max solubility of C: in α-ferrite 0.025% (at 723 °C), in austenite 2.0–2.11% (at 1147 °C), in δ-ferrite 0.1%.
- Phases/micro-constituents: ferrite (soft, ductile, BCC), austenite (FCC, non-magnetic, stable above 723 °C in plain steel), cementite (Fe_3C, 6.67% C, hardest & brittle, orthorhombic), pearlite (lamellar ferrite + cementite – 87.5% ferrite), ledeburite (eutectic mixture).
- Steels: < 2% C. Hypoeutectoid < 0.8% (ferrite + pearlite), eutectoid 0.8% (100% pearlite), hypereutectoid 0.8–2% (pearlite + cementite). Cast iron: 2–6.67% C (practically 2–4%).
- Critical temperatures: A_1 (lower critical, 723 °C), A_3 (upper for hypoeutectoid), A_cm (for hypereutectoid).

★ **PYQ-TREND:** Eutectoid at 723 °C & 0.8% C giving pearlite; eutectic at 1147 °C & 4.3% C giving ledeburite. Cementite 6.67% C. Max carbon in austenite 2%. Peritectic reaction equation. Steel/cast iron carbon limits. Austenite is FCC and non-magnetic. Lever rule for phase fractions.

✔ **TOPPER TRICK:** Lever rule: % pearlite in a 0.4% C steel ≈ (0.4 − 0.025)/(0.8 − 0.025) × 100 ≈ 48% (≈ 50%). For quick work, % pearlite ≈ (%C/0.8) × 100.

14.3 Steels and Cast Irons

- Plain carbon steels: dead mild (< 0.15% C – wires, sheets), low carbon/mild (0.15–0.3% C – structural, bolts), medium (0.3–0.6% – rails, axles, shafts, gears), high carbon (0.6–1.5% – tools, springs, files, razors).
- Alloying effects: Ni – toughness, corrosion resistance, austenite stabiliser. Cr – hardness, corrosion resistance (> 12% stainless), carbide former. Mn – hardenability, deoxidiser, counters S (hot shortness); Hadfield steel 12–14% Mn (wear resistant – crusher jaws, rails crossings, work-hardening). Mo – reduces temper embrittlement, high-temp strength. W – hot hardness (tool steels). V – grain refinement, hot hardness. Si – deoxidiser, electrical (transformer) steel, spring steel. Co – only element that decreases hardenability (in HSS raises hot hardness). S & P – impurities (S → hot shortness, P → cold shortness); S & Pb added for free cutting.

- Stainless steels: ferritic (Cr only, magnetic), martensitic (Cr + higher C, hardenable – cutlery, surgical), austenitic (18-8: 18% Cr, 8% Ni, non-magnetic, most corrosion resistant – kitchen, chemical), duplex, precipitation hardening (17-4 PH). Sensitisation (weld decay) – Cr carbide at grain boundaries in 18-8 steel at 500–800 °C; prevented with low C (304L) or stabilisers (Ti, Nb – 321, 347).
- Designation IS: 40C8 → 0.40% C, 0.8% Mn (approx. 0.6–0.9). AISI 1040 → plain carbon 0.40% C. Alloy steel 35Ni1Cr60: 0.35% C, 1% Ni, 0.6% Cr.

Cast iron	Graphite form / properties / uses
Grey CI	Graphite flakes; good damping & machinability, compressive strength, brittle; machine beds, engine blocks
White CI	Carbon as cementite (fast cooling, low Si); very hard & brittle; wear parts; base for malleable
Malleable CI	Temper carbon (rosettes) by annealing white CI; ductile; pipe fittings, chains
Nodular (SG/ductile) CI	Spheroidal graphite by Mg/Ce; ductility & strength near steel; crankshafts
Chilled CI	Hard white surface, grey core
Alloy CI (Ni-hard, Ni-resist)	Abrasion/corrosion resistance

★ **PYQ-TREND:** 18-8 stainless composition (Cr-Ni). Hadfield manganese steel work-hardens. Mg/Ce for SG iron. Grey CI has flake graphite and best damping. Malleable CI by annealing white CI. Hot shortness by S, cold shortness by P. Cr > 12% for stainless. Co decreases hardenability. IS designation 40C8.

14.4 Non-Ferrous Metals & Alloys

Alloy	Composition	Use
Brass	Cu + Zn (yellow brass 70/30 – cartridge brass; Muntz metal 60/40)	Cartridges, fittings
Bronze	Cu + Sn (phosphor bronze adds P; Al bronze; gun metal 88Cu-10Sn-2Zn)	Bearings, springs, gears
German silver (nickel silver)	Cu + Zn + Ni (no silver)	Utensils, decorative
Monel	Ni (~67%) + Cu (~30%)	Corrosion resistance – marine, chemical
Inconel	Ni + Cr + Fe	High temperature – furnace, jet parts
Nichrome	Ni 80 + Cr 20	Heating elements
Duralumin	Al + 4% Cu + Mg + Mn	Aircraft (age hardenable)
Y-alloy	Al + Cu + Ni + Mg	Pistons, cylinder heads (high temp)
Babbitt (white metal)	Sn + Sb + Cu (or Pb based)	Bearings
Solder (soft)	Sn + Pb (63/37 eutectic)	Electrical joints
Invar	Fe + 36% Ni	Near-zero thermal expansion – measuring tapes, clocks
Constantan	Cu 55 + Ni 45	Thermocouples, resistors

Alloy	Composition	Use
Magnalium	Al + Mg	Light structures
Stellite	Co + Cr + W	Cutting tools, valve seats
Alnico	Al + Ni + Co + Fe	Permanent magnets
Hindalium	Al + Mg + Mn + Cr + Si	Utensils
Gun metal	Cu 88 + Sn 10 + Zn 2	Bearings, valves
Bell metal	Cu 80 + Sn 20	Bells

★ **PYQ-TREND:** Brass = Cu + Zn; bronze = Cu + Sn – the most common facts question. Duralumin composition & aircraft use. German silver has no silver. Invar's low expansion. Monel = Ni + Cu. Babbitt for bearings. Y-alloy for pistons. Gun metal composition.

14.5 Heat Treatment

- Heat treatment: controlled heating & cooling of solid metal to alter properties without changing shape.

Process	Procedure	Purpose / result
Full annealing	Heat 30–50 °C above A_3 (hypo) or A_1 (hyper), slow furnace cooling	Softening, ductility, machinability, relieve stresses, coarse pearlite
Process annealing	Below A_1 (550–650 °C)	Restore ductility between cold-working operations (recrystallisation)
Spheroidising	Long heating just below A_1	Spheroidal cementite – best machinability of high-C steels
Stress relieving	Below A_1, slow cooling	Remove residual stresses (welding/casting)
Normalising	Heat above A_3/A_cm, air cooling	Grain refinement, uniform fine pearlite; stronger & harder than annealed
Hardening	Heat above A_3 (hypo) or A_1 (hyper), quench in water/oil/brine	Martensite (BCT, hardest & brittle)
Tempering	Reheat hardened steel below A_1 (150–650 °C), cool	Reduce brittleness, relieve stress, gain toughness; low (150–250 °C – tools), medium (350–450 °C – springs), high (500–650 °C – shafts, gears – sorbite)
Austempering	Quench to above Ms, hold isothermally → bainite	Toughness without tempering, less distortion
Martempering (marquenching)	Quench to just above Ms, hold till uniform, then air cool → martensite	Reduces cracking/distortion
Patenting	Austenitise, quench in lead bath ~500 °C → fine pearlite	High-strength wire

- Martensite: supersaturated solid solution of carbon in α-iron; BCT structure; forms by diffusionless shear transformation; hardness depends mainly on carbon content.
- TTT (isothermal transformation) diagram – C-curve; critical cooling rate = minimum rate to avoid nose and get 100% martensite. Products: coarse pearlite (high T) → fine pearlite →

upper bainite → lower bainite → martensite (below Ms).

- Quenching media (severity, decreasing): brine > water > oil > molten salt > air.
- Hardenability: depth to which steel can be hardened – Jominy end-quench test. Increased by most alloying elements (Mn, Cr, Mo, Ni) and larger grain size; decreased by Co.
- Retained austenite: reduced by sub-zero (cryogenic) treatment.

Surface hardening

Process	Medium / temp	Notes
Pack carburising	Charcoal + $BaCO_3$ energiser, 900–950 °C	Low C steel; case 0.5–2 mm
Gas carburising	CH_4/propane	Mass production
Liquid carburising (cyaniding)	Sodium cyanide bath ~850 °C	C + N; thin case
Carbonitriding	Gas: hydrocarbons + NH_3 (~850 °C)	Thin hard case
Nitriding	NH_3 at 500–550 °C (below A_1); no quench needed	Hardest case, least distortion; alloy steels with Al, Cr, Mo (Nitralloy)
Flame hardening	Oxy-acetylene flame + quench	Medium carbon steel; large parts
Induction hardening	High-frequency current (skin effect) + quench	Medium carbon steel; gears, crankshafts; case depth ∝ 1/√f

★ **PYQ-TREND:** Normalising = air cooling; annealing = furnace cooling. Martensite is BCT and formed by diffusionless transformation. Jominy test for hardenability. Nitriding is done below the critical temperature with no quench and least distortion. Austempering produces bainite. Induction hardening depth decreases with frequency. Tempering reduces brittleness. Severity of quench order.

✖ **COMMON MISTAKE:** Carburising adds carbon to low-carbon steel. Flame and induction hardening add nothing; they need a medium-carbon steel that already has enough carbon to harden.

14.6 Mechanical Testing

Test	Notes
Tensile (UTM)	Gauge length $L_0 = 5.65\sqrt{A_0}$ (IS); % elongation, % reduction in area
Brinell hardness (BHN)	10 mm hardened steel/carbide ball, 3000 kgf (steel), 500 kgf (soft); BHN = P/(curved area); UTS (MPa) ≈ 3.45 × BHN
Rockwell	Depth of penetration; B scale (1/16" ball, 100 kgf – soft), C scale (diamond cone 120°, 150 kgf – hard steels); minor load 10 kgf
Vickers (VHN/HV)	Square-based diamond pyramid 136°; all materials; micro-hardness
Knoop	Elongated pyramid; micro-hardness, thin layers, brittle materials
Shore scleroscope	Rebound height – dynamic hardness
Mohs scale	Scratch, 1 (talc) to 10 (diamond)
Izod impact	Cantilever specimen, V-notch facing striker; 2 mm notch; 120 J

Test	Notes
Charpy impact	Simply supported specimen, notch away from striker; U or V notch; 300 J
Fatigue	R.R. Moore rotating beam – S–N curve
Creep	Constant load at high temperature: primary, secondary (steady minimum rate – design), tertiary
NDT	Dye penetrant (surface cracks – non-porous), magnetic particle (surface/subsurface – ferromagnetic only), eddy current (conductive), ultrasonic (internal flaws, thickness), radiography (X/γ rays – internal, porosity)

★ **PYQ-TREND:** Rockwell C uses diamond cone; Vickers 136° pyramid. Izod (cantilever) vs Charpy (simply supported). Creep stages – secondary used for design. NDT method selection (magnetic particle only for ferromagnetic). Mohs scale extremes. Brinell–UTS relation.

14.7 Polymers, Ceramics and Composites

- Thermoplastics (soften on heating, recyclable; linear chains): PE, PP, PVC, PS, nylon, PTFE (Teflon – lowest μ, non-stick), PMMA (acrylic/Perspex – transparent), PC, ABS. Thermosets (cross-linked, set permanently): Bakelite (phenol formaldehyde), epoxy, polyester, urea formaldehyde, melamine. Elastomers: natural rubber, neoprene, silicone; vulcanisation with sulphur.
- Polymer processing: injection moulding (thermoplastics, mass production), extrusion (pipes, films), blow moulding (bottles), compression & transfer moulding (thermosets), thermoforming, calendering (sheets), rotational moulding (large hollow – tanks).
- Glass transition temperature Tg: amorphous polymer changes from glassy to rubbery.
- Ceramics: high hardness, high melting point, brittle, low conductivity: alumina, SiC, Si_3N_4, zirconia.
- Composites: matrix (polymer, metal, ceramic) + reinforcement (fibres: glass, carbon, Kevlar; particles; laminates). Rule of mixtures: Ec = Ef Vf + Em Vm (longitudinal). Concrete (particulate), wood (natural fibre composite), RCC, GFRP, CFRP.

★ **PYQ-TREND:** Bakelite is a thermoset. PTFE (Teflon) lowest friction. Vulcanisation adds sulphur. Injection moulding for thermoplastics, compression moulding for thermosets. Blow moulding for bottles. Rule of mixtures. Wood is a natural composite.

✎ **DESCRIPTIVE Q:** Draw the iron–carbon equilibrium diagram (steel portion) and explain the eutectoid reaction and microstructure of hypo- and hyper-eutectoid steels on slow cooling.

On slow cooling, austenite (γ, FCC) of 0.8% C at 723 °C transforms by the eutectoid reaction γ → α + Fe_3C into pearlite, a lamellar mixture of ferrite and cementite. In a hypoeutectoid steel (< 0.8% C), proeutectoid ferrite starts to form at the A_3 line and grows until 723 °C. The remaining austenite, now at 0.8% C, transforms to pearlite, so the final structure is ferrite + pearlite. In a hypereutectoid steel (0.8–2% C), proeutectoid cementite precipitates along austenite grain boundaries from the A_cm line down to 723 °C, then the remaining austenite becomes pearlite. The final structure is pearlite + network of cementite, which is harder and more brittle. Phase proportions come from the lever rule.

CHAPTER 15

Metrology, Limits & Fits, Industrial Engineering

15.1 Measurement Basics

- Accuracy: closeness to the true value. Precision: repeatability (closeness among repeated readings). Sensitivity: output change per unit input. Resolution: smallest detectable change. Readability, range, span, hysteresis, drift, calibration, traceability.
- Errors: gross (human), systematic (instrumental, environmental, observational – correctable), random (statistical – reduced by averaging).
- Standard temperature for measurement: 20 °C. Line standard (scale, yard, metre bar) vs end standard (slip gauges, length bars) vs wavelength standard (metre defined by light: distance travelled by light in 1/299,792,458 s).
- Abbe's principle: the measuring axis should be in line with the scale axis to avoid cosine/parallax errors (micrometer obeys; vernier caliper does not).

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Vernier least count = 1 MSD − 1 VSD = value of smallest main-scale division / number of vernier divisions
Micrometer least count = pitch / number of circular scale divisions (0.5/50 = 0.01 mm)
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★ **PYQ-TREND:** Accuracy vs precision. Least count of vernier (0.02 mm) and micrometer (0.01 mm). Abbe's principle. Standard measuring temperature 20 °C. Systematic errors are correctable.

15.2 Linear & Angular Measurement Instruments

- Slip gauges (Johansson gauge blocks): wringing; grades 00 (calibration), 0 (inspection), 1, 2 (workshop); normal set M-87 or M-112; protector slips of 2.5 mm.
- Dial gauge (dial indicator): comparative; rack & pinion.
- Comparators: mechanical (Johansson Mikrokator – twisted strip; Sigma), optical, electrical (LVDT), pneumatic (Solex – high magnification, non-contact, bores), electronic.
- Angular: sine bar (sin θ = h/L; not reliable > 45°; length between roller centres 100/200/250 mm), bevel protractor (least count 5'), angle gauges (13 pieces combine to measure in steps of 3"), autocollimator (small angular deviations, straightness, flatness), clinometer, spirit level.
- Surface plate (CI or granite), V-block, height gauge, telescopic gauge, bore gauge.
- Optical flat & monochromatic light: flatness by interference fringes; each fringe = λ/2 height difference.
- CMM (coordinate measuring machine): 3D measurement with probe.

★ **PYQ-TREND:** Sine bar unreliable above 45°. Wringing of slip gauges. Optical flat fringe = λ/2. Autocollimator for small angles and flatness. LVDT is an electrical comparator. Bevel protractor least count 5 minutes. Pneumatic comparator for bore measurement.

15.3 Limits, Fits and Tolerances

- Basic size, actual size, limits (max & min), tolerance = upper limit − lower limit, allowance = intentional difference between max material limits of mating parts (min clearance or max interference), deviation (upper/lower) from basic size, zero line.
- Hole basis system (hole lower deviation = 0, H): preferred because holes are made with standard fixed-size tools (drills, reamers) and shafts are easier to vary. Shaft basis (h).
- Unilateral tolerance (one direction – preferred for interchangeable manufacturing) vs bilateral (±).
- Indian/ISO system: 18 (IT01, IT0, IT1–IT16) grades; 28 fundamental deviations (A–ZC for holes, a–zc for shafts).

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Standard tolerance unit i = 0.45 ∛D + 0.001 D (µm, D in mm = geometric mean of diameter step)
IT7 = 16i, IT8 = 25i, IT9 = 40i, IT10 = 64i, IT11 = 100i (R5 series from IT6 = 10i)
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Fit	Condition	Examples
Clearance	Min shaft size < min hole size (always gap)	H7/g6 sliding, H8/f7 running, H11/c11 loose
Transition	May give clearance or interference	H7/k6, H7/n6 – gears, pulleys on shafts with keys
Interference	Max shaft > min hole (always overlap)	H7/p6 press fit, H7/s6 shrink fit – bushes, wheels on axles

- Selective assembly: parts graded into groups and matched – high accuracy at lower cost (ball bearings).
- Interchangeability: any part from a lot fits the mating part without fitting/adjustment – essential for mass production.
- Limit gauges (Taylor's principle): GO gauge checks maximum material condition with full form; NO-GO gauge checks the other limit with a single dimension (point contact). Plug gauge for holes, ring/snap gauge for shafts.
- Gauge maker's tolerance ≈ 10% of work tolerance; wear allowance applied to GO gauge.

★ **PYQ-TREND:** Hole basis is preferred. Taylor's principle of gauging. H7/p6 interference fit. Allowance vs tolerance definition. Number of IT grades (18) and fundamental deviations (28). Selective assembly. GO gauge checks the maximum metal limit.

✔ **TOPPER TRICK:** The hole always carries the capital letter. H is the basic hole (EI = 0). If the shaft letter comes earlier in the alphabet (a–h) the fit is clearance; js–n is transition; p–zc is interference (with H holes).

15.4 Surface Finish, Screw Thread & Gear Measurement

- Surface roughness parameters: Ra (CLA – centre line average, arithmetic mean deviation – most used), Rq (RMS), Rz (ten-point height), Rt/Rmax (peak to valley). Ra ≈ Rq/1.11 for sinusoid.
- Lay: direction of predominant surface pattern; waviness (larger spacing) vs roughness (finer). Sampling length (cut-off) typically 0.8 mm.
- Instruments: Taylor–Hobson Talysurf (stylus, electronic), Tomlinson surface meter (mechanical-optical), profilometer.
- Screw thread: effective (pitch) diameter by two-wire/three-wire method (best wire size d = (p/2) sec(θ/2)); floating carriage micrometer; thread pitch by pitch measuring machine; tool maker's microscope; profile projector.

- Gear: tooth thickness by gear tooth vernier caliper (chordal thickness at chordal addendum), constant chord method, base tangent (span) method (David Brown); Parkinson gear tester (composite rolling test).

★ **PYQ-TREND:** CLA = Ra. Talysurf uses a stylus. Three-wire method for effective diameter; best wire size. Gear tooth vernier for chordal thickness. Profile projector & toolmaker's microscope for thread form.

15.5 Production Planning, Forecasting & Inventory

- Productivity = output/input. Production types: job (low volume, high variety, general-purpose machines, process layout), batch, mass/flow (high volume, special-purpose machines, product/line layout), continuous (process industries).
- Plant layouts: process (functional – job shops), product (line – mass production), fixed-position (ships, aircraft – product remains stationary), cellular/group (GT), combination.
- Forecasting: qualitative (Delphi – expert consensus, market survey) and quantitative (moving average, weighted moving average, exponential smoothing $F_{t+1} = F_t + \alpha(D_t - F_t)$, regression/least squares). Smoothing constant α: high α → more responsive.
- Production planning & control (PPC) functions: routing (path), scheduling (time), dispatching (release orders), follow-up/expediting, loading.
- Sequencing: Johnson's rule (n jobs on 2 machines – minimise makespan): pick the smallest time; if on machine 1 place first, if on machine 2 place last. Priority rules: SPT (shortest processing time – minimises mean flow time), EDD (earliest due date – minimises max lateness), FCFS, critical ratio.

Inventory control

EOQ (Wilson): $Q^* = \sqrt{(2DC_o/C_h)}$

At EOQ: ordering cost = holding cost; total cost $TC = \sqrt{(2DC_oC_h)}$; number of orders $N = D/Q^*$; cycle time $= Q^*/D$

EPQ (production at rate p, demand rate d): $Q^* = \sqrt{(2DC_o/(C_h(1 - d/p)))}$

Reorder point ROP = demand during lead time + safety stock

- ABC analysis: A – ~10% items, ~70% value (tight control); B – ~20% items, ~20% value; C – ~70% items, ~10% value. VED (vital, essential, desirable – criticality, spares). FSN (fast, slow, non-moving). XYZ (value of stock held). SDE (scarce, difficult, easy – procurement). HML (unit price).
- MRP (material requirements planning) – dependent demand, uses BOM, MPS and inventory records. MRP-II (manufacturing resource planning), ERP. JIT (Just-in-time – Toyota, Kanban cards, zero inventory). Lean manufacturing, Kaizen (continuous improvement), Poka-yoke (mistake-proofing), 5S (Sort, Set in order, Shine, Standardise, Sustain), SMED (single minute exchange of dies).

★ **PYQ-TREND:** EOQ formula & that at EOQ ordering cost equals carrying cost – very frequent. If demand is 4 times, EOQ doubles. ABC analysis is based on annual consumption value. VED for spare parts. Johnson's rule. Kanban relates to JIT. Fixed-position layout for ships. Process layout for job shops. Exponential smoothing formula.

15.6 Work Study

- Work study = method study (how – reduce work content; improve methods) + work measurement (how long – standard time).

- Method study steps (SREDIM): Select, Record, Examine, Develop, Install (define), Maintain.
- Recording techniques: outline process chart, flow process chart (man/material/equipment type), two-handed (operator) chart, multiple activity chart (man-machine), flow diagram, string diagram (movement of workers – distances), travel chart, SIMO chart (micro-motion, therbligs), cyclegraph & chronocyclegraph (light paths), memo-motion study.

Symbol	Activity
○ (circle)	Operation
□ (square)	Inspection
⇒ (arrow)	Transport
D	Delay (temporary storage)
▽ (triangle)	Storage (permanent)

- Therbligs (Gilbreth): 17/18 basic motion elements (search, select, grasp, transport empty/loaded, hold, release load, position, pre-position, inspect, assemble, disassemble, use, unavoidable delay, avoidable delay, plan, rest).
- Principles of motion economy (Barnes): related to human body, workplace layout, tools & equipment.
- Work measurement: time study (stopwatch – repetitive short cycle jobs), work sampling (activity sampling/ratio-delay – long cycle/group activities; Tippett), PMTS (predetermined motion time systems – MTM, work factor), standard data, analytical estimating.

```
Normal time = observed time × rating factor (%/100)
Standard time = normal time × (1 + allowance fraction) (allowances: personal, fatigue, delay, contingency)
Work sampling (95% confidence): N = 4p(1 − p)/L² for absolute accuracy L; N = 4(1 − p)/(S²p) for relative accuracy S
```

- Rating: comparing the worker's pace with the normal pace (Westinghouse system considers skill, effort, conditions, consistency). Normal pace benchmark: walking at 4.8 km/h (3 mph), dealing a pack of 52 cards in 0.5 min.
- Ergonomics: fitting the job to the worker. Anthropometry.

★ **PYQ-TREND:** SREDIM steps. Process chart symbols. Therblig developed by Gilbreth; father of scientific management F.W. Taylor. String diagram for movement. Standard time = normal time + allowances. Work sampling for long-cycle jobs. SIMO chart for micro-motion. Rating factor definition.

15.7 Quality Control

- SQC: process control (control charts) and acceptance sampling.

Chart	Type	For
$\bar{X}$ and R chart	Variables	Process mean and dispersion (small samples)
$\bar{X}$ and σ chart	Variables	Large samples
p chart	Attributes	Fraction defective (variable sample size allowed)
np chart	Attributes	Number defective (constant sample size)

Chart	Type	For
c chart	Attributes	Number of defects per unit (Poisson) – e.g., defects in cloth, castings
u chart	Attributes	Defects per unit with variable size

Control limits: mean ± 3σ (99.73% of points within); $\bar{X}$ chart: UCL = $\bar{\bar{X}}$ + $A_2\bar{R}$; R chart: UCL = $D_4\bar{R}$, LCL = $D_3\bar{R}$

p chart: $\bar{p} \pm 3\sqrt{\bar{p}(1-\bar{p})/n}$; c chart: $\bar{c} \pm 3\sqrt{\bar{c}}$

Process capability $C_p = (USL - LSL)/(6\sigma)$; $C_{pk} = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$

- Acceptance sampling: OC curve (probability of acceptance vs lot quality); AQL (acceptable quality level – producer's risk α, type I error, usually 5%), LTPD (lot tolerance percent defective – consumer's risk β, type II error, usually 10%). Single, double, multiple, sequential sampling plans. AOQL – maximum average outgoing quality.
- TQM, Deming's PDCA cycle (Plan-Do-Check-Act), Juran trilogy, Crosby (zero defects), Ishikawa (fishbone/cause & effect diagram), Pareto (80/20), Six Sigma (3.4 DPMO; DMAIC), ISO 9000 (quality management), ISO 14000 (environment), Quality circles (Japan).
- Reliability: R(t) = e^(−λt) (constant failure rate); MTBF = 1/λ; series $R = \Pi R_i$; parallel $R = 1 - \Pi(1 - R_i)$. Bath-tub curve: infant mortality (decreasing), useful life (constant), wear-out (increasing).

★ **PYQ-TREND:** c-chart for number of defects; p-chart for fraction defective. ±3σ limits (99.73%). Producer's risk α (type I) vs consumer's risk β (type II). Six Sigma = 3.4 defects per million. Ishikawa diagram. PDCA by Deming. Bath-tub curve regions. Reliability of series/parallel systems.

15.8 Operations Research & Project Management

- Linear programming: objective function + constraints; graphical method (2 variables), simplex method (slack, surplus, artificial variables; Big-M, two-phase). Duality. Degeneracy, unbounded, infeasible, alternate optima.
- Transportation problem: initial solution by NWCR (north-west corner rule), least cost, Vogel's approximation method (VAM – best initial); optimality by MODI (u-v) or stepping stone. Non-degenerate basic feasible solution needs m + n − 1 allocations. Balanced if supply = demand.
- Assignment problem: Hungarian method (n × n); special case of transportation (highly degenerate).
- Queuing (M/M/1): λ arrival rate, μ service rate, ρ = λ/μ < 1.

$L_s = \lambda/(\mu - \lambda)$; $L_q = \lambda^2/(\mu(\mu - \lambda))$; $W_s = 1/(\mu - \lambda)$; $W_q = \lambda/(\mu(\mu - \lambda))$; Little's law $L = \lambda W$

- Game theory: saddle point (maximin = minimax) → pure strategy; value of game. Simulation: Monte Carlo (random numbers).
- Replacement: items deteriorating gradually (replace when average annual cost is minimum) vs sudden failure (group replacement).

PERT & CPM

CPM	PERT
Deterministic, single time estimate	Probabilistic, three time estimates
Activity oriented	Event oriented

CPM	PERT
Repetitive projects (construction)	R&D, new projects
Time–cost trade-off (crashing)	Probability of completion

```
PERT expected time te = (to + 4tm + tp)/6; variance σ² = ((tp − to)/6)² (β-distribution)
Probability of completion: Z = (T_s − T_e)/σ_cp (normal distribution along critical path)
Total float = LS − ES = LF − EF; free float = ES(succ) − EF; independent float = ES(succ) − LF(pred) − duration
```

- Critical path: longest path through the network; zero total float; determines project duration. Dummy activity: zero time, shows dependency only (dotted arrow).
- Crashing: reduce project duration by spending more on critical activities with lowest cost slope (crash cost − normal cost)/(normal time − crash time).
- Fulkerson's rule for numbering events. Gantt chart (bar chart) – Henry Gantt; Milestone chart.

★ **PYQ-TREND:** PERT uses β-distribution & 3 time estimates; CPM deterministic. te formula. Critical path has zero float & is the longest path. Dummy activity. VAM gives the best initial solution. m + n − 1 rule. Hungarian method for assignment. Little's law. Crashing cost slope.

15.9 Engineering Economics & Management Facts

- Break-even analysis: BEP (units) = Fixed cost / (Selling price − Variable cost per unit) = F/contribution. Margin of safety = (actual sales − BEP sales)/actual sales. P/V ratio = contribution/sales.
- Depreciation: straight line D = (C − S)/n; declining balance; sum-of-years' digits; sinking fund.
- Value engineering/analysis (L.D. Miles, GE): value = function/cost.
- Father of scientific management: F.W. Taylor (functional foremanship, time study). Henri Fayol: 14 principles of management (administrative management). Gilbreth: motion study. Elton Mayo: Hawthorne studies (human relations). Maslow: hierarchy of needs. Herzberg: two-factor theory. McGregor: Theory X & Y.

★ **PYQ-TREND:** Break-even point formula. Value = function/cost. Taylor = scientific management; Fayol = 14 principles; Gilbreth = motion study & therbligs; Gantt chart by Henry Gantt.

CHAPTER 15B

Mechanical Measurements & Instrumentation

15B.1 Generalised Measurement System

- Elements: primary sensing element (transducer/sensor) → variable conversion element → variable manipulation element → data transmission → data presentation (indicator/recorder).
- Static characteristics: accuracy, precision, sensitivity, linearity, hysteresis, threshold, resolution, dead zone, drift, static error, repeatability, range & span.
- Dynamic characteristics: speed of response, measuring lag, fidelity, dynamic error. Zero-order (potentiometer – instantaneous), first-order (thermometer – time constant τ; reaches 63.2% of step in τ), second-order (spring-mass-damper instruments – natural frequency, damping ratio ~0.6–0.7 optimal).
- Transducers: active (self-generating – thermocouple, piezoelectric, photovoltaic, tachogenerator) vs passive (need external power – strain gauge, RTD, thermistor, LVDT, capacitive). Primary vs secondary; analog vs digital.
- Null-type instruments (more accurate, e.g., potentiometer, Wheatstone bridge balance) vs deflection type (faster, e.g., PMMC meter).

★ **PYQ-TREND:** First-order system reaches 63.2% in one time constant. Thermocouple and piezoelectric are active transducers. Optimal damping ratio ~0.7 for second-order instruments. Null method is more accurate.

15B.2 Temperature Measurement

Instrument	Principle	Range / notes
Liquid-in-glass (mercury)	Thermal expansion	−39 to 357 °C (alcohol for low temps)
Bimetallic strip	Differential expansion of two metals	Thermostats, −30 to 550 °C
Pressure thermometer	Expansion of gas/liquid/vapour in a bulb	Remote indication
Thermocouple	Seebeck effect (EMF from junctions at different T)	Type K (chromel–alumel) up to ~1250 °C; J (iron–constantan); T (copper–constantan, cryogenic); S/R (Pt–Rh, up to 1600 °C); B up to 1800 °C
RTD	Resistance increases with T (platinum Pt-100: 100 Ω at 0 °C)	−200 to 850 °C; most accurate, linear
Thermistor	Semiconductor; NTC – resistance falls with T	High sensitivity, non-linear, narrow range
Optical pyrometer	Brightness comparison with a filament (disappearing filament)	> 700 °C; furnaces

Instrument	Principle	Range / notes
Radiation pyrometer	Total radiation (Stefan–Boltzmann) focused on thermopile	High temperatures, non-contact
Infrared thermometer	IR emission	Non-contact

- Thermocouple laws: law of intermediate metals (a third metal at the same temperature adds no EMF – allows meters in circuit), law of intermediate temperatures ($E(T_1,T_3) = E(T_1,T_2) + E(T_2,T_3)$), law of homogeneous circuit. Cold-junction compensation required.
- Thermopile: thermocouples in series for higher output.
- Peltier effect (heat absorbed/evolved at a junction when current flows) and Thomson effect (along a conductor with temperature gradient) are related thermoelectric effects.

★ **PYQ-TREND:** Thermocouple works on the Seebeck effect. Pt-100 RTD is the most accurate & linear. Thermistor has NTC and high sensitivity. Optical pyrometer for very high temperatures, non-contact. Law of intermediate metals. Chromel–alumel is type K.

15B.3 Pressure Measurement

- Manometers (U-tube, inclined, well-type, micromanometer) – low pressures, primary standards.
- Elastic gauges: Bourdon tube (C-type, spiral, helical – most common, gauge pressure), diaphragm (corrosive/low pressure), bellows (low pressure, differential).
- Dead-weight piston gauge – primary calibration standard for pressure gauges.
- Electrical: strain-gauge pressure transducers, piezoelectric (dynamic/rapidly varying pressure – engine combustion), capacitive, LVDT-based.
- Vacuum (low-pressure) gauges: McLeod gauge (compression type – standard for calibrating others, $\sim 10^{-4}$ mm Hg), Pirani gauge (thermal conductivity of gas – hot wire resistance), thermocouple gauge, ionisation gauge (very high vacuum $\sim 10^{-10}$ mm Hg), Knudsen gauge (molecular momentum).
- High pressure: Bridgman gauge (resistance of manganin wire changes with pressure).

★ **PYQ-TREND:** McLeod gauge used for calibrating vacuum gauges. Pirani works on thermal conductivity. Ionisation gauge for very low pressures. Bridgman gauge for very high pressures. Piezoelectric transducers for dynamic pressure. Dead-weight tester for calibration.

15B.4 Flow, Level, Speed and Force Measurement

- Flow (obstruction type): orifice, venturi, flow nozzle, Dall tube – $Q \propto \sqrt{\Delta p}$. Variable area: rotameter (float in tapered tube – $Q \propto$ height). Velocity type: Pitot tube, hot-wire/hot-film anemometer (King's law – turbulence measurement), laser Doppler anemometer (LDA – non-intrusive), cup anemometer (wind), current meter (rivers).
- Other flow meters: turbine (rotor speed $\propto Q$), positive displacement (domestic water/gas meters – nutating disc, lobed rotor), electromagnetic (Faraday's law – conductive liquids, no obstruction, slurries), ultrasonic (transit-time & Doppler), vortex-shedding (Strouhal number), Coriolis mass flow meter (true mass flow).
- Level: sight glass, float, displacer, bubbler (purge), capacitance, ultrasonic, radar, differential pressure, radioactive (nucleonic).
- Speed: tachometers – mechanical (centrifugal, revolution counter), electrical (DC/AC tachogenerator, eddy-current drag cup as in speedometers), contactless (stroboscope –

flashes synchronised to make rotating object appear stationary; photoelectric/optical; magnetic pickup).

- Force/torque: load cells (strain gauge, hydraulic, pneumatic), proving ring (calibration standard), dynamometers (absorption: Prony brake, eddy current, hydraulic; transmission: torsion meter), spring balance.
- Displacement: LVDT (linear variable differential transformer – AC excitation, two secondaries in series opposition; output ∝ displacement; null at centre), RVDT (rotary), potentiometer, capacitive, optical encoders (incremental/absolute).
- Vibration: accelerometers (piezoelectric – natural frequency much higher than measured; seismic instruments), vibrometer (natural frequency much lower than measured).

★ **PYQ-TREND:** Electromagnetic flowmeter works on Faraday's law and needs conductive fluid. Hot-wire anemometer for turbulence. Rotameter is a variable-area meter. Stroboscope for contactless speed. LVDT principle (mutual induction, differential output). Proving ring calibrates force. Accelerometer natural frequency ≫ measured; vibrometer ≪ measured. Coriolis meter gives mass flow.

15B.5 Strain Measurement

- Electrical resistance strain gauge: $R = \rho L/A$; under strain both L and A change. Gauge factor $GF = (\Delta R/R)/\varepsilon \approx 2$ for metallic foil gauges (constantan), 50–200 for semiconductor gauges (high sensitivity but temperature sensitive).

```
ΔR/R = GF × ε; bridge output (quarter bridge) ≈ (V/4) × GF × ε
```

- Wheatstone bridge configurations: quarter (1 active), half (2 active – temperature compensation with dummy gauge), full (4 active – max sensitivity, 4× quarter bridge output).
- Rosettes (rectangular 0–45–90°, delta 0–60–120°) measure strains in three directions to compute principal strains when directions are unknown.
- Temperature compensation: dummy gauge in adjacent bridge arm; self-temperature-compensated gauges.
- Other: photoelasticity (birefringent models under polarised light – fringe patterns show stress), brittle coatings, Moiré, digital image correlation.

★ **PYQ-TREND:** Gauge factor of metallic gauges ≈ 2. Strain rosette needed when principal directions are unknown. Dummy gauge for temperature compensation. Full bridge gives max output. Photoelasticity uses polarised light.

15B.6 Control Systems Basics

- Open-loop (no feedback – washing machine timer, traffic lights) vs closed-loop (feedback – thermostat AC, cruise control, governor).
- Controller actions: on-off (two position), proportional (P – offset error remains), integral (I – eliminates steady-state error, may cause oscillation), derivative (D – anticipates, improves stability/damping), PID combined.
- Transfer function: Laplace output/input with zero initial conditions. First-order: $K/(\tau s + 1)$. Second-order: $\omega_n^2/(s^2 + 2\zeta\omega_n s + \omega_n^2)$.
- Stability: all poles in the left half of the s-plane. Routh–Hurwitz criterion, root locus, Bode plot (gain & phase margins), Nyquist.

- Hydraulic & pneumatic systems: actuators (cylinders single/double acting, motors), valves (directional control 4/3, 5/2 etc., pressure relief, flow control), accumulators, FRL unit (filter–regulator–lubricator) in pneumatics.

★ **PYQ-TREND:** Feedback distinguishes closed-loop systems. Integral action removes steady-state error. Derivative action improves stability. FRL unit in pneumatics. A 4/3 valve has 4 ports & 3 positions.

✎ **DESCRIPTIVE Q:** Explain the working of an LVDT with a neat sketch and list its advantages.

An LVDT has one primary coil and two identical secondary coils wound symmetrically on a hollow former, with a movable soft-iron core attached to the object whose displacement is measured. The primary is excited by AC (1–10 kHz). At the null position the core couples equally to both secondaries. They are connected in series opposition, so the induced EMFs cancel and the output is zero. When the core moves one way, coupling with one secondary rises and with the other falls. The differential output voltage then increases linearly with displacement, and its phase (0° or 180°) shows direction. Advantages: frictionless (no contact) so long life and infinite resolution, high sensitivity and linearity over the range, rugged, low hysteresis, electrical isolation. Limitations: sensitive to stray magnetic fields, needs AC excitation and demodulation, and has limited dynamic response due to core mass.

CHAPTER 16

Solved Concept Numericals (PYQ-Pattern)

These problems use the numerical patterns SSC JE repeatedly sets. Each is solved step by step so you learn the method, not the option. Values are illustrative.

16.1 Mechanics & SOM

Problem 1 – Resultant of two forces

Two forces of 10 N and 10 N act at a point with 120° between them. Find the resultant.

$R = 2P\cos(\theta/2) = 2 \times 10 \times \cos 60° = 10\ N$

Concept: two equal forces at 120° give a resultant equal to either force. At 90°: R = P√2; at 60°: R = P√3.

Problem 2 – Lami's theorem

A 100 N lamp hangs from two strings making 30° and 60° with the ceiling. Find the tensions. The angle between the strings is 90°. The angle between T_1 (30° string) and the weight is 120°; between T_2 and the weight, 150°.

$T_1/\sin 150° = T_2/\sin 120° = 100/\sin 90° \rightarrow T_1 = 50\ N,\ T_2 = 86.6\ N$

Concept: the string closer to vertical (60° with ceiling) carries more tension.

Problem 3 – Projectile

A ball is projected at 20 m/s at 30°. Find time of flight, max height and range (g = 10 m/s²).

$T = 2u\sin\theta/g = 2\ s;\ H = u^2\sin^2\theta/(2g) = 5\ m;\ R = u^2\sin 2\theta/g = 34.64\ m$

Problem 4 – Elongation of a bar

A steel rod 2 m long, 20 mm diameter, carries 31.4 kN tension. E = 200 GPa. Find elongation.

$A = \pi/4 \times 20^2 = 314\ mm^2;\ \sigma = 31400/314 = 100\ MPa;\ \delta = \sigma L/E = 100 \times 2000 / 200000 = 1\ mm$

Problem 5 – Thermal stress

A steel rail 10 m long is fixed at both ends at 20 °C. Find the stress at 70 °C ($\alpha = 12 \times 10^{-6}$ /°C, E = 200 GPa).

$\sigma = E\alpha\Delta T = 200000 \times 12 \times 10^{-6} \times 50 = 120\ MPa\ (compressive)$

Concept: the answer does not depend on length; 10 m is a distractor.

Problem 6 – Elastic constants

E = 200 GPa and G = 80 GPa. Find μ and K.

$\mu = E/(2G) - 1 = 200/160 - 1 = 0.25;\ K = E/(3(1 - 2\mu)) = 200/1.5 = 133.3\ GPa$

Problem 7 – SS beam with UDL

A simply supported beam of span 4 m carries a UDL of 10 kN/m. Find max BM and max SF.

$R_A = R_B = wL/2 = 20\ kN = max\ SF;\ M_{max} = wL^2/8 = 10 \times 16 / 8 = 20\ kN\cdot m$ (at mid-span, where SF = 0)

Problem 8 – Bending stress

A rectangular beam 100 mm × 200 mm (deep) carries M = 20 kN·m. Find max bending stress.

$Z = bd^2/6 = 100 \times 200^2/6 = 6.667 \times 10^5\ mm^3$; $\sigma = M/Z = 20 \times 10^6 / 6.667 \times 10^5 = 30$ MPa

Problem 9 – Shear stress in beam

The same beam carries SF = 40 kN. Find max shear stress.

$\tau_{avg} = V/A = 40000/20000 = 2$ MPa; $\tau_{max} = 1.5 \times 2 = 3$ MPa (at NA)

Problem 10 – Torsion & power

A solid shaft transmits 50 kW at 300 rpm. Allowable τ = 60 MPa. Find diameter.

$T = 60P/(2\pi N) = 60 \times 50000/(2\pi \times 300) = 1591.5\ N\cdot m = 1.5915 \times 10^6\ N\cdot mm$

$d^3 = 16T/(\pi\tau) = 16 \times 1.5915 \times 10^6 / (\pi \times 60) = 1.351 \times 10^5 \rightarrow d \approx 51.3$ mm → adopt 55 mm

Problem 11 – Principal stresses

σx = 80 MPa, σy = 20 MPa, τxy = 40 MPa. Find principal stresses and max shear stress.

centre = 50; radius = $\sqrt{(30^2 + 40^2)} = 50$; $\sigma_1 = 100$ MPa, $\sigma_2 = 0$, $\tau_{max} = 50$ MPa

Problem 12 – Euler column

A column (both ends hinged) has P_cr = 100 kN. Find P_cr if both ends are fixed, and if one end is fixed and the other free.

fixed-fixed: Le = L/2 → 4 × 100 = 400 kN; fixed-free: Le = 2L → 100/4 = 25 kN

Problem 13 – Thin cylinder

A boiler shell of 1 m diameter has internal pressure 1.2 MPa. Allowable hoop stress 120 MPa. Find thickness.

$t = pd/(2\sigma) = 1.2 \times 1000 / 240 = 5$ mm (longitudinal stress = 60 MPa)

Problem 14 – Spring stiffness

Two springs of stiffness 20 N/mm and 30 N/mm. Find equivalent stiffness in series and in parallel.

series k = (20 × 30)/(20 + 30) = 12 N/mm; parallel k = 50 N/mm

16.2 TOM & Machine Design

Problem 15 – Degrees of freedom

A planar mechanism has 6 links, 7 lower pairs and no higher pair. Find DOF.

$F = 3(6 - 1) - 2 \times 7 = 15 - 14 = 1$ → constrained mechanism (e.g., Watt/Stephenson 6-bar)

Problem 16 – Belt tension ratio

A flat belt has μ = 0.3 and angle of lap 180°. Find T_1/T_2.

$T_1/T_2 = e^{(0.3\pi)} = e^{0.942} \approx 2.57$

Problem 17 – Max power belt

The max allowable belt tension is 1500 N and the mass is 0.5 kg/m. Find the speed for max power.

$v = \sqrt{(T_{max}/(3m))} = \sqrt{(1500/1.5)} = 31.6$ m/s

Problem 18 – Gear train

A compound train: driver A (20 T) meshes with B (60 T); C (25 T) on the same shaft as B meshes with D (75 T). A runs at 900 rpm. Find speed of D.

```
N_D = 900 × (20/60) × (25/75) = 100 rpm
```

Problem 19 – Flywheel

A flywheel has a mass of 500 kg and radius of gyration 1 m. Mean speed 300 rpm, Cs = 0.02. Find ΔE.

```
ω = 2π × 300/60 = 31.42 rad/s; ΔE = mk²ω²Cs = 500 × 1 × 987 × 0.02 = 9870 J ≈ 9.87 kJ
```

Problem 20 – Watt governor

Find the height of a Watt governor at 60 rpm.

```
h = 895/N² = 895/3600 = 0.249 m ≈ 249 mm
```

Problem 21 – Natural frequency

A mass causes a static deflection of 2.5 mm on a spring. Find the natural frequency.

```
fn = 0.4985/√0.0025 = 0.4985/0.05 = 9.97 Hz ≈ 10 Hz
```

Problem 22 – Bearing life

A ball bearing has a rated life of 8000 h at load P. Find the life at load 2P.

```
L ∝ 1/P³ → L = 8000/8 = 1000 h
```

Problem 23 – Rivet diameter

Find the rivet diameter for a 16 mm plate by Unwin's formula.

```
d = 6√16 = 24 mm
```

16.3 Thermal Engineering

Problem 24 – Carnot engine

An engine works between 527 °C and 27 °C. Find the Carnot efficiency.

```
η = 1 − 300/800 = 0.625 = 62.5%
```

Problem 25 – Refrigerator / heat pump

A Carnot refrigerator works between −23 °C and 27 °C. Find COP_R and COP_HP.

```
COP_R = 250/(300 − 250) = 5; COP_HP = 300/50 = 6 (= COP_R + 1)
```

Problem 26 – Otto cycle

Find the air-standard efficiency of an Otto cycle with r = 8 (γ = 1.4).

```
η = 1 − 1/8^0.4 = 1 − 1/2.297 = 0.5647 = 56.5%
```

Problem 27 – Isothermal work

1 kg of air expands isothermally at 300 K from 1 m³ to 2 m³. Find work and heat.

```
W = mRT ln(V₂/V₁) = 1 × 0.287 × 300 × 0.693 = 59.7 kJ = Q (ΔU = 0)
```

Problem 28 – Nozzle velocity

The enthalpy drop in a nozzle is 200 kJ/kg (inlet velocity negligible). Find exit velocity.

```
V = 44.72√200 = 44.72 × 14.14 = 632.5 m/s
```

Problem 29 – Entropy change of mixing (heat transfer)

1000 kJ of heat flows from a reservoir at 1000 K to one at 500 K. Find ΔS_universe.

```
ΔS = −1000/1000 + 1000/500 = −1 + 2 = +1 kJ/K (irreversible, positive)
```

Problem 30 – IC engine power

A 4-stroke single-cylinder engine: bore 100 mm, stroke 120 mm, IMEP 6 bar, 1500 rpm. Find IP.

```
A = π/4 × 0.1² = 7.854 × 10⁻³ m²; n = 1500/2 = 750 per min
IP = pm L A n / 60 = 6 × 10⁵ × 0.12 × 7.854 × 10⁻³ × 750 / 60 = 7069 W ≈ 7.07 kW
```

Problem 31 – Mechanical efficiency

IP = 50 kW, FP = 10 kW. Find BP and ηm.

```
BP = 40 kW; ηm = 40/50 = 80%
```

Problem 32 – Refrigeration capacity

A VCR system has RE = 150 kJ/kg and 2 TR capacity. Find the refrigerant flow rate.

```
ṁ = 2 × 211/150 = 2.81 kg/min
```

Problem 33 – Heat conduction through wall

A wall 0.2 m thick, k = 1 W/mK, area 10 m², faces at 40 °C and 20 °C. Find heat flow.

```
Q = kAΔT/L = 1 × 10 × 20/0.2 = 1000 W
```

Problem 34 – Critical radius

A pipe is insulated with material of k = 0.1 W/mK; h = 10 W/m²K. Find the critical radius.

```
r_c = k/h = 0.1/10 = 0.01 m = 10 mm
```

Problem 35 – LMTD

In a counter-flow exchanger, hot fluid goes 150 → 90 °C and cold fluid 30 → 70 °C. Find LMTD.

```
ΔT₁ = 150 − 70 = 80; ΔT₂ = 90 − 30 = 60; LMTD = 20/ln(80/60) = 20/0.2877 = 69.5 °C
```

Problem 36 – Radiation, Wien's law

The sun's surface is at ~5800 K. Find the wavelength of maximum emission.

```
λ_max = 2898/5800 = 0.5 μm (visible region)
```

Problem 37 – Compressor intermediate pressure

A 2-stage compressor runs from 1 bar to 16 bar with perfect intercooling. Find intermediate pressure.

```
p₂ = √(1 × 16) = 4 bar (ratio 4 per stage)
```

16.4 Fluid Mechanics & Machines

Problem 38 – Pressure head conversion

Find the pressure head of 1 atm in metres of water and of an oil (S = 0.8).

```
water: 101325/9810 = 10.33 m; oil: 10.33/0.8 = 12.9 m
```

Problem 39 – Capillary rise

Find the capillary rise of water in a 1 mm glass tube (σ = 0.073 N/m, θ = 0).

```
h = 4σ/(ρgd) = 4 × 0.073/(1000 × 9.81 × 0.001) = 0.0298 m ≈ 30 mm
```

Problem 40 – Total pressure on gate

A vertical rectangular gate 2 m wide × 3 m deep has its top edge at the water surface. Find the total force and centre of pressure.

```
F = ρg A h̄ = 9810 × 6 × 1.5 = 88.29 kN; h* = 2h/3 = 2 m below surface
```

Problem 41 – Continuity & Bernoulli

Water flows in a pipe that reduces from 200 mm to 100 mm diameter. Velocity in the larger section is 1 m/s. Find velocity in the smaller section.

```
V₂ = V₁(D₁/D₂)² = 1 × 4 = 4 m/s
```

Problem 42 – Laminar flow head loss

Oil (μ = 0.1 Pa·s, ρ = 900) flows at 1 m/s in a 50 mm pipe, 100 m long. Check the regime and find h_f.

```
Re = 900 × 1 × 0.05/0.1 = 450 → laminar; f = 64/450 = 0.142
h_f = fLV²/(2gD) = 0.142 × 100 × 1/(2 × 9.81 × 0.05) = 14.5 m of oil
```

Problem 43 – Darcy friction loss

Water flows at 2 m/s in a 0.2 m pipe, 500 m long, f = 0.02. Find h_f.

```
h_f = 0.02 × 500 × 4 /(2 × 9.81 × 0.2) = 10.2 m
```

Problem 44 – Jet force

A 50 mm water jet at 20 m/s strikes a fixed flat plate normally. Find the force.

```
a = 1.963 × 10⁻³ m²; F = ρaV² = 1000 × 1.963 × 10⁻³ × 400 = 785 N
```

Problem 45 – Specific speed of turbine

A turbine develops 10,000 kW under 100 m head at 500 rpm. Find Ns and identify the type.

```
Ns = N√P/H^(5/4) = 500 × 100 / 316.2 = 158 → Francis turbine (medium specific speed)
```

Problem 46 – Pelton jet velocity

Find the jet velocity under a 400 m head (Cv = 0.98).

```
V = 0.98 √(2 × 9.81 × 400) = 0.98 × 88.6 = 86.8 m/s; bucket speed ≈ 0.46 × 88.6 ≈ 40.7 m/s
```

Problem 47 – Pump affinity

A centrifugal pump delivers 100 L/s at 20 m head with 30 kW at 1000 rpm. Find Q, H and P at 1500 rpm.

```
Q = 100 × 1.5 = 150 L/s; H = 20 × 2.25 = 45 m; P = 30 × 3.375 = 101 kW
```

Problem 48 – Reciprocating pump slip

Single-acting: D = 150 mm, L = 300 mm, N = 50 rpm, actual Q = 4 L/s. Find % slip.

```
Q_th = (π/4)(0.15)² × 0.3 × 50/60 = 4.42 L/s; slip = (4.42 − 4)/4.42 = 9.5%
```

16.5 Production & IE

Problem 49 – Chvorinov's rule

A 100 mm cube solidifies in 6 min. Find the solidification time of a 200 mm cube of the same metal and mould.

```
V/A ∝ side → (V/A) doubles → t ∝ (V/A)² → 4 × 6 = 24 min
```

Problem 50 – Blanking force

Blank a 50 mm diameter disc from 2 mm sheet with $\tau = 300$ MPa.

$F = \pi d t \tau = \pi \times 50 \times 2 \times 300 = 94.2$ kN

Problem 51 – Max draft in rolling

Roll radius is 300 mm and $\mu = 0.1$. Find the maximum possible draft.

$\Delta h_{max} = \mu^2 R = 0.01 \times 300 = 3$ mm

Problem 52 – Merchant shear angle

Rake angle $\alpha = 10°$, friction angle $\beta = 40°$. Find the shear angle.

$\varphi = 45 + \alpha/2 - \beta/2 = 45 + 5 - 20 = 30°$

Problem 53 – Taylor tool life

For HSS, $n = 0.125$. Tool life is 60 min at 30 m/min. Find tool life at 40 m/min.

$T_2 = T_1(V_1/V_2)^{(1/n)} = 60 \times (0.75)^8 = 60 \times 0.1001 = 6$ min

Problem 54 – Turning time

Turn a 100 mm dia bar 200 mm long at 300 rpm with 0.2 mm/rev feed in one pass. Find the machining time.

$t = L/(fN) = 200/(0.2 \times 300) = 3.33$ min; cutting speed $V = \pi \times 100 \times 300/1000 = 94.2$ m/min

Problem 55 – Indexing

Find the indexing for 36 divisions on a 40:1 dividing head.

40/36 = 1 1/9 → 1 full turn + 2 holes in an 18-hole circle (or 3 in 27)

Problem 56 – EOQ

Annual demand is 10,000 units, ordering cost Rs 200 per order, holding cost Rs 4 per unit per year. Find EOQ and the number of orders.

$Q^* = \sqrt{(2 \times 10000 \times 200/4)} = \sqrt{10^6} = 1000$ units; orders = 10 per year

Problem 57 – PERT

to = 4, tm = 6, tp = 14 days. Find te and σ.

$te = (4 + 24 + 14)/6 = 7$ days; $\sigma = (14 - 4)/6 = 1.67$ days

Problem 58 – Standard time

Observed time 0.8 min, rating 125%, allowances 10%. Find standard time.

$NT = 0.8 \times 1.25 = 1.0$ min; $ST = 1.0 \times 1.10 = 1.1$ min

Problem 59 – Break-even

Fixed cost Rs 1,00,000; selling price Rs 50; variable cost Rs 30 per unit. Find BEP.

$BEP = 100000/(50 - 30) = 5000$ units

Problem 60 – Reliability

Three components, each with R = 0.9. Find system reliability in series and in parallel.

series: $0.9^3 = 0.729$; parallel: $1 - 0.1^3 = 0.999$

CHAPTER 16B

Solved Concept Numericals – Set 2

A second set with slightly more steps, the kind that appears as 2–3 mark numericals in Paper-II and as tricky Paper-I questions.

16B.1 Mechanics & SOM

Problem 61 – Ladder friction

A uniform 5 m ladder weighing W rests against a smooth wall with its foot 3 m from the wall on rough ground. Find the minimum μ at the floor for equilibrium. Height reached = 4 m. Wall reaction R_w is horizontal; floor gives N = W and friction F = R_w. Take moments about the foot: R_w × 4 = W × 1.5 → R_w = 0.375W.

```
μ_min = F/N = 0.375
```

Problem 62 – Body on an incline

A 100 N block sits on a 30° incline with μ = 0.2. Find the force parallel to the plane needed to push it up.

```
P = W(sinθ + μcosθ) = 100(0.5 + 0.2 × 0.866) = 100 × 0.673 = 67.3 N
```

Problem 63 – Composite centroid

An L-section is made of two rectangles: 100 × 20 (horizontal flange at the bottom) and 20 × 80 (vertical, sitting on the flange at the left). Find ȳ from the bottom.

```
A₁ = 2000, y₁ = 10; A₂ = 1600, y₂ = 20 + 40 = 60
ȳ = (2000 × 10 + 1600 × 60)/(3600) = (20000 + 96000)/3600 = 32.2 mm
```

Problem 64 – Parallel axis

Find the MOI of a 100 × 200 mm rectangle about an axis 50 mm below its base.

```
I_G = 100 × 200³/12 = 6.667 × 10⁷; h = 100 + 50 = 150 mm
I = I_G + Ah² = 6.667 × 10⁷ + 20000 × 22500 = 6.667 × 10⁷ + 4.5 × 10⁸ = 5.167 × 10⁸ mm⁴
```

Problem 65 – Composite bar

A steel rod (A = 500 mm², E = 200 GPa) and a copper tube (A = 1000 mm², E = 100 GPa) of equal length share a load of 150 kN. Find stresses. Equal strain: σs/200 = σc/100 → σs = 2σc. Load: 500σs + 1000σc = 150000 → 1000σc + 1000σc = 150000.

```
σc = 75 MPa; σs = 150 MPa
```

Problem 66 – Strain energy

Find the strain energy in a bar 1 m long, area 100 mm², under 20 kN (E = 200 GPa).

```
U = P²L/(2AE) = (20000)² × 1000 / (2 × 100 × 200000) = 4 × 10¹¹ / 4 × 10⁷ = 10,000 N·mm = 10 J
```

Problem 67 – Overhanging beam contraflexure

A beam of span 6 m (supports at A and B) overhangs 2 m beyond B and carries UDL 10 kN/m over the whole 8 m. Find the reactions and the point of contraflexure. Moments about A: R_B ×

6 = 10 × 8 × 4 → R_B = 53.33 kN; R_A = 80 − 53.33 = 26.67 kN. Between A and B at x from A: M = 26.67x − 5x². M = 0 at x = 5.33 m (other than A).

```
Point of contraflexure 5.33 m from A (0.67 m left of B)
```

Problem 68 – Hollow shaft torque

A hollow shaft (D = 100 mm, d = 60 mm) has allowable τ = 80 MPa. Find the torque capacity.

```
T = (π/16)τ(D⁴ − d⁴)/D = (π/16) × 80 × (10⁸ − 1.296 × 10⁷)/100 = 15.708 × 8.704 × 10⁵ ≈ 1.367 × 10⁷ N·mm = 13.67 kN·m
```

Problem 69 – Angle of twist

A solid shaft of 50 mm diameter, 1 m long, carries T = 1 kN·m (G = 80 GPa). Find the twist.

```
J = π × 50⁴/32 = 6.136 × 10⁵ mm⁴; θ = TL/(GJ) = 10⁶ × 1000/(80000 × 6.136 × 10⁵) = 0.0204 rad = 1.17°
```

Problem 70 – Stresses on an oblique plane

A bar under σ = 100 MPa uniaxial tension. Find σn and τ on a plane at 30° to the cross-section.

```
σn = σcos²θ = 100 × 0.75 = 75 MPa; τ = (σ/2) sin2θ = 50 × 0.866 = 43.3 MPa
```

Problem 71 – Thin cylinder volume change

A thin cylinder: d = 1 m, t = 10 mm, p = 2 MPa, E = 200 GPa, μ = 0.3. Find the volumetric strain.

```
εv = (pd/4tE)(5 − 4μ) = (2 × 1000/(4 × 10 × 200000))(5 − 1.2) = 2.5 × 10⁻⁴ × 3.8 = 9.5 × 10⁻⁴
```

Problem 72 – Rankine column

A short column has σc = 320 MPa, A = 4000 mm², λ = 100, a = 1/7500. Find the Rankine load.

```
P = σcA/(1 + aλ²) = 320 × 4000/(1 + 10000/7500) = 1.28 × 10⁶/2.333 = 548.6 kN
```

16B.2 TOM & Machine Design

Problem 73 – Quick return ratio

In a crank & slotted lever QRM, crank length is 100 mm and the distance between centres is 200 mm. Find the time ratio.

```
cos(α/2) = r/c = 0.5 → α/2 = 60° → return angle α = 120°
Ratio = (360 − 120)/120 = 2
```

Problem 74 – Open belt length

Pulleys of 400 mm and 200 mm diameter are 1 m apart. Find the open belt length.

```
L = π(r₁ + r₂) + 2x + (r₁ − r₂)²/x = π(300) + 2000 + 100²/1000 = 942.5 + 2000 + 10 = 2952.5 mm
```

Problem 75 – Belt power

In a belt drive T₁ = 2000 N, T₂ = 800 N, v = 15 m/s. Find power.

```
P = (T₁ − T₂)v = 1200 × 15 = 18 kW
```

Problem 76 – Epicyclic train

In an epicyclic train the sun has 40 T and the ring 100 T. The ring is fixed and the sun turns at 700 rpm. Find arm speed. Planet teeth = (100 − 40)/2 = 30. Tabular method: arm fixed → sun +1, ring −40/100. Add y: ring = y − 0.4x = 0; sun = x + y = 700 → x = 500, y = 200.

```
Arm speed N_arm = N_sun × T_s/(T_s + T_r) = 700 × 40/140 = 200 rpm (same direction)
```

Problem 77 – Clutch torque

A single-plate clutch (both sides effective) has r_1 = 150 mm, r_2 = 100 mm, W = 4 kN, μ = 0.3. Find torque (uniform wear).

```
T = n μ W (r₁ + r₂)/2 = 2 × 0.3 × 4000 × 125 = 300,000 N·mm = 300 N·m
```

Problem 78 – Damped vibration

A system has m = 10 kg, k = 4000 N/m, c = 100 N·s/m. Find ζ and ωd.

```
ωn = √(4000/10) = 20 rad/s; c_c = 2mωn = 400; ζ = 0.25; ωd = 20√(1 − 0.0625) = 19.36 rad/s
```

Problem 79 – Gear contact

Two 20° gears of module 5 mm have 20 and 40 teeth. Find centre distance and pitch circle diameters.

```
d₁ = 100 mm, d₂ = 200 mm; centre distance = (100 + 200)/2 = 150 mm; circular pitch = πm = 15.7 mm
```

Problem 80 – Riveted joint efficiency

Plate t = 10 mm, rivet d = 20 mm, pitch p = 60 mm, single-riveted lap joint. σt = 100, τ = 80, σc = 160 MPa. Find efficiency.

```
Pt = (60 − 20) × 10 × 100 = 40 kN; Ps = (π/4)(400) × 80 = 25.1 kN; Pc = 20 × 10 × 160 = 32 kN
Solid plate strength = 60 × 10 × 100 = 60 kN; η = 25.1/60 = 41.9% (governed by shear)
```

Problem 81 – Fillet weld length

A plate is joined by a double parallel fillet weld of size 10 mm carrying 100 kN. τ = 80 MPa. Find the weld length (each side).

```
P = 2 × 0.707 × s × l × τ → l = 100000/(2 × 0.707 × 10 × 80) = 88.4 mm (+ 12.5 mm for start/stop ≈ 100 mm)
```

Problem 82 – Soderberg design

A steel bar has σm = 100 MPa and σa = 60 MPa. Syt = 400 MPa, Se = 200 MPa. Find FOS (Soderberg).

```
1/n = σa/Se + σm/Syt = 0.3 + 0.25 = 0.55 → n = 1.82
```

Problem 83 – Bearing selection

A ball bearing must carry 2 kN for 100 million revolutions. Find the required dynamic capacity C.

```
C = P × L^(1/3) = 2 × 100^(1/3) = 2 × 4.64 = 9.28 kN
```

16B.3 Thermal & HT

Problem 84 – Polytropic work

0.1 m^3 of air at 1 bar is compressed polytropically (n = 1.3) to 8 bar. Find V_2 and work.

```
V₂ = 0.1 × (1/8)^(1/1.3) = 0.1 × 0.2016 = 0.0202 m³
W = (p₂V₂ − p₁V₁)/(n − 1) = (800 × 0.0202 − 100 × 0.1)/0.3 = (16.13 − 10)/0.3 = 20.4 kJ (done on gas)
```

Problem 85 – Adiabatic temperature rise

Air at 300 K is compressed isentropically through a pressure ratio of 8. Find T_2.

```
T₂ = 300 × 8^(0.4/1.4) = 300 × 8^0.2857 = 300 × 1.811 = 543 K
```

Problem 86 – Diesel cycle

Diesel cycle with r = 16 and ρ = 2 (γ = 1.4). Find efficiency.

```
r^0.4 = 3.031; (ρ^γ − 1)/(γ(ρ − 1)) = (2.639 − 1)/1.4 = 1.171
η = 1 − 1.171/3.031 = 1 − 0.386 = 61.4%
```

Problem 87 – Brayton cycle

Find the efficiency of a Brayton cycle with pressure ratio 10.

```
η = 1 − 1/10^(0.2857) = 1 − 1/1.931 = 48.2%
```

Problem 88 – Steam quality

Steam at 10 bar has h = 2500 kJ/kg. At 10 bar, hf = 762.6, hfg = 2013.6 kJ/kg. Find the dryness fraction.

```
x = (2500 − 762.6)/2013.6 = 0.863
```

Problem 89 – Rankine cycle (simplified)

Turbine inlet h_3 = 3200, exit h_4 = 2200, pump exit h_2 = 200, condensate h_1 = 192 (kJ/kg). Find η and SSC.

```
W_T = 1000; W_P = 8; Q₁ = 3200 − 200 = 3000; η = 992/3000 = 33.1%; SSC = 3600/992 = 3.63 kg/kWh
```

Problem 90 – Volumetric efficiency of compressor

Clearance ratio 0.05, pressure ratio 7, n = 1.3. Find ηv.

```
ηv = 1 + C − C(7)^(1/1.3) = 1.05 − 0.05 × 4.46 = 1.05 − 0.223 = 82.7%
```

Problem 91 – VCR from enthalpies

h_1 = 1450, h_2 = 1650, h_3 = h_4 = 350 kJ/kg (ammonia). Find COP and heat rejected per kg.

```
RE = 1100; W = 200; COP = 5.5; Q_cond = 1300 kJ/kg
```

Problem 92 – Psychrometry

Air at 1.013 bar has a partial pressure of vapour of 0.02 bar. Find specific humidity.

```
ω = 0.622 × 0.02/(1.013 − 0.02) = 0.01244/0.993 = 0.01253 kg/kg dry air
```

Problem 93 – Composite wall

A brick wall (L = 0.2 m, k = 0.8) has plaster (L = 0.02 m, k = 0.4) on one side. h_in = 10, h_out = 20 W/m²K, ΔT = 30 °C. Find U and q.

```
1/U = 0.1 + 0.25 + 0.05 + 0.05 = 0.45 → U = 2.22 W/m²K; q = 2.22 × 30 = 66.7 W/m²
```

Problem 94 – Lumped cooling

A steel ball (ρ = 7800, c = 460, r = 5 mm, k = 40) is cooled with h = 100 W/m²K. Check lumped validity and find the time constant.

```
Lc = r/3 = 1.667 mm; Bi = 100 × 0.001667/40 = 0.0042 < 0.1 → valid
τ = ρcLc/h = 7800 × 460 × 0.001667/100 = 59.8 s
```

Problem 95 – Radiation between plates

Two large parallel plates at 800 K and 400 K, both ε = 0.8. Find net radiation per m².

```
q = σ(T₁⁴ − T₂⁴)/(2/ε − 1) = 5.67 × 10⁻⁸ × (4.096 × 10¹¹ − 2.56 × 10¹⁰)/1.5 = 5.67 × 10⁻⁸ × 3.84 × 10¹¹/1.5 = 14.5 kW/m²
```

Problem 96 – Heat exchanger NTU

A condenser has UA = 2000 W/K and cooling water C = 1000 W/K. Find effectiveness.

```
NTU = 2; ε = 1 − e^(−2) = 0.865
```

16B.4 Fluids, Machines, Production

Problem 97 – Manometer

A U-tube mercury manometer connected to a water pipe shows a 0.2 m deflection (other limb open, centre of pipe 0.1 m above the mercury level in the pipe-side limb). Find pipe pressure.

```
p + 9810 × 0.1 = 13.6 × 9810 × 0.2 → p = 26683 − 981 = 25.7 kPa (gauge)
```

Problem 98 – Venturimeter

A 200 × 100 mm venturimeter shows a 0.5 m water-head difference. Cd = 0.98. Find Q.

```
A₁ = 0.03142, A₂ = 0.007854; Q = Cd A₁A₂√(2gh)/√(A₁² − A₂²)
= 0.98 × 2.467 × 10⁻⁴ × 3.132 / 0.03042 = 0.0249 m³/s ≈ 25 L/s
```

Problem 99 – Buoyancy

A wooden block (S = 0.6) of 1 m × 0.5 m × 0.4 m (height) floats in water. Find the depth of immersion.

```
0.6 × 0.4 = 1 × h → h = 0.24 m
```

Problem 100 – Metacentric height

For the block above floating with the 0.4 m side vertical, find GM (rolling about the long axis).

```
I = 1 × 0.5³/12 = 0.01042 m⁴; V = 1 × 0.5 × 0.24 = 0.12 m³; BM = 0.0868 m
BG = 0.2 − 0.12 = 0.08 m; GM = 0.0868 − 0.08 = 0.0068 m > 0 → stable (barely)
```

Problem 101 – Water hammer

Water flows at 2 m/s in a rigid pipe and the valve closes suddenly (C = 1400 m/s). Find the pressure rise.

```
Δp = ρVC = 1000 × 2 × 1400 = 2.8 MPa
```

Problem 102 – Froude model

A 1:25 ship model is tested. The prototype speed is 10 m/s. Find model speed and the force scale.

```
Vm = Vp/√25 = 2 m/s; force ratio = Lr³ = 15625
```

Problem 103 – Pelton power

Q = 0.5 m³/s, H = 300 m, ηo = 0.85. Find the power.

```
P = ηo ρgQH = 0.85 × 9810 × 0.5 × 300 = 1.25 MW
```

Problem 104 – Centrifugal pump head

Impeller outlet D₂ = 0.4 m, N = 1450 rpm, Vw₂ = 25 m/s, radial entry. Find Euler head and manometric efficiency if Hm = 50 m.

$u_2 = \pi \times 0.4 \times 1450/60 = 30.4$ m/s; $H_e = V_{w2}u_2/g = 25 \times 30.4/9.81 = 77.4$ m; $\eta_{mano} = 50/77.4 = 64.6\%$

Problem 105 – Riser design

A casting is a 100 × 100 × 100 mm cube. A cylindrical riser with H = D must take 1.25 times the solidification time of the casting. Find D.

(V/A) cube $= 100/6 = 16.67$ mm; cylinder $H = D$: $V/A = D/6$

$t \propto (V/A)^2 \rightarrow (D/6)^2 = 1.25 \times 16.67^2 \rightarrow D/6 = 18.63 \rightarrow D = 111.8$ mm

Problem 106 – Shear angle from chip ratio

In orthogonal cutting t = 0.5 mm, t_c = 1.0 mm, α = 10°. Find φ.

$r = 0.5$; $\tan\varphi = 0.5\cos10°/(1 - 0.5\sin10°) = 0.4924/0.9132 = 0.539 \rightarrow \varphi = 28.3°$

Problem 107 – Cutting power

Fc = 1000 N, V = 120 m/min. Find the cutting power.

$P = 1000 \times 2 = 2$ kW

Problem 108 – Drilling time

Drill a 20 mm hole 40 mm deep at 500 rpm and 0.2 mm/rev feed. Find time.

$t = (L + 0.3D)/(fN) = (40 + 6)/(0.2 \times 500) = 0.46$ min

Problem 109 – Deep drawing blank

Find the blank diameter for a cup of d = 50 mm, h = 30 mm (sharp corners).

$D = \sqrt{(d^2 + 4dh)} = \sqrt{(2500 + 6000)} = 92.2$ mm; draw ratio $1.84 < 2 \rightarrow$ single draw OK

Problem 110 – Bending of a sheet

Find the bend allowance for a 90° bend with R = 5 mm, t = 2 mm, k = 0.33.

$BA = \alpha(R + kt) = (\pi/2)(5 + 0.66) = 8.89$ mm

Problem 111 – Tolerance grade

Find IT7 for a diameter in the 30–50 mm step.

$D = \sqrt{(30 \times 50)} = 38.73$; $i = 0.45 \times 3.383 + 0.0387 = 1.561$ µm; $IT7 = 16i = 25$ µm

Problem 112 – Sine bar

A 200 mm sine bar is used to set 30°. Find the slip gauge height.

$h = L \sin\theta = 200 \times 0.5 = 100$ mm

Problem 113 – Johnson's rule

Jobs (M1, M2): A(5,2), B(1,6), C(9,7), D(3,8), E(10,4). Find the sequence. Smallest times: B (1 on M1) → first; A (2 on M2) → last; D (3 on M1) → second; E (4 on M2) → second last; C remains.

Sequence: B – D – C – E – A

Problem 114 – Exponential smoothing

Forecast for May = 100, actual May = 120, α = 0.2. Find the June forecast.

$F_{June} = 100 + 0.2(120 - 100) = 104$

Problem 115 – M/M/1 queue

λ = 8/h, µ = 10/h. Find Ls, Lq, Ws.

```
Ls = 8/2 = 4; Lq = 64/(10 × 2) = 3.2; Ws = 1/2 h = 30 min
```

Problem 116 – Work sampling observations

An activity is expected to occupy 25% of the time. How many observations are needed for ±5% relative accuracy at 95% confidence?

```
N = 4(1 − p)/(S²p) = 4 × 0.75/(0.0025 × 0.25) = 4800
```

Problem 117 – p-chart limits

$\bar{p}$ = 0.04, n = 100. Find the control limits.

```
σ = √(0.04 × 0.96/100) = 0.0196; UCL = 0.04 + 0.0588 = 0.0988; LCL = 0.04 − 0.0588 < 0 → 0
```

Problem 118 – Process capability

USL = 10.05, LSL = 9.95 mm, σ = 0.01 mm. Find Cp.

```
Cp = 0.1/(6 × 0.01) = 1.67 → capable
```

Problem 119 – Probability of project completion

Critical path te = 40 days, σ = 4 days. Find the probability of finishing in 44 days.

```
Z = (44 − 40)/4 = 1 → P = 84.1%
```

Problem 120 – Straight-line depreciation

A machine costs Rs 1,00,000, has salvage Rs 10,000 and a 9-year life. Find annual depreciation.

```
D = (100000 − 10000)/9 = Rs 10,000 per year
```

CHAPTER 16C

Solved Concept Numericals – Set 3 (Concept Twisters)

These are "ratio and proportion" twisters. SSC likes them because they test understanding, not plug-in. Each one shows the shortcut a topper uses.

16C.1 Ratio-Type Twisters – SOM & TOM

Problem 121 – Doubling diameter of a rod

If a rod's diameter is doubled with load and length unchanged, what happens to stress and elongation?

```
σ ∝ 1/d² → σ becomes 1/4; δ ∝ 1/d² → δ becomes 1/4
```

Problem 122 – Beam depth doubled

How does the bending strength of a rectangular beam change if its depth is doubled? Its stiffness?

```
Z = bd²/6 → strength ×4; I = bd³/12 → stiffness ×8, deflection becomes 1/8
```

Problem 123 – Span doubled

For a SS beam with UDL (same w), what happens to max BM and max deflection if the span doubles?

```
M ∝ L² → ×4; δ ∝ L⁴ → ×16
```

Problem 124 – Shaft diameter doubled

How do torque capacity and torsional stiffness change if the shaft diameter doubles?

```
T ∝ d³ → ×8; GJ/L ∝ d⁴ → ×16
```

Problem 125 – Hollow vs solid shaft (same outer diameter)

A hollow shaft has d/D = 0.5. Compare its strength and weight with a solid shaft of the same D.

```
Strength ratio = 1 − (d/D)⁴ = 1 − 0.0625 = 0.9375 (only 6.25% less)
Weight ratio = 1 − (d/D)² = 0.75 (25% lighter) → this is why hollow shafts are preferred
```

Problem 126 – Column length doubled

What happens to the Euler load if the length of a column doubles?

```
P ∝ 1/L² → becomes 1/4
```

Problem 127 – Column diameter doubled

What happens to the Euler load if a circular column's diameter doubles?

```
P ∝ I ∝ d⁴ → ×16
```

Problem 128 – Spring wire diameter doubled

For a close-coiled spring with D and n fixed, what happens to stiffness if the wire diameter doubles?

```
k ∝ d⁴ → ×16
```

Problem 129 – Spring coils halved

What happens to stiffness if the number of coils is halved?

```
k ∝ 1/n → ×2
```

Problem 130 – Natural frequency with mass quadrupled

How does the natural frequency change if the mass becomes 4 times?

```
ωn ∝ 1/√m → becomes 1/2
```

Problem 131 – Governor speed doubled

How does the height of a Watt governor change if speed doubles?

```
h ∝ 1/N² → becomes 1/4
```

Problem 132 – Flywheel speed doubled

For the same Cs and I, how does ΔE change if the speed doubles?

```
ΔE ∝ ω² → ×4 (so a flywheel at higher speed can be lighter: I ∝ 1/ω²)
```

Problem 133 – Centrifugal tension

How does centrifugal tension change if belt speed doubles?

```
Tc = mv² → ×4
```

Problem 134 – Bearing load reduced

If the load on a roller bearing halves, how does its life change?

```
L ∝ (1/P)^(10/3) → ×2^(3.33) ≈ ×10
```

16C.2 Ratio-Type Twisters – Thermal & HT

Problem 135 – Carnot with both temperatures raised

A Carnot engine works between 600 K and 300 K. Both temperatures rise by 100 K. New efficiency?

```
Old: 1 − 300/600 = 50%; New: 1 − 400/700 = 42.9% → efficiency falls
```

✔ **TOPPER TRICK:** Lowering the sink temperature by ΔT raises η more than raising the source temperature by the same ΔT, because $\partial\eta/\partial T_2 = -1/T_1$ is larger in magnitude than $\partial\eta/\partial T_1 = T_2/T_1^2$.

Problem 136 – Refrigerator COP with smaller temperature lift

A Carnot refrigerator works between 250 K and 300 K. The condenser temperature drops to 275 K. New COP?

```
Old COP = 250/50 = 5; New = 250/25 = 10 (doubles)
```

Problem 137 – Otto efficiency for r = 4 and 16

Compare Otto efficiencies (γ = 1.4) for r = 4 and r = 16.

```
r = 4: 1 − 1/4^0.4 = 1 − 1/1.741 = 42.6%; r = 16: 1 − 1/3.031 = 67%
```

Problem 138 – Ideal gas heated at constant pressure

Air is heated at constant pressure from 300 K to 600 K. How do volume and density change?

```
V ∝ T → doubles; ρ ∝ 1/T → halves
```

Problem 139 – Heat flow when thickness doubles

Steady conduction through a slab: how does Q change if thickness doubles and k halves?

```
Q ∝ k/L → (1/2)/(2) = 1/4
```

Problem 140 – Radiation when T doubles

How does a black body's emissive power change when absolute temperature doubles?

```
E ∝ T⁴ → ×16; λ_max halves (Wien)
```

Problem 141 – Fin with longer length

For an infinitely long fin, does doubling the length change heat transfer?

```
No — Q = √(hPkA)θ₀ is independent of length for a long fin (tanh(mL) → 1)
```

Problem 142 – Critical radius check

A wire of 2 mm radius is covered with insulation (k = 0.2 W/mK, h = 20 W/m²K). Will adding insulation increase heat loss?

```
r_c = k/h = 0.01 m = 10 mm > 2 mm → yes, heat loss increases until the outer radius reaches 10 mm
```

Problem 143 – Rankine with lower condenser pressure

State qualitatively what happens to turbine work, η, exhaust dryness and heat rejected when condenser pressure is lowered.

```
W_T ↑, η ↑, dryness ↓ (more moisture), heat rejected per kg ↑ only slightly (temperature of rejection ↓)
```

Problem 144 – Nozzle exit velocity with double enthalpy drop

If the enthalpy drop doubles, what is the new velocity?

```
V ∝ √Δh → ×√2 = ×1.414
```

16C.3 Ratio-Type Twisters – Fluids & Machines

Problem 145 – Pipe diameter doubled (same Q, turbulent, Darcy)

For the same discharge and f, how does head loss change if pipe diameter doubles?

```
h_f ∝ Q²/D⁵ → becomes 1/32
```

Problem 146 – Laminar flow, diameter doubled (same Q)

How does the pressure drop in laminar flow change if diameter doubles at the same discharge?

```
Δp ∝ Q/D⁴ (Hagen–Poiseuille) → becomes 1/16
```

Problem 147 – Capillary tube diameter halved

What happens to capillary rise when tube diameter halves?

```
h ∝ 1/d → doubles
```

Problem 148 – Notch head doubled

How does discharge change when head doubles over a rectangular weir and over a V-notch?

```
Rectangular: Q ∝ H^1.5 → ×2.83; V-notch: Q ∝ H^2.5 → ×5.66
```

Problem 149 – Pump speed increased by 20%

How do Q, H and P change if pump speed increases by 20%?

```
Q ×1.2; H ×1.44; P ×1.728
```

Problem 150 – Turbine head doubled (same turbine)

For the same turbine, how do speed, discharge and power change if head doubles (unit quantities constant)?

```
N ∝ √H → ×1.414; Q ∝ √H → ×1.414; P ∝ H^1.5 → ×2.83
```

Problem 151 – Jet velocity doubled

How does the force of a jet on a fixed plate change if jet velocity doubles?

```
F = ρaV² → ×4
```

Problem 152 – Reynolds model of a pipe

A 1:10 model of a pipe uses the same fluid. Find the velocity ratio Vm/Vp for Reynolds similarity.

```
VmLm = VpLp → Vm/Vp = 10 (model runs 10 times faster)
```

Problem 153 – Froude model time scale

For a 1:100 spillway model, a flood rises in 1 hour in the prototype. How long does it take in the model?

```
Tr = √Lr → Tm = 60/10 = 6 min
```

Problem 154 – Drag at double speed

For constant C_D, how does drag change when speed doubles? Power?

```
Drag ∝ V² → ×4; power = D·V ∝ V³ → ×8
```

16C.4 Ratio-Type Twisters – Production & IE

Problem 155 – Taylor equation with n = 0.5

How does tool life change if cutting speed doubles (n = 0.5)?

```
T ∝ V^(−1/n) = V^(−2) → becomes 1/4
```

Problem 156 – Surface roughness with feed doubled

How does peak-to-valley roughness change if feed doubles?

```
h = f²/(8r) → ×4
```

Problem 157 – Nose radius doubled

How does roughness change if nose radius doubles?

```
h ∝ 1/r → halves
```

Problem 158 – Casting size doubled

How does solidification time change if all dimensions of a casting double?

```
V/A doubles → t ∝ (V/A)² → ×4
```

Problem 159 – Blanking force for double thickness

How does blanking force change if sheet thickness doubles?

```
F = Ltτ → doubles
```

Problem 160 – EOQ with demand ×4

How does EOQ change if annual demand becomes 4 times?

```
Q* ∝ √D → doubles
```

Problem 161 – EOQ with holding cost ×4

How does EOQ change if holding cost becomes 4 times?

```
Q* ∝ 1/√Ch → halves
```

Problem 162 – Reliability of a 2-out-of-2 vs 1-out-of-2 system

Two units each have R = 0.8. Compare series and parallel.

```
Series: 0.64; Parallel: 1 − 0.04 = 0.96
```

Problem 163 – Work sampling accuracy

How many times the observations are needed if the required accuracy is halved (made twice as tight)?

```
N ∝ 1/S² → ×4
```

Problem 164 – Rolling – roll radius doubled

How does maximum draft change if roll radius doubles (same μ)?

```
Δh_max = μ²R → doubles
```

Problem 165 – Welding heat input

How does heat input per unit length change if travel speed doubles at the same V and I?

```
H = ηVI/v → halves
```

16C.5 Multi-Step Numericals (Paper-II Style)

Problem 166 – Beam design

A simply supported timber beam of 4 m span carries 6 kN/m UDL. σ_allow = 10 MPa and d = 2b. Find b and d.

```
M = wL²/8 = 6 × 16/8 = 12 kN·m = 12 × 10⁶ N·mm
Z = M/σ = 1.2 × 10⁶ mm³; bd²/6 = b(2b)²/6 = 2b³/3 = 1.2 × 10⁶ → b³ = 1.8 × 10⁶ → b = 121.6 mm, d = 243 mm
```

Problem 167 – Shaft under bending and torsion

A shaft carries M = 3 kN·m and T = 4 kN·m. τ_allow = 50 MPa. Find d by maximum shear stress theory.

```
Te = √(3² + 4²) = 5 kN·m = 5 × 10⁶ N·mm
d³ = 16Te/(πτ) = 16 × 5 × 10⁶/(π × 50) = 5.093 × 10⁵ → d = 79.9 ≈ 80 mm
```

Problem 168 – Engine test

A 4-stroke, 4-cylinder petrol engine (bore 80 mm, stroke 100 mm) at 3000 rpm has BP = 30 kW and fuel consumption 9 kg/h (CV 44 MJ/kg). Find bsfc and brake thermal efficiency.

```
bsfc = 9/30 = 0.3 kg/kWh
Heat input = 9 × 44000/3600 = 110 kW; ηbth = 30/110 = 27.3%
```

Problem 169 – Swept volume and CR

A cylinder has bore 100 mm, stroke 120 mm and clearance volume 120 cc. Find swept volume and compression ratio.

```
Vs = (π/4) × 10² × 12 = 942.5 cc; r = (942.5 + 120)/120 = 8.85
```

Problem 170 – Heat exchanger outlet

Hot oil (C_h = 2000 W/K) enters at 150 °C and cold water (C_c = 4000 W/K) at 30 °C, ε = 0.6. Find Q and outlet temperatures.

```
C_min = 2000; Q_max = 2000 × 120 = 240 kW; Q = 0.6 × 240 = 144 kW
T_h,out = 150 − 144000/2000 = 78 °C; T_c,out = 30 + 144000/4000 = 66 °C
```

Problem 171 – Pipe network (series)

Two pipes in series: 1000 m of 0.3 m diameter and 500 m of 0.2 m diameter. Find the equivalent length of 0.3 m pipe (same f).

```
Le = L₁ + L₂(D₁/D₂)⁵ = 1000 + 500 × (1.5)⁵ = 1000 + 500 × 7.594 = 4797 m
```

Problem 172 – Turbine selection

A site has 30 m head and 20 m³/s flow. η = 0.9 and N = 150 rpm. Find power and Ns, and select the turbine.

```
P = 0.9 × 9810 × 20 × 30 = 5297 kW; Ns = 150√5297/30^1.25 = 150 × 72.8/70.2 = 155.5
Ns ≈ 155 → Francis turbine (medium specific speed)
```

Problem 173 – Pump power

A pump lifts 50 L/s through a total head of 30 m with overall efficiency 75%. Find the input power.

```
P = ρgQH/η = 9810 × 0.05 × 30/0.75 = 19.6 kW
```

Problem 174 – Merchant force analysis

Orthogonal cutting: Fc = 1200 N, Ft = 500 N, α = 10°, φ = 30°. Find friction force, normal force and μ.

```
F = Fc sinα + Ft cosα = 208.4 + 492.4 = 700.8 N
N = Fc cosα − Ft sinα = 1181.8 − 86.8 = 1095 N; μ = 0.64
Fs = Fc cosφ − Ft sinφ = 1039 − 250 = 789 N
```

Problem 175 – EOQ with total cost

D = 12000/yr, Co = Rs 150, unit cost Rs 20, carrying 15% per year. Find EOQ and total inventory cost (excluding purchase).

```
Ch = 0.15 × 20 = 3; Q* = √(2 × 12000 × 150/3) = √1.2 × 10⁶ = 1095 units
TC = √(2 × 12000 × 150 × 3) = Rs 3286 per year
```

Problem 176 – CPM network

Activities (duration): A(3), B(4) after A, C(2) after A, D(5) after B and C, E(1) after C. Find the critical path and duration. Paths: A–B–D = 3 + 4 + 5 = 12; A–C–D = 3 + 2 + 5 = 10; A–C–E = 6.

```
Critical path A–B–D, duration 12 days. Total float of C = LF − EF = 7 − 5 = 2 days
```

CHAPTER 17

Last-Minute Revision: High-Yield Facts

17.1 Units, Constants & Conversions

Quantity	Value
g	9.81 m/s²
Universal gas constant	8.314 kJ/kmol·K
R (air)	0.287 kJ/kg·K
cp / cv (air)	1.005 / 0.718 kJ/kg·K, γ = 1.4
Stefan–Boltzmann σ	5.67×10^{-8} W/m²K⁴
Wien's constant	2898 μm·K
Latent heat of steam at 1 atm	2257 kJ/kg
Latent heat of ice	335 kJ/kg
cp of water	4.187 kJ/kg·K
1 TR	3.5167 kW = 211 kJ/min
1 atm	101.325 kPa = 760 mm Hg = 10.33 m water
1 hp	746 W (British), 735.5 W (metric)
1 kWh	3600 kJ
1 poise / 1 stoke	0.1 Pa·s / 10^{-4} m²/s
E steel / CI / Al	200 / 100–120 / 70 GPa
Critical point water	221.2 bar, 374.15 °C
Triple point water	0.01 °C, 0.611 kPa
Speed of sound in air (20 °C)	≈ 343 m/s
Density mercury	13,600 kg/m³

17.2 Number Facts SSC Loves

Fact	Value
Le for fixed-free column	2L
τ_max/τ_avg rectangle / circle	1.5 / 1.33
Hoop/longitudinal stress (thin cylinder)	2
Critical Re (pipe / plate)	2000 (2300) / 5×10^5
Laminar f (Darcy)	64/Re
KE correction factor laminar pipe	2
Max power transmission through pipe	h_f = H/3, η = 66.7%
Max efficiency – jet on series of flat plates	50%
Pelton jet ratio / speed ratio	≈ 12 / 0.46

Fact	Value
Critical pressure ratio air / superheated / saturated steam	0.528 / 0.546 / 0.577
Parsons turbine reaction	50%
CR SI / CI	6–10 / 14–22
Stoichiometric A/F petrol	14.7
Drill point angle	118°
Thread angle metric / Whitworth / Acme	60° / 55° / 29°
Pressure angle of gears	14.5° / 20°
Min teeth (no interference, 20°)	18 (≈ 17)
Eutectoid / eutectic carbon	0.8% / 4.3%
Eutectoid / eutectic temperature	723 °C / 1147 °C
APF of FCC / BCC	0.74 / 0.68
Rockwell C indenter	120° diamond cone, 150 kgf
Vickers indenter	136° diamond pyramid
Six Sigma defects	3.4 per million
Control limits	±3σ (99.73%)
Lumped analysis valid	Bi < 0.1
Fin worthwhile	Effectiveness > 2
Vibration isolation	ω/ωn > √2
Self-locking machine/screw	η < 50%
Max range of projectile	at 45°
Grain size number of grinding wheel (coarse)	10–24
Shrinkage allowance of CI	~10 mm/m (least)

17.3 "Who Said What" (Laws & Persons)

Name	Associated with
Hooke	Stress ∝ strain
Euler	Column buckling; turbine equation
Rankine	Column formula (Rankine–Gordon); max principal stress theory; Rankine cycle
Tresca / Guest	Max shear stress theory
von Mises / Hencky	Distortion energy theory
Mohr	Stress circle
Castigliano	Strain energy theorem for deflection
Maxwell	Reciprocal deflection theorem; thermodynamic relations
Grashof	Four-bar rotatability; Grashof number
Kennedy (Aronhold)	Three I-centres theorem
Kutzbach / Grübler	Mobility criterion
Taylor (F.W.)	Tool life equation; scientific management
Merchant	Shear angle relation & force circle
Chvorinov	Solidification time

Name	Associated with
Bernoulli	Energy equation in fluids
Reynolds	Laminar/turbulent transition
Darcy–Weisbach	Pipe friction loss
Hagen–Poiseuille	Laminar pipe flow
Pascal	Equal pressure transmission
Archimedes	Buoyancy
Prandtl	Boundary layer; mixing length
Froude / Weber / Mach	Dimensionless numbers
Buckingham	π-theorem
Pelton / Francis / Kaplan	Turbines
Carnot	Ideal heat engine cycle
Clausius / Kelvin-Planck	Second law statements
Joule	Mechanical equivalent of heat; free expansion experiment
Otto / Diesel / Brayton	Cycles
Fourier / Newton / Stefan–Boltzmann	Conduction / convection / radiation laws
Kirchhoff / Wien / Planck	Radiation laws
Nusselt / Prandtl / Biot	Dimensionless numbers in HT
Gilbreth	Motion study, therbligs
Gantt	Bar chart
Wilson (Harris)	EOQ formula
Johnson	Sequencing rule
Deming / Juran / Ishikawa / Crosby	Quality gurus (PDCA / trilogy / fishbone / zero defects)
Shewhart	Control charts
Lami	Three concurrent forces in equilibrium
Varignon	Principle of moments
D'Alembert	Inertia force – dynamic equilibrium
Unwin	Rivet diameter $6\sqrt{t}$
Lewis	Gear tooth bending strength
Wahl	Spring curvature correction
Soderberg / Goodman / Gerber	Fatigue criteria
Stribeck / McKee	Bearing friction curves
Sommerfeld / Petroff	Journal bearing equations

17.4 Paired Comparisons (Frequent Twisters)

A	B
Governor – mean speed (long term)	Flywheel – cyclic fluctuation (within cycle)
Toughness – area under whole curve	Resilience – area up to elastic limit
Hot working – above recrystallisation T	Cold working – below recrystallisation T
Blanking – clearance on punch	Punching – clearance on die

A	B
TIG – non-consumable W electrode	MIG – consumable wire
DCSP – deeper penetration	DCRP – cleaning action, thin sheet
Annealing – furnace cooling	Normalising – air cooling
Austempering – bainite	Martempering – martensite with less distortion
Impulse turbine – pressure constant over moving blades	Reaction turbine – pressure drops in moving blades too
Pelton – high head, low Ns	Kaplan – low head, high Ns
Centrifugal pump – priming needed	Reciprocating pump – self-priming
Pumps in series – head adds	Pumps in parallel – discharge adds
Viscosity of liquid ↓ with T	Viscosity of gas ↑ with T
Droplet $\Delta p = 4\sigma/d$	Soap bubble $\Delta p = 8\sigma/d$
Streamline/pathline differ (unsteady)	Same (steady)
Stream function – any 2D incompressible flow	Velocity potential – only irrotational flow
Up milling – chip thickness 0 → max	Down milling – chip thickness max → 0
Jig – guides tool	Fixture – holds work only
Hole basis – H (preferred)	Shaft basis – h
PERT – probabilistic, event-oriented	CPM – deterministic, activity-oriented
p chart – fraction defective	c chart – number of defects
Producer's risk α – type I error	Consumer's risk β – type II error
SI knocking – end of combustion	CI knocking – beginning of combustion
Octane – iso-octane & n-heptane	Cetane – cetane & α-methyl naphthalene
Quantity governing (SI)	Quality governing (CI)
Fire-tube – Cochran, Lancashire, Locomotive	Water-tube – B&W, La Mont, Benson
Mountings – safety (safety valve, fusible plug)	Accessories – efficiency (economiser, air preheater)
Dropwise condensation – higher h	Filmwise – used for design
Parallel flow HX	Counter flow HX – higher LMTD, more effective
Brass – Cu + Zn	Bronze – Cu + Sn
Thermoplastic – recyclable (PVC, PE)	Thermoset – permanent (Bakelite, epoxy)

17.5 Final 24-Hour Checklist

- Revise every yellow formula box in Chapters 2, 5, 10 and 11 (highest yield).
- Revise every blue PYQ-Trend box once, quickly.
- Revise the tables of 17.2 and 17.4 twice.
- Don't start new topics. Sleep 7+ hours. Carry admit card and ID, and reach the centre early.

✔ **TOPPER TRICK:** In the exam, attempt the facts-based questions first (≈ 30 s each), then formula plug-ins, then the lengthy numericals. With negative marking (−0.25 per wrong answer in recent SSC JE notifications – confirm in yours), skip only questions you have no idea about. Eliminating even two options makes a guess worthwhile.

CHAPTER 18

PYQ Concept Recall Bank – Part A

Each row is a concept SSC JE has examined repeatedly, written as a short question with a crisp answer. Cover the right column and test yourself. This is active recall, not MCQ practice.

A1. Engineering Mechanics

Concept question	Answer
What is the principle of transmissibility?	A force can be shifted along its line of action without changing its external effect on a rigid body
What is the resultant of two equal forces P at 120°?	P
When is the resultant of two forces maximum?	When they are collinear and in the same direction (θ = 0°)
What does Lami's theorem need?	Exactly three coplanar, concurrent forces in equilibrium
What does Varignon's theorem state?	The moment of the resultant equals the sum of moments of the components
Can a couple be balanced by a single force?	No – only by an equal and opposite couple
What is the angle of repose?	Equal to the angle of friction φ
Does friction depend on contact area?	No (dry friction is independent of apparent area)
Least force to drag a body on a rough horizontal plane?	W sin φ, applied at angle φ to the horizontal
Condition for a self-locking screw?	φ > α (friction angle exceeds helix angle); η < 50%
Max efficiency of a screw jack?	(1 − sin φ)/(1 + sin φ)
Centroid of a triangle from base?	h/3
Centroid of a semicircular area from diameter?	4r/(3π)
CG of a solid cone from base?	h/4
CG of a solid hemisphere from flat face?	3r/8
MOI of a rectangle about base?	$bd^3/3$
MOI of a triangle about centroidal axis ∥ base?	$bh^3/36$
Polar MOI of a solid circle?	$\pi d^4/32$
What is the perpendicular axis theorem valid for?	Plane laminae only: Izz = Ixx + Iyy
Mass MOI of a solid sphere about diameter?	$2mR^2/5$
Projection angle for maximum range?	45°
Relation between H and R at θ = 45°?	H = R/4
Apparent weight in a lift accelerating upward?	m(g + a)
Apparent weight in free fall?	Zero
What does D'Alembert's principle introduce?	An inertia force (−ma) to convert dynamics into statics
Coefficient of restitution for a perfectly plastic impact?	e = 0
Equal masses in elastic head-on collision?	They exchange velocities

Concept question	Answer
Bounce height for coefficient e?	e^2 × drop height
Which rolls down an incline fastest?	Solid sphere (smallest k^2/r^2)
Condition for a machine to be reversible?	η > 50%
Velocity ratio of the 1st system of pulleys (n movable)?	2^n
Velocity ratio of a Weston differential pulley?	2D/(D − d)
Unit of impulse?	N·s (same as momentum)
Rotational KE?	½ $I\omega^2$
Equivalent stiffness of springs in series?	$1/k = 1/k_1 + 1/k_2$

A2. Strength of Materials

Concept question	Answer
Where does necking begin on the σ–ε curve?	At the ultimate stress point
What is toughness?	Energy absorbed up to fracture (area under the full σ–ε curve)
What is the modulus of resilience?	Strain energy per unit volume up to elastic limit, $\sigma^2/2E$
Relation between E, G and μ?	E = 2G(1 + μ)
Relation between E, K and μ?	E = 3K(1 − 2μ)
Max Poisson's ratio for isotropic materials?	0.5
Poisson's ratio of cork?	≈ 0
Independent elastic constants for an isotropic material?	2
Elongation of a bar under self-weight?	WL/2AE (half that due to the same end load)
Thermal stress in a fully restrained bar?	EαΔT, independent of length and area
Stress under a suddenly applied load?	2P/A
Elongation of a tapered circular rod?	$4PL/(\pi E d_1 d_2)$
BMD shape under UDL?	Parabolic
SFD shape under UVL?	Parabolic
Where is BM maximum?	Where SF is zero or changes sign
What is a point of contraflexure?	Where BM changes sign
Max BM, SS beam, UDL?	$wL^2/8$
Max deflection, SS beam, central load?	$WL^3/48EI$
Max deflection of a cantilever with UDL?	$wL^4/8EI$
Deflection ratio fixed : SS beam under UDL?	1 : 5
Prop reaction of a propped cantilever under UDL?	3wL/8
Bending equation?	$M/I = \sigma/y = E/R$
Section modulus of a rectangle?	$bd^2/6$
Strongest rectangular beam from a log: d/b?	√2
τ_max/τ_avg for a rectangle?	1.5
τ_max/τ_avg for a circle?	4/3
Torsion equation?	$T/J = \tau/r = G\theta/L$

Concept question	Answer
Power transmitted by a shaft?	$2\pi NT/60$
Hollow vs solid shaft of the same weight?	Hollow is stronger and stiffer
Equivalent torque under combined loading?	$\sqrt{(M^2 + T^2)}$
Planes of max shear relative to principal planes?	At 45°
Mohr's circle for pure shear?	Centre at origin, radius τ
Failure theory for ductile materials?	Distortion energy (von Mises)
Shape of Tresca in σ_1–σ_2 space?	Hexagon
Shear yield by von Mises?	0.577 σy
Le for a column with both ends fixed?	L/2
Buckling load ratio fixed-fixed : fixed-free?	16 : 1
Middle-third rule?	Load within b/6 of the centroid causes no tension
Core of a circular section?	Circle of diameter d/4
Hoop vs longitudinal stress in a thin cylinder?	Hoop = 2 × longitudinal
Stress in a thin sphere?	$pd/4t$
When is a cylinder thin?	$d/t > 20$
Where is hoop stress max in a thick cylinder?	At the inner surface
Stiffness of a close-coiled spring?	$Gd^4/(8D^3n)$
Stiffness when a spring is cut in half?	Doubles
Stress in a leaf spring?	$3WL/(2nbt^2)$

A3. Theory of Machines

Concept question	Answer
Is a ball bearing a lower or higher pair?	Higher pair (point contact)
Is a nut and bolt a lower or higher pair?	Lower pair (screw pair)
What is a cam & follower pair?	Higher pair, force-closed
Grübler's criterion for planar mechanisms?	$3n - 2j - 4 = 0$
Kutzbach equation?	$F = 3(n - 1) - 2j - h$
Grashof condition?	$s + l \leq p + q$
Inversions of a 4-bar chain?	Beam engine, coupling rod of locomotive, Watt's indicator
Whitworth QRM is an inversion of?	Single slider crank chain (crank fixed)
Elliptical trammel is an inversion of?	Double slider crank chain
Scotch yoke produces?	Simple harmonic motion
Oldham coupling is used for?	Parallel shafts with small lateral offset
Exact straight-line mechanism?	Peaucellier, Hart, Scott-Russell
Steering used in automobiles?	Ackermann
Condition for correct steering?	$\cot \varphi - \cot \theta = c/b$
Number of I-centres for n links?	$n(n - 1)/2$
Kennedy's theorem?	Three I-centres of three bodies are collinear
Coriolis component?	$2\omega v$ (sliding on a rotating link)
Which theory is used for clutch design?	Uniform wear

Concept question	Answer
Friction torque of a flat pivot (uniform wear)?	μWR/2
Belt tension ratio (flat)?	e^(μθ)
Condition for max power in a belt drive?	Tc = T_max/3
Does centrifugal tension affect power?	No
Crossed belt length depends on?	Sum of pulley radii
What is creep in belts?	Relative motion due to unequal elastic stretching
Law of gearing?	Common normal passes through the pitch point
Module?	D/T
Standard addendum?	1 module
Min teeth on pinion for a 20° involute (rack)?	18 (17)
Contact ratio requirement?	> 1
Gear to eliminate axial thrust?	Herringbone (double helical)
Gear for non-intersecting perpendicular shafts with high reduction?	Worm & worm wheel
Gear in automobile differential?	Hypoid / bevel (epicyclic train)
Function of an idler?	Changes direction, not speed ratio
Height of a Watt governor?	895/N² m
Isochronous governor?	Same speed at all radii – infinitely sensitive
Hunting?	Continuous oscillation of speed due to over-sensitivity
Spring-loaded governor?	Hartnell
Flywheel controls?	Cyclic speed fluctuation within a cycle
Max fluctuation of energy?	Iω²Cs
Condition for dynamic balance?	Σmr = 0 and Σmrl = 0
Frequency of secondary unbalanced force?	Twice crank speed
Effects of partial balancing of locomotives?	Hammer blow, swaying couple, variation of tractive effort
Follower motion best for high speed?	Cycloidal
Follower motion with infinite acceleration?	Uniform velocity
Natural frequency from static deflection?	0.4985/√δ Hz
Critical damping coefficient?	2√(km)
Damping ratio used in gun recoil?	Critically damped (ζ = 1)
Frequency ratio for isolation?	> √2
Whirling speed equals?	Natural transverse frequency of the shaft

A4. Machine Design

Concept question	Answer
FOS for ductile materials is based on?	Yield strength
Kt for a small circular hole in a plate?	3
Notch sensitivity of cast iron?	≈ 0 (insensitive)
Endurance limit of steel ≈?	0.5 × UTS
Most conservative fatigue line?	Soderberg

Concept question	Answer
Goodman line joins?	Se and Sut
Process that improves fatigue life?	Shot peening
Unwin's formula?	d = 6√t
Riveting operations for leak-proofing?	Caulking and fullering
Throat of a fillet weld?	0.707 × leg
Thread for power transmission with max efficiency?	Square
Lead screw thread of a lathe?	Acme (29°)
Thread for unidirectional thrust?	Buttress
Metric thread angle?	60°
Initial bolt tension (Rotscher)?	2840d N
Key allowing axial movement?	Feather key
Self-aligning key?	Woodruff key
Width of a sunk key?	d/4
Length of key for strength equal to shaft?	≈ 1.5d
Coupling for angular misalignment?	Universal (Hooke's)
What is an axle mainly subjected to?	Bending
Ball bearing life equation?	L = (C/P)³ million rev
L_{10} means?	90% bearings survive
Bearing for combined radial + axial loads?	Taper roller
Bearing metal of Sn–Sb–Cu?	Babbitt
Is the Sommerfeld number dimensional?	No (dimensionless)
Lewis equation assumption?	Tooth is a cantilever with load at tip
Most common surface failure of gears?	Pitting
Spring index range?	4–12
What is surge in springs?	Resonance of coils with the exciting frequency
Cotter joint used for?	Axial loads (piston rod & crosshead)
Knuckle joint pin is in?	Double shear
Preferred number series ratio R10?	1.25 (10^(1/10))

A5. Thermodynamics

Concept question	Answer
Is a turbine an open or closed system?	Open
Is pressure intensive or extensive?	Intensive
Is heat a property?	No – path function
Basis of temperature measurement?	Zeroth law
Temperature equal in °C and °F?	−40
Work done in free expansion?	Zero
Polytropic index for an isothermal process?	n = 1
Which curve is steeper on p–V: isothermal or adiabatic?	Adiabatic (γ times)
PMM-1 violates?	First law

Concept question	Answer
PMM-2 violates?	Second law (Kelvin–Planck)
Internal energy of an ideal gas depends on?	Temperature only
Throttling process?	Isenthalpic ($h_1 = h_2$)
Joule–Thomson coefficient of an ideal gas?	Zero
Carnot efficiency depends on?	Reservoir temperatures only
COP_HP − COP_R?	1
Reversible adiabatic process?	Isentropic
Entropy of an isolated system?	Never decreases
Which is steeper on T–s: isochor or isobar?	Isochor
Unavailable energy?	$T_0\Delta S$
Gouy–Stodola theorem?	$I = T_0\ \Delta S_gen$
Critical point of water?	221.2 bar, 374.15 °C
Latent heat at critical point?	Zero
Dryness measurement for very wet steam?	Combined separating & throttling calorimeter
Degrees of freedom at the triple point?	0
Van der Waals constant 'a' accounts for?	Intermolecular attraction
Compressibility factor for an ideal gas?	1
Clausius–Clapeyron equation relates?	Saturation slope to latent heat
Third law?	Entropy of a perfect crystal at 0 K is zero
Intermediate temperature for equal efficiency of engines in series?	$\sqrt{(T_1T_2)}$
cp − cv for an ideal gas?	R
γ for monatomic gas?	1.67

CHAPTER 19

PYQ Concept Recall Bank – Part B

B1. IC Engines & Gas Cycles

Concept question	Answer
Otto cycle efficiency depends on?	Compression ratio only (for given γ)
Order of efficiency for the same CR?	Otto > Dual > Diesel
Order of efficiency for the same max pressure & temperature?	Diesel > Dual > Otto
Effect of cut-off ratio on Diesel efficiency?	Efficiency decreases as cut-off increases
Cycle with efficiency equal to Carnot?	Stirling and Ericsson (with regenerator)
Gas turbine cycle?	Brayton (Joule)
Effect of regeneration in a gas turbine?	Increases efficiency, not specific work
Governing in SI engines?	Quantity governing
Governing in CI engines?	Quality governing
CR range of a diesel engine?	14–22
Firing order of a 4-cylinder in-line engine?	1-3-4-2
Most efficient scavenging?	Uniflow
Friction power test for CI engines only?	Willan's line
Friction power test for multi-cylinder engines?	Morse test
Reference fuels for octane number?	Iso-octane (100) & n-heptane (0)
Reference fuels for cetane number?	Cetane (100) & α-methyl naphthalene (0)
Cause of diesel knock?	Long ignition delay
Effect of high CR on knock in SI / CI?	Increases SI knock / reduces CI knock
Stoichiometric A/F for petrol?	14.7:1
Mixture needed at cold start?	Rich (choke)
Purpose of the condenser in battery ignition?	Prevents arcing at contact breaker points
Plug for high-compression engines?	Cold plug
Two-stroke lubrication?	Petroil (oil mixed with petrol)
Device that keeps engine temperature optimum?	Thermostat
Pollutant reduced by EGR?	NOx
Catalyst metals in a three-way converter?	Pt, Pd, Rh
BS-VI implemented in India?	April 2020
Main constituent of CNG?	Methane
Turbocharger is driven by?	Exhaust gases
Supercharging is more suitable for?	CI engines
Engine needing glow plugs?	IDI diesel (cold start)
Ramjet can start from rest?	No
Propulsion that works in vacuum?	Rocket

B2. RAC

Concept question	Answer
1 TR equals?	3.5 kW (211 kJ/min)
Process in the expansion valve?	Throttling (isenthalpic)
Effect of subcooling?	Increases RE and COP
Effect of lowering evaporator temperature?	COP decreases
Expansion device in domestic refrigerators?	Capillary tube
Expansion valve maintaining constant superheat?	Thermostatic expansion valve
Refrigerant R-717?	Ammonia
Refrigerant R-744?	CO_2
Refrigerant used in LiBr–water systems?	Water
Metal not to be used with ammonia?	Copper (and its alloys)
ODP of R-134a?	Zero
Halocarbon leak detection?	Halide torch
Third fluid in an Electrolux refrigerator?	Hydrogen
Air refrigeration cycle?	Bell–Coleman (reversed Brayton)
Why is air refrigeration used in aircraft?	Low weight per TR (despite low COP)
Thermoelectric refrigeration effect?	Peltier
Saturated air: relation of DBT, WBT, DPT?	All equal
Sensible heating on the psychrometric chart?	Horizontal line (ω constant)
Evaporative cooling follows?	Constant WBT line
Specific humidity formula?	$0.622\ pv/(p - pv)$
Chemical dehumidification follows?	Constant enthalpy line (DBT rises)
Comfort condition?	21–25 °C DBT, 40–60% RH
Bypass factor?	$(t_2 - t_ADP)/(t_1 - t_ADP)$
Dry ice?	Solid CO_2 (−78.5 °C)

B3. Steam, Power Plant & Compressors

Concept question	Answer
Rankine pump work is small because?	Water has small specific volume
Main purpose of reheating?	Increase dryness at turbine exhaust
Regeneration increases?	Cycle efficiency (mean temperature of heat addition)
Boiler that is supercritical and drumless?	Benson
Babcock & Wilcox boiler type?	Water-tube, externally fired
Cochran boiler type?	Vertical, fire-tube, multi-tubular
Lancashire boiler?	Horizontal, two flue tubes, fire-tube
Function of the fusible plug?	Melts to extinguish fire when water level falls too low
Economiser heats?	Feedwater
Air preheater heats?	Combustion air

Concept question	Answer
Is a safety valve a mounting or an accessory?	Mounting
Equivalent evaporation is from and at?	100 °C (2257 kJ/kg)
Critical pressure ratio for superheated steam?	0.546
Mach number at the throat of a choked nozzle?	1
Effect of supersaturation on discharge?	Increases discharge
Parsons turbine?	50% reaction
Curtis turbine?	Velocity compounded impulse
Rateau turbine?	Pressure compounded impulse
Optimum blade speed ratio for single-stage impulse?	cos α / 2
Condenser used in large power plants?	Surface condenser
Moderator in Indian PHWRs?	Heavy water
Control rod materials?	Cadmium, boron
Device removing fly ash?	Electrostatic precipitator
Coal with highest carbon?	Anthracite
Compression process needing minimum work?	Isothermal
Intermediate pressure in 2-stage compression?	$\sqrt{(p_1 p_3)}$
Effect of clearance on volumetric efficiency?	Decreases it
Unstable low-flow phenomenon in centrifugal compressors?	Surging
Compressor with no internal compression?	Roots blower

B4. Heat Transfer

Concept question	Answer
Metal with highest thermal conductivity?	Silver
How does k of gases change with temperature?	Increases
Thermal diffusivity?	k/(ρc)
Critical radius of insulation for a cylinder?	k/h
Critical radius for a sphere?	2k/h
Heat loss at critical radius?	Maximum
Fins should be placed on which side?	The side with lower h
Fin effectiveness for fins to be justified?	> 2
Lumped analysis valid when?	Bi < 0.1
Characteristic length of a sphere?	r/3
Nusselt number signifies?	Dimensionless convective heat transfer coefficient
Prandtl number signifies?	Momentum diffusivity / thermal diffusivity
Pr of liquid metals?	Very low (≈ 0.01)
Natural convection is governed by?	Grashof (and Prandtl) number
Nu for laminar flow, constant wall temperature?	3.66
Nu for laminar flow, constant heat flux?	4.36
Dittus–Boelter exponent of Pr for heating?	0.4

Concept question	Answer
Boiling regime with max h?	Nucleate boiling
Condensation with higher h?	Dropwise
Kirchhoff's law?	α = ε (at thermal equilibrium)
Wien's law?	λ_max T = 2898 μm·K
Effect of n radiation shields?	Heat transfer reduced to 1/(n + 1)
View factor of a flat or convex surface to itself?	Zero
More effective heat exchanger type?	Counter flow
Effectiveness when one fluid changes phase?	1 − e^(−NTU)
Car radiator flow arrangement?	Cross flow

B5. Fluid Mechanics

Concept question	Answer
Viscosity of liquids with rising temperature?	Decreases
Viscosity of gases with rising temperature?	Increases
Example of a shear-thinning fluid?	Blood, paint
Example of a Bingham plastic?	Toothpaste
Example of a dilatant fluid?	Starch suspension, quicksand
Pressure inside a soap bubble?	8σ/d
Capillary rise is proportional to?	1/d
Mercury in glass shows?	Capillary depression (convex meniscus)
Why is mercury used in barometers?	High density, low vapour pressure
Centre of pressure lies?	Below the centroid
CP of a vertical rectangle with top at the surface?	2h/3 depth
Stability condition of a floating body?	M above G
Stability condition of a submerged body?	B above G
Metacentric height?	I/V − BG
Streamline = pathline when?	Steady flow
Velocity potential exists for?	Irrotational flow only
Free vortex relation?	Vr = constant
Free surface in a forced vortex?	Paraboloid
Bernoulli assumptions?	Steady, incompressible, inviscid, along a streamline
Cd of a venturimeter?	0.95–0.98
Cd of an orifice?	≈ 0.62
Pitot tube measures?	Stagnation pressure (point velocity)
V-notch discharge ∝?	H^(5/2)
Borda's mouthpiece (running free) Cd?	0.5
Laminar friction factor?	64/Re
Avg/max velocity in laminar pipe flow?	1/2
Shear stress in laminar pipe flow is max at?	Wall
Critical Reynolds number for pipes?	2000
Condition for max power through a pipe?	h_f = H/3

Concept question	Answer
Loss at sudden enlargement?	$(V_1 - V_2)^2/2g$
Device protecting a penstock against water hammer?	Surge tank
Boundary layer separation occurs with?	Adverse pressure gradient
Laminar BL thickness growth?	$\delta \propto \sqrt{x}$
Swing of a spinning ball is due to?	Magnus effect
Model law for ships?	Froude
Model law for pipe flow / submarines?	Reynolds
Discharge scale in Froude models?	Lr^2.5
Number of π terms?	n − m
Across a normal shock, what stays constant?	Stagnation temperature
Most economical rectangular channel?	b = 2y

B6. Hydraulic Machines

Concept question	Answer
Turbine for high head?	Pelton
Turbine for low head?	Kaplan
Turbine for medium head?	Francis
Specific speed of a turbine?	N√P/H^(5/4)
Specific speed of a pump?	N√Q/H^(3/4)
Turbine with adjustable runner blades?	Kaplan
Is a draft tube used with a Pelton wheel?	No
Function of a draft tube?	Recover exit KE and allow the turbine to sit above tail race
Pelton jet ratio?	≈ 12
Purpose of the Pelton bucket splitter?	Divides the jet into two halves
Efficiency of jet on a series of flat plates?	50% at u = V/2
Where does cavitation occur in reaction turbines?	Runner outlet (draft tube inlet)
Where does cavitation occur in centrifugal pumps?	Impeller eye (inlet)
Why is priming needed?	Head developed by air is negligible
Vane shape used in centrifugal pumps?	Backward-curved
Power vs speed (affinity law)?	$P \propto N^3$
Pumps in series give?	More head
When can negative slip occur?	Long suction, short delivery, high speed
Function of an air vessel?	Uniform flow, reduced acceleration head, saves work
Hydraulic ram works on?	Water hammer
Device that stores hydraulic energy?	Accumulator
Device with a stator for torque multiplication?	Torque converter
Synchronous speed?	120f/p

B7. Manufacturing

Concept question	Answer
Negative pattern allowance?	Shake (rapping) allowance
Metal with least shrinkage allowance?	Cast iron
Pattern for mass production on machines?	Match plate
Support for cores inside moulds?	Chaplets
Chvorinov's rule?	$t = B(V/A)^2$
Optimum riser?	Cylinder with H = D
Sprue is tapered to avoid?	Aspiration
Casting for turbine blades?	Investment casting
Die casting for Zn?	Hot chamber
Die casting for Al?	Cold chamber
Casting for pipes without cores?	True centrifugal
Defect from two streams not fusing?	Cold shut
Defect from low collapsibility?	Hot tear
TIG electrode?	Non-consumable tungsten
Polarity for deeper penetration?	DCSP (electrode −ve)
Welding of rails?	Thermit / flash butt
Welding of car bodies?	Spot welding
Flame for brass?	Oxidising
Max temperature of oxy-acetylene flame?	≈ 3200 °C
Acetylene is dissolved in?	Acetone
Brazing filler melting temperature?	Above 450 °C
Hot working is done above?	Recrystallisation temperature
Max draft in rolling?	$\mu^2 R$
Mill using backup rolls?	Four-high
Process for seamless tubes?	Mannesmann piercing
Clearance in blanking is given on?	Punch
Clearance in punching is given on?	Die
Die doing several operations at one station?	Compound die
Process for collapsible tubes?	Impact extrusion
Merchant's equation?	$2\varphi + \beta - \alpha = 90°$
Chip type with BUE?	Continuous with built-up edge (low speed, ductile)
Chip from cast iron?	Discontinuous
Taylor's equation?	$VT^n = C$
n for HSS / carbide?	≈ 0.125 / 0.25
Wear that decides tool life?	Flank wear
Crater wear mechanism?	Diffusion
HSS composition?	18-4-1 (W-Cr-V)
Why isn't diamond used on steel?	Carbon diffuses into iron
Theoretical roughness?	$f^2/8r$
Lathe tapers by tailstock set-over are?	Long, small, external
Twist drill point angle?	118°

Concept question	Answer
Jig vs fixture?	Jig guides the tool; fixture only holds the work
Location principle?	3-2-1
Milling with better finish?	Down (climb) milling
Simple indexing?	40/N
Gear hobbing cannot cut?	Internal gears
Wheel for hard materials?	Soft grade
Abrasive for cast iron?	Silicon carbide
Process for cylinder bores?	Honing
NTM for glass?	USM
NTM based on Faraday's law?	ECM
NTM needing vacuum?	EBM
Dielectric in EDM?	Kerosene / deionised water
G02 means?	Circular interpolation, clockwise
M03 means?	Spindle on, clockwise
Sintering temperature?	Below melting point of main constituent

B8. Materials, Metrology & IE

Concept question	Answer
APF of BCC?	0.68
Austenite structure?	FCC
Martensite structure?	BCT
Eutectoid composition?	0.8% C at 723 °C
Cementite carbon content?	6.67%
Composition of 18-8 stainless steel?	18% Cr, 8% Ni
Hadfield steel contains?	12–14% Mn
Graphite in SG iron is produced by?	Mg or Ce addition
Brass?	Cu + Zn
Bronze?	Cu + Sn
Alloy for measuring tapes?	Invar
Cooling in normalising?	Air
Product of austempering?	Bainite
Test for hardenability?	Jominy end-quench
Case hardening without quenching?	Nitriding
Rockwell C indenter?	Diamond cone
Izod specimen is?	Cantilever
NDT only for ferromagnetic materials?	Magnetic particle
Is Bakelite a thermoplastic or thermoset?	Thermoset
Least count of a micrometer?	0.01 mm
Sine bar is unreliable above?	45°
Preferred fit system?	Hole basis

Concept question	Answer
Taylor's principle of gauging?	GO checks max material with full form; NO-GO checks the other limit, single element
H7/p6 is?	Interference fit
Ra is also called?	CLA (centre line average)
EOQ?	√(2DCo/Ch)
At EOQ?	Ordering cost = carrying cost
ABC analysis is based on?	Annual consumption value
VED analysis is used for?	Spare parts (criticality)
Method study steps?	SREDIM
Standard time?	Normal time + allowances
Developer of therbligs?	Gilbreth
Chart for number of defects?	c-chart
Control limits?	±3σ
Producer's risk?	α (type I error)
Six Sigma DPMO?	3.4
PERT distribution?	β-distribution
PERT expected time?	(to + 4tm + tp)/6
Float on the critical path?	Zero
Best initial solution for the transportation problem?	Vogel's approximation method
Method for the assignment problem?	Hungarian method
Little's law?	L = λW
Break-even point?	F/(S − V)
Father of scientific management?	F.W. Taylor

CHAPTER 20

Paper-II Style Model Answers

Paper-II rewards structured answers: definition → sketch (describe it in words if you can't draw) → principle/derivation → applications → advantages and limitations. Each model answer below is written that way. Learn the structure, not the sentences.

20.1 Strength of Materials & Mechanics

✎ **DESCRIPTIVE Q:** Derive the expression for the elongation of a uniformly tapering circular bar under axial load P.

Let the diameter vary linearly from d_1 to d_2 over length L. At distance x from the larger end, $d_x = d_1 - kx$, with $k = (d_1 - d_2)/L$. The elongation of an elemental length dx is $d\delta = P\,dx / (E\cdot(\pi/4)d_x^2)$. Integrate from 0 to L: $\delta = (4P/\pi E)\int dx/(d_1 - kx)^2 = (4P/\pi Ek)[1/(d_1 - kx)]$ from 0 to L $= (4P/\pi Ek)(1/d_2 - 1/d_1) = (4P/\pi Ek)\cdot(d_1 - d_2)/(d_1d_2)$. Substitute k: $\delta = 4PL/(\pi Ed_1d_2)$. Check: when $d_1 = d_2 = d$, $\delta = PL/(AE)$, as expected.

✎ **DESCRIPTIVE Q:** Derive the expression for strain energy stored in a body due to an impact load falling from height h.

Let a load P fall through h onto a collar at the end of a bar of length L and area A, producing extension δ and instantaneous stress σ. Work done by the load = $P(h + \delta)$. Strain energy stored $= (\sigma^2/2E)AL$. With $\delta = \sigma L/E$: $P(h + \sigma L/E) = \sigma^2 AL/(2E)$. This is a quadratic in σ: $(AL/2E)\sigma^2 - (PL/E)\sigma - Ph = 0$. Solving: $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$. Special cases: h = 0 (sudden load) gives $\sigma = 2P/A$. If $h \gg \delta$, then $\sigma \approx \sqrt{(2EPh/(AL))}$.

✎ **DESCRIPTIVE Q:** Explain the assumptions of Euler's theory of columns and its limitations.

Assumptions: the column is initially perfectly straight and the load is exactly axial; the material is homogeneous, isotropic and obeys Hooke's law; the cross-section is uniform; the self-weight is ignored; the column fails only by buckling; and the length is large compared with the lateral dimensions (direct stress is negligible). Limitations: Euler's formula gives crippling stress π^2E/λ^2, which exceeds the crushing stress for short columns, so it is valid only when λ exceeds the limiting slenderness ratio (≈ 80 for mild steel). Real columns have initial crookedness and load eccentricity, so they fail below Euler's load. The Rankine–Gordon formula covers both short and long columns empirically.

✎ **DESCRIPTIVE Q:** Draw the SFD and BMD for a simply supported beam carrying a UDL w over span L and locate the maximum BM.

Reactions: $R_A = R_B = wL/2$. At distance x from A: $SF = wL/2 - wx$, a straight line from $+wL/2$ to $-wL/2$ that is zero at $x = L/2$. $BM = (wL/2)x - wx^2/2$, a parabola that is zero at both supports. Since $dM/dx = SF = 0$ at $x = L/2$, the maximum BM there is $wL/2\cdot L/2 - w(L/2)^2/2 = wL^2/8$ (sagging). There is no point of contraflexure.

✎ **DESCRIPTIVE Q:** Explain the principle of virtual work / work–energy principle with an application.

Work–energy principle: the net work done by all forces on a body equals its change in kinetic energy, $W_{net} = \frac{1}{2}m(v_2^2 - v_1^2)$. It is derived by integrating Newton's second law, $F = m\,v\,dv/ds$, along the path. Application: finding the stopping distance of a vehicle on a rough road. Friction work $\mu mg \cdot s = \frac{1}{2}mv^2$, so $s = v^2/(2\mu g)$. Mass does not affect it. Advantages: it avoids computing acceleration, and it handles variable forces when the work is known.

20.2 Theory of Machines & Machine Design

✎ **DESCRIPTIVE Q:** Derive the ratio of belt tensions for a flat belt, $T_1/T_2 = e^{(\mu\theta)}$.

Take a belt element subtending angle $\delta\theta$ at the pulley centre, with tensions T and T + δT at its ends. Normal reaction R; friction μR acts along the belt. Radial equilibrium: $R = (T + \delta T)\sin(\delta\theta/2) + T\sin(\delta\theta/2) \approx T\,\delta\theta$ (neglecting second-order terms). Tangential equilibrium: $\mu R = (T + \delta T)\cos(\delta\theta/2) - T\cos(\delta\theta/2) \approx \delta T$. Hence $\delta T/T = \mu\,\delta\theta$. Integrating from T_2 to T_1 over the arc of contact θ: $\ln(T_1/T_2) = \mu\theta$, so $T_1/T_2 = e^{(\mu\theta)}$. For V-belts the normal reaction is increased by the wedge action ($2R\sin\beta$), giving $e^{(\mu\theta\,\mathrm{cosec}\,\beta)}$.

✎ **DESCRIPTIVE Q:** What is interference in involute gears? State the methods to avoid it.

The involute profile exists only outside the base circle. If the addendum of the mating gear extends beyond the interference point (the tangency point of the line of action with the base circle), its tip contacts the non-involute flank below the base circle. This is interference: the tip digs into the flank, removes material (undercutting) and weakens the tooth. Avoid it by: (1) increasing the number of teeth on the pinion (≥ 18 for 20°, 32 for 14.5°); (2) increasing the pressure angle; (3) using stub teeth with reduced addendum; (4) profile shifting (positive correction on the pinion); (5) undercutting the flank during generation (but this weakens the tooth).

✎ **DESCRIPTIVE Q:** Explain the working of a centrifugal governor (Porter) and derive the expression for its height.

A Porter governor has two balls of mass m on arms pivoted at the spindle, with lower links joined to a central sleeve of mass M that slides on the spindle. When speed rises, centrifugal force throws the balls outward and lifts the sleeve. The sleeve, through a lever, reduces the fuel or steam supply, and the reverse happens when speed falls. For equal arm and link lengths, take moments about the instantaneous centre of the lower link (ignoring friction): $m\omega^2 r \cdot h = mg \cdot r + (M/2)g \cdot 2r$. This gives $h = (m + M)g/(m\omega^2)$, i.e., $h = (m + M)/m \times 895/N^2$ (m, N in rpm). A Watt governor is the special case M = 0: $h = 895/N^2$.

✎ **DESCRIPTIVE Q:** Explain why rolling contact bearings are preferred in some applications and sliding bearings in others.

Rolling bearings (ball, roller) have low starting friction, need little lubrication and maintenance, take combined loads, are standardised and easily replaced, and keep accurate shaft alignment. They are preferred in automobiles, machine tool spindles and electric motors. However, they are noisy at high speed, have finite fatigue life, are sensitive to shock and need more radial space. Sliding (journal) bearings with a hydrodynamic film run silently, damp vibration, carry very heavy and shock loads, have practically unlimited life with proper lubrication, and are compact radially. They suit turbines, large generators, IC engine crankshafts and connecting rods. Their drawbacks are high starting friction and the need for continuous lubrication.

✎ **DESCRIPTIVE Q:** Explain the design procedure for a cotter joint (socket and spigot).

Failure modes considered, with load P: (1) tension in the rod, d = √(4P/πσt); (2) tension across the slot of the spigot, P = [(π/4)d_2^2 − d_2t]σt; (3) tension across the slot of the socket, P = [(π/4)(d_1^2 − d_2^2) − (d_1 − d_2)t]σt; (4) shear of the cotter, P = 2bt·τ; (5) crushing of spigot on cotter, P = d_2tσc; (6) crushing of socket collar, P = (d_4 − d_2)tσc; (7) shear of the spigot end, P = 2a d_2 τ; (8) shear of the socket end, P = 2(d_4 − d_2)cτ; (9) bending of the cotter. Empirical proportions are first assumed: t ≈ 0.31d, d_2 ≈ 1.21d, d_1 ≈ 1.75d. Each mode is then checked and the dimensions are increased where a stress exceeds the allowable value.

20.3 Thermal Engineering

✎ **DESCRIPTIVE Q:** Derive the air-standard efficiency of the Otto cycle.

Processes: 1–2 isentropic compression; 2–3 heat addition at constant volume, Q_1 = mcv(T_3 − T_2); 3–4 isentropic expansion; 4–1 heat rejection at constant volume, Q_2 = mcv(T_4 − T_1). Efficiency η = 1 − Q_2/Q_1 = 1 − (T_4 − T_1)/(T_3 − T_2). For isentropic processes with compression ratio r = V_1/V_2 = V_4/V_3: T_2/T_1 = T_3/T_4 = r^(γ−1). Hence T_3 − T_2 = r^(γ−1)(T_4 − T_1), and η = 1 − 1/r^(γ−1). Efficiency rises with r but is limited by knock in SI engines. It does not depend on heat added.

✎ **DESCRIPTIVE Q:** Explain the working of a four-stroke diesel engine with a valve timing diagram.

Suction: the inlet valve opens (about 10–20° before TDC) and the piston moves down drawing in air only; the valve closes 25–50° after BDC to use the inertia ram effect. Compression: both valves closed; air is compressed (CR 14–22) to 30–45 bar and about 600 °C. Power: fuel is injected starting 10–15° before TDC, self-ignites after a short delay, and burns roughly at constant pressure; the gases expand and push the piston down. Exhaust: the exhaust valve opens 30–50° before BDC (blow-down) and the piston expels gases, closing 10–15° after TDC. Near TDC both valves are open (valve overlap) to scavenge residual gases. One power stroke occurs every two crank revolutions, so a flywheel is needed.

✎ **DESCRIPTIVE Q:** Explain the effects of superheating, condenser pressure and boiler pressure on the Rankine cycle using a T–s diagram.

Superheating raises the mean temperature of heat addition, increasing efficiency and work output. It also moves the expansion line to the right, so exhaust steam is drier and turbine blades erode less. Lowering condenser pressure lowers the mean temperature of heat rejection, which increases work and efficiency, but exhaust moisture rises and air leakage into the condenser increases. Raising boiler pressure at fixed maximum temperature raises the mean temperature of heat addition (higher η), but the expansion line shifts left and exhaust dryness drops. So high pressure is combined with reheating to keep exhaust quality above about 0.88.

✎ **DESCRIPTIVE Q:** Explain the working of a vapour absorption refrigeration system (aqua-ammonia) and compare it with VCR.

Low-pressure ammonia vapour from the evaporator is absorbed by weak aqua-ammonia solution in the absorber; heat of absorption is removed by cooling water. A small pump raises the strong solution to condenser pressure through a heat exchanger into the generator. There heat (steam, waste heat or solar) drives off ammonia vapour. An analyser and rectifier remove water vapour from it. Pure ammonia condenses in the condenser, is throttled and evaporates in the evaporator to produce cooling. The weak solution returns to the absorber through a pressure-reducing valve. Comparison: VAR uses heat energy and a tiny pump in place of a compressor, so it has few moving parts, runs quietly and suits waste heat and large capacities.

Its COP (0.5–0.7) is lower than VCR (2–5), and it is bulkier with a higher initial cost.

✎ **DESCRIPTIVE Q:** Explain modes of heat transfer and derive the heat flow equation through a composite plane wall.

Conduction is molecular energy transfer in a medium without bulk motion (Fourier's law). Convection is energy transfer between a surface and a moving fluid (Newton's law of cooling). Radiation is electromagnetic emission needing no medium (Stefan–Boltzmann law). For a composite wall of layers L_1, L_2, L_3 with conductivities k_1, k_2, k_3 and convection h_i, h_o on the two faces, the same heat Q flows through each resistance in series. Q = (T_i − T_o) / [1/(h_iA) + $L_1/(k_1A)$ + $L_2/(k_2A)$ + $L_3/(k_3A)$ + 1/(h_oA)], and the overall coefficient is U = 1/[1/h_i + ΣL/k + 1/h_o]. Interface temperatures follow from Q multiplied by each partial resistance.

20.4 Fluid Mechanics & Machines

✎ **DESCRIPTIVE Q:** Derive Hagen–Poiseuille's equation for laminar flow in a circular pipe.

Take a coaxial fluid cylinder of radius r and length L. Pressure force $(-dp/dx)\pi r^2 L$ balances the shear force $\tau \cdot 2\pi rL$, so $\tau = (-dp/dx)r/2$: shear varies linearly from zero at the axis to maximum at the wall. With $\tau = -\mu\, du/dr$, integrating and using u = 0 at r = R gives $u = (-dp/dx)(R^2 - r^2)/(4\mu)$, a parabolic profile with u_max at the centre. Discharge $Q = \int u \cdot 2\pi r\, dr = (\pi/8\mu)(-dp/dx)R^4$, so mean velocity V = u_max/2. Rearranging: $\Delta p = 32\mu VL/D^2$, and head loss $h_f = 32\mu VL/(\rho g D^2)$, which corresponds to f = 64/Re.

✎ **DESCRIPTIVE Q:** Define specific speed of a turbine, derive its expression and explain its significance.

Specific speed is the speed of a geometrically similar turbine that produces unit power under unit head. For similar turbines, u ∝ √H and u ∝ DN, so D ∝ √H/N. Discharge Q ∝ area × velocity ∝ $D^2\sqrt{H}$. Power P ∝ QH ∝ $D^2H^{(3/2)} \propto H^{(5/2)}/N^2$. For the specific turbine (P = 1, H = 1, N = Ns), the proportionality constant gives $P = (Ns^2/N^2)H^{(5/2)}$, so $Ns = N\sqrt{P}/H^{(5/4)}$. Significance: it classifies turbines (Pelton low, Francis medium, Kaplan high), predicts performance of prototypes from models, and guides selection for a given head and power.

✎ **DESCRIPTIVE Q:** Explain cavitation in hydraulic machines – causes, effects and remedies.

Cavitation is the formation of vapour bubbles where local static pressure falls below the liquid's vapour pressure, and their violent collapse when they reach higher-pressure zones. Causes: high suction lift, high speed, high water temperature, low atmospheric pressure (altitude), and poor blade design. It occurs at the runner outlet of reaction turbines and the impeller eye of pumps. Effects: pitting and erosion of metal surfaces (very high local pressures on collapse), noise and vibration, and a sudden drop in head, discharge and efficiency. Remedies: keep pressure above vapour pressure by limiting suction head and keeping NPSH available above NPSH required; set turbines lower (Thoma's σ above critical); use cavitation-resistant materials (stainless steel, bronze) and polished surfaces; avoid overloading and excess speed.

✎ **DESCRIPTIVE Q:** Describe the working of a centrifugal pump and explain the importance of priming.

The impeller, keyed to a motor-driven shaft, rotates inside a volute casing. Liquid enters the eye axially, is turned radially outward by backward-curved vanes, and gains kinetic and pressure energy through centrifugal (forced-vortex) action. The volute's gradually increasing area converts much of the velocity head into pressure before the liquid reaches the delivery pipe. Priming fills the suction pipe, casing and impeller with liquid before starting. Head

developed is H = Vw_2u_2/g regardless of fluid, so pressure rise ∝ ρ. With air the pressure developed is about 1/800 of that with water, which cannot lift water up the suction pipe. A foot valve (non-return) keeps the pump primed; self-priming designs and jet-assisted priming are alternatives.

20.5 Production & Industrial Engineering

✎ **DESCRIPTIVE Q:** Explain the various allowances provided on patterns with reasons.

(1) Shrinkage allowance: the pattern is made oversize to compensate for liquid, solidification and solid contraction (CI ≈ 1%, steel ≈ 2%). Contraction rulers are used. (2) Machining allowance: extra metal on surfaces to be machined, larger for rough castings, CI, big parts and internal surfaces. (3) Draft allowance: a 1–3° taper on vertical faces so the pattern can be withdrawn without damaging the mould; internal surfaces get more. (4) Distortion allowance: the pattern is deliberately distorted in the opposite sense for long, thin or U-shaped castings that warp on cooling. (5) Shake allowance: rapping enlarges the cavity, so the pattern is made slightly smaller. It is the only negative allowance.

✎ **DESCRIPTIVE Q:** Compare TIG and MIG welding.

TIG (GTAW) uses a non-consumable tungsten electrode with separate filler rod if needed and an inert gas shield (argon/helium). It gives high-quality, spatter-free welds on thin sheets and non-ferrous metals (Al, Mg, stainless). AC is used for aluminium for its cleaning action and DCSP for steels. Deposition rate is low and it needs skilled operators. MIG (GMAW) uses a continuously fed consumable bare wire that is both electrode and filler, with inert or CO_2 shielding (MAG) and DCRP with a constant-voltage source. It is faster, has higher deposition, is easy to automate and suits thicker sections and production welding. It produces some spatter and the equipment is costlier.

✎ **DESCRIPTIVE Q:** Explain Taylor's tool life equation and factors affecting tool life.

Taylor's experiments showed that cutting speed and tool life follow $VT^n = C$. Here n depends mainly on tool material (≈ 0.1–0.15 for HSS, 0.2–0.5 for carbides, 0.5–0.7 for ceramics), and C is the speed giving 1 min tool life. On log–log axes it is a straight line of slope −1/n. The extended form $VT^nf^a d^b = C$ includes feed and depth of cut. Factors: cutting speed (strongest influence), feed, depth of cut, tool material and geometry (rake, clearance, nose radius), work material (hardness, microstructure, inclusions), cutting fluid, rigidity of machine and set-up, and type of cut (continuous or interrupted).

✎ **DESCRIPTIVE Q:** Derive the EOQ formula and state its assumptions.

Assumptions: constant and known annual demand D; instantaneous replenishment; constant lead time; no shortages; constant ordering cost Co per order and holding cost Ch per unit per year; no quantity discounts. With order quantity Q: number of orders = D/Q, so annual ordering cost = (D/Q)Co. Average inventory = Q/2, so annual holding cost = (Q/2)Ch. Total cost TC = (D/Q)Co + (Q/2)Ch. Setting $dTC/dQ = -DCo/Q^2 + Ch/2 = 0$ gives $Q^* = \sqrt{(2DCo/Ch)}$. At Q*, ordering cost equals holding cost, and $TC_{min} = \sqrt{(2DCoCh)}$.

✎ **DESCRIPTIVE Q:** Compare PERT and CPM.

CPM (Critical Path Method, DuPont, 1957) is deterministic with one time estimate per activity. It is activity-oriented, suits repetitive projects with known durations (construction, maintenance) and emphasises time–cost trade-off (crashing). PERT (Program Evaluation and Review Technique, US Navy Polaris, 1958) is probabilistic with three estimates (optimistic, most likely,

pessimistic) fitted to a β-distribution, te = (to + 4tm + tp)/6. It is event-oriented, suits R&D and first-of-a-kind projects and gives the probability of meeting a target date using the normal distribution along the critical path. Both use network diagrams, forward and backward passes, floats and the critical path (zero total float).

CHAPTER 20B

Paper-II Model Answers – Set 2

20B.1 Short-Note Answers (5-mark style)

✎ **DESCRIPTIVE Q:** Write a short note on creep.

Creep is slow, time-dependent permanent deformation under constant load, significant above about 0.4 Tm (absolute melting temperature). A creep curve (strain vs time at constant load and temperature) shows instantaneous elastic strain, then primary creep (decreasing rate due to strain hardening), secondary or steady-state creep (constant minimum rate, a balance between hardening and recovery – used for design), and tertiary creep (accelerating rate from necking and void formation, ending in rupture). Creep matters in turbine blades, boiler tubes, steam pipes, IC engine valves and lead pipes (lead creeps at room temperature). Creep-resistant materials have high melting points, coarse grains (fewer grain boundaries), solid-solution and precipitate strengthening; single-crystal nickel superalloys are used for turbine blades.

✎ **DESCRIPTIVE Q:** Write a short note on fatigue failure.

Fatigue is failure under repeated or fluctuating stresses well below the ultimate (often below yield) strength after many cycles. It has three stages: crack initiation at stress raisers (surface scratches, notches, inclusions), crack propagation (beach marks or striations on the fracture surface), and sudden final fracture of the reduced section (rough, granular zone). It causes most failures of machine parts such as shafts, springs, gears and aircraft components. The S–N curve and endurance limit describe resistance. Improve fatigue life by reducing stress concentration (fillets), improving surface finish, inducing compressive residual stress (shot peening, surface rolling, nitriding), avoiding corrosion and choosing suitable materials.

✎ **DESCRIPTIVE Q:** Write a short note on cavitation and NPSH.

Cavitation itself is explained in the first Paper-II answers chapter. NPSH (net positive suction head) is the absolute pressure head at the pump suction above the vapour pressure head: NPSH_A = p_atm/ρg − h_s − h_fs − p_v/ρg. NPSH_A must exceed NPSH_R (given by the manufacturer) to avoid cavitation. Increase NPSH_A by lowering the pump (reducing suction lift), using a larger suction pipe with fewer fittings (less friction), cooling the liquid (lower vapour pressure) or pressurising the suction tank.

✎ **DESCRIPTIVE Q:** Write a short note on knocking in SI engines.

Knocking (detonation) is spontaneous auto-ignition of the last portion of unburnt charge (end gas) before the normal flame front reaches it. It happens because the end gas is compressed and heated by the expanding burnt gases ahead of the flame. The sudden pressure rise sets up pressure waves that hit the cylinder walls, producing a metallic pinging sound, loss of power, overheating, and damage to pistons, rings and spark plug electrodes. It is promoted by high CR, high intake temperature and pressure, advanced spark timing, low-octane fuel, large bore, lean mixtures and low speed. It is controlled by using high-octane fuel or anti-knock additives, retarding the spark (knock sensors), lowering CR, compact combustion chambers with central

spark plugs, more turbulence, cooling the end-gas region and EGR.

✎ **DESCRIPTIVE Q:** Write a short note on the boundary layer and its separation.

When a real fluid flows past a surface, the no-slip condition makes velocity zero at the wall. A thin layer forms in which velocity rises to the free-stream value U and viscous effects are important: this is the boundary layer. Its thickness δ (where u = 0.99U) grows downstream. It starts laminar ($\delta \propto \sqrt{x}$) and becomes turbulent beyond $Re_x \approx 5 \times 10^5$ on a flat plate, with a thin laminar sub-layer remaining near the wall. Under an adverse pressure gradient (pressure increasing downstream, as at the rear of a cylinder or in a diffuser), the slow fluid near the wall loses momentum and stops; where $(\partial u/\partial y)$ at the wall = 0, the flow separates and reverse flow forms a wake. This greatly raises pressure drag. Separation is controlled by streamlining, suction or blowing of the boundary layer, slotted wings, vortex generators, rotating cylinders and turbulent tripping (golf ball dimples).

✎ **DESCRIPTIVE Q:** Write a short note on hydraulic jump.

A hydraulic jump is the sudden rise of the water surface when supercritical flow (Fr > 1, shallow and fast) changes to subcritical flow (Fr < 1, deep and slow) in an open channel, for example at the foot of a spillway or below a sluice gate. It involves violent turbulence and rollers. Much energy is dissipated, which is useful for protecting the downstream channel bed from scour. It also raises the water level for irrigation, mixes chemicals in water treatment and aerates water. Sequent depths: $y_2/y_1 = \tfrac{1}{2}[-1 + \sqrt{(1 + 8Fr_1^2)}]$. Energy loss = $(y_2 - y_1)^3/(4y_1y_2)$.

✎ **DESCRIPTIVE Q:** Write a short note on psychrometric chart.

A psychrometric chart is a graphical representation of the properties of moist air at a given barometric pressure (usually 1.01325 bar). Dry bulb temperature is on the horizontal axis and specific humidity (with vapour pressure) on the vertical axis. The saturation curve (100% RH) is the upper boundary, with relative humidity curves below it. Wet bulb temperature lines slope down to the right and nearly coincide with constant-enthalpy lines (enthalpy scale on the diagonal). Specific volume lines are steeper. Dew point is read horizontally at the saturation curve. Two independent properties locate a state, and the chart shows AC processes as lines: sensible heating and cooling horizontal, humidification and dehumidification vertical, evaporative cooling along WBT lines, and mixing along a straight line between states.

✎ **DESCRIPTIVE Q:** Write a short note on CNC machines.

A CNC (computer numerical control) machine tool is controlled by a dedicated microcomputer that reads a part program (G and M codes) and drives the axes with servo or stepper motors through ball screws, using feedback from encoders or linear scales in closed-loop systems. Advantages: high accuracy and repeatability, complex contours by interpolation, reduced set-up and lead time, flexibility (just change the program), less operator skill needed, unmanned operation, easy integration with CAD/CAM. Limitations: high initial and maintenance cost and the need for trained programmers. Developments: machining centres with automatic tool changers and pallet changers, DNC networks, adaptive control, and multi-axis (5-axis) machines.

✎ **DESCRIPTIVE Q:** Write a short note on Just-in-Time (JIT).

JIT, developed at Toyota (Taiichi Ohno), produces and delivers only what is needed, when it is needed and in the quantity needed. It aims at eliminating the seven wastes (overproduction, waiting, transport, over-processing, inventory, motion, defects). Tools: pull system with Kanban cards, small lot sizes and set-up reduction (SMED), level (heijunka) scheduling, a few reliable

suppliers close to the plant, total productive maintenance, quality at the source (jidoka, poka-yoke), multi-skilled workers and continuous improvement (kaizen). Benefits: low inventory and floor space, shorter lead times, better quality and exposure of hidden problems. Risks: vulnerability to supply disruptions and the need for stable demand.

20B.2 Explain-with-Diagram Answers

✎ **DESCRIPTIVE Q:** Explain the working of a two-stroke petrol engine and compare it with a four-stroke engine.

In a crankcase-scavenged two-stroke engine, the piston itself acts as the valve through ports. Upward stroke: the piston compresses the charge above it, and its rising skirt uncovers the inlet port so a fresh mixture is drawn into the sealed crankcase by the partial vacuum. Near TDC the spark ignites the charge. Downward stroke: the burning gases push the piston down (power), and the mixture in the crankcase is compressed slightly. The exhaust port is uncovered first (blow-down), then the transfer port, through which compressed crankcase charge flows into the cylinder and pushes out the exhaust (scavenging, aided by the deflector on the piston crown). So one power stroke occurs every revolution. Compared with four-stroke: simpler, lighter, cheaper, no valve gear, more uniform torque and a lighter flywheel, and theoretically twice the power for the same size. But it has lower thermal and volumetric efficiency, higher fuel consumption due to short-circuiting of fresh charge, higher HC emissions, higher lubricating oil consumption (petroil) and more wear and noise. So it suits mopeds, chainsaws and large marine diesels (uniflow).

✎ **DESCRIPTIVE Q:** Explain the working of a Pelton wheel with a velocity triangle and derive the condition for maximum efficiency.

Water from a high-head reservoir is carried by the penstock to a nozzle with a spear valve, forming a high-velocity jet $V_1 = C_v\sqrt{(2gH)}$. The jet strikes the splitter of double-cup buckets on the wheel periphery (speed u), is divided into two halves, and is deflected back through about 165°. Inlet triangle: V_1 and u are collinear, so relative velocity $V_{r1} = V_1 - u$ and $V_{w1} = V_1$. Outlet: $V_{r2} = kV_{r1}$ at bucket angle φ to the reverse direction, so $V_{w2} = V_{r2}\cos\varphi - u$ (opposite direction). Work per kg = $(V_{w1} + V_{w2})u = (V_1 - u)(1 + k\cos\varphi)u$. Hydraulic efficiency $\eta_h = 2(V_1 - u)(1 + k\cos\varphi)u/V_1^2$. Setting $d\eta_h/du = 0$ gives $V_1 - 2u = 0$, so $u = V_1/2$, and $\eta_{h,max} = (1 + k\cos\varphi)/2$. With $k = 1$ and $\varphi = 0$ this becomes 100% in theory. In practice $u \approx 0.46V_1$ and the overall efficiency is about 85–90%.

✎ **DESCRIPTIVE Q:** Explain the working of a steam power plant with a layout and state the function of each component.

Coal from the storage yard is crushed and pulverised and burnt in the boiler furnace with air preheated in the air preheater. The forced-draught fan supplies air; the induced-draught fan and chimney remove flue gases after the ESP removes fly ash. Water from the feed pump passes through the economiser (heated by flue gas), then the boiler drum and water walls (evaporation), and the superheater, producing high-pressure superheated steam. Steam expands in the HP turbine, returns to the reheater, then expands through the IP and LP turbines, which drive the alternator. Exhaust steam condenses in a surface condenser cooled by water from the cooling tower. Condensate is pumped through LP feed heaters, the deaerator (removes dissolved O_2 and CO_2) and HP feed heaters (bled steam, regeneration) back to the boiler. Ash handling and coal handling plants, water treatment (to prevent scaling), and switchyard complete the plant. Overall efficiency is about 33–40%.

✎ **DESCRIPTIVE Q:** Describe the shaping machine quick-return mechanism (crank and slotted lever) with a sketch.

A bull gear driven by the motor carries a crank pin with a sliding block that fits in the slot of a slotted lever. The lever pivots at its lower end and its upper end connects to the ram, which carries the tool. As the bull gear rotates uniformly, the block slides in the slot and oscillates the lever, which reciprocates the ram. Extreme positions occur when the crank is perpendicular to the slotted lever. The crank turns through the larger angle (360° − α) during the cutting stroke and the smaller angle α during the return stroke, so the return is quicker (ratio = (360° − α)/α, typically about 3:2). Stroke length is changed by altering the crank radius (the crank pin position on the bull gear), and stroke position by moving the ram clamp. Idle return time is reduced, which increases productivity.

✎ **DESCRIPTIVE Q:** Explain the iron–carbon diagram based heat treatment of hardening and tempering with its objectives.

Hardening: a hypoeutectoid steel is heated 30–50 °C above A_3 (above A_1 for hypereutectoid), held to form homogeneous austenite, and quenched in water, oil or brine faster than the critical cooling rate. Austenite transforms without diffusion into martensite, a supersaturated BCT solution of carbon in iron. It is very hard and strong but brittle and highly stressed, sometimes with retained austenite. Objectives: high hardness and wear resistance for tools, dies, gears and bearings. Tempering: the hardened steel is reheated below A_1 and cooled. Martensite decomposes into tempered martensite (fine carbides in ferrite). Hardness drops slightly, while toughness and ductility rise a lot and internal stresses are relieved. Low-temperature tempering (150–250 °C) is for cutting tools, medium (350–450 °C) for springs, high (500–650 °C, giving sorbite) for shafts, axles and connecting rods. Hardening plus high-temperature tempering is called hardening and tempering (H&T or Q&T).

✎ **DESCRIPTIVE Q:** Explain the principle and working of electric discharge machining.

EDM removes metal by a rapid series of controlled electric sparks between a shaped tool electrode (usually the cathode – copper, brass or graphite) and the workpiece (anode), both submerged in a dielectric fluid (kerosene, transformer oil or deionised water). A pulsed DC supply (RC relaxation circuit or transistorised pulse generator) charges the gap (0.01–0.5 mm). When the voltage exceeds the dielectric's breakdown strength, a plasma channel forms, with local temperatures of 8000–12000 °C, which melts and vaporises a tiny crater on both surfaces. The dielectric quenches and flushes away the debris and deionises the gap for the next spark. A servo system keeps the gap constant as the tool feeds in. MRR depends on spark energy ($\frac{1}{2}CV^2$) and frequency; finish improves with lower energy and higher frequency. Applications: dies, moulds, complex cavities, small holes in hard conductive materials (hardened steel, carbides). Limitations: the work must be conductive, the tool wears, MRR is slow, and a recast layer and HAZ form.

20B.3 Compare-and-Contrast Answers

✎ **DESCRIPTIVE Q:** Compare impulse and reaction turbines (steam/water).

Impulse: all pressure energy is converted to kinetic energy in nozzles. Moving blades receive a high-velocity jet at constant pressure and only change its direction. Blades are symmetrical. Partial admission is possible. Casing has no hydraulic function. Examples: De Laval and Curtis steam turbines, Pelton wheel. Reaction: pressure drops in both fixed and moving blades. The fluid accelerates through moving blades too, producing a reaction force. Blades are

aerofoil-shaped and form nozzles. Full admission is needed, and leakage across blade tips matters (balance piston for axial thrust). In water turbines the runner is full of water and a draft tube is used. Examples: Parsons steam turbine, Francis and Kaplan. Reaction turbines need more stages but are more efficient at their design point.

✎ **DESCRIPTIVE Q:** Compare casting, forging and machining as manufacturing routes for a crankshaft.

Casting (nodular CI): low cost, near-net shape, complex shapes, good damping. But it has lower strength and fatigue resistance and possible porosity. Used for low-to-medium duty cars. Forging (medium-carbon or alloy steel, closed die): continuous grain flow following the contour gives high strength, toughness and fatigue life. The die cost is high and it needs finish machining. Used for high-performance and diesel engines. Machining from solid bar: no tooling cost and flexible, but the grain flow is cut, so fatigue strength is lower, and there is much material waste and long machining time. Used only for prototypes and single pieces. Forged crankshafts are preferred for heavy duty and cast SG iron for economy.

✎ **DESCRIPTIVE Q:** Compare annealing and normalising.

Both heat the steel into the austenite range (annealing about 30–50 °C above A_3/A_1; normalising about 40–50 °C above A_3/A_cm). Annealing cools slowly in the furnace, giving coarse pearlite, lowest hardness, maximum ductility and machinability, and full stress relief. It takes longer and costs more. Normalising cools in still air, giving fine pearlite and finer grain, higher strength and hardness, and a more uniform structure. It is quicker and cheaper and is used after forging or casting to refine grains, often as a preliminary to hardening. Hypereutectoid steels are not fully annealed above A_cm because a brittle cementite network would form; they are spheroidised instead.

APPENDIX

Appendix A – One-Place Formula Sheet

Every formula box from the theory chapters, collected chapter by chapter for the final revision. Read it with a pen and rewrite any formula you hesitate on.

Engineering Mechanics

$R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$

$\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$ (α = angle of R with P)

$\theta = 0° \rightarrow R = P+Q$ (maximum); $\theta = 180° \rightarrow R = |P-Q|$ (minimum); $\theta = 90° \rightarrow R = \sqrt{(P^2+Q^2)}$

Two equal forces P at angle θ: $R = 2P\cos(\theta/2)$

$P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$ (α = angle between Q and R, etc.)

$\mu = F/N = \tan\varphi$ (φ = angle of friction)

Angle of repose $\alpha = \varphi$ (max inclination at which a body rests on its own)

Body on incline (no applied force), tendency to slide down: equilibrium if $\tan\theta \le \mu$

Pull P parallel to plane to move up: $P = W(\sin\theta + \mu\cos\theta)$; to move down: $P = W(\mu\cos\theta - \sin\theta)$

Least force to drag a body on a horizontal plane: $P_{min} = W\sin\varphi$, applied at angle φ to the horizontal

$\bar{x} = \Sigma(A_i x_i)/\Sigma A_i$; $\bar{y} = \Sigma(A_i y_i)/\Sigma A_i$ (subtract cut-out areas)

Parallel axis theorem: $I_{AB} = I_G + A h^2$

Perpendicular axis theorem (plane lamina): $I_{zz} = I_{xx} + I_{yy}$ (polar MOI J)

Radius of gyration $k = \sqrt{(I/A)}$

$v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$; $s_{nth} = u + a(2n-1)/2$

Projectile: $T = 2u\sin\theta/g$; $H = u^2\sin^2\theta/(2g)$; $R = u^2\sin2\theta/g$; $R_{max} = u^2/g$ at $\theta = 45°$

Complementary angles (θ, $90°-\theta$) give equal range. Path is parabola: $y = x\tan\theta - gx^2/(2u^2\cos^2\theta)$

Spring energy = $\frac{1}{2}kx^2$; springs in series $1/k = 1/k_1 + 1/k_2$; in parallel $k = k_1 + k_2$

Rolling body KE = $\frac{1}{2}mv^2(1 + k^2/r^2)$; acceleration down incline $a = g\sin\theta/(1 + k^2/r^2)$

Impulse = $F\cdot\Delta t = \Delta(mv)$; Linear momentum $p = mv$

Coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$ = velocity of separation / velocity of approach

Loss of KE in direct impact = $m_1m_2(u_1-u_2)^2(1-e^2) / [2(m_1+m_2)]$

Strength of Materials

$E = 2G(1 + \mu)$

$E = 3K(1 - 2\mu)$

$E = 9KG/(3K + G)$; $\mu = (3K - 2G)/(6K + 2G)$

Volumetric strain (rod, axial load): $\varepsilon_v = \varepsilon(1 - 2\mu)$; (rectangular block) $\varepsilon_v = \varepsilon_x + \varepsilon_y + \varepsilon_z$

Volumetric strain of a sphere = 3 × diametral strain; of a cylinder = $2\varepsilon_d + \varepsilon_L$

Elongation $\delta = PL/(AE)$

Stepped bar: $\delta = (P/E)\,\Sigma(L_i/A_i)$

Tapered circular bar (d_1 to d_2): $\delta = 4PL/(\pi E d_1 d_2)$

Tapered flat bar (width $b_1 \to b_2$, thickness t): $\delta = PL \ln(b_1/b_2) / [E t (b_1 - b_2)]$

Bar under self-weight (γ = weight density): $\delta = \gamma L^2/(2E) = WL/(2AE)$ (half of equal end load)

Cone under self-weight: $\delta = \gamma L^2/(6E)$

Bar of uniform strength: $A = A_0 e^{(\gamma x/\sigma)}$

Free expansion $\delta = \alpha L \Delta T$; if fully restrained, $\sigma = E\alpha\Delta T$ (independent of length and area)

Partial yielding of support by Δ: $\sigma = E(\alpha L \Delta T - \Delta)/L$

Composite bar heated ($\alpha_1 > \alpha_2$): material 1 in compression, material 2 in tension; $\sigma_1 A_1 = \sigma_2 A_2$

Strain energy $U = P^2L/(2AE) = \sigma^2/(2E) \times$ Volume

Modulus of resilience = $\sigma_e^2/(2E)$ (strain energy per unit volume at elastic limit)

Shear strain energy = $\tau^2/(2G) \times$ Volume; torsion $U = T^2L/(2GJ)$; bending $U = \int M^2 dx/(2EI)$

Sudden load: $\sigma = 2P/A$ (twice the gradual stress)

Impact (falling from h): $\sigma = (P/A)[1 + \sqrt{(1 + 2AEh/(PL))}]$

Slope at free end of cantilever: W at end $\theta = WL^2/(2EI)$; UDL $\theta = wL^3/(6EI)$

Slope at support of SS beam: point at centre $\theta = WL^2/(16EI)$; UDL $\theta = wL^3/(24EI)$

Bending equation: $M/I = \sigma/y = E/R$

Section modulus $Z = I/y_{max}$; $\sigma_{max} = M/Z$

$\tau = V \cdot A \cdot \bar{y}/(I \cdot b)$ ($A\bar{y}$ = first moment of area above the level)

$T/J = \tau/r = G\theta/L$

Solid shaft: $J = \pi d^4/32$; $T = (\pi/16) \tau d^3$; $Z_p = \pi d^3/16$

Hollow shaft: $J = \pi(D^4 - d^4)/32$; $T = (\pi/16) \tau (D^4 - d^4)/D$

Power: $P = 2\pi NT/60$ (W, with N in rpm and T in N·m)

Torsional stiffness = $T/\theta = GJ/L$

Combined bending and torsion: equivalent torque $Te = \sqrt{(M^2 + T^2)}$; equivalent BM $Me = \frac{1}{2}[M + \sqrt{(M^2 + T^2)}]$

Close-coiled helical spring: $\tau = 8WD/(\pi d^3) \times K_w$; deflection $\delta = 8WD^3n/(Gd^4)$; stiffness $k = Gd^4/(8D^3n)$

Wahl factor $K_w = (4C-1)/(4C-4) + 0.615/C$, spring index $C = D/d$

$\sigma_{1,2} = (\sigma x + \sigma y)/2 \pm \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]}$

$\tau_{max} = \sqrt{[((\sigma x - \sigma y)/2)^2 + \tau xy^2]} = (\sigma_1 - \sigma_2)/2$

Principal plane angle: $\tan 2\theta p = 2\tau xy/(\sigma x - \sigma y)$

Normal stress on oblique plane: $\sigma n = (\sigma x + \sigma y)/2 + (\sigma x - \sigma y)/2 \cos 2\theta + \tau xy \sin 2\theta$

Euler's critical load: $P_{cr} = \pi^2 E I_{min} / Le^2$

Slenderness ratio $\lambda = Le/k_{min}$

Rankine–Gordon: $1/P = 1/Pc + 1/Pe \to P = \sigma c A / (1 + a \lambda^2)$; $a = \sigma c/(\pi^2 E)$

Thin cylinder: hoop stress $\sigma h = pd/(2t)$; longitudinal $\sigma l = pd/(4t)$ ($\sigma h = 2\sigma l$)

Thin sphere: $\sigma = pd/(4t)$

Hoop strain $\varepsilon h = (pd/4tE)(2 - \mu)$; longitudinal strain $\varepsilon l = (pd/4tE)(1 - 2\mu)$

Volumetric strain (cylinder) = $(pd/4tE)(5 - 4\mu)$; sphere = $(3pd/4tE)(1 - \mu)$

Max in-plane shear (cylinder) = $(\sigma h - \sigma l)/2 = pd/(8t)$

Leaf spring (semi-elliptic, n leaves): $\sigma = 3WL/(2nbt^2)$; $\delta = 3WL^3/(8Enbt^3)$

Carriage spring with full-length leaves is a beam of uniform strength.

Theory of Machines

Kutzbach (planar): $F = 3(n - 1) - 2j - h$ (j = lower pairs, h = higher pairs)

Grübler (F = 1, only lower pairs): $3n - 2j - 4 = 0$

Kinematic chain relation: $l = 2p - 4$; $j = (3/2)l - 2$

Quick-return ratio = time of cutting / time of return = $(360° - \alpha)/\alpha$ (α = return-stroke crank angle)

Crank & slotted lever: stroke length = 2 × slotted lever length × (crank radius / distance between centres)

Velocity of a point = ω × distance from I-centre

Acceleration components: centripetal (radial) $a_r = \omega^2 r = v^2/r$; tangential $a_t = \alpha r$

Coriolis component = $2\omega v$ (v = sliding velocity of slider on rotating link); direction: rotate v by 90° in the sense of ω

Piston velocity $v_p \approx \omega r(\sin\theta + \sin 2\theta/(2n))$

Piston acceleration $a_p \approx \omega^2 r(\cos\theta + \cos 2\theta/n)$

Angular velocity of connecting rod $\omega_c = \omega \cos\theta/n$

Velocity ratio $N_2/N_1 = d_1/d_2$ (with thickness t: $(d_1+t)/(d_2+t)$; with slip s%: × $(1 - s/100)$)

Ratio of tensions (flat belt): $T_1/T_2 = e^{(\mu\theta)}$

V-belt / rope (groove angle 2β): $T_1/T_2 = e^{(\mu\theta \operatorname{cosec} \beta)}$

Power $P = (T_1 - T_2)v$

Centrifugal tension $T_c = m v^2$ (m = mass per metre)

Max power condition: $T_c = T_{max}/3$, $v_{opt} = \sqrt{(T_{max}/(3m))}$

Initial tension $T_0 = (T_1 + T_2)/2$ (neglecting T_c); with T_c: $(T_1 + T_2 + 2T_c)/2$

Open belt length $L = \pi(r_1+r_2) + 2x + (r_1-r_2)^2/x$; Crossed belt $L = \pi(r_1+r_2) + 2x + (r_1+r_2)^2/x$

Module $m = D/T$ (mm); circular pitch $p_c = \pi m$; diametral pitch $P_d = T/D = 1/m$ (per mm or per inch)

Addendum = 1m; dedendum = 1.25m (20° full depth); clearance = 0.25m; whole depth = 2.25m

Velocity ratio $= \omega_1/\omega_2 = T_2/T_1 = D_2/D_1$

Path of contact $= \sqrt{(Ra^2 - R^2\cos^2\varphi)} + \sqrt{(ra^2 - r^2\cos^2\varphi)} - (R + r)\sin\varphi$

Arc of contact = path of contact / $\cos\varphi$; contact ratio = arc of contact / circular pitch

Min teeth on pinion to avoid interference (rack): $T_{min} = 2a_w / \sin^2\varphi \rightarrow 32$ for 14.5°, 18 for 20° (≈17)

Epicyclic tabular method: arm fixed → give +1 rotation to sun; add y to all, x to sun.

For sun–planet–ring: $T_{ring} = T_{sun} + 2T_{planet}$

Torque balance: $T_{in} + T_{out} + T_{fixed} = 0$

Coefficient of fluctuation of speed $C_s = (\omega_1 - \omega_2)/\omega_{mean}$

Max fluctuation of energy $\Delta E = I \omega^2 C_s = m k^2 \omega^2 C_s = 2E \cdot C_s$

Coefficient of fluctuation of energy $C_e = \Delta E$ / (work done per cycle)

Rim stress in flywheel $\sigma = \rho v^2$ (v = peripheral speed)

Natural frequency (undamped, longitudinal): $\omega_n = \sqrt{(k/m)}$; $f_n = (1/2\pi)\sqrt{(g/\delta)} = 0.4985/\sqrt{\delta}$ (δ in m)

Simple pendulum $T = 2\pi\sqrt{(L/g)}$; compound pendulum $T = 2\pi\sqrt{((k_G^2 + h^2)/(gh))}$

Damping ratio $\zeta = c/c_c$; critical damping $c_c = 2\sqrt{(km)} = 2m\omega_n$

Damped frequency $\omega_d = \omega_n \sqrt{(1 - \zeta^2)}$

Logarithmic decrement $\delta = \ln(x_1/x_2) = 2\pi\zeta/\sqrt{(1 - \zeta^2)}$

Transmissibility: isolation only when $\omega/\omega_n > \sqrt{2}$

Whirling (critical) speed of shaft = natural transverse frequency; $N_c = 60/(2\pi)\cdot\sqrt{(g/\delta)}$

Torsional vibration: $\omega_n = \sqrt{(q/I)}$, $q = GJ/L$

Two-rotor system: node located such that $l_A \cdot I_A = l_B \cdot I_B$

Machine Design

Se = ka·kb·kc·kd·ke·Se' (Marin factors: surface, size, reliability, temperature, misc.)

σ_mean = (σmax + σmin)/2; σ_amp = (σmax − σmin)/2; stress ratio R = σmin/σmax

Tearing resistance Pt = (p − d) t σt

Shearing resistance Ps = n (π/4) d² τ (×2 for double shear; practice uses 1.875 for double shear in boiler code)

Crushing resistance Pc = n d t σc

Efficiency η = least of (Pt, Ps, Pc) / (p t σt)

Unwin's formula: d = 6√t (mm)

Margin m = 1.5d; pitch p ≥ 2d (min)

Transverse fillet (single): P = 0.707 s l σt; double transverse: P = 1.414 s l σt

Parallel fillet (shear): P = 1.414 s l τ (double)

Eccentric weld: primary shear τ_1 = P/A, secondary τ_2 = P·e·r/J; resultant by vector addition

Stress area: A = (π/4)·[(dp + dc)/2]² ≈ (π/4)dc²

Initial tightening load (Rotscher): Pi = 2840·d (N, d in mm)

Torque to tighten: T = 0.2 P d

Bolt of uniform strength – shank dia reduced to core dia (or axial hole) to increase shock-absorbing capacity

Key in shear: T = l w τ (d/2); in crushing: T = l (t/2) σc (d/2)

Square key: equal strength in shear and crushing if σc = 2τ; length for equal strength to shaft ≈ 1.571d (≈1.5d)

ASME code: d³ = (16/πτ) √[(Km M)² + (Kt T)²]

Max shear stress theory: τ_max = (16/πd³) √(M² + T²)

Max normal stress theory: σ_max = (16/πd³)[M + √(M² + T²)]

Sommerfeld number S = (r/c)² (μN/p) (dimensionless; N in rps)

Bearing characteristic number = μN/p; Petroff's equation: f = 2π²(μN/p)(r/c)

Rolling bearing life L_{10} = (C/P)^k million rev; k = 3 (ball), 10/3 (roller)

Equivalent load P = X V Fr + Y Fa

Engineering Thermodynamics

T(K) = t(°C) + 273.15; t(°F) = 1.8 t(°C) + 32; −40 °C = −40 °F

1 bar = 10^5 Pa = 100 kPa; 1 atm = 1.01325 bar = 760 mm Hg = 10.33 m of water

Absolute pressure = gauge + atmospheric; vacuum = atmospheric − absolute

Displacement work W = ∫p dV (only for quasi-static process in closed system)

Enthalpy H = U + pV; dh = cp dT, du = cv dT (ideal gas)

cp − cv = R; γ = cp/cv; cv = R/(γ − 1); cp = γR/(γ − 1)

Air: R = 0.287 kJ/kg·K, cp = 1.005, cv = 0.718, γ = 1.4; universal $\bar{R}$ = 8.314 kJ/kmol·K

$h_1 + V_1^2/2 + g\,z_1 + q = h_2 + V_2^2/2 + g\,z_2 + w$ (per kg)

Heat engine η = W/Q_1 = 1 − Q_2/Q_1; Carnot η = 1 − T_2/T_1

Refrigerator COP_R = Q_2/W = $T_2/(T_1 - T_2)$; Heat pump COP_{HP} = Q_1/W = $T_1/(T_1 - T_2)$

COP_{HP} = COP_R + 1

dS = δQ_rev/T; Clausius inequality ∮δQ/T ≤ 0 (= 0 reversible, < 0 irreversible)

ΔS = m cv ln(T_2/T_1) + m R ln(V_2/V_1) = m cp ln(T_2/T_1) − m R ln(p_2/p_1)

Isothermal ΔS = mR ln(V_2/V_1); Isochoric ΔS = m cv ln(T_2/T_1); Isobaric ΔS = m cp ln(T_2/T_1)

Polytropic: $\Delta S = m\, c_n \ln(T_2/T_1)$, where polytropic specific heat $c_n = c_v (n - \gamma)/(n - 1)$

Mixing / heat transfer $\Delta S_{universe} = Q(1/T_{cold} - 1/T_{hot}) > 0$

Dryness fraction $x = m_{vapour}/(m_{vapour} + m_{liquid})$

$h = h_f + x\, h_{fg}$; $v = v_f + x\, v_{fg} \approx x\, v_g$; $s = s_f + x\, s_{fg}$

Superheated: $h = h_g + c_{ps}(T_{sup} - T_{sat})$; degree of superheat $= T_{sup} - T_{sat}$

$pV = mRT = nR^{-}T$; $R = R^{-}/M$

Van der Waals: $(p + a/v^2)(v - b) = R^{-}T$ (a – intermolecular attraction, b – finite molecular volume)

Compressibility factor $Z = pv/RT$; $Z = 1$ for ideal gas; $Z_c \approx 0.27$ (most gases), vdW predicts 0.375

Air-Standard Cycles & IC Engines

Mean effective pressure (Otto): $MEP = p_1\, r\, [(\alpha - 1)(r^{(\gamma-1)} - 1)] / [(\gamma - 1)(r - 1)]$

MEP = net work / swept volume

Brayton: optimum r_p for max work $= (T_{max}/T_{min})^{(\gamma/(2(\gamma-1)))}$; then $T_2 = T_4 = \sqrt{(T_1 T_3)}$

Work ratio (Brayton) $= 1 - (T_1/T_3)\cdot r_p^{((\gamma-1)/\gamma)}$

Indicated power $IP = p_m\, L\, A\, n\, k / 60$ ($n = N/2$ for 4-stroke, N for 2-stroke; k = cylinders)

Brake power $BP = 2\pi NT/60$

Friction power $FP = IP - BP$; mechanical efficiency $\eta_m = BP/IP$

Brake thermal efficiency $\eta_{bth} = BP/(\dot{m}_f \times CV)$; indicated thermal $\eta_{ith} = IP/(\dot{m}_f \times CV)$

Volumetric efficiency η_v = actual volume of air inducted / swept volume (at intake conditions)

bsfc $= \dot{m}_f / BP$ (kg/kWh)

Swept volume $V_s = (\pi/4)D^2L$; clearance volume V_c; $r = (V_s + V_c)/V_c$

Mean piston speed $= 2LN/60$

Automobile Engineering Essentials

Total resistance $R = \mu_r W + \tfrac{1}{2}\rho C_d A V^2 + W \sin\theta$

Power required $= R \times V$; tractive effort $= T_{engine} \times G \times \eta_t / r_{wheel}$

Refrigeration & Air Conditioning

1 TR (ton of refrigeration) = 3.5167 kW ≈ 211 kJ/min = 12,660 kJ/h (≈ 50 kcal/min)

COP_{ref} = Refrigerating effect / Work input; Carnot $COP = T_L/(T_H - T_L)$

Relative COP = actual COP / theoretical COP

Bell–Coleman $COP = 1/(r_p^{((\gamma-1)/\gamma)} - 1)$

Refrigerating effect $RE = h_1 - h_4$ ($h_4 = h_3$); Work $W = h_2 - h_1$; $COP = (h_1 - h_4)/(h_2 - h_1)$

Heat rejected in condenser $= h_2 - h_3 = RE + W$

Refrigerant mass flow $\dot{m} = 211 \times TR / RE$ (kg/min, RE in kJ/kg)

Max COP of VAR $= [(T_g - T_o)/T_g] \times [T_e/(T_o - T_e)]$ (heat engine × refrigerator)

$\omega = 0.622\, pv/(p - pv)$ kg/kg of dry air

$\varphi = pv/ps$; $\mu = \varphi(p - ps)/(p - pv)$

Enthalpy of moist air $h = 1.005\, t + \omega(2500 + 1.88\, t)$ kJ/kg dry air

Sensible heat factor $SHF = SH/(SH + LH)$; bypass factor $BPF = (t_2 - t_{ADP})/(t_1 - t_{ADP})$; contact factor $= 1 - BPF$

Steam Engineering, Power Plants & Compressors

$\eta_{Rankine} = [(h_3 - h_4) - (h_2 - h_1)] / (h_3 - h_2)$

Pump work $w_p = v_f (p_2 - p_1)$ (small; often neglected)

Specific steam consumption SSC = 3600 / w_{net} (kg/kWh, w_{net} in kJ/kg)

Work ratio = $w_{net} / w_{turbine}$ (close to 1 for Rankine – an advantage over gas cycles)

Chimney draught $h_w = 353\ H\ (1/T_a - ((m_a+1)/m_a)(1/T_g))$ mm of water (H in m)

Equivalent evaporation = $m_s(h - h_f)/2257$ (from and at 100 °C)

Boiler efficiency = $m_s(h - h_{fw}) / (m_f \times CV)$

Exit velocity $V = 44.72 \sqrt{(h_1 - h_2)}$ (Δh in kJ/kg)

Critical pressure ratio $p_c/p_1 = (2/(n+1))^{(n/(n-1))}$

Blade (diagram) efficiency – impulse: $\eta_b = 2(V_b/V_1)(\cos\alpha - V_b/V_1)(1 + k \cdot \cos\beta_2/\cos\beta_1)$

Optimum blade speed ratio – single-stage impulse: $V_b/V_1 = \cos\alpha/2$; max $\eta = \cos^2\alpha$

Parsons (50% reaction): $V_b/V_1 = \cos\alpha$; max $\eta = 2\cos^2\alpha/(1 + \cos^2\alpha)$

Degree of reaction R = $\Delta h_{moving} / \Delta h_{stage}$

Work done per kg = $V_b (V_{w1} \pm V_{w2})$ (+ when the whirl components are in opposite directions)

Vacuum efficiency = actual vacuum / ideal vacuum (barometric pressure – partial pressure of steam)

Condenser efficiency = $(t_{w,out} - t_{w,in}) / (t_{sat} - t_{w,in})$

Work (single stage, polytropic, no clearance) $W = (n/(n-1))\ p_1V_1\ [(p_2/p_1)^{((n-1)/n)} - 1]$

Isothermal work $W_{iso} = p_1V_1 \ln(p_2/p_1)$ (minimum work)

Isothermal efficiency = isothermal work / actual work

Volumetric efficiency $\eta_v = 1 + C - C\ (p_2/p_1)^{(1/n)}$ (C = clearance ratio V_c/V_s)

Multistage with perfect intercooling: optimum intermediate pressure $p_2 = \sqrt{(p_1p_3)}$ (2-stage); pressure ratio per stage = $(p_{final}/p_1)^{(1/stages)}$; work equal in each stage

Thrust $F = \dot{m}\ (V_j - V_a)$; propulsive efficiency $\eta_p = 2V_a/(V_a + V_j)$

Heat Transfer

$\sigma = 5.67 \times 10^{-8}$ W/m²·K⁴

Plane wall: $Q = kA(T_1 - T_2)/L$; thermal resistance $R = L/(kA)$

Convection resistance $R = 1/(hA)$; radiation equivalent $h_r = \varepsilon\sigma(T_1^2 + T_2^2)(T_1 + T_2)$

Composite wall (series): $Q = \Delta T_{overall} / \Sigma R$

Cylinder: $R = \ln(r_2/r_1)/(2\pi kL)$; Sphere: $R = (r_2 - r_1)/(4\pi k\ r_1\ r_2)$

Overall heat transfer coefficient: $1/U = 1/h_1 + \Sigma(L/k) + 1/h_2$

Critical radius of insulation: cylinder $r_c = k/h$; sphere $r_c = 2k/h$

$m = \sqrt{(hP/(kA_c))}$

Infinitely long fin: $Q = \sqrt{(hPkA_c)}\ \theta_0$

Insulated tip: $Q = \sqrt{(hPkA_c)}\ \theta_0 \tanh(mL)$

Fin efficiency η_f = actual heat transfer / heat transfer if entire fin were at base temperature; long fin $\eta = 1/(mL)$

Fin effectiveness ε = $Q_{with\ fin} / Q_{without\ fin} = \sqrt{(kP/(hA_c))}$; use fins only if $\varepsilon > 2$

Biot number $Bi = hL_c/k_{solid}$ ($L_c = V/A_s$); lumped analysis valid if $Bi < 0.1$

Lumped: $(T - T_\infty)/(T_0 - T_\infty) = \exp(-hA_s\ t/(\rho Vc)) = \exp(-Bi \cdot Fo)$

Fourier number $Fo = \alpha t/L_c^2$ (dimensionless time)

Time constant $\tau = \rho Vc/(hA_s)$

Laminar flow in a tube, fully developed: Nu = 3.66 (constant wall temperature), 4.36 (constant heat flux)

Dittus–Boelter (turbulent tube flow): Nu = 0.023 Re^0.8 Pr^n; n = 0.4 heating, 0.3 cooling

Flat plate laminar: Nu_x = 0.332 Re_x^0.5 Pr^(1/3); δ = 5x/√Re_x; average Nu = 0.664 Re^0.5 Pr^(1/3)

Critical Re: flat plate 5 × 10^5; pipe 2300

α + ρ + τ = 1

Kirchhoff's law: at thermal equilibrium α = ε

Wien's displacement law: λ_max T = 2898 μm·K

Planck's law gives spectral distribution; total emissive power $E_b = \sigma T^4$

Two infinite parallel plates: $q = \sigma(T_1^4 - T_2^4)/(1/\varepsilon_1 + 1/\varepsilon_2 - 1)$

Small body in large enclosure: $q = \varepsilon_1 \sigma A_1 (T_1^4 - T_2^4)$

n radiation shields (all same ε) between plates: heat transfer reduced to 1/(n + 1)

$Q = U A \Delta T_{LMTD}$; $LMTD = (\Delta T_1 - \Delta T_2)/\ln(\Delta T_1/\Delta T_2)$

Effectiveness ε = Q_actual/Q_max; Q_max = C_min (T_h,in − T_c,in)

NTU = UA/C_min; capacity ratio C = C_min/C_max

Parallel flow: ε = [1 − e^(−NTU(1+C))]/(1 + C); Counter flow: ε = [1 − e^(−NTU(1−C))]/[1 − C e^(−NTU(1−C))]

Fluid Mechanics

Density ρ (kg/m³; water 1000); specific weight w = ρg (water 9810 N/m³); specific gravity S = ρ/ρ_water; specific volume = 1/ρ

Newton's law of viscosity: τ = μ (du/dy)

Kinematic viscosity ν = μ/ρ (m²/s; stoke = 1 cm²/s = 10^{-4} m²/s)

Dynamic viscosity unit: N·s/m² = Pa·s = 10 poise; 1 poise = 0.1 Pa·s; water at 20 °C ≈ 1 cP = 10^{-3} Pa·s

Surface tension (N/m): pressure inside droplet $\Delta p = 4\sigma/d$; soap bubble $\Delta p = 8\sigma/d$ (two surfaces); liquid jet $\Delta p = 2\sigma/d$

Capillary rise $h = 4\sigma \cos\theta/(\rho g d)$ (water–glass θ ≈ 0 → rise; mercury–glass θ ≈ 130–140° → depression)

1 atm = 101.325 kPa = 760 mm Hg = 10.33 m of water

Total pressure on plane surface $F = \rho g A \bar{h}$ ($\bar{h}$ = depth of centroid)

Centre of pressure (vertical/inclined): $h^* = \bar{h} + I_G \sin^2\theta / (A \bar{h})$

Metacentric height GM = I/V − BG (I = least second moment of the waterplane area, V = displaced volume)

Experimentally: GM = w x / (W tanθ)

Time period of rolling: $T = 2\pi \sqrt{(k^2/(g \cdot GM))}$

Continuity (incompressible): $A_1V_1 = A_2V_2$; 3D: $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$

Acceleration: $a = \partial V/\partial t$ (local) + $V \cdot \nabla V$ (convective)

Rotation $\omega z = \frac{1}{2}(\partial v/\partial x - \partial u/\partial y)$; vorticity = 2ω; irrotational if ωz = 0

Velocity potential φ: $u = -\partial\phi/\partial x$, $v = -\partial\phi/\partial y$ (exists only for irrotational flow); stream function ψ: $u = -\partial\psi/\partial y$, $v = \partial\psi/\partial x$ (sign convention varies; exists for any 2D incompressible flow)

Bernoulli: $p/(\rho g) + V^2/(2g) + z$ = constant (pressure head + velocity head + datum head)

Momentum equation: $\Sigma F = \dot{m} (V_2 - V_1)$

Force on pipe bend: $Fx = \dot{m}(V_1 - V_2\cos\theta) + p_1A_1 - p_2A_2\cos\theta$

Time to empty a tank through orifice (from H_1 to H_2): $T = 2A(\sqrt{H_1} - \sqrt{H_2})/(Cd\ a \sqrt{(2g)})$

Hagen–Poiseuille (pipe): $\Delta p = 32\mu VL/D^2$; head loss $h_f = 32\mu VL/(\rho g D^2)$

Velocity profile parabolic: $u = u_{max} (1 - r^2/R^2)$; $V_{avg} = u_{max}/2$

Shear stress varies linearly: $\tau = (-dp/dx)\, r/2$ (zero at centre, max at wall)

Friction factor $f = 64/Re$ (Darcy); $f' = 16/Re$ (Fanning)

Flow between fixed parallel plates: $V_{avg} = (2/3)\, u_{max}$; $\Delta p = 12\mu VL/t^2$

Couette flow (one plate moving): linear velocity profile

Stokes' law terminal velocity: $V = D^2(\rho s - \rho)g/(18\mu)$

Darcy–Weisbach: $h_f = f\, L\, V^2/(2gD) = f\, L\, Q^2/(12.1\, D^5)$

Blasius (smooth pipe, $Re < 10^5$): $f = 0.316/Re^{0.25}$

Chezy: $V = C\sqrt{(m\, i)}$; hydraulic mean depth $m = A/P$ (= D/4 for full pipe)

Power transmitted through pipe $P = \rho g Q(H - h_f)$; maximum when $h_f = H/3$ ($\eta_{max} = 66.67\%$)

Nozzle diameter for max power at end of pipe: $d = (D^5/(8fL))^{1/4}$

Water hammer pressure rise (sudden closure, rigid pipe): $\Delta p = \rho VC$; $C = \sqrt{(K/\rho)}$; critical time $2L/C$

Laminar (Blasius): $\delta = 5x/\sqrt{Re_x}$; $\delta^* = 1.72x/\sqrt{Re_x}$; $\theta = 0.664x/\sqrt{Re_x}$; local $Cf = 0.664/\sqrt{Re_x}$

Turbulent: $\delta = 0.37x/Re_x^{1/5}$

Linear velocity profile: $\delta^*/\delta = 1/2$, $\theta/\delta = 1/6$

Drag $F_D = C_D\, (\frac{1}{2}\rho V^2)\, A$; Lift $F_L = C_L\, (\frac{1}{2}\rho V^2)\, A$

Speed of sound $C = \sqrt{(\gamma RT)} = \sqrt{(K/\rho)}$; air at 20 °C ≈ 343 m/s; Mach angle $\sin\alpha = 1/M$

Stagnation temperature $T_0 = T(1 + (\gamma-1)M^2/2)$

Chezy $V = C\sqrt{(mi)}$; Manning $V = (1/n)\, R^{2/3}\, S^{1/2}$

Specific energy $E = y + V^2/(2g)$; critical depth (rectangular) $y_c = (q^2/g)^{1/3}$; $E_{min} = 1.5\, y_c$

Froude number $Fr = V/\sqrt{(gy)}$: $Fr < 1$ subcritical, $= 1$ critical, > 1 supercritical

Hydraulic Machines

Force of jet on fixed vertical flat plate: $F = \rho a V^2$

Moving vertical plate (single): $F = \rho a(V - u)^2$; series of plates (wheel): $F = \rho a V(V - u)$

Fixed curved vane, jet deflected by angle: $F = \rho a V^2(1 + \cos\theta)$ (θ measured from opposite direction; for semicircular vane $F = 2\rho a V^2$)

Efficiency of series of flat plates: $\eta = 2u(V - u)/V^2$; max $\eta = 50\%$ at $u = V/2$

Series of curved vanes (Pelton bucket, 180°): max $\eta = 100\%$ at $u = V/2$ (theoretically); with deflection φ: $\eta_{max} = (1 + \cos\varphi)/2$

Single moving plate: max efficiency at $u = V/3$, $\eta_{max} = 8/27$ (29.6%)

Specific speed of turbine $N_s = N\sqrt{P}\, /\, H^{5/4}$

Specific speed of pump $N_s = N\sqrt{Q}\, /\, H^{3/4}$

Unit quantities: $N_u = N/\sqrt{H}$; $Q_u = Q/\sqrt{H}$; $P_u = P/H^{3/2}$

Jet velocity $V = C_v\, \sqrt{(2gH)}$ ($C_v \approx 0.98$)

Bucket speed $u = \varphi\sqrt{(2gH)}$, speed ratio $\varphi \approx 0.43–0.47$

Jet ratio $m = D/d$ (≈ 12 for good design, min 6); number of buckets $z = 15 + m/2$ (Taygun)

Work per kg $= (V - u)(1 + k\cos\varphi)\, u$ (k = bucket friction factor)

Draft tube efficiency $= (V_2^2 - V_3^2 - 2g \cdot h_f)/V_2^2$

Work per kg (radial entry $V_{w1} = 0$): $W = V_{w2}\, u_2\, /\, g$ (head)

Manometric efficiency $\eta_{mano} = g\, Hm\, /\, (V_{w2}\, u_2)$

Minimum starting speed: $N = 120\, \eta_{mano}\, V_{w2}\, D_2\, /\, (\pi\, (D_2^2 - D_1^2))$ (from $u_2^2/2g - u_1^2/2g \geq Hm$)

Static head $Hs = hs + hd$; manometric head $Hm = Hs + h_{fs} + h_{fd} + Vd^2/2g$

NPSH $= (p_{atm}/\rho g) - hs - h_{fs} - (p_v/\rho g)$ (must exceed NPSH required to avoid cavitation)

Theoretical discharge: single-acting Q = ALN/60; double-acting Q = 2ALN/60

Slip = Q_th − Q_act; % slip = (1 − Cd) × 100. Negative slip occurs with long suction pipe, short delivery, high speed (Q_act > Q_th)

Power = ρg Q (hs + hd)/1000 kW

Pressure head due to acceleration: ha = (L/g)(A/a)ω²r cosθ

Manufacturing I – Casting, Welding, Forming

Permeability number = V H / (p A t) (V = 2000 cm³ air, H height of specimen, p pressure in g/cm², A area, t time in min)

Sprue taper (continuity): $A_1/A_2 = \sqrt{(h_2/h_1)}$

Gating ratio sprue : runner : ingate = 1:2:2 (unpressurised, Al) or 1:0.75:0.5 (pressurised, ferrous)

Pouring time (top gate) t = A_m h_m / (A_g √(2g h_t)); bottom gate t = 2A_m(√h_t − √(h_t − h_m))/(A_g√(2g))

Chvorinov's rule: solidification time t_s = B (V/A)^n, n ≈ 2

Heat input H = η V I / v (J/mm); arc length ≈ electrode diameter

Arc voltage V = a + b·l (Ayrton equation); arc power maximum when the arc characteristic and power source characteristic are tangent

Max reduction (draft) Δh_max = μ²R; bite condition μ ≥ tan α; contact length L = √(RΔh)

Neutral (no-slip) point: roll velocity = strip velocity; forward slip ahead of it

Roll separating force F ≈ L w σ_avg; torque per roll T ≈ F·L/2; power (both rolls) = 2 × T × ω

Forging force (cylinder, sticking/friction): F = σ_f π r² (1 + 2μr/(3h))

Extrusion force F = A_0 σ_f ln(A_0/A_f) × k (ideal); extrusion ratio R = A_0/A_f

Max reduction per pass in wire drawing (ideal, no friction): ln(A_0/A_f) = 1 → 63%

Blanking/punching force F = L t τ (L = perimeter, τ = shear strength)

Clearance (per side) c = 0.0032 t √τ (Indian texts) — typically 2–8% of t

Deep drawing: blank diameter D = √(d² + 4dh); limiting draw ratio LDR ≈ 2 (max D/d)

Manufacturing II – Machining & Machine Tools

Chip thickness ratio r = t/t_c = L_c/L = V_c/V (< 1); cutting ratio

Shear angle: tan φ = r cos α / (1 − r sin α)

Merchant's theory: 2φ + β − α = 90° (β = friction angle)

Shear strain γ = cot φ + tan(φ − α); velocity of shear Vs = V cos α / cos(φ − α); chip velocity Vc = V sin φ / cos(φ − α)

Friction coefficient μ = (Fc tan α + Ft)/(Fc − Ft tan α)

Cutting power P = Fc·V; specific cutting energy u = Fc/(b·t)

Taylor's tool life equation: $V T^n = C$

Extended Taylor: $V T^n$ f^a d^b = C

Optimum tool life for min cost: T_o = (1/n − 1)(C_t/C_m + T_c); for max production rate: T_o = (1/n − 1) T_c

Optimum speed for minimum cost < optimum speed for maximum production (V_mc < V_mp)

Cutting speed V = πDN/1000 (m/min)

Machining time t = L/(fN) per pass

Material removal rate MRR = π D_avg d f N ≈ V f d

Taper turning: tan(α/2) = (D − d)/(2L); tailstock set-over = L_total (D − d)/(2L)

Thread cutting: change gear ratio = pitch of job / pitch of lead screw = driver/driven

Surface roughness (theoretical, nose radius r): $h = f^2/(8r)$

Drilling time $t = (L + 0.3D)/(fN)$; $MRR = (\pi D^2/4)\, f\, N$

Feed per minute $fm = fz \times Z \times N$

Approach (slab milling) $= \sqrt{d(D - d)}$; machining time $t = (L + \text{approach})/fm$

Max chip thickness (up milling) $\approx 2fz\sqrt{(d/D)}$

Shaper cutting time per stroke: $L(1 + m)/(1000V)$ where m = return/cutting time ratio

Specific energy in grinding is very high (due to size effect & ploughing, negative rake grits)

Engineering Materials & Heat Treatment

BCC: $a = 4r/\sqrt{3}$; FCC: $a = 2\sqrt{2}\ r = 4r/\sqrt{2}$

Metrology, Limits & Fits, Industrial Engineering

Vernier least count = 1 MSD − 1 VSD = value of smallest main-scale division / number of vernier divisions

Micrometer least count = pitch / number of circular scale divisions (0.5/50 = 0.01 mm)

Standard tolerance unit $i = 0.45\sqrt[3]{D} + 0.001\,D$ (μm, D in mm = geometric mean of diameter step)

IT7 = 16i, IT8 = 25i, IT9 = 40i, IT10 = 64i, IT11 = 100i (R5 series from IT6 = 10i)

EOQ (Wilson): $Q^* = \sqrt{(2DC_o/C_h)}$

At EOQ: ordering cost = holding cost; total cost $TC = \sqrt{(2DC_oC_h)}$; number of orders $N = D/Q^*$; cycle time $= Q^*/D$

EPQ (production at rate p, demand rate d): $Q^* = \sqrt{(2DC_o/(C_h(1 - d/p)))}$

Reorder point ROP = demand during lead time + safety stock

Normal time = observed time × rating factor (%/100)

Standard time = normal time × (1 + allowance fraction) (allowances: personal, fatigue, delay, contingency)

Work sampling (95% confidence): $N = 4p(1 - p)/L^2$ for absolute accuracy L; $N = 4(1 - p)/(S^2p)$ for relative accuracy S

Control limits: mean ± 3σ (99.73% of points within); $\bar{X}$ chart: $UCL = \bar{\bar{X}} + A_2\bar{R}$; R chart: $UCL = D_4\bar{R}$, $LCL = D_3\bar{R}$

p chart: $\bar{p} \pm 3\sqrt{(\bar{p}(1 - \bar{p})/n)}$; c chart: $\bar{c} \pm 3\sqrt{\bar{c}}$

Process capability $Cp = (USL - LSL)/(6\sigma)$; $Cpk = \min[(USL - \mu)/3\sigma, (\mu - LSL)/3\sigma]$

$Ls = \lambda/(\mu - \lambda)$; $Lq = \lambda^2/(\mu(\mu - \lambda))$; $Ws = 1/(\mu - \lambda)$; $Wq = \lambda/(\mu(\mu - \lambda))$; Little's law $L = \lambda W$

PERT expected time $te = (to + 4tm + tp)/6$; variance $\sigma^2 = ((tp - to)/6)^2$ (β-distribution)

Probability of completion: $Z = (T_s - T_e)/\sigma_{cp}$ (normal distribution along critical path)

Total float = LS − ES = LF − EF; free float = ES(succ) − EF; independent float = ES(succ) − LF(pred) − duration

Mechanical Measurements & Instrumentation

$\Delta R/R = GF \times \varepsilon$; bridge output (quarter bridge) $\approx (V/4) \times GF \times \varepsilon$

APPENDIX

Appendix B – PYQ-Trend Concept Map

All PYQ-Trend boxes in one place, chapter-wise. These are the concepts SSC JE has tested repeatedly; revise this list in the last week.

Engineering Mechanics

★ Transmissibility applies to rigid bodies only. Definition of a newton or a dyne and their conversion. Scalar vs vector identification (work and energy are scalar; moment is vector; pressure is scalar).

★ Two equal forces at 120° give R equal to each force. Two equal forces at 60° give R = √3 P. The resultant of two forces is maximum when they are collinear and like. "Magnitude of resultant of 3 N and 4 N at right angles = 5 N" is a typical conceptual check.

★ Lami's theorem is valid only for three concurrent forces. A couple's moment is independent of the reference point. Number of reactions at fixed, hinged and roller supports. Varignon's theorem relates to moments.

★ Angle of repose = angle of friction. Efficiency of a self-locking screw is less than 50%. Force of friction is independent of the area of contact. The least force to drag a body on a rough horizontal plane is W sin φ. Coefficient of friction depends on the nature of surfaces.

★ CG of a solid cone at h/4 from base, solid hemisphere at 3r/8, semicircle at 4r/3π, triangle at h/3. These four come almost every year.

★ I of triangle about base $bh^3/12$ vs about centroid $bh^3/36$. MOI of a square about its diagonal equals that about the centroidal axis. Polar MOI of circle $\pi d^4/32$. Mass MOI of solid sphere $2mR^2/5$.

★ Maximum range at 45°; at maximum range H = R/4. Equal range for 30° and 60°. Time to reach max height = u sinθ/g. Centripetal acceleration $\omega^2 r$ directed towards centre.

★ D'Alembert principle: inertia force. Apparent weight in a lift. Rolling-without-slipping order. Unit of power and conversion of hp. Spring constants in series/parallel.

★ Value of e for perfectly elastic and plastic impact. Equal masses with elastic head-on collision exchange velocities. Bounce height $h_1 = e^2 h$. Momentum conserved in all impacts.

★ Condition for reversibility (η > 50%) is asked almost every year. Velocity ratio of the first/second/third system of pulleys and of the differential pulley block.

Strength of Materials

★ Sequence of points on the stress–strain diagram. "Necking starts at ultimate stress". Toughness = area under the whole curve; resilience = area up to elastic limit. Lüders bands occur at yield in mild steel. Gold is the most malleable metal.

★ $E = 2G(1+\mu)$ and $E = 3K(1-2\mu)$ are asked every cycle. Number of independent elastic constants for isotropic material = 2. If μ = 0.5, volume change is zero. Relation: when E = 2G, μ = 0; when K = E, μ = 1/3.

★ Elongation due to self-weight is half that of the same load at the end. Thermal stress $E\alpha\Delta T$ does not depend on cross-section. In a steel–copper composite bar when heated, copper (higher α) is in compression. Elongation of a tapered rod $4PL/(\pi E d_1 d_2)$.

★ Suddenly applied load gives double stress. Proof resilience vs modulus of resilience (per unit volume). Strain energy stored in solid shaft in torsion = $(\tau^2/4G) \times$ volume; hollow =

$(\tau^2/4G)(D^2+d^2)/D^2 \times$ volume.

★ SF/BM shapes for UDL & UVL. Max BM $wL^2/8$ and deflection $5wL^4/384EI$ for SS UDL. Ratio of central deflection of fixed to SS beam with UDL = 1:5; with central point load = 1:4. Prop reaction 3wL/8. Point of contraflexure in overhanging beam.

★ Strongest beam from a log: $d/b = \sqrt{2}$. Z for rectangular and circular sections. Neutral axis passes through centroid. A square beam with diagonal vertical is weaker (Z less) than with a side horizontal.

★ Ratio τ_{max}/τ_{avg} = 1.5 for rectangle and 4/3 for circle – asked repeatedly. In a triangular section max shear stress is at mid-height, not at the NA.

★ Power transmitted $2\pi NT/60$. Hollow shaft stronger than solid of same weight. Equivalent torque & bending moment. Torque transmission ratio of hollow to solid shafts. Spring stiffness $\propto d^4/(D^3n)$. Cutting a spring in half doubles stiffness.

★ Angle between principal planes = 90°; between principal plane and max shear plane = 45°. Mohr's circle for pure shear has centre at origin. Brittle material in torsion fails on a 45° helicoidal surface; ductile fails on plane perpendicular to the axis.

★ Theory best for ductile = distortion energy (von Mises). Tresca hexagon, von Mises ellipse. Shear yield = 0.577 σy (von Mises).

★ Le for four end conditions (most-repeated SOM question). Ratio of buckling loads of fixed-fixed to fixed-free = 16. Middle-third rule ($e \leq b/6$). Core of a circular section d/4.

★ Hoop stress is twice the longitudinal stress. Thin cylinder criterion $d/t > 20$. For a sphere, stress pd/4t. Lamé's theorem for thick cylinders. Volumetric strain $(5-4\mu)$ expression.

★ Stiffness of springs in series/parallel; hoop stress in a rotating ring ρv^2.

Theory of Machines

★ Lower vs higher pair examples (ball bearing is a higher pair; nut & bolt is a lower pair). Number of inversions. Grübler/Kutzbach DOF calculation. Grashof condition. A cam & follower is a higher, force-closed pair.

★ Whitworth & crank-slotted lever QRM are inversions of the single slider crank chain. Elliptical trammel, Scotch yoke, Oldham coupling come from double slider chain. Peaucellier gives exact straight line. Ackermann steering condition. Hooke's joint used for intersecting shafts.

★ Number of I-centres for a 4-bar = 6, 6-link = 15. Kennedy's theorem (three centres collinear). Coriolis component $2\omega v$ appears in quick-return mechanisms. Direction of Coriolis.

★ Uniform wear vs uniform pressure torque; uniform wear used for design since it is conservative. Number of friction surfaces in a multi-plate clutch.

★ Max power transmitted when Tc = T_max/3. Centrifugal tension does not affect power. Crossed belt length depends on sum of radii. Creep definition. T_1/T_2 for V-belt is greater than flat belt (wedge action). Initial tension formula.

★ Self-energising vs self-locking brakes. Prony and rope brake are absorption type. Torsion dynamometer is transmission type.

★ Module definition and standard addendum (1 module). Minimum teeth to avoid interference (32 & 18). Contact ratio > 1. Herringbone eliminates axial thrust. Hypoid used in differential. Cycloidal teeth used in watches.

★ Purpose of idler gear. Reverted train condition. Epicyclic train applications (differential). Ring teeth relation.

★ Height of Watt governor $\propto 1/N^2$. Isochronous governor is infinitely sensitive. Hunting definition. Hartnell is spring loaded. Governor controls mean speed, flywheel controls cyclic

fluctuations – favourite comparison question.

★ $\Delta E = I\omega^2 Cs$. Function of flywheel vs governor. Four-stroke engine needs a heavier flywheel than two-stroke. Rim stress ρv^2.

★ Conditions for static vs dynamic balance. Hammer blow, swaying couple, tractive effort variation. Secondary force frequency = 2ω. Complete reciprocating balance is impossible by rotating masses.

★ Cycloidal motion has no abrupt acceleration change → used at high speeds. Uniform velocity gives infinite acceleration. Pressure angle definition. Mushroom follower used in IC engine valves.

★ $fn = 0.4985/\sqrt{\delta}$. Critical damping definition and its use in recoil mechanisms. Vibration isolation requires frequency ratio > $\sqrt{2}$. Log decrement. Whirling speed equals natural transverse frequency.

Machine Design

★ Kt = 3 for a small hole in an infinite plate. Notch sensitivity range 0–1. Cast iron is notch-insensitive. Renard series R10 ratio = 10^(1/10).

★ Soderberg is the most conservative line (joins Se and Syt). Goodman joins Se and Sut. Shot peening improves fatigue life. Endurance limit ≈ 0.5 × UTS for steels. Aluminium has no distinct endurance limit.

★ Unwin's formula $d = 6\sqrt{t}$. Efficiency definition. Caulking & fullering for leak-proofing. Margin = 1.5d. Double-shear area of rivet.

★ Throat = 0.707 × leg. Strength of fillet weld is based on throat area. Parallel fillet weld is designed for shear.

★ Square thread most efficient for power transmission; Acme used in lathe lead screw; buttress for one-way thrust. Metric thread angle 60°, Whitworth 55°. Designation M meaning. Initial tension 2840d.

★ Woodruff key is self-aligning. Feather key allows axial movement. Key length ≈ 1.5d. Coupling for lateral misalignment = Oldham; angular = universal. Width of rectangular key = d/4.

★ Axle subjected mainly to bending. ASME code combined shock factors. Equivalent torque concept.

★ Life formula exponent 3 for ball bearing (life $\propto 1/P^3$). Sommerfeld number is dimensionless. Babbitt metal composition. Taper roller for combined loads. L_{10} meaning (90% reliability).

★ Lewis equation assumes the tooth is a cantilever beam with full load at the tip. Pitting is surface fatigue. Spring index range. Surge in springs.

★ Cotter joint transmits axial load only and is used for piston rod & crosshead. Knuckle joint pin is in double shear. Taper of cotter. Self-locking condition $\varphi > \alpha$.

Engineering Thermodynamics

★ Intensive vs extensive identification. Heat & work as path functions. Zeroth law is the basis of temperature measurement. Temperature at which °C = °F (−40). Triple point of water 273.16 K. Turbine/compressor as open systems.

★ Work in free expansion is zero. Work in isothermal process. Value of n for each process. Area under p–V curve = work; under T–s curve = heat. On p–V, the adiabatic curve is steeper than the isothermal (slope γ times).

★ Throttling is isenthalpic. Nozzle velocity $44.72\sqrt{\Delta h}$. cp − cv = R. U of an ideal gas depends only on T. PMM-1 violates first law. Joule–Thomson coefficient zero for ideal gas.

★ Kelvin–Planck vs Clausius statements. Carnot efficiency depends only on temperatures. COP_HP − COP_R = 1. Intermediate temperature for equal efficiency $\sqrt{(T_1T_2)}$. Carnot cycle not practical because isothermal heat transfer is too slow and adiabatic processes need very fast motion.

★ Reversible adiabatic = isentropic. Entropy of universe always increases for real processes. Slope of isochor vs isobar on T–s. Entropy change for isothermal expansion. Third law statement. Unit of entropy kJ/K.

★ Unavailable energy = $T_0\Delta S$. Gouy–Stodola theorem. Gibbs function used for chemical reactions at constant p, T.

★ Critical pressure 221.2 bar & temperature 374.15 °C. Latent heat zero at critical point. Throttling calorimeter for dryness. Triple point DOF = 0. Mollier = h–s chart.

★ Van der Waals constants meaning. Ideal behaviour at low p, high T. Z for ideal gas = 1. Value of R for air 287 J/kg·K. Dalton's law.

★ Clausius–Clapeyron relates latent heat and saturation slope. Maxwell relations derived from Gibbs/Helmholtz functions. cp = cv for water at 4 °C.

Air-Standard Cycles & IC Engines

★ Otto efficiency formula and its dependence only on r. Order of efficiencies for same r and for same max pressure. Stirling/Ericsson efficiency equals Carnot (with regenerator). Brayton cycle for gas turbines. Optimum pressure ratio for max work. Regeneration effect.

★ SI uses quantity governing, CI uses quality governing. CR ranges for SI & CI. Valve overlap meaning. Firing order of 4-cylinder engine 1-3-4-2. Two-stroke engine needs a lighter flywheel. Uniflow scavenging most efficient.

★ Willan's line for CI engines; Morse test for multi-cylinder SI engines. IP formula with n = N/2 for four-stroke. Definitions of ηm, ηv. bsfc units. Supercharging increases power and is more suitable for CI engines (SI gets detonation).

★ Octane number reference fuels (iso-octane & n-heptane), cetane number (cetane & α-methyl naphthalene). Factors which reduce knocking in SI increase it in CI. Stoichiometric A/F 14.7. Detonation in SI occurs at the end of combustion; diesel knock occurs at the start.

★ Delay period causes diesel knock. Glow plug used in IDI engines. CRDI full form. Higher CR used to reduce delay.

★ Purpose of condenser in battery ignition (prevents arcing at contact points). Choke gives rich mixture during cold start. Petroil lubrication in two-stroke engines. Magneto ignition is better at high speed. Thermostat function. SAE 20W meaning. Cold plug for high-compression engines.

★ EGR controls NOx. Three-way catalyst metals. Main constituent of CNG = methane, LPG = propane + butane. BS-VI from 2020. Biodiesel is produced by transesterification.

★ Turbocharger is driven by exhaust gases. Supercharging is more suitable for CI engines. Roots blower is a positive displacement supercharger.

Automobile Engineering Essentials

★ Monocoque/integral construction for cars. Air resistance ∝ V^2. Components of the chassis. Ladder frame for heavy vehicles.

★ Diaphragm spring clutch in modern cars. Wet multi-plate clutch in motorcycles. Centrifugal clutch in mopeds. Torsional damper springs in the clutch plate. Purpose of pedal free play.

★ Synchromesh avoids gear clash using cone synchronisers. Constant mesh uses dog clutches. Epicyclic gears in automatic transmissions. Overdrive ratio < 1. Double declutching

needed in sliding mesh.

★ Function of the differential. Slip joint on the propeller shaft. CV joints in FWD. Hypoid gear needs EP oil. Full-floating axle shafts carry only torque. Crown wheel & pinion in the final drive.

★ Castor gives directional stability & self-righting. Camber definition. Toe-in purpose. Rack & pinion in cars. Ackermann condition. Under-steer vs over-steer. KPI reduces steering effort.

★ Disc brakes resist fade better. Hydraulic brakes work on Pascal's law. ABS prevents wheel locking and retains steering control. Air brakes for heavy vehicles. Leading shoe is self-energising. Brake fade definition.

★ MacPherson strut for front suspension. Shock absorber dissipates energy. Anti-roll bar reduces body roll. Radial tyres have lower rolling resistance. Tyre designation decoding. Leaf springs in heavy vehicles.

★ Starter motor is a DC series motor. Battery cell voltage 2 V (12 V = 6 cells). Hydrometer checks electrolyte specific gravity. Alternator needs rectifier diodes. Oxygen sensor for closed-loop A/F control. Regenerative braking in EVs.

Refrigeration & Air Conditioning

★ 1 TR = 3.5 kW (211 kJ/min). Air refrigeration used in aircraft because of low weight per TR, despite low COP. Carnot COP decreases as the evaporator temperature decreases.

★ Throttling in expansion valve is isenthalpic. Capillary tube in domestic refrigerators. Subcooling increases COP. Lowering evaporator temperature reduces COP. TEV maintains constant superheat. Centrifugal compressors for large capacity plants. COP from p–h chart values.

★ Refrigerant numbering (R-717 = NH_3, R-744 = CO_2, R-718 = water). Ammonia is used in cold storage, not with copper. R-134a has zero ODP. Halide torch for Freon leak. Montreal protocol relates to ozone.

★ Electrolux refrigerator uses hydrogen as a third fluid and has no moving parts. In LiBr–water system water is the refrigerant. Ideal COP of VAR formula. Generator replaces compressor's high-pressure side function.

★ Peltier effect in thermoelectric refrigeration. Vortex tube has no moving parts. Dry ice temperature −78.5 °C.

★ Order DBT > WBT > DPT; all equal at saturation. Specific humidity formula 0.622 pv/(p − pv). Sensible heating = horizontal line. Evaporative cooling follows constant WBT. Dew point definition. Comfort conditions. Bypass factor definition. Chemical dehumidification follows constant enthalpy.

Steam Engineering, Power Plants & Compressors

★ Pump work is small due to low specific volume of water. Effect of superheat, reheat, regeneration on efficiency and dryness. Reheating mainly increases dryness of exhaust steam. Regeneration increases efficiency. SSC definition. Combined cycle uses HRSG.

★ Fire-tube vs water-tube examples (B&W is water-tube; Cochran/Lancashire/Locomotive are fire-tube). Benson is supercritical and drumless. Fusible plug function. Mountings vs accessories (economiser is an accessory; safety valve is a mounting). Economiser heats feedwater; air preheater heats air. Loeffler avoids scale. Equivalent evaporation 2257 kJ/kg.

★ Critical pressure ratio values 0.528 (air), 0.546 (superheated), 0.577 (saturated). At throat Mach = 1. Supersaturated flow increases discharge. Velocity formula $44.72\sqrt{\Delta h}$.

★ Parsons turbine is 50% reaction. Optimum speed ratio cos α/2 for impulse, cos α for 50% reaction. Curtis = velocity compounded; Rateau = pressure compounded. De Laval is simple impulse. Degree of reaction definition.

★ Surface condenser used in thermal power plants. Vacuum efficiency definition. Approach and range of cooling tower. Hyperbolic shape for natural draught.

★ Moderator & control rod materials. PHWR uses heavy water. ESP for fly ash. Load factor and diversity factor definitions (diversity factor > 1). Anthracite has the highest carbon content. Proximate analysis components.

★ Isothermal compression requires minimum work. Intermediate pressure $\sqrt{(p_1 p_3)}$. Effect of clearance on volumetric efficiency. Surging and choking in centrifugal compressors. FAD definition. Roots blower has no internal compression.

★ Ramjet has no moving parts and cannot start from rest. Rocket carries oxidiser. Propulsive efficiency formula. Turbofan has higher propulsive efficiency.

Heat Transfer

★ Units of k, h, σ. Highest conductivity metal = silver. Fourier's law. k of gases increases with temperature. Radiation needs no medium. Thermal diffusivity definition.

★ Critical radius k/h (cylinder) and 2k/h (sphere) – very frequently asked. At critical radius heat loss is maximum. Series/parallel resistance concept. Overall U formula.

★ Fins placed on the side with lower heat transfer coefficient. Fin effectiveness should be > 2. Long fin heat flow formula. Corrected length.

★ Lumped capacity valid for Bi < 0.1. Biot number uses solid k; Nusselt uses fluid k. Characteristic length of sphere r/3.

★ Physical meaning of Nu, Pr, Gr, St (very frequently asked). Pr of liquid metals very low. Nu = 3.66/4.36. Grashof number in natural convection. Dittus-Boelter exponent n = 0.4 for heating. Prandtl number of air ≈ 0.7.

★ Dropwise condensation gives higher h than filmwise. Nucleate boiling regime has max h. Leidenfrost point marks start of stable film boiling. Burnout point = critical heat flux.

★ Kirchhoff's law α = ε. Wien's law constant 2898 μm·K. Radiation shields reduce heat flow to 1/(n+1). Shape factor reciprocity and summation rules. Black body emissivity 1. Stefan constant value.

★ Counter flow is more effective. LMTD formula. Effectiveness ε = 1 − e^(−NTU) when one fluid changes phase. NTU definition. Radiator of a car is a cross-flow heat exchanger.

Fluid Mechanics

★ Viscosity variation with temperature for liquids vs gases (asked almost every year). Non-Newtonian examples (blood – shear-thinning, toothpaste – Bingham). Pressure in soap bubble 8σ/d vs droplet 4σ/d. Capillary rise formula and ∝ 1/d. Unit of kinematic viscosity stoke. Why mercury is used in barometers.

★ Centre of pressure lies below centroid. Depth 2h/3 for a rectangular gate with top at surface. Piezometer limitations. Bourdon gauge measures gauge pressure. Inclined manometer for low pressures.

★ Stability conditions for floating vs submerged bodies. Metacentric height formula I/V − BG. Larger GM means a shorter rolling period. Centre of buoyancy = centroid of displaced volume.

★ Streamline vs pathline vs streakline (equal in steady flow). Continuity equation for incompressible flow. Stream function exists for both rotational & irrotational flows; velocity

potential exists only for irrotational. Free vs forced vortex examples. Paraboloid in forced vortex.

★ Bernoulli assumptions. Venturimeter Cd > orificemeter Cd. Divergent cone angle of venturi 5–7° (to avoid separation). Pitot tube measures stagnation (total) pressure. Cd = Cc × Cv. V-notch discharge ∝ H^(5/2), rectangular ∝ H^(3/2). Borda mouthpiece Cd 0.5. KE correction factor = 2 for laminar.

★ f = 64/Re for laminar flow (Darcy). V_avg = u_max/2 in pipe, (2/3)u_max between plates. Shear stress max at wall, zero at centre. Stokes' law drag 3πμDV. Critical Re 2000.

★ Darcy formula. Max power transmission when h_f = H/3 – asked very often. Loss at sudden enlargement formula. Pipes in series vs parallel. Water hammer and surge tank. Equivalent pipe (Dupuit equation). Exit loss = V²/2g. Hydraulic mean depth D/4.

★ Separation occurs with adverse pressure gradient. Displacement thickness definition. δ ∝ √x for laminar. Magnus effect. Bluff bodies – pressure drag dominates. Golf ball dimples reduce drag. Order δ > δ* > θ.

★ Froude model law for ships/spillways; Reynolds for pipes/submarines. Discharge ratio Lr^2.5 in Froude models. Number of π-terms = n − m. Meaning of Weber and Mach numbers. Mach angle. Across normal shock stagnation temperature constant.

★ Most economical rectangular channel b = 2y. Hydraulic jump converts supercritical to subcritical flow and dissipates energy. Critical depth relation.

Hydraulic Machines

★ Max efficiency of a series of flat plates = 50% at u = V/2. Single moving plate u = V/3. Force on fixed flat plate ρaV². Semicircular vane doubles force.

★ Pelton for high head, Kaplan for low head – the most asked HM question. Specific speed formula (with H^(5/4)). Specific speed ranges. Kaplan is axial-flow reaction. Modern Francis is mixed-flow.

★ Jet ratio ~12. Number of buckets formula. Splitter divides jet. Deflection angle 165°. Casing has no hydraulic function. Spear valve controls discharge.

★ Kaplan has adjustable runner blades. Draft tube is used only in reaction turbines. Draft tube converts KE to pressure energy. Cavitation definition and Thoma's coefficient. Wicket gates regulate flow in Francis.

★ Kaplan efficiency at part load is high. Synchronous speed formula. Iso-efficiency curves. Runaway speed definition.

★ Priming is needed for centrifugal pumps. Backward-curved vanes are used. Affinity laws (P ∝ N³). Pumps in series for head, parallel for discharge. Manometric efficiency formula. Cavitation & NPSH. Centrifugal pump started with delivery valve closed. Volute casing converts KE into pressure.

★ Negative slip conditions. Purpose of air vessel. Reciprocating pump does not need priming (self-priming) but has pulsating flow. Discharge formula. Acceleration head max at start/end of stroke.

★ Hydraulic ram works on water hammer. Accumulator stores energy. Intensifier increases pressure. Torque converter has a stator; fluid coupling has none. Gear pump is a positive displacement pump.

Manufacturing I – Casting, Welding, Forming

★ Shake allowance is negative. Draft allowance purpose. CI has the least shrinkage. Match-plate pattern for mass production. Chaplets support cores. Colour code of patterns.

Sweep pattern for bells/large symmetrical castings.

★ Chvorinov's rule ($t \propto (V/A)^2$) – most frequent casting question. Riser with H = D. Sprue tapered to avoid aspiration. Bentonite as binder. Permeability number formula. Collapsibility prevents hot tears. Gating ratio pressurised vs unpressurised.

★ Hot chamber for Zn (low MP), cold chamber for Al. Investment casting for turbine blades, best finish. True centrifugal casting for pipes (no core). Slush casting for hollow ornaments. Shell moulding (Croning process).

★ Cold shut vs misrun. Blow hole due to low permeability/high moisture. Hot tear due to poor collapsibility. Scab from sand expansion. Swell due to soft ramming.

★ TIG uses non-consumable tungsten electrode, MIG uses consumable wire. DCSP gives deeper penetration. AC used for aluminium in TIG. SAW only in flat position. Atomic hydrogen welding principle. Electrode coating functions.

★ Neutral flame for most metals; oxidising for brass; carburising for high-carbon steel/Monel. Spot welding for car bodies; seam for tanks; flash butt for rails. Thermit welding for rails. Soldering vs brazing temperature 450 °C. Acetylene dissolved in acetone. Electron beam needs vacuum. Friction welding for dissimilar metals.

★ Hot vs cold working definitions via recrystallisation temperature. Max draft μ^2R. Four-high mill backup rolls. Mannesmann for seamless tubes. Clearance on punch for blanking and on die for punching – very frequent. Blanking force $Lt\tau$. Compound vs progressive die. Impact extrusion for collapsible tubes. Coining vs embossing. Thread rolling gives stronger threads. Lead worked at room temperature is hot working.

★ Sintering is done below melting point. Self-lubricating bearings by PM. Cemented carbide with cobalt binder. Atomisation for powder production.

Manufacturing II – Machining & Machine Tools

★ Merchant's equation $2\varphi + \beta - \alpha = 90°$ – asked every cycle. Chip thickness ratio < 1. BUE formation at low speed in ductile materials. Discontinuous chips in cast iron. Negative rake for carbide tools. Maximum heat goes to chips. Orthogonal vs oblique cutting difference.

★ Taylor's equation and n values for HSS vs carbide. Flank wear decides tool life; crater wear due to diffusion. Diamond not used for ferrous metals. HSS composition 18-4-1. Free-cutting steel contains S and Pb. Cutting speed most affects tool life. CI is machined dry.

★ Lead screw has Acme threads. Size of lathe specification. Tailstock set-over method for long tapers. Knurling is a forming operation. Peak-to-valley roughness $f^2/8r$. Cutting speed formula. Steady rest vs follower rest. Four-jaw chuck for irregular jobs.

★ Point angle 118°. Radial drilling machine for heavy jobs. Countersinking vs counterboring. Jig guides tool, fixture does not. 3-2-1 principle. Chisel edge causes large thrust. Trepanning definition.

★ Up vs down milling – climb milling gives better finish but needs backlash eliminator. Simple indexing 40/N. Hobbing cannot cut internal gears; gear shaping can. Straddle milling uses two side cutters. Number of cutters in a set = 8.

★ In a shaper the tool reciprocates; in a planer the workpiece reciprocates. Broaching for internal keyways and splines. Quick return ratio in shaper.

★ Grinding wheel marking order – almost every year. Soft wheel for hard material. SiC for cast iron, Al_2O_3 for steel. Glazing vs loading; dressing vs truing. Centreless grinding regulating wheel. Honing for engine cylinder bores. Lapping gives the best finish.

★ USM for glass/ceramics; ECM based on Faraday's law, no tool wear; EDM dielectric kerosene; EBM requires vacuum; LBM doesn't. AJM does not use oxygen (fire hazard). ECM

highest MRR among NTMs. Non-conductive materials can't be machined by EDM/ECM. Magnetostriction in USM.

★ G02/G03 meanings, M03/M05/M06. Z-axis is along the spindle. DNC definition. FDM/SLA are additive manufacturing. Group technology coding systems.

★ Optimum speed for max production is higher than for minimum cost. Feed affects surface finish the most ($h = f^2/8r$).

Engineering Materials & Heat Treatment

★ APF of FCC/HCP 0.74, BCC 0.68. Number of atoms per unit cell. Coordination numbers. Examples of BCC/FCC/HCP metals. Allotropic temperatures of iron. Edge vs screw dislocation Burgers vector. Hall–Petch relation.

★ Eutectoid at 723 °C & 0.8% C giving pearlite; eutectic at 1147 °C & 4.3% C giving ledeburite. Cementite 6.67% C. Max carbon in austenite 2%. Peritectic reaction equation. Steel/cast iron carbon limits. Austenite is FCC and non-magnetic. Lever rule for phase fractions.

★ 18-8 stainless composition (Cr-Ni). Hadfield manganese steel work-hardens. Mg/Ce for SG iron. Grey CI has flake graphite and best damping. Malleable CI by annealing white CI. Hot shortness by S, cold shortness by P. Cr > 12% for stainless. Co decreases hardenability. IS designation 40C8.

★ Brass = Cu + Zn; bronze = Cu + Sn – the most common facts question. Duralumin composition & aircraft use. German silver has no silver. Invar's low expansion. Monel = Ni + Cu. Babbitt for bearings. Y-alloy for pistons. Gun metal composition.

★ Normalising = air cooling; annealing = furnace cooling. Martensite is BCT and formed by diffusionless transformation. Jominy test for hardenability. Nitriding is done below the critical temperature with no quench and least distortion. Austempering produces bainite. Induction hardening depth decreases with frequency. Tempering reduces brittleness. Severity of quench order.

★ Rockwell C uses diamond cone; Vickers 136° pyramid. Izod (cantilever) vs Charpy (simply supported). Creep stages – secondary used for design. NDT method selection (magnetic particle only for ferromagnetic). Mohs scale extremes. Brinell–UTS relation.

★ Bakelite is a thermoset. PTFE (Teflon) lowest friction. Vulcanisation adds sulphur. Injection moulding for thermoplastics, compression moulding for thermosets. Blow moulding for bottles. Rule of mixtures. Wood is a natural composite.

Metrology, Limits & Fits, Industrial Engineering

★ Accuracy vs precision. Least count of vernier (0.02 mm) and micrometer (0.01 mm). Abbe's principle. Standard measuring temperature 20 °C. Systematic errors are correctable.

★ Sine bar unreliable above 45°. Wringing of slip gauges. Optical flat fringe = λ/2. Autocollimator for small angles and flatness. LVDT is an electrical comparator. Bevel protractor least count 5 minutes. Pneumatic comparator for bore measurement.

★ Hole basis is preferred. Taylor's principle of gauging. H7/p6 interference fit. Allowance vs tolerance definition. Number of IT grades (18) and fundamental deviations (28). Selective assembly. GO gauge checks the maximum metal limit.

★ CLA = Ra. Talysurf uses a stylus. Three-wire method for effective diameter; best wire size. Gear tooth vernier for chordal thickness. Profile projector & toolmaker's microscope for thread form.

★ EOQ formula & that at EOQ ordering cost equals carrying cost – very frequent. If demand is 4 times, EOQ doubles. ABC analysis is based on annual consumption value. VED for spare parts. Johnson's rule. Kanban relates to JIT. Fixed-position layout for ships. Process layout for job shops. Exponential smoothing formula.

★ SREDIM steps. Process chart symbols. Therblig developed by Gilbreth; father of scientific management F.W. Taylor. String diagram for movement. Standard time = normal time + allowances. Work sampling for long-cycle jobs. SIMO chart for micro-motion. Rating factor definition.

★ c-chart for number of defects; p-chart for fraction defective. ±3σ limits (99.73%). Producer's risk α (type I) vs consumer's risk β (type II). Six Sigma = 3.4 defects per million. Ishikawa diagram. PDCA by Deming. Bath-tub curve regions. Reliability of series/parallel systems.

★ PERT uses β-distribution & 3 time estimates; CPM deterministic. te formula. Critical path has zero float & is the longest path. Dummy activity. VAM gives the best initial solution. m + n − 1 rule. Hungarian method for assignment. Little's law. Crashing cost slope.

★ Break-even point formula. Value = function/cost. Taylor = scientific management; Fayol = 14 principles; Gilbreth = motion study & therbligs; Gantt chart by Henry Gantt.

Mechanical Measurements & Instrumentation

★ First-order system reaches 63.2% in one time constant. Thermocouple and piezoelectric are active transducers. Optimal damping ratio ~0.7 for second-order instruments. Null method is more accurate.

★ Thermocouple works on the Seebeck effect. Pt-100 RTD is the most accurate & linear. Thermistor has NTC and high sensitivity. Optical pyrometer for very high temperatures, non-contact. Law of intermediate metals. Chromel–alumel is type K.

★ McLeod gauge used for calibrating vacuum gauges. Pirani works on thermal conductivity. Ionisation gauge for very low pressures. Bridgman gauge for very high pressures. Piezoelectric transducers for dynamic pressure. Dead-weight tester for calibration.

★ Electromagnetic flowmeter works on Faraday's law and needs conductive fluid. Hot-wire anemometer for turbulence. Rotameter is a variable-area meter. Stroboscope for contactless speed. LVDT principle (mutual induction, differential output). Proving ring calibrates force. Accelerometer natural frequency ≫ measured; vibrometer ≪ measured. Coriolis meter gives mass flow.

★ Gauge factor of metallic gauges ≈ 2. Strain rosette needed when principal directions are unknown. Dummy gauge for temperature compensation. Full bridge gives max output. Photoelasticity uses polarised light.

★ Feedback distinguishes closed-loop systems. Integral action removes steady-state error. Derivative action improves stability. FRL unit in pneumatics. A 4/3 valve has 4 ports & 3 positions.